Proteomics CoPYlot

The complete manual

Applies to Proteomics CoPYlot 1.0.0 and the CoPYlot DataViewer 1.0.0

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Jonas Marx
AG Funktionelle Proteomanalyse

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PART I

Getting started



1 CoPYlot, and how to read this manual

What the tool is, what it is made of, who it is written for, and how the chapters that follow are arranged.

AT A GLANCE

What it is	A browser interface over a Python analysis backend, for quantitative proteomics and phosphoproteomics.
Two applications	The analysis tool, which takes an upload and works on it, and the DataViewer, which is read-only over a dataset somebody else prepared.
Two packages	The same analysis functions again for a Python notebook, as the package `proteomicscopylot`, and for R, as a script of R functions.
Data levels	Protein, phosphosite, phosphoprotein, peptide, precursor.
Read first	This chapter, then Getting started → Starting CoPYlot, then Getting started → The vocabulary.
Conventions	At the end of this chapter, and used without further comment everywhere else.

1.1 What the tool is

CoPYlot takes the quantitative table a search engine produces and carries it through annotation, filtering, normalization, imputation, quality control, statistics, phosphosite analysis and reporting without the table leaving the application. The interface is a React application in the browser; the arithmetic is Python — pandas, NumPy, SciPy, scikit-learn — served by a FastAPI backend, which draws static figures with Matplotlib and Seaborn and interactive ones with Plotly.

Wide-format exports are read from CSV, TSV, TXT, XLSX and Parquet. Spectronaut and DIA-NN column names are recognised on sight; any other export is usable by naming its sample columns during annotation, which is the only thing the recognition would have done for you.

The uploaded table, the annotation that describes it and every setting derived from them belong to whoever uploaded them. Nothing is written back to the file you uploaded, at any

point in this manual.

1.2 The two applications

Both are built from the same source and differ in which build was made.

- The **analysis application** is the whole tool: upload, annotate, filter, impute, collapse, analyse, compare, report. Everything this manual describes is in it.
- The **DataViewer** is read-only over datasets that were configured before it started. Its Data section keeps only **Upload**, which previews rather than accepts, and **Annotation**, which shows the annotation rather than edits it; **Imputation**, **Distribution**, **ID Translator**, **Colors** and **Restore** are absent, as are **Replicate & Variance Projection** (a what-if projection, not a power calculation), **Sequence Coverage**, **Kinase Library**, **Peptide Collapse** and the **Text** and **Report** pages of Summary →. A whole section disappears when the published project has no dataset that would fill it. What remains is every plot the data supports.

Either application can also be built with the assistant enabled, which adds a chat panel beside the analysis and nothing else. It runs against a model through Ollama by default; that server may be local or remote. A cloud provider can be configured instead, and the tool asks for consent before the first such request and states what is sent with it. The assistant is described in Part XII.

1.3 The notebook packages

The same analysis is installable as a Python package, `proteomicscopylot`, with `pip install ".[bio]"` from the source (`-e` for development); the `bio` extra is needed only for reading FASTA files. It carries `notebook_functions.py` for Python and `notebook_functions.R` for R. The R functions are a script loaded with `source()`, not an installable R package. The core workflow uses the same function names in both, which is why the function reference that closes every chapter names one function rather than two: where the two implementations differ, this manual marks which is which.

The package also carries the bundled example dataset and the Gene Ontology reference table the enrichment pages use, and a worked notebook and a worked R script ship beside it. `load_example_data(name)` reads the bundled HEK293 example — `"protein"`, `"phospho"`, `"phosphoprotein"`, `"protein_metadata"` or `"phospho_metadata"` — and `example_data_path(name)` gives its file path; both exist under the same names in R.

> **NOTE**
>
> Variance-stabilizing normalization is not reimplemented here: it is computed by the Bioconductor vsn package, so it is offered only where R is present. Settings → reports whether Rscript was found and where.

1.4 Who this manual is for

It is written for two readers, and every chapter serves both in the same order. The first is running a dataset through the tool for the first time and needs to know what a page is for, what it needs before it will draw anything, and what the result should look like. The second knows the tool and has a number to check: which matrix it was computed from, which formula, what happened to the missing values. The first reader reads the opening block and the first section; the second reads *How it is computed* and the table of columns. Neither has to read the other's part to trust the answer.

What the manual assumes is a quantitative proteomics or phosphoproteomics table and the experimental design that produced it. It does not assume Python, R, or any statistics beyond a mean and a standard deviation; where a method needs more than that, the chapter states the procedure in full rather than naming it and moving on.

1.5 How this manual is arranged

One chapter per page of the application, in the order the sidebar lists them: Data, then the analysis sections in the order a dataset passes through them, then the plot appearance panel, the assistant and external tools, and administration. Two appendices close it: the keyboard shortcuts and the glossary, and a single alphabetical list of every package function named anywhere in the manual.

1.5.1 The shape of every chapter

- **At a glance** — where the page is, what it needs before it will draw, and what it produces.
- **What the plot answers** — the question somebody brings to the page, and an annotated figure with a key.
- **Controls** — one table for the controls on the page and one for those in the plot appearance panel, each with its default.

- **How it is computed** — the arithmetic in full, then the edge cases.
- **The table below the plot**, where the page has one, column by column.
- **Exporting**, and the package functions behind the page.

A chapter that is not a plot — this one, the vocabulary, administration — keeps the opening block, the controls and the function reference, and replaces the middle with whatever that page actually is.

1.5.2 Conventions

- **Control names** are set like this and are spelled exactly as the interface spells them, including its American **Color** where the surrounding prose says colour.
- Values, file names and column headers are set like this.
- Section → Page means: choose that section in the sidebar, then that tab.
- Missing values are called **NA** throughout, never *NaN* and never *blank*.
- *Feature* is the general word for a row — protein, phosphosite, phosphoprotein, peptide or precursor — and is used wherever a sentence is true of all of them. *Sample* is a column; *condition* is a group of samples.
- A coral box marks something that is easy to get wrong; a blue box is worth knowing but never load-bearing; an amber box records a release in which numbers or labels changed.

1.6 Where else to look

The **Manual** button at the right of the tab row opens the manual for the page you are on: the same chapter as here, published on its own, so there is no second copy of the text that could disagree with this one. The changelog that ships with the application records what changed in which release and is the right place for a question of the form *when did this stop doing that*; this manual describes the release on its title page and no other. The two notebook packages document themselves in their docstrings, and the worked examples beside them are the fastest way to see a whole analysis as code.

Where an analysis starts

Function	What it produces
info	Prints what the package is. The one-line check that the installation works
read_data	Reads a quantitative table from CSV, TSV, TXT, XLSX or Parquet and normalises its column names

2
Starting CoPYlot

The two ways the application is run, what each of them needs, and how to stop it again.

AT A GLANCE

How it runs	As a container you host yourself, or as a standalone executable you start on your own machine. Both serve the same application to a browser.
What it is	One process: a FastAPI backend that also serves the built React interface. There is no separate server to start.
Default address	The executable listens on the loopback interface only. The analysis application defaults to port 8000, the viewer launcher to 8001; the container listens on 10000 unless PORT says otherwise.
Browser	Any current browser: Chrome, Edge, Firefox and Safari all run the interface. It is served as a static bundle and needs no plug-in and no extension.
Access control	Off for a loopback bind, and required before the application will listen on anything else.
Stopping it	Close the window the executable opened, or stop the container. There is no button inside the application that ends it.

CHANGED IN the React release

The application described by the previous user manual was a Streamlit build: an executable that opened a fixed local address and was closed with a **Quit App** button in its sidebar. None of that exists now. The interface is a React application served by a FastAPI backend; the sidebar lists analysis sections rather than application commands, and there is no **Quit App** button to look for. The port the old manual named is not the port anything listens on today.

2.1 Which of the two you want

The container is for an instance more than one person uses, or for one that has to run unattended: it pins every dependency, it brings R and the vsn package with it, and it refuses to start without the credentials file that protects it. The executable is for one person on one machine: the bundled application needs no separate Python or Node installation. Its analysis runs on that computer. AI providers and external services may send selected information elsewhere; review their settings before using confidential data.

Neither is a demonstration version of the other. They are the same application, built the same way, and a save state written by one restores in the other.

2.2 The container

1. Take a copy of the source, and from its root create the local credentials files. Three templates are checked in — one for the analysis tool, one for the viewer portal and one for viewer QR links — and each is copied into a local/ directory under its own .local.json name. Every template ships with a placeholder digest of zeros, which is not a password and will not authenticate anybody.
2. Replace each placeholder with a real salted digest. The source includes a one-line helper that prompts for the password and prints the value to store under password_pbkdf2_sha256, so the password itself never reaches the shell history. Keep the resulting files out of version control.
3. Build and start the stack with Docker Compose. The analysis stack and the viewer stack are separate compose files, and a host can run either or both.
4. Open the address that whoever deployed the instance published, and sign in with the password that was issued with it.

The image is built in two stages: the frontend bundle is compiled first, then a Python image is assembled around it with the locked runtime dependencies, R and the Bioconductor vsn package, and the metric-compatible Liberation and DejaVu font families that the figure font list offers. The build fails rather than shipping a font menu whose entries silently fall back to something else. The container runs as an unprivileged user on a read-only filesystem, with a writable temporary filesystem for the figures it draws.

> **EASY TO GET WRONG**
>
> The analysis and viewer interfaces are two different builds of the same source, and which one a bundle is cannot be seen once it is built. The launcher therefore checks the marker the build writes and refuses to serve an analysis bundle from a viewer process. It is not being pedantic: a viewer serving the analysis bundle would put

> Upload, Imputation and Restore on screen over read-only data, with nothing visible to say that the read-only gating is not running.

2.2.1 Listening beyond loopback

The launcher binds to the loopback interface unless told otherwise, and it refuses a non-loopback bind outright unless both of two things are true: an access-control file is configured and enabled, and session isolation is on. The refusal is the point — an unauthenticated instance reachable from the network is an unauthenticated instance with somebody else's proteomics data in it. An explicit override exists for development, prints a warning when it is used, and is not for data anybody cares about. Behind a reverse proxy, the address of the proxy itself must be declared as trusted, or the login limiter counts every attempt as coming from the proxy.

2.3 The executable

1. Obtain the executable for the application you want. Each variant is a single file: the analysis application, the DataViewer, and an assistant-enabled build of each. If it arrives compressed, unpack it first, and keep the file where it is meant to live — it resolves the data it ships with relative to itself. The executables are published as releases on the project's GitHub repository; a reader who would rather not run a downloaded binary can host the application themselves from the same source, which is the subject of the next section.
2. Start it. On Windows, double-click it; on macOS, open it from the context menu the first time rather than by double-clicking.
3. Answer the first-run security warning. An executable that is not signed by a certificate your machine already trusts is held back by the operating system until you confirm it, once.
4. A console window opens and stays open. That window *is* the application; the browser only displays it.
5. About a second and a half later the default browser opens on the application. If it does not, the console window prints the address it is listening on.

The executable carries the built interface, the backend, the Gene Ontology reference table and the bundled example dataset inside itself. It does not carry R, so variance-stabilizing normalization is offered only if R and the vsn package are already installed on the machine; Settings → reports what was found.

> **NOTE**
>
> A build whose file name ends in ai starts with the assistant enabled. Renaming a file therefore changes what it does, which is worth knowing before renaming one.

2.3.1 Stopping it

Close the console window, or interrupt it from the keyboard. Closing the browser tab does not stop the application: the tab is a client, and the process goes on listening. Stopping it does not automatically save the analysis. Temporary files on disk are not a restorable session, so save a state first if you intend to come back to the same analysis. That is the subject of Getting started → Sessions.

2.4 What the launcher takes

Both routes start the same launcher, and both accept the same switches. They are listed here because a deployment note that says *start it in viewer mode on this project* is otherwise unreadable.

Switches on the launcher

Control	What it does	Default	Where
--host	Which interface to listen on. A non-loopback value is refused unless authentication and session isolation are both enabled.	127.0.0.1	launcher
--port	Which port to listen on. The viewer launcher sets this to 8001 unless it is given one.	8000	launcher
--viewer	Starts the DataViewer instead of the analysis application.	off	launcher
--project	Which configured viewer project to open. Only with --viewer; the project name may also be given on its own.	none	launcher
--viewer-config	An explicit viewer configuration file, instead of the search through the known locations.	none	launcher
--list-projects	Prints the viewer projects it can find and exits.	off	launcher
--no-browser	Starts without opening a browser. The container always uses this.	off	launcher

Switches on the launcher (continued)

Control	What it does	Default	Where
--allow-unsafe-public	Acknowledges an unauthenticated non-loopback bind and starts anyway, with a warning. Not for real data.	off	launcher

2.5 Leaving and coming back

An analysis lives in the memory of the running application and in the tab you are working in. Refreshing the browser starts a new analysis, and the application asks the browser to warn you first whenever anything has changed since the last successful save — although the browser writes the wording of that dialog, and may decline to show it at all, so it is a safety net rather than a guarantee.

Before you leave, therefore, save a state. It is one file, it holds the settings and the annotation rather than the measurements, and it is what lets tomorrow's session continue today's. Getting started → Sessions says exactly what it contains and what it does not.

What starts what

Function	What it produces
launch.py	The launcher itself: validates the bind address and the frontend bundle, then serves the application
launch_tool.py	Entry point of the analysis executable
launch_viewer.py	Entry point of the viewer executable; adds --viewer and port 8001 unless they are given

3
Finding your way around

The parts of the window, what lives where, and the few things that are not where you would expect.

AT A GLANCE

Sidebar	Twelve sections, left. Open by default on a wide window, hidden behind a **Menu** button on a narrow one.
Tab row	The pages of the current section, across the top, with the **Manual** button at its right.
The page	A column of cards: the dataset level, the options, the plot, the table. Nothing is hidden in a drawer.
Appearance panel	A further card between the options and the plot, shown only when advanced mode is on (**Ctrl** + **Shift** + **E**).
Assistant	A panel on the right, in assistant-enabled builds only. Closed until it is asked for.
Opens on	Data → Upload, every time.

3.1 The window

3.1.1 Reading it

- The sidebar. Twelve sections, in the order a dataset passes through them. It opens by itself on a window wide enough for it and is reached through the **Menu** button otherwise, where it also closes again as soon as a section is chosen.
- The tab row: the pages of the section you are in. Changing section keeps the tab you were last on in the section you left. **Analysis** and **Settings** have no tabs and no tab row. Above it, when there is more than one of either, sit the **Dataset** and **Metadata** selectors, which switch the whole window between uploaded dataset packages and between metadata profiles.
- The **Manual** button, at the right of the tab row. It opens the manual for the page you are on, in place, on the pages that have one.

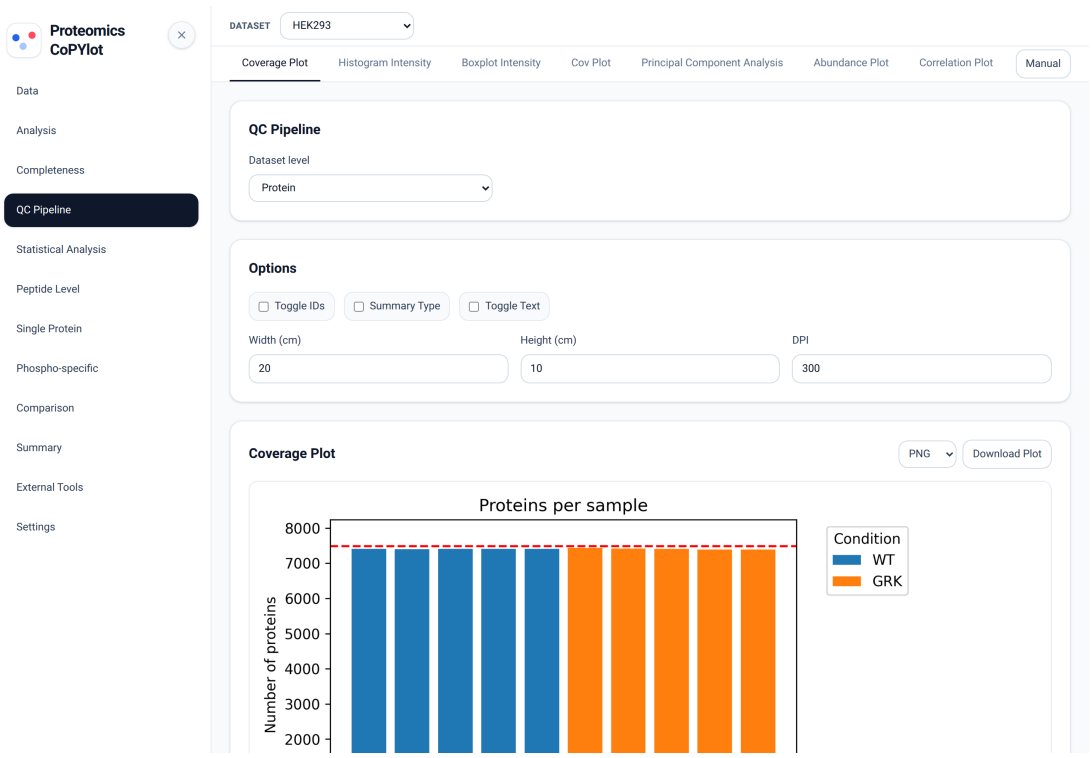


Figure 3.1. The main window with the HEK293 example project loaded, on Quality control ⟶ Coverage. The sidebar is on the left, the tab row for the section across the top with the **Manual** button at its right end, and the page itself is a stack of cards: the dataset level, the options, then the plot with its format selector and **Download Plot**.

- The **Dataset level** card: which of the uploaded levels every card below it applies to.
- The **Options** card: every control that changes what is plotted. Switches first, in a row that wraps; then the selectors and fields, up to three abreast; then the three size fields.
- The plot, with the download format selector and **Download Plot** in its header — or **Open Interactive Plot** where the page draws a Plotly figure instead of an image.
- The table under the plot, with **Download CSV** in its header. Pages without a table end at the plot.

> **WORTH KNOWING**
>
> The options are *above* the plot, not beside it. Anything set in that card takes effect on the figure below as soon as it is changed, with one exception: **Width (cm)**, **Height (cm)** and **DPI** set the rendering size; the browser scales the result to fit the page.

3.2 The sections, in order

- **Data** — upload, annotation, imputation, distribution, the ID translator, the condition colours and the save states. Part II.
- **Analysis** — the analysis canvas: a set of global controls and a stack of tiles, each tile a volcano plot with the boxplots and lineplots of the features selected in it. It has no tabs.
- **Completeness** — where the missing values are. Part IV.
- **QC Pipeline** — coverage, intensity distributions, variability, PCA, abundance and correlation. Part III.
- **Statistical Analysis** — volcano, ANOVA, over-representation, pathway heatmap and the replicate and variance projection, a what-if projection, not a power calculation. Part V.
- **Peptide Level** — retention time, modifications, missed cleavages and sequence coverage. Part VI.
- **Single Protein** — one protein at a time. Part VII.
- **Phospho-specific** — sites, their coverage and distribution, the S/T/Y split, kinase-substrate enrichment and phosphoprotein regulation. Part VIII.
- **Comparison** — Pearson correlation and Venn diagrams between datasets. Part IX.
- **Summary** — the tables, the methods text and the report. Part X.
- **External Tools** — peptide collapse and GONetView. Part XII.
- **Settings** — theme, advanced plot options, and what the application found on the machine. Part XIII.

In the DataViewer the list is shorter, and shorter again for a project without phosphosite or peptide data: a section whose data is not published is not shown. Which pages are gated is in Administration → The DataViewer.

3.3 Controls that belong to the window

Every chapter lists the controls of its own page. These belong to the frame around it and are therefore described once, here.

Controls on this page

Control	What it does	Default	Where
Menu	Opens the sidebar when the window is too narrow to keep it open.	open on a wide window	page
Manual	Opens the manual for the page you are on.	—	page

Controls on this page (continued)

Control	What it does	Default	Where
Dataset	Which uploaded dataset package the whole window works on. Shown only when there are at least two.	the active package	page
Metadata	Which metadata profile is in force. Shown only when there are at least two; on Data → Annotation the profiles derived automatically from the others are left out of the list.	the active profile	page
Dataset level	Which uploaded dataset every card below applies to: Protein, Phospho, Phosphoprotein, and on the coverage plot also Peptide and Precursor. Only levels that exist are offered.	the first available level	page
Data Source	Which matrix of that level to use: Data, Filtered Data or Imputed Data. Only the ones that exist are offered, and on a few pages the same control is labelled **Data source**.	the most processed matrix available	page
Width (cm), Height (cm), DPI	The size and resolution of the *downloaded* figure. The browser scales the rendered figure to fit the page. On a page that offers a Normal and an Interactive plot they belong to Normal and are hidden for Interactive.	per page	page
Download Plot	Downloads the figure in the format chosen beside it: PNG, JPG, WEBP, SVG or PDF.	PNG	page
Open Interactive Plot	Replaces the download on pages drawing an interactive figure: opens it in its own tab. The picture of an interactive plot is saved with the camera button in its own toolbar.	—	page
Download CSV	Downloads the table under the plot exactly as shown.	—	page

3.3.1 Two keyboard shortcuts

Ctrl + **Shift** + **E** turns advanced mode on and off from anywhere. **Ctrl** + **S** writes the save state, to the file it was last written to if there is one. Both read the physical key rather than the letter printed on it, so they are the same two shortcuts on a QWERTZ or AZERTY keyboard. Quick save is absent from the DataViewer, which has no session of its own to save. The full list is in Appendix A.

3.4 The plot appearance panel

When advanced mode is on, a card headed **Advanced Plot Options** appears between the options and the plot. It styles the plot you are looking at and nothing else: one collapsible section per element of the figure — font, title, the two axes, legend, frame and grid, background, and whichever of points, boxes, bars, site classes, Venn sets, value labels and colour scale that plot actually has — plus a section for renaming and recolouring conditions and samples.

It is where a control you remember from an older manual has usually gone. **Bar Color**, **Dot Color**, **Dot Size**, **Text Size**, the class colours of the phosphosite pages, and the switches that showed or hid a title or a legend are all in this panel now, not beside the plot. Each plot chapter repeats only the entries particular to that plot; the panel itself is Part XI.

> **WORTH KNOWING**
>
> The same switch is in Settings → under **Plot Options**, as **Advanced plot option mode**. The shortcut is **Ctrl** + **Shift** + **E**: not **Ctrl** + **Shift** + **A**, which the browser answers first with its own tab search.

3.5 Data levels, and why a tab is missing

Dataset level offers only what has been uploaded, and a page draws nothing until one of them is chosen. Protein and Phospho appear when a dataset of that kind has been uploaded and annotated; Phosphoprotein appears once a phosphosite dataset has been aggregated to protein level, or once a phosphoprotein table has been uploaded directly. Peptide and Precursor come from the long-format report rather than from a wide matrix, and are offered only on the pages that can read it — the coverage plot, and the peptide section.

A whole section can be missing for the same reason: without a phosphosite or phosphoprotein dataset there is nothing for **Phospho-specific** to draw, and in the DataViewer that section is not shown at all rather than shown empty.

3.6 The analysis canvas

Analysis brings several existing views onto one page. Use + **Add Plot** to add a volcano, a control-adjusted volcano, an individual feature boxplot, a feature line plot or an ORA list tile. Tiles can be reordered, collapsed or removed. The canvas is a workspace for following features between views, not a separate scientific pipeline.

Start with one volcano and check its level, data source, condition order, test and thresholds. Click a point to select its feature. A boxplot set to **Active volcano selection** then shows that feature's replicate values; a line plot can follow the active feature, a multi-selection, a named list or manual identifiers. This is useful when a striking volcano point needs to be checked against the actual measurements.

With several volcanoes, the source matters. Selections are linked to the comparison that produced them; selecting a point in another tile changes that context. Two tiles with different sources or conditions need not agree, even though they use the same calculation as their standalone pages. Check their settings before treating a difference as a contradiction.

The list sidebar stores named selections. **Add Selected** copies the current selection into the active list; unions, intersections and set differences combine lists. List membership is a selection, not a new test. An ORA tile still needs an appropriate background, and a list selected after looking at the results is exploratory evidence.

In an AI-enabled build, explicitly ask for a protein, a particular phosphosite, or all sites on a protein. These are distinct requests. A feature excluded from the chosen volcano comparison cannot acquire a point merely because it was requested or highlighted.

Canvas figures can be exported where their tile offers it. There is no separate canvas notebook download; use the corresponding standalone plot for a notebook replay. Save the analysis state to retain the workspace.

3.7 The assistant

In an assistant-enabled build, a button at the bottom right opens a panel down the right-hand side of the window. It holds the conversation, the model in use, and a collapsed block of settings: the provider, the model, and — for a cloud provider — the consent that must be given before anything is sent. The analysis keeps its full width while the panel is open on a wide window.

The assistant can move you between pages and set controls for you, which is why it is described here and not only in its own chapter: a control that changes without you having touched it was changed by the assistant, and it is in Part XII.

4
The vocabulary

The words this manual uses precisely, and what each of them means here.

AT A GLANCE

A row	A *feature*: a protein, phosphosite, phosphoprotein, peptide or precursor.
A column	A *sample*. A group of samples is a *condition*, and both come from the annotation.
Data levels	Protein, phosphosite, phosphoprotein, peptide, precursor.
Data sources	Data, Filtered Data, Imputed Data — the same features, three stages along.
Scale	Most intensity analyses use a log2 or log-like matrix; CV and abundance summaries require linear values. Missingness is determined after the declared input scale is handled.
Defined here	The boxplot: median, quartiles, IQR, the 1.5×IQR whiskers and what falls outside them. Every later chapter refers back to this page rather than repeating it.

4.1 Why this chapter exists

Each of these words means something slightly different in each laboratory, and a number computed by this tool only means anything against the definition the tool used. What follows is that definition, and nothing here is a convention this manual invented for the occasion: each one is what the code does.

4.2 The terms

4.2.1 Feature

A row of the matrix, whatever the matrix is: a protein group, a localized phosphosite, a phosphoprotein, a peptide or a precursor. *Feature* is used wherever a sentence is true of all of them, and the level is named wherever it is not. A count of features is therefore a count in the unit named by the plot. Matrix analyses work with rows; coverage summaries may count distinct identifiers or the union quantified across samples. Peptides and precursors are different counting units.

4.2.2 Sample and original sample name

A sample is one column of measurements: one injection, one run. Inside the tool a sample carries a label of the form condition_n, numbered within its condition rather than across the experiment, so control_2 is the second control and not the second sample of the file. The name the column had in the uploaded file is the *original sample name*; it is kept, and several pages offer to print it as a second line under the label or as an extra column in the table.

4.2.3 Condition

A group of samples that are replicates of one another. Conditions come from the annotation, not from the file, and every plot that colours by condition uses the same colours in the same order for the whole session. That order is the metadata order — the order in which the annotation lists the conditions — everywhere: conditions are never sorted alphabetically, and default condition choices are taken from the start of that order. A condition with one sample is legal and produces a bar, a point and a box wherever the arithmetic allows it; it produces no standard deviation, no CV and no error bar, because there is nothing to take a spread over.

4.2.4 The annotation

The annotation is the table that says which columns are samples and which condition each belongs to, and it is what Data → Annotation produces. Nothing downstream will draw before it exists.

It carries more than the mapping. The same page settles whether the values are already on the $\log_2$ scale, which normalization to apply, and which completeness filter to keep a feature by; the result is stored with the dataset, so every later page works from one annotated matrix rather than from its own reading of the file. An uploaded metadata file may carry up to five categorical grouping columns; each becomes a *metadata profile*, one of which is in force at any time, and a pair_id column is kept as pairing structure rather than treated as a grouping. Profiles are the subject of Data → Annotation.

4.2.5 Data level

Which matrix is on screen.

- **Protein** — one row per protein group, as the upload defines it.

- **Phosphosite** — one row per localized site. Labelled Phospho in the **Dataset level** selector.
- **Phosphoprotein** — one row per protein, aggregated from its sites by Data → Annotation, or uploaded directly as a phosphoprotein table. Which aggregation produced it is remembered, because one of the four — robust_site_median — yields ratios rather than abundances, and the plots label their axes accordingly.
- **Peptide** and **precursor** — read from the long-format report rather than from a wide matrix, and offered only on the pages that can use it.

4.2.6 Data source

How far along the pipeline the matrix is. Three exist, and a selector offers only those that do.

- Data — the annotated matrix: on the $\log_2$ scale, with the chosen normalization applied, and nothing removed.
- Filtered Data — the same matrix after the completeness filter, so fewer rows and the same columns.
- Imputed Data — the filtered matrix after imputation, so the same rows and columns, with missing cells filled where the selected method has enough information. Inspect diagnostics: imputation is not a promise that every matrix becomes complete.

A panel opens on the most processed matrix that exists — imputed before filtered before unfiltered — because that is the one the pipeline was configured to produce. It is a default and not a coercion: a panel restoring a saved state keeps the choice that was saved, even where a more processed matrix now exists.

> EASY TO GET WRONG
>
> **Dataset level** and **Data Source** are two different questions and sit next to each other in the same card. The level is *which features*: proteins, sites, phosphoproteins. The source is *how far along*: as annotated, filtered, imputed. Changing the level changes what a row is; changing the source changes how many rows there are and which values were filled. Check the source label rather than inferring it from a figure's appearance. Individual plots can exclude additional unusable rows, such as incomplete or constant features in PCA.

4.2.7 Quantified, and missing

A value is quantified if the cell holds a finite number. Everything else is NA, and NA is the only kind of missing this tool recognises: there is no second convention, no sentinel value, no zero that means absent.

That takes work at the boundary, and the work happens once, when the annotation is built. In a linear upload a zero means *not detected*, so zeros in the sample columns become NA before the transform — $\log_2 0$ has no value to put in the cell. In an upload that is already on the $\log_2$ scale, zero is a real measurement and is kept; what is dropped there is $\pm\infty$, which

is what some exports write for a non-detection and which would otherwise survive every completeness count and produce an infinite fold change. Both conversions are confined to the declared sample columns, so a numeric annotation column of zeros is untouched.

4.2.8 Replicate

Two samples are replicates when the annotation puts them in the same condition. The tool makes no distinction between technical and biological replicates and applies none: a condition of six is treated as six replicates whatever they are, and it is the annotation, not the arithmetic, that decides whether that is the right claim. Where an analysis needs the pairing between two conditions rather than the grouping within one — a paired test — it uses the pair_id column of the metadata.

Grouping technical runs under one label does not make them independent biological replicates. Define the experimental unit before testing; repeated injections of one specimen must not be interpreted as additional subjects.

4.3 The shared phosphosite localisation cutoff

At phosphosite level, **Localization probability cutoff** selects the sites shown across QC, completeness, statistics, comparison, single-feature and phospho-specific views. It is one session-wide choice, not a separate filter for each figure. It starts at 0, which retains all sites, including missing probabilities. Above 0 only sites with a usable probability at least the cutoff are retained; a table without that column cannot satisfy a positive cutoff.

This selection happens when an analysis reads the matrix. It does not recompute normalization, the completeness filter or imputation. Consequently, changing it can change the tested feature set and its adjusted p-values, without re-estimating the processing that created the matrix. Record it alongside the selected data source. The phosphosite-to-phosphoprotein collapse has its own cutoff, recorded at creation; changing the session cutoff does not retroactively rebuild an existing collapsed matrix.

4.4 The $\log_2$ scale

Most intensity plots and statistical comparisons use transformed values. On a log2 scale, multiplicative changes become additive differences: values of 8 and 9 correspond to a twofold ratio on the linear scale. A logarithm often reduces right skew but does not guarantee normality. Test assumptions concern each feature's observations or residuals, not the distribution made by pooling all proteins. CV and abundance summaries convert to linear values for their respective calculations; VSN produces a variance-stabilised, log-like scale rather than an exact log2 transform at every intensity.

THE TRANSFORM

For an upload on the linear scale, zeros in the sample columns are set to NA and the transform is applied cell by cell:

$$y_{ij} = \log_2 x_{ij}.$$

For an upload already on the $\log_2$ scale the values are taken as they are, with non-finite cells set to NA.
Which of the two applies is detected rather than assumed: the column names are read for a log2 token, and the value ranges are voted on. Where the two disagree, or where neither is conclusive, the application refuses to guess and asks for the scale to be stated explicitly.
Normalization — none, median centering, quantile normalization, or VSN — precedes the completeness filter. VSN uses a linear-scale input internally, rather than applying its model directly to log2 values. Record the chosen method because it affects the scale and interpretation of subsequent results.

4.5 How a dataset moves through the tool

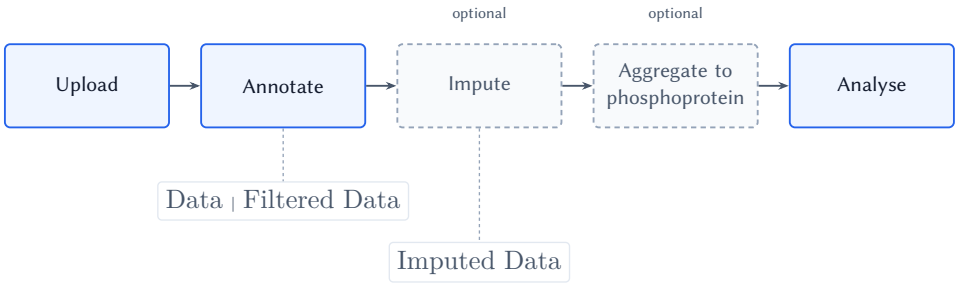


Figure 4.1. The path a dataset takes. The two solid steps are compulsory and in that order; the two dashed ones are not. Under a step is the matrix it leaves behind, and those three names are what every data source selector in the application offers. Upload leaves none: a file that has not been annotated is not yet a matrix.

1. **Upload** a wide quantitative table per level, and the long-format report if there is one. Nothing is interpreted yet.
2. **Annotate** it: which columns are samples, which condition each belongs to, the $\log_2$ scale, the normalization and the completeness filter. This produces Data and Filtered Data in one step.
3. **Impute**, optionally, to produce Imputed Data. Check the source selector on downstream pages. An imputed value is a model-based replacement, not an additional measurement.
4. **Aggregate** phosphosites to phosphoprotein level, optionally, choosing the aggregation

and whether to aggregate the imputed sites or the observed ones.

5. **Analyse**: every section from Completeness onwards reads one of those matrices and writes nothing back to any of them.

Only the first two steps are compulsory, and they are compulsory in that order.

4.6 The boxplot, defined once

Several pages draw boxplots, and they all draw the same one. It is defined here; each of those chapters states only what its own boxes contain — the intensities of one sample, the mean intensities of one condition, the CV values of one condition, the intensities of one protein across the samples of one condition.

THE STANDARD BOXPLOT

At the centre of each box lies the **median**, the 50th percentile of the values in it. The box itself spans the **interquartile range**, from the 25th percentile Q_1 to the 75th percentile Q_3, so it covers the central half of the distribution:

$$\mathrm{IQR} = Q_3 - Q_1.$$

From the edges of the box, **whiskers** extend to the most extreme values that still fall within 1.5 times the IQR. The lower whisker reaches the smallest value greater than or equal to

$$Q_1 - 1.5 \times \mathrm{IQR},$$

and the upper whisker reaches the largest value less than or equal to

$$Q_3 + 1.5 \times \mathrm{IQR}.$$

The whiskers therefore end on observed values, not on those two bounds.
Any value beyond a whisker is an **outlier**. Where a page offers **Toggle Outliers**, outliers are drawn as individual markers when it is on and are omitted from the drawing when it is off — omitted from the *drawing* only: they remain in the data, in the quartiles and in the table under the plot.

Compare medians, IQRs and tails between samples that the experimental design makes comparable. A displaced or unusually wide box is a prompt to check sample identity, loading, processing and biological differences, not a diagnosis on its own. Log transformation does not require a symmetric distribution. Outlier markers are values beyond a drawing rule, not automatically errors to remove; normalisation can make boxes similar by construction without demonstrating that the samples are equivalent.

The steps these terms name

Function	What it produces
read_data	Reads the uploaded table and normalises its column names
log2_transform_data	Zeros to NA, then $\log_2$, on the sample columns only
filter_data	The completeness filter, per condition, in at least one condition, or overall
impute_values	Fills the missing values, producing Imputed Data
transform_phosprot	Aggregates phosphosites to phosphoprotein level

5 Sessions, saving and restoring

What the tool remembers, for how long, what a save state contains, and what a restored one brings back.

AT A GLANCE

Where	Data → Restore, and **Ctrl** + **S** from anywhere.
Where state lives	In the memory of the running application, one set per browser tab. Uploaded files and temporary artifacts may also be stored on the application host; this is not an automatic backup.
What ends it	Refreshing the tab, closing the application, or leaving it untouched past the idle limit the deployment sets.
The file	One compressed document with the extension .copylotstate: settings, annotations, metadata and the imputed matrix.
Not in the file	The uploaded measurements. An ordinary state records where they were read from; a portable state does not, and asks for them on restore.
Never written to	Your dataset files. Saving reads them, and restoring reads them; neither writes to them.

5.1 What a session is

Everything the tool knows about your analysis — the uploaded tables, the annotation, the filter, the imputed matrix, the state of every control on every page — lives in the memory of the running application and in the browser tab you are working in. Two tabs are two sessions: each has its own datasets, its own annotations and its own caches, so two projects can be open side by side without either seeing the other.

That has three consequences worth stating plainly. Refreshing the tab starts a new analysis rather than reloading the old one. Stopping the application ends every session it was holding. And a session nobody touches is eventually discarded — after a day, unless whoever deployed the instance chose otherwise — to give its memory back.

The application asks the browser to warn you before a refresh, a close or a navigation when-

ever anything has changed since the last successful save. The browser writes the wording of that dialog and may decline to show it at all, particularly on a phone, so treat it as a safety net and not as the mechanism. The mechanism is saving a state.

> **NOTE**
>
> The identifier in a plot or download address is a routing label, not a credential: it tells the backend which tab's data to draw, and another browser still has to authenticate. No cookie and no routing label is ever written into a save state.

5.2 Saving a state

On Data → Restore

Control	What it does	Default	Where
Save State...	Writes a state file, asking where to put it. Once a file has been chosen the button becomes **Update Save State** and writes to that file again.	—	page
Save State As...	Writes a state file to a new location and makes that the file **Update Save State** and **Ctrl** + **S** will write to next.	—	page
Save Portable State...	Writes a state file with the dataset paths left out, for another machine or another person.	—	page
Open Save File...	Chooses an existing state file, both as the one to restore below and as the one to update later.	—	page
Restore	Applies the selected state file. Disabled until a file is selected, and disabled again while any missing dataset is still unanswered.	—	page

The file is one gzip-compressed JSON document named for the moment it was written, with the extension .copylotstate. It holds, for every uploaded dataset package:

- the file name of every dataset at every level, and — in an ordinary state — the absolute path it was read from;
- every metadata profile, with the metadata table itself, which is small enough to travel inside the file;
- every annotation: whether the values were already on the $\log_2$ scale and whether that was detected or stated, the normalization, the complete filter configuration, the imputation

settings, and which aggregation produced a phosphoprotein matrix;

- the imputed matrix, in full, because nothing on disk holds it and imputation draws random numbers;
- the peptide metadata table, and the condition colours;
- the interface state: which section and which tab you were on, whether advanced mode was on, the appearance settings of every plot you have styled, and the control values every page has stored for itself.

It also records the version of the application that wrote it, which is what lets a restore say something useful when the versions differ.

5.2.1 What it does not hold

Not the measurements. A state file carries the settings that turn your data into an analysis; the data itself stays in the files you uploaded, and an ordinary state simply records where those files were. Nothing in saving or restoring writes to, alters, moves or replaces them.

There is one consequence of that worth knowing in advance. A dataset that arrived through the browser rather than from a path on the server has no path to record, so the application writes a copy of it into a temporary directory and records *that* path instead. It is a copy, made so that the state has something to point at; the original is untouched. But a temporary directory is a temporary directory, and when the machine clears it the state will ask for the file again, exactly as a portable state does.

The conversation with the assistant is not saved either, and neither is any figure: a plot is recomputed when the page that draws it is opened.

5.2.2 Ordinary or portable

Use an ordinary state to come back to your own analysis on your own machine: it finds its data by itself and restores without a question. Use a portable state to send the analysis somewhere else — another machine, a colleague, a supplementary archive. The portable state leaves out the absolute paths, which are meaningless elsewhere and would otherwise disclose the layout of your filesystem, and asks whoever restores it to supply the dataset files.

5.3 Restoring a state

1. Choose the state file, either in the file field or with **Open Save File...**.
2. Press **Restore**. The file is decoded, its format checked and its datasets looked for.
3. If any dataset cannot be found, a **Missing Files** panel lists each one with the path that

was expected and the name it had. Supply a file for each, then press **Restore** again. Nothing has been changed at this point — a restore that is missing a dataset applies none of itself.

4. The application reports how many datasets and annotations were restored, and rehydrates the interface: the section, the tab, the appearance settings and the control values.

WHAT A RESTORE ACTUALLY DOES

Each dataset is read again — from the recorded path, or from the file you supplied in its place — and the annotation is then *recomputed* from it with the saved settings: the same $\log_2$ decision, the same normalization, the same completeness filter. The result is a session built the same way the original was, not a picture of it.
The imputed matrix is the exception, and is taken from the file rather than recomputed, because imputation draws random values and rerunning it would give different numbers under the same settings. It is discarded, with a warning, if its sample columns no longer match the metadata being restored.

A restore always reports one notice: saved settings were applied, and anything the file does not mention takes the current defaults. Where the file was written by a different version of the application, a second notice names both versions. Both are worth reading rather than dismissing, and the second one is the subject of the warning below.

EASY TO GET WRONG

A save state stores settings, not results, so a restored session recomputes everything with *today's* code. Where a release has changed what a computation means, an old state will reproduce the old settings and potentially a different answer. Do not infer compatibility from a successful restore alone. Record the application version and build, retain the original inputs and exported results, and check the processing and contrast direction when comparing releases. The current volcano convention is condition1 minus condition2.

5.3.1 Edge cases

- A state restored into a session that is not local to the machine running the application ignores the saved paths entirely and asks for every dataset to be uploaded, whatever the state contains. No path from the file is opened or even tested.
- A state whose format marker is not the one this release writes is refused outright rather than partially applied.
- The size of a state file, of an individual replacement dataset and of the whole restore is bounded; a legitimate project that exceeds a bound needs the operator to raise it, not a smaller analysis.
- Restoring in the DataViewer is not possible: the section is not there. A viewer session shows what was published for it and has no state of its own to save.

- A state saved while a setting was at its default, and restored after that default has changed, comes back at the *new* default: what is not in the file is not restored from it.

Behind this page

Function	What it produces
export_state_blob	Builds the state document and compresses it
inspect_missing_files	Looks for every dataset the state names, and reports the ones it cannot find
apply_restore_state	Reads the datasets back, recomputes the annotations and installs the session
migrate_restore_ui_state	Applies the disclosed migrations to interface state written by an earlier release

PART II

Data



6 Uploading data

Getting a search-engine output into the tool, and what it has to look like.

AT A GLANCE

Where	Data → Upload
Accepts	.csv, .tsv, .txt, .xlsx, .parquet. Anything else is refused by name before it is read.
Levels	Protein, phospho, phosphoprotein and peptide are uploaded separately, one file each. A phosphoprotein matrix can also be built here from an uploaded phospho file rather than supplied — see *Phosphosites to phosphoproteins*.
Shape	Protein, phospho and phosphoprotein are wide: one row per feature, one column per sample. Peptide is the long report as the search engine wrote it.
Does not	Search, normalise, filter, impute or decide which columns are samples. That is the annotation's work.
Replaces	Uploading a level again discards the annotation and the uploaded metadata of that level.

6.1 What you can upload

The tool reads a table, not a raw file. What is uploaded is the output of a search engine — a protein-group report, a collapsed phosphosite report, a precursor or peptide report — exported to one of five file types. The extension decides how it is parsed, and nothing else does:

- .csv is read as comma-separated text.
- .tsv and .txt are read as tab-separated text. A .txt file that is really comma-separated arrives as one column and is refused for that reason.
- .xlsx is read as a workbook, first worksheet only.
- .parquet is read in batches through its own schema.

Three of the four dataset types are wide matrices: one row per feature, one column per sample, plus whatever annotation columns the report carries. The peptide type is different — it is the long report, one row per measurement, and the tool stores only the path to it rather than loading it into the session, so it has no preview and no row count on this page.

> **WORTH KNOWING**
>
> Nothing about the upload is level-specific except which of the four slots the table lands in. A phosphosite report uploaded as **Protein** is accepted; it will simply be described as protein data everywhere afterwards.

6.1.1 What the tool does to the table

Four things happen to every wide table between the file and the session, in this order, and all four are visible in the preview underneath:

1. Column names are stripped of surrounding whitespace.
2. Eight column families are renamed to the canonical names the rest of the tool uses (below). A name that is already canonical, or whose target is already present in the table, is left alone.
3. Repeated column names are made unique by appending .1, .2 and so on, so that two columns called Genes do not silently become one.
4. Every column that is not numeric is stored as text.

6.2 Columns the tool looks for

The report does not have to use these names — the aliases on the right are recognised and renamed on the way in. The name on the left is what appears in the preview and in every later table.

Columns renamed on upload

Column	What it holds
File.Name	The run. Recognised from Run or R.FileName.
ProteinNames	The protein group. Recognised from PG.ProteinNames, PG.ProteinGroups, ProteinIds or ProteinNames.
GeneNames	The gene. Recognised from Genes, GeneNames or PG.Genes.
Stripped.Sequence	The unmodified peptide sequence. Recognised from PEP.StrippedSequence.
Modified.Sequence	The modified peptide sequence. Recognised from EG.ModifiedSequence.
Precursor.Quantity	The precursor quantity. Recognised from FG.Quantity.
RT	Retention time. Recognised from EG.ApexRT.
Predicted.RT	Predicted retention time. Recognised from EG.RTPredicted.

Matching ignores case, spaces and punctuation, so PG.Protein Names, pg_proteinnames and PG.ProteinNames are all the same name. Any other column is kept exactly as it is written.

6.2.1 Columns a phospho file needs for collapsing

A phosphosite table can be uploaded without them, but it cannot be collapsed to phosphoproteins unless it carries all five of UPD_seq, PTM_localization, Protein_group, Gene_group and PTM_Collapse_key, and unless each Protein_group maps to exactly one Gene_group. The collapse refuses the table by name when one is missing.

6.2.2 Columns a peptide report needs

The peptide pages look for a run column — File.Name, or the first column whose name contains *file* — a stripped-sequence column, and Precursor.Quantity for anything that pivots a quantity. A missing one is reported when the page that needs it is opened, not on upload, because the file is never read here.

6.3 Uploading

1. Choose the level in **Dataset type**. The choice decides which of the four slots the table fills, and nothing else.
2. Choose the file. The browser's file dialogue is restricted to the five accepted extensions.
3. Press **Upload**. A wide table is parsed immediately, so a file that cannot be read is refused now rather than three pages later.
4. Check the preview: the row and column counts, and whether the sample columns are the ones you expect and carry numbers rather than text.
5. Go to Data → Annotation and assign the sample columns to conditions. Until that is done the tool knows the table but not the experiment.

When the browser exposes the file's own location on disk, that path is stored and the file is read from where it is. Otherwise a copy is written to a working directory and its path is stored instead. Either way the path is shown as Restore Path on the **Current datasets** card, and it is what a saved analysis is restored from.

> EASY TO GET WRONG
>
> A file that uploads cleanly has not been understood. Upload decides nothing about which columns are samples — the annotation does, and it is where a mistyped or unselected column becomes a missing replicate. Two things are worth checking in

the preview before moving on: that every run you expect has a column, and that its cells are numbers. A quantity column exported with a decimal comma parses as text, and every value in it is read as not detected; the annotation reports it as a warning, but only once the column has been assigned to a condition.

6.3.1 When a file is refused

The parser's own wording is replaced by the cause wherever the cause is known. The refusals that occur in practice:

- **Unsupported file type.** The extension is not one of the five.
- **A NUL byte where text was expected.** Almost always a file saved as UTF-16, or truncated mid-write. Re-save it as UTF-8 text; it is refused rather than repaired because the C parser would otherwise truncate the header at that byte and rename a column silently.
- **Not valid UTF-8.** Re-export, or re-save, as UTF-8.
- **Rows do not all have the same number of columns.** A ragged export, or a stray delimiter inside an unquoted field.
- **No columns to parse.** The file is empty, or its first line is not the header.
- **Named** .xlsx **but not a valid workbook.** Usually a .xls or a CSV renamed by hand.
- **No data rows.** The table parsed, and the header is all there is.
- **Only one column.** A report needs an identifier column and at least one measurement column. This is what a saved web page, or the wrong separator, looks like from here.

The default upload ceiling is 64 GiB, with at most 20,000 columns and 10 billion cells. A separate parsed-table memory guard normally uses a quarter of detected physical/container memory, capped at 128 GiB; it falls back to 1 GiB when capacity cannot be detected. Administrators can change these limits, and a hosting proxy may impose a smaller upload limit.

These are safety ceilings, not a promise that a file of that size can be analysed on every computer. Compressed Excel and Parquet files can expand considerably. Parsing and later analyses also need working memory beyond the stored table. For large reports, prefer a suitable tabular export and enough server RAM; raising a limit does not make the analysis out-of-core.

6.4 Controls

Controls on this page

Control	What it does	Default	Where
Dataset type	Which level the file is: Protein, Phospho, Phosphoprotein or Peptide. Also selects which dataset the preview below describes.	Protein	page
File	The file to upload. Restricted to the five accepted extensions.	none	page
Upload	Reads and stores the file. Disabled until a file is chosen.	—	page
Show Dataset Manager	Reveals the three controls below. The choice is remembered between visits.	hidden	page
Selected dataset	Which dataset package is active. A package holds one slot per level, so a second experiment can be loaded beside the first without overwriting it.	Data	page
New dataset	Creates an empty package under the name given and makes it active.	empty	page
Rename selected dataset	Renames the active package. The annotation and metadata profiles attached to it follow the name.	the active package	page

Every plot, table and test in the session reads the active package. The level selector on a plot page offers only the levels that package actually holds.

> NOTE
>
> In viewer mode the upload form and the dataset manager are not drawn at all. What remains is a **Dataset level preview** selector over the datasets the instance was configured with.

6.5 What the upload replaces

Uploading a protein, phospho or phosphoprotein file into a slot that already holds one replaces it, and discards two things that were derived from the old table: the annotation for that level, in every metadata profile, and — for protein and phospho — the uploaded metadata for that level. Both are discarded rather than re-applied because a metadata file matches samples by column name, and the new table's columns may be different ones. Uploading a peptide file clears the peptide metadata of the active package in the same way.

Nothing else is touched: the other three levels, the other packages and the condition palette all survive.

6.6 Example data

The Python package ships a real HEK293 experiment under proteomicscopylot/example_data: ten samples, five wild type and five GRK, run as both a whole-proteome and a phosphoproteome arm. It is what example.ipynb and example.R run on, and the files upload here like any others — a protein-group report of 7,490 rows, a phosphosite report of 11,916 rows, a phosphoprotein matrix of 2,301 rows, and a sample-to-condition table for each arm that can be uploaded as metadata on the annotation page. The protein report carries PG.Log2Quantity, so its quantities are already on the $\log_2$ scale and the annotation detects that without being told. In a notebook, load_example_data("protein") reads it, and the same function takes "phospho", "phosphoprotein", "protein_metadata" and "phospho_metadata", in Python and in R.

There is no button that loads them: the example files are uploaded on this page like any other dataset, which is deliberate — trying the tool on them is the same procedure as using it on your own data, so nothing learnt on the example has to be unlearnt afterwards.

6.7 Exporting

Nothing is exported from this page. The table as uploaded is downloadable once it has been annotated, from Data → Annotation, as <level>_data_full.csv.

Behind this page

Function	What it produces
read_dataframe	Reads an uploaded table and refuses one that is not a table
get_extension	Accepts or refuses the file type by name
rename_cols	Renames the eight recognised column families to their canonical names
make_columns_unique	Deduplicates repeated column names
sanitize_dataframe	Stores every non-numeric column as text
save_table_dataset	Puts the parsed table in the active package
save_peptide_path	Stores the peptide report's path rather than its contents

7 What is in the dataset

The matrices currently loaded, their dimensions, and every step that has been applied to them.

AT A GLANCE

Where	Data → Upload for what was loaded; Data → Annotation for what has been done to it; Data → Imputation for what imputation is in force.
Shows	Raw file names, the loaded and processed matrices, and the processing history of each.
Levels	Protein, phospho, phosphoprotein and peptide, each described separately.
Downloads	The matrix as uploaded, the filtered matrix and the metadata table, each as CSV.
Answers	Which matrix a number elsewhere in the tool came from.

7.1 The three blocks

Three questions have to be answerable at any point in a session: what was loaded, what is being shown, and what has been done to it in between. This chapter is those three, in that order.

A QUICK IDENTITY CHECK

Before interpreting a result, check the active package, the dataset's filename and dimensions, and the sample names in its preview. Then check the annotation and the plot's data source. A table can be loaded correctly while the wrong metadata profile or processed matrix is selected.

7.1.1 The raw file names

The **Current datasets** panel carries one card per level — Protein, Phospho, Phosphoprotein, Peptide — and an empty card says so rather than disappearing. For the three wide levels the card gives:

On a dataset card

Column	What it holds
Filename	The name of the file that was uploaded, truncated in the card and shown in full on hover.
Format	Its extension: csv, tsv, txt, xlsx or parquet.
Rows	Rows in the table as uploaded — features, before any filtering.
Columns	Columns in the table as uploaded — samples plus annotation columns, not samples alone.
Restore Path	Where the table is being read from on disk: the file's own location when the browser exposed it, otherwise the copy the tool wrote. Not stored yet when neither exists.

The peptide card gives only Filename and Path, because a peptide report is referenced rather than loaded.

The names of the individual runs are the sample column names of the report, and they are visible in two places: as the header row of the preview under the panel, and as the sample column of the metadata table once the dataset has been annotated. For a peptide report they are the unique values of the run column, which the peptide pages read directly from the file.

7.1.2 The datasets displayed

The preview below the panel shows the first twenty rows of the level selected in **Dataset type**, exactly as uploaded — not transformed, not filtered, not imputed. Long headers are wrapped and long cells truncated at twenty characters, with the full value on hover.

> **CHANGED IN version 1.0.0**
>
> The older manual describes this second block as showing the $\log_2$-transformed datasets. It does not, and on this page it never did: the $\log_2$ matrix does not exist until an annotation has been generated, because the transform needs to know which columns are samples. The transformed matrices are on the annotation page.

A session holds up to three matrices per level, and they are a ladder — each is built from the one above it:

The three matrices

Column	What it holds
Data	The uploaded table on the analysis scale: $\log_2$, and normalised if a normalisation was chosen. Built when the annotation is generated.
Filtered Data	Data with the rows that failed the completeness filter removed. Same columns, fewer rows.
Imputed Data	Filtered Data with its missing values filled where the selected method can estimate replacements. Exists only after imputation has been run and stored.

The annotation page previews twenty rows of **Filtered Data** and thirty metadata rows. Each preview has a full CSV download. The imputation page previews the stored imputed matrix. Annotation downloads use these names, with the selected level in place of level:

- Uploaded table: level_data_full.csv.
- Filtered matrix: level_filtered_data_full.csv.
- Metadata: level_metadata_full.csv.

> **WORTH KNOWING**
>
> Most plot pages carry a **Data source** selector over exactly these three, and it defaults to the most processed matrix that exists. That is the usual reason two pages disagree about how many features there are.

7.2 The processing summary

7.2.1 Annotation, filtering and normalisation

Generating an annotation writes a four-card summary on Data → Annotation:

Result Summary

Column	What it holds
Source Rows	Rows in the uploaded table — the same number as Rows on the dataset card.
Metadata Rows	Samples that were annotated. Fewer than the dataset has columns, and the difference is the annotation columns plus anything unassigned.
Conditions	Distinct condition names in the metadata.
Filtered Rows	Rows that survived the filter. Source Rows minus this is what filtering cost.

Under the cards, an amber band records what was decided rather than chosen. It carries, when each applies: that the $\log_2$ scale was auto-detected and on what evidence; that columns held values which look like measurements but do not parse as numbers, and were therefore read as missing; that assigned columns or uploaded metadata rows named samples the dataset does not have and were ignored, with how many were annotated instead; and that metadata was auto-generated from date-like column names. The normalisation that was applied is recorded in the same list and is the one entry the page does not display, because the **Normalization** control beside it already says so.

The filter itself is not summarised in prose: the **Filter Settings** controls hold whatever was used for the annotation in force, because they are reloaded from it when the page opens.

7.2.2 Imputation

Data → Imputation carries an **Applied imputation** card describing the run that is stored — not the preview above it, which re-computes as the controls are moved. It names the method and its settings in a sentence, and gives Rows, Number of values, Number imputed and Imputed %. When nothing has been stored for that level, it says so, which is the difference between a preview and an applied imputation.

7.2.3 Collapse

When a phosphoprotein matrix was built by collapsing phosphosites, the mode used is recorded in three places: in the generated file name, phosprot_aggregated_<source>_<mode>.csv, visible on the phosphoprotein dataset card; in the first warning of the phosphoprotein annotation, which also names the source matrix and, for robust_site_median, the coverage rule; and in the annotation itself, where the plots read it to decide whether their axis says log2 Intensity or log2 Ratio to Site Median.

> **EASY TO GET WRONG**
>
> Rows on the dataset card and **Filtered Rows** in the result summary answer different questions, and a number quoted from the wrong one is the most common source of a figure legend that does not match its figure. The card counts the file. The summary counts what a plot drawing Filtered Data will actually see. Neither counts features quantified in at least one sample, which is a third number again and is on the coverage page.

> **NOTE**
>
> Quality control is the one part of the tool with no data-source choice: the other QC surfaces read Data: annotated, normalised when requested, but not completeness-filtered or imputed. They do not necessarily show raw instrument intensities. Principal component analysis is the deliberate exception and keeps a selector, because it cannot use incomplete features and choosing its input is a real decision.

7.3 Packages and profiles

Two selectors change what every block above describes, and both are hidden until asked for.

Controls on this page

Control	What it does	Default	Where
Selected dataset	The active dataset package: one set of four level slots, with its own annotation. Under **Show Dataset Manager** on Data → Upload.	Data	page
Selected metadata	The active metadata profile: one sample-to-condition map over the same data. Under **Show Annotation Manager** on Data → Annotation.	Metadata	page
Dataset type	Which level the preview and the dataset cards describe.	Protein	page

A package is a second experiment loaded beside the first; a profile is a second way of grouping the same samples. Changing either changes the matrices, the dimensions and the processing history shown everywhere, without anything being re-uploaded.

The three blocks the old manual showed on one screen — the raw file names, the matrices and the processing history — are deliberately on three tabs now, each beside the controls that produce it.

7.4 Exporting

Everything on this page is a description of something downloadable elsewhere. The uploaded, filtered and metadata tables download from the annotation page as CSV, at full length rather than at preview length. The imputed matrix downloads from the imputation page. The dataset cards themselves are not exported; the report generator restates their contents in its methods text.

Behind this page

Function	What it produces
get_all_current_datasets	The four dataset cards

Behind this page

Function	What it produces
select_matrix	Resolves Data, Filtered Data or Imputed Data for a level
qc_matrix	The one matrix quality control describes, equal to Data
get_annotation	The stored annotation: metadata, matrices, filter settings, imputation and collapse mode
sample_label_frame	Maps each original sample name to its condition_n label

8
Annotating samples

Telling the tool which samples belong together, which is what makes every later plot mean something.

AT A GLANCE

Where	Data → Annotation
Produces	The metadata table, the $\log_2$ matrix and the filtered matrix, for one level at a time.
Needed by	Conditions, colours, sample labels, filtering, imputation, every statistical test and every summary plot.
Two routes	Assign columns to conditions by hand, or upload a metadata file. An uploaded file wins.
Optional	No. Until a level is annotated it has no $\log_2$ matrix, and most pages have nothing to draw.
Pairing	Stored here, as pair_id; edited on the pages that run paired tests.

8.1 What the annotation is

The annotation is a table with one row per sample: the name of the dataset column, and the condition that column belongs to. Everything the tool knows about the experiment is in it. Generating it does three things at once — it writes that table, it builds the $\log_2$ matrix from the columns the table names, and it filters that matrix by the settings beside the button.

The metadata table

Column	What it holds
sample	The dataset column, exactly as it is spelled in the uploaded file. This is the *original sample name*.
condition	The condition the sample belongs to. Case and spelling are taken as given: Control and control are two conditions.
pair_id	One identifier shared by the two samples of a pair. Present only once a pairing has been saved, or once an uploaded file supplied the column.

A sample appears exactly once. Assigning one column to two conditions is refused by name, and so is a metadata file that lists a sample twice — identically, because a file that repeats a row was assembled wrong, and with different values, because the file then states two incompatible facts and row order would decide which one wins.

8.1.1 What the samples are called afterwards

Plots do not use the original sample name. Each sample is relabelled condition_n, where *n* is its 1-based position among the samples of its own condition in metadata order — so the fourth sample of WT is WT_4 whether it is the fourth or the ninth column of the file. Numbering restarts in every condition. The original name is carried alongside, and **Toggle IDs** on a plot page adds it under the label on a second line; the tables that list samples carry it as Original_Name.

8.2 Annotating by hand

1. Choose the level in **Dataset level**. Only levels that the active package actually holds are offered.
2. In **Manual Condition Mapping**, type a **Condition name** and select its **Sample columns** from the list. The list is every column of the uploaded table, so annotation columns are offered too and must simply not be selected.
3. **Add condition** for the next one. **Remove** deletes a condition block and its assignments.
4. Check **Dataset is already log2 transformed**. It is set for you as soon as the first samples are assigned, and is only guessed at until you touch it — see below.
5. Set the filter, and press **Generate Annotation**.

A name copied with a trailing space still matches, whitespace ignored. Anything that still does not match is dropped, and the warning band names it and states how many samples were annotated instead.

> **IF NOTHING IS ASSIGNED**
>
> Generating with no conditions is not an error if the column names carry dates. Of the columns containing an eight-digit string beginning 20, those whose name also says quantity, intensity, abundance, area or lfq become the samples when there are any; otherwise every dated column becomes a sample except the dated per-run statistics — q-values, p-values, counts, NrOf columns, identified, scores and PEP. All of them go

> into one condition called sample, and the warning band says the metadata was auto-generated. It exists so a new dataset can be looked at before it is described, not so a test can be run against it.

8.3 Annotating from a file

Metadata file takes the same five file types as a dataset. It needs a sample column — recognised as sample, sample_name, data_column_name, data column name, data_column, column_name or column, and otherwise taken to be the first column of the file — and at least one further column to group by. A column called condition is used as the condition if one is present; otherwise the first grouping column is.

Each grouping column becomes its own *metadata profile*, named after the column, and a file with several of them therefore describes the same samples several ways at once — by treatment, by timepoint, by replicate batch — with automatic combinations generated on top. At most five grouping columns are accepted, and at most five manual profiles exist at a time; the annotation manager shows the count.

While a metadata file is active for a level it supplies the conditions, and the manual mapping below is kept but not used. The page says so at the top of the mapping block.

Rows naming a sample the dataset does not have are dropped rather than refused — a shared metadata file legitimately covers more runs than are loaded — and the warning band names them, because three mistyped names silently turn a 6-vs-6 design into 6-vs-3. Rows with a blank sample or condition are dropped, a trailing .0 is stripped from a sample name because that is what a spreadsheet does to a numeric run identifier, and a file matching no column at all is refused.

> **WORTH KNOWING**
>
> Uploading a metadata file for the phospho level also annotates the phosphoprotein level, which shares it. There is no separate phosphoprotein metadata upload.

8.4 The log2 question

The tool has to know whether the uploaded quantities are already on the $\log_2$ scale, because it will otherwise take the logarithm of a logarithm. **Dataset is already log2 transformed** states the answer. Until it is clicked, it is answered by detection over the assigned sample columns only:

HOW THE SCALE IS DETECTED

Column names are weak evidence and are trusted only when *every* assigned column carries log2 as a token, and never over a clear contradiction from the values.
The values are read next. If at least 0.1% of them are negative, the data are $\log_2$. If the 1st and 99th percentiles both fall in $[0, 2]$, no answer is given: that range is equally consistent with ratios and with transformed data. Otherwise each column votes — linear if its 95th percentile exceeds 50 or its 99th exceeds 100, $\log_2$ if its median is at most 35, its 95th percentile at most 45 and its 99th at most 60 — and a majority of the voting columns decides.
When nothing decides, generating the annotation is refused with the reason, and the checkbox must be set by hand.

Whatever the outcome, it is stated in the warning band with the evidence behind it. Setting the checkbox yourself removes the guess, and the detection stops running for that dataset.

8.5 Pairing samples

A paired test needs to know which sample of one condition corresponds to which sample of the other — the same donor, the same passage, the same day. That correspondence is stored in the annotation as pair_id: two samples carry the same identifier when they are a pair.

It is edited in the **Pair samples** panel, which appears on the pages that can run a paired analysis — statistical analysis, the phospho-specific pages, Kinase Library and the analysis canvas — rather than here, because the two conditions being paired are chosen there. Each row of the panel is one sample of the first condition with a selector for its partner in the second, and each sample can be used once: a sample already chosen is disabled in every other row. **Leave unpaired (excluded)** drops a sample from the comparison, and the footer counts what has been excluded on each side. Two pairs are the minimum.

Save pairing writes the identifiers into the annotation, and into the uploaded metadata if there is one; until it is pressed the panel says so and the analysis will not run on the mapping. An identifier the file supplied is kept, and one the panel has to invent is called ui-pair-1, ui-pair-2 and so on. Identifiers are shared across conditions, so with more than two conditions saving a mapping also fixes those samples' partners in later comparisons; the panel warns that it will.

Regenerating the annotation preserves the pairing, but only when the sample-to-condition map came back identical. Move one sample to another condition and the identifiers are dropped, because a pairing across a design that has changed is no longer a pairing.

8.6 Controls

Controls on this page

Control	What it does	Default	Where
Dataset level	Which level to annotate: Protein, Phospho, Phosphoprotein or Peptide. Only loaded levels are listed.	the first loaded level	page
Dataset is already log2 transformed	States the scale of the uploaded quantities. Set by detection until it is clicked.	on, then detected	page
Metadata file	A sample-to-condition table to upload. Same file types as a dataset.	none	page
Upload Metadata	Reads the file and splits its grouping columns into metadata profiles.	—	page
Condition name	The name of one condition. Used verbatim as the condition label everywhere.	empty	page
Sample columns	The dataset columns belonging to that condition. Multiple selection. Called **Peptide file names** at peptide level.	none	page
Add condition	Adds an empty condition block.	—	page
Remove	Deletes one condition block and its assignments.	—	page
Generate Annotation	Builds the metadata table, the $\log_2$ matrix and the filtered matrix. **Generate Peptide Annotation** at peptide level.	—	page
Filter method, Minimum valid values, Scope, Normalization	The completeness filter applied while generating. Described in *Filtering*. At phosphoprotein level no **Normalization** control is shown: phosphoprotein data is never normalized at that level.	see there	page
Show Annotation Manager	Reveals the four controls below. The choice is remembered between visits.	hidden	page
Selected metadata	The active metadata profile. Switching it switches the conditions every page sees, without re-uploading anything.	Metadata	manager
New metadata profile	Creates an empty profile. At most five manual profiles exist at once, and the count is shown under the field.	empty	manager
Rename selected metadata	Renames the active profile.	the active profile	manager
Delete	Deletes the active profile and every automatic combination built from it. Asks first.	—	manager

EASY TO GET WRONG

A condition has two names and they are not the same thing. The one typed here is the condition itself: it decides which samples are compared, which colour they share, what the statistics call the groups, and it changes every page at once. The one in the plot appearance panel under **Condition names** is a display override — it changes the text printed on one figure and nothing else. Rename here to change the experiment; rename there to fix a label for a poster.

8.7 What the page shows afterwards

Result Summary, the warning band, and previews of the filtered matrix and the metadata table with a **Download Full CSV** button each — all described in *What is in the dataset*. At phospho level, **Aggregate to Phosphoprotein** follows; see *Phosphosites to phosphoproteins*.

Behind this page

Function	What it produces
compute_annotation	The manual route: metadata, $\log_2$ matrix, filtered matrix
compute_annotation_from_metadata	The same, from an uploaded metadata table
build_metadata_from_conditions	Turns condition blocks into the metadata table
normalize_uploaded_metadata	Validates an uploaded metadata file against the loaded dataset
split_uploaded_metadata_profiles	One metadata profile per grouping column
detect_log2_transformed	Decides the scale of the assigned sample columns
sample_label_frame	Maps each original sample name to its condition_n label
update_annotation_pairing	Writes pair_id from the pairing panel
paired_sample_columns	Resolves the two matched sample orders for a paired test

9 Filtering

Dropping features that were not measured often enough to be worth carrying, and the ways of counting that.

AT A GLANCE

Where	Data → Annotation, under **Filter Settings**. Applied when **Generate Annotation** is pressed.
Two methods	A count of valid values, or a percentage of the samples being counted.
Three scopes	Per group, in at least one group, and overall.
Counts	Non-missing entries in the $\log_2$ matrix. A row is kept whole or dropped whole.
Default	At least N values, minimum 0, scope Each group — which keeps everything.
Produces	Filtered Data. The unfiltered matrix stays as Data.

9.1 Why filter at all

A feature measured in one sample of six carries one number and five gaps. That is not enough to estimate its variability in both conditions, and a test may be unavailable. Filtering states the minimum completeness required for the intended analysis; it does not make remaining values independent or guarantee a reliable variance estimate. Imputing sparse rows can make a comparison calculable while leaving it strongly dependent on the imputation model. Choose and report the filter before interpreting the resulting hits.

Filtering is the decision about how much evidence a feature has to carry before it is allowed into that arithmetic. It is a scientific decision and the tool does not make it: the shipped default keeps every row.

9.2 The two methods

Each scope is applied to a requirement, and the requirement can be stated two ways.

- **At least N values** — **Minimum valid values** is a non-negative integer, and the row is kept where the count of non-missing entries reaches it. This is the setting the worked examples below use.
- **Percentage** — **Minimum completeness (%)** is a number between 0 and 100, and the row is kept where the *fraction* of non-missing entries reaches it. With three samples in a condition, the default 70 % means valid/3 $\geq$ 0.7, which three whole samples can only satisfy at 3: two give 0.67.

A percentage outside 0–100, or a negative count, is refused. A requirement that keeps nothing is not refused — it is a legitimate answer to a strict request — but every page that reads the filtered matrix will then say so rather than quietly substituting the unfiltered one.

9.3 The three scopes

9.3.1 Per group

The **Each group** scope requires that *every* condition contains at least the required number of valid values. One condition failing is enough to discard the row.

FILTERING PER GROUP, MINIMUM 3

Feature	A1	A2	A3	B1	B2	B3
1	1.34	2.45	1.42	NA	3.43	3.44
2	1.34	2.45	1.42	5.32	3.43	3.44

Six samples across two conditions, three each. For feature 1, condition A has three valid values and condition B has two, so the requirement is met in A and not in B: the row is discarded. For feature 2 both conditions reach three, and the row is kept.

9.3.2 In at least one group

The **At least one group** scope requires the number to be reached in *some* condition. The other conditions are not consulted.

FILTERING IN AT LEAST ONE GROUP, MINIMUM 3

Feature	A1	A2	A3	B1	B2	B3
1	1.34	2.45	1.42	NA	3.43	3.44
2	1.34	NA	1.42	NA	3.43	3.44

For feature 1, condition A has three valid values, so the row is kept — even though condition B has only two. For feature 2 neither condition reaches three, each holding two, so the row is discarded.

9.3.3 Overall

The **Overall** scope ignores the conditions and counts across all annotated samples at once. Feature 1 of either table has five valid values out of six and passes any requirement up to 5; feature 2 of the second table has four. This scope is the right one when the conditions are not the structure that matters — a time course read as a series, or a dataset annotated as a single group — and the wrong one when they are, because it lets a feature measured in one condition only survive on that condition's values alone.

THE ARITHMETIC

Let x_{ij} be the value of feature i in sample j of the $\log_2$ matrix, and let C be the set of samples of one condition. The count for that condition is

$$v_{i,C} = \sum_{j \in C} \mathbf{1}\left[x_{ij} \neq \mathrm{NA}\right],$$

and the condition passes when $v_{i,C} \geq N$, or, under the percentage method, when $v_{i,C}/|C| \geq p/100$. Feature i is kept when every condition passes (**Each group**) or when any condition passes (**At least one group**). Under **Overall** the same count is taken once over all annotated samples.

Only NA counts as missing. The count is taken on the $\log_2$ matrix, after normalisation, so what is counted is what the analysis will actually see: a linear intensity of zero became NA at the transform, and a value that could not be read as a number became NA on the way in.

9.4 Controls

Controls on this page

Control	What it does	Default	Where
Filter method	At least N values or Percentage.	At least N values	page
Minimum valid values	The count each scope has to reach. Shown for the count method only. Non-negative integer.	0	page
Minimum completeness (%)	The fraction each scope has to reach. Shown for the percentage method only. Between 0 and 100.	70	page
Scope	Each group, At least one group or Overall.	Each group	page
Normalization	None, Median centering, Quantile normalization, and VSN where the R package it delegates to is installed. Applied across the annotated samples *before* filtering; VSN uses linear values internally. Not offered at phosphoprotein level.	None	page
Generate Annotation	Applies the settings and rebuilds Filtered Data.	—	page

The controls are reloaded from the annotation in force when the page opens, so what they show is what was used, not what was last typed. Changing one changes nothing until **Generate Annotation** is pressed again.

EASY TO GET WRONG

Each group requires the chosen completeness in every group; **At least one group** also retains features detected well on only one side. Neither decides whether an undetected feature is truly absent. With no observed values in one condition, an ordinary two-group test cannot estimate that condition's variation. Imputing low values makes the comparison depend on that missingness assumption. Choose the rule from the study design, not to obtain a desirable p-value distribution, and inspect the missing-value pages alongside the volcano.

9.5 Choosing a normalization

Normalization addresses systematic differences between sample quantities; filtering addresses whether a feature has enough observed values. They are different decisions. CoPYlot normalizes across the annotated samples before applying the completeness filter, not separately within each condition.

- **None** leaves the uploaded quantities unchanged apart from the declared scale conversion and missing-value handling. Choose it when the input is already appropriately normalized or when that is the analysis you intend; it does not undo upstream normalization.
- **Median centering** shifts each sample's log2 values until sample medians share the median of those medians. If two medians are 20 and 22, their common target is 21: one column gains 1, the other loses 1. Relative differences within a column are preserved. This assumes the broad shift is unwanted; genuine global abundance changes can be removed along with loading differences.
- **Quantile normalization** aligns the ranked distributions across samples. It is stronger than aligning medians: similar ranks are assigned a common target distribution. Use it only when broadly comparable distributions are a defensible assumption. Matching distributions afterwards are a consequence of the method, not an independent demonstration of good data.
- **VSN** estimates calibration and a variance-stabilizing transformation through Bioconductor. Its output is log-like, especially at high intensity, but differences at low intensity are not exact ordinary log2 ratios. It requires the R runtime and package; the standalone Python notebook normalization helper does not implement VSN. Use the supported R or application route.

Normalization does not impute missing observations. Retain the chosen method in the report and distinguish normalization done here from any normalization already performed by the search software.

9.6 What filtering does not do

It does not change a value, and it does not fill one in: every entry that survives is the entry that went in. It does not touch the columns — a sample is never dropped by filtering, only a feature.

It also does not replace the unfiltered matrix. Both remain, as the first two rungs of the data-source ladder: Data is the $\log_2$ and normalised matrix in full, Filtered Data is the same matrix with rows removed, and a plot page's **Data source** selector chooses between them. Imputation, when it runs, starts from the filtered matrix.

> NOTE
>
> Phosphoprotein annotations do not get their own filter settings: they inherit the phospho level's, because the phosphoprotein matrix was built from the phosphosite one. They inherit its normalisation too, as already applied, and no second normalisation is performed.

9.7 Further reading

The vignette *Missing value handling* of the Bioconductor package DEP (Bioconductor 3.22) illustrates how completeness rules select different features. Huber's introduction to VSN, the vignette of the Bioconductor package vsn (Bioconductor 3.22), explains calibration, variance stabilization and the generalized-log scale; the method itself is Huber et al. 2002, doi:10.1093/bioinformatics/18.suppl_1.S96.

Behind this page

Function	What it produces
filter_data	Applies the method and scope to the $\log_2$ matrix
compute_annotation	Normalises, transforms and then filters, in that order
choose_best_source	Resolves which matrix an analysis reads, and refuses to substitute unfiltered data when the filter kept nothing

10
Phosphosites to phosphoproteins

Summarising the phosphosites of one protein into a single phosphoprotein value, and the four ways of doing it.

AT A GLANCE

Where	Data → Annotation, under **Aggregate to Phosphoprotein**. Shown only while the dataset level is Phospho.
Produces	The phosphoprotein data level, as a dataset in the active package with its own annotation.
Methods	robust_site_median (the default), mean, sum_ignore_na, sum_propagate_na.
Needs	A generated phospho annotation, and the five collapse columns in the phospho table.
Scale	Every method returns $\log_2$ values. robust_site_median returns a ratio to each site's own median, not an abundance.
Inherits	The phospho metadata and filter settings. No second normalisation is applied.

CHANGED IN version 1.0.0

A fifth method, sum_mean_impute, is described with a worked table in the older manual. It does not exist and never ran: neither implementation of transform_phosprot accepts the name, and passing it is refused. If you want missing sites filled before summing, impute the phosphosite data first and then collapse with Imputed phospho as the source — which is the same intention, performed by the imputation you chose rather than by an unconditional condition mean.

The default method, robust_site_median, is documented here for the first time. Where a hosted instance's configuration still names sum_mean_impute, the value is not understood and is ignored; the effect is descriptive only, and those phosphoprotein values are labelled as intensities rather than by their aggregation.

10.1 What collapsing is for

A phosphosite table answers a question about sites. Some questions are about an aggregate of the measured sites on a protein and need one summary per protein per sample. Collapsing produces that summary. It is not total protein abundance, a direct measurement of phosphorylation occupancy, or proof that every site changed in the same direction.

What is lost is the thing the site-level data was for. Two sites on one protein can move in opposite directions, and a summary can obscure or partly cancel those changes. A phosphoprotein value is therefore a convenience for protein-level comparisons, not a replacement for the site-level result, and a regulated protein should always be read back at site level before it is believed.

10.2 Before any method runs

WHAT IS FIXED FIRST

The phospho table must carry `UPD_seq`, `PTM_localization`, `Protein_group`, `Gene_group` and `PTM_Collapse_key`, and each `Protein_group` must map to exactly one `Gene_group`. A missing column or a violated mapping is refused by name.

Sites are then split into eligible and ineligible. A site is *eligible* when it has at least one finite value across the samples; a site with no value anywhere occupies a row and takes no part in any aggregation. The eligible set is fixed once, before the method is chosen, and it is what `site_num_eligible` counts. Fixing it is what makes `sum_propagate_na` usable at all: without it, one dead site would make its whole protein missing in every sample.

When the values being aggregated are imputed, eligibility and the coverage diagnostics are still taken from the non-imputed matrix, so the exported coverage columns describe real observations rather than filled ones.

Both scales are then prepared from the supplied one: the $\log_2$ matrix that `robust_site_median` works on, and the linear intensities the other three work on. Non-numeric values and infinities become missing on both.

10.3 The four methods

10.3.1 robust_site_median

The default. Every site is first centred on its own across-sample median, so that a site's absolute intensity — which depends on how well that particular peptide flies — drops out and only its movement remains. The protein's value in a sample is then the median of its centred sites, which is what makes it robust: one site behaving unlike the rest moves the median by nothing.

ROBUST_SITE_MEDIAN — WORKED ON THE LOG2 MATRIX

Site	A1	A2	A3	B1	B2	B3	median
ABC_Y211	20.0	NA	21.0	22.0	22.0	23.0	22.0
ABC_S344	18.0	NA	19.0	20.0	NA	21.0	19.5
ABC_T500	15.0	NA	16.0	NA	NA	18.0	16.0
each site minus its own median							
ABC_Y211	−2.0	NA	−1.0	0.0	0.0	1.0	
ABC_S344	−1.5	NA	−0.5	0.5	NA	1.5	
ABC_T500	−1.0	NA	0.0	NA	NA	2.0	
sites with a value	3	0	3	2	1	3	
median of the centred sites	−1.5	NA	−0.5	0.25	0.0	1.5	
ABC	−1.5	NA	−0.5	0.25	NA	1.5	

Three eligible sites, so a sample needs values in at least $\lceil 3 \times 0.5 \rceil = 2$ of them. A1, A3 and B3 have all three and keep their median. B1 has two, which is enough: the median of 0.0 and 0.5 is 0.25. B2 has one, which is not, so its computed 0.0 is discarded and the value is missing. A2 has none. The requirement is always at least 1, so a protein with a single eligible site is not excluded by the rule.

EASY TO GET WRONG

The numbers this method produces are ratios, not abundances. They are centred on zero by construction, they are negative for half the samples, and comparing two proteins' values says nothing — both are measured against their own sites. Plots know this and label the axis log2 Ratio to Site Median instead of log2 Intensity; a figure of these values carrying an intensity axis was made with a different method, or from a file whose collapse method was not recorded.

10.3.2 The three arithmetic methods

The other three combine the eligible sites on the linear intensity scale and differ only in what they do about a missing site. All three then take $\log_2$ of the result, so the stored matrix is on the same scale for every method; an aggregate that came out as zero or negative cannot be transformed and is set to missing instead, with the count reported in the annotation warning.

The worked tables share one protein with two eligible sites:

SUM_IGNORE_NA — ADD WHAT IS THERE

Site	A1	A2	A3	B1	B2	B3
ABC_Y211	1	NA	3	NA	4	6
ABC_S344	2	NA	4	3	NA	5
ABC, linear	3	NA	7	3	4	11
ABC, stored	1.58	NA	2.81	1.58	2.00	3.46

Missing sites are ignored and only the present values contribute, so a protein with partial measurements still gets a value. B1 sums one site and B3 sums two, and nothing in the result says which. The sample is missing only when every eligible site is.

SUM_PROPAGATE_NA — ADD ONLY WHEN ALL ARE THERE

Site	A1	A2	A3	B1	B2	B3
ABC_Y211	1	NA	3	NA	4	6
ABC_S344	2	NA	4	3	NA	5
ABC, linear	3	NA	7	NA	NA	11
ABC, stored	1.58	NA	2.81	NA	NA	3.46

Any eligible site missing in a sample makes that sample's total missing, so B1 and B2 — which sum_ignore_na kept — are dropped. Every value that survives is the sum of the same site set, which is what makes two of them comparable, and the cost is paid in missing values.

MEAN — AVERAGE THE SITES QUANTIFIED IN EVERY SAMPLE

Site	A1	A2	A3	B1	B2	B3
ABC_Y211	1	NA	3	NA	4	6
ABC_S344	2	NA	4	3	NA	5
ABC_T500	2	1	2	4	4	8
ABC, linear	2	1	2	4	4	8
ABC, stored	1.00	0.00	1.00	2.00	2.00	3.00

This table adds a third site to the shared two. Only sites quantified in every sample enter the average — here ABC_T500 alone, since the other two are missing in A2 and in B1 or B2 — and the linear mean over those sites is taken in each sample. Every sample is therefore the average of the same sites, so a change in which sites happened to be observed cannot reverse a fold change. A protein without any such site is missing in every sample: with only the two shared sites, ABC would have no value at all. When the collapse source is Imputed phospho, imputed values count as quantified.

EASY TO GET WRONG

Use robust_site_median unless there is a reason not to. It is the only method whose

value does not change when a site is added or lost, because centring removes the site's own level before combining. sum_ignore_na moves when the observed site set moves, which in a phospho dataset it does between samples — exactly the comparison the collapse was made for. mean avoids that by using only the sites quantified in every sample, and pays for it in proteins that have no such site. sum_propagate_na is stable for the opposite reason and loses the samples that pay for it. Whichever is chosen, the exported site_observed and site_coverage columns say how much of the protein each sample actually saw, and they are worth reading before a phosphoprotein result is reported.

10.4 Controls

Controls on this page

Control	What it does	Default	Where
Aggregation mode	robust_site_median, sum_propagate_na, sum_ignore_na or mean. The line under the selector describes the one chosen.	robust_site_median	page
Source data	Non-imputed phospho, or Imputed phospho when an imputation has been run and stored for the phospho level. The non-imputed option uses the filtered matrix where one exists.	Non-imputed phospho	page
Site localization probability cutoff	Which sites enter this collapse, from 0 to 1. Initially follows the shared phosphosite cutoff unless explicitly edited. Above zero, sites without a probability are excluded. Stored with the resulting matrix.	shared cutoff	page
Aggregate to Phosphoprotein	Runs the collapse and stores the result as the phosphoprotein dataset, replacing any previous one.	—	page

The button appears only once phospho annotation has been generated; until then the panel says so. The result is stored as phosprot_aggregated_<source>_<mode>.csv — the file name is where the method is recorded on the dataset card — and is annotated immediately from the phospho metadata, with the phospho filter settings inherited and normalisation forced to none, because the phosphosite matrix was already normalised before the collapse.

> **NOTE**
>
> Changing the shared phosphosite cutoff later does not rebuild an existing phosphoprotein matrix. To change the sites contributing to that matrix, run the collapse again with the intended collapse cutoff.
> Phosphoprotein data cannot be imputed directly; the attempt is refused with that reason. Impute at phosphosite level and collapse with Imputed phospho instead, so that the filling happens where the measurements are.

10.5 The columns it writes

Columns in this table

Column	What it holds
Phosphoprotein	The Protein_group the sites belong to. The row identifier at this level.
Gene_group	Its gene, carried over unchanged.
site_num	Sites of this protein in the input table.
site_num_eligible	Of those, the ones with a value in at least one sample — the fixed set the methods work on.
PTM_Collapse_keys	The collapse keys of its sites, unique and sorted, semicolon separated.
PTM_localization	Its sites' localisation values, semicolon separated in table order.
UPD_seq	Its sites' sequences, unique and sorted.
<sample>	One column per sample, in metadata order: the collapsed $\log_2$ value.
site_observed_min	Over the samples, the smallest number of its eligible sites observed in any one of them.
site_observed_mean	The same, averaged over samples.
site_coverage_min	The same as a fraction of site_num_eligible: the worst sample's coverage.
site_coverage_mean	The mean coverage across samples.
site_complete_samples	Samples in which every eligible site was observed.
site_observed___<sample>	Eligible sites observed in that sample.
site_coverage___<sample>	The same as a fraction.

Behind this page

Function	What it produces
transform_phosprot	Builds the phosphoprotein matrix and its coverage diagnostics

Behind this page

Function	What it produces
aggregate_from_phospho	Chooses the source matrix, runs the collapse and records what it did
filter_data	Re-applies the inherited phospho filter to the collapsed matrix

Collapse

11 Imputation

Filling missing values so that methods which cannot cope with them can run, and the price of doing it.

AT A GLANCE

Where	Data → Imputation
Data levels	Protein and phosphosite. Phosphoproteins cannot be imputed here at all.
Needs	An annotated dataset. The matrix imputed is the filtered one if filtering has been run, otherwise the annotated $\log_2$ matrix.
Two methods	**Quantile**, the default, draws around a low quantile of the observed values, in the spirit of MinProb (Lazar et al. 2016); **Numerical** is the Perseus downshift (Tyanova et al. 2016), drawing around the mean shifted down by a multiple of the standard deviation.
Two scopes	Sample-wise, one draw distribution per sample, or global, one for the whole matrix.
Stores	Imputed Data, an additional matrix a panel with a **Data Source** selector can be pointed at.
Reproducible	Keep the seed, input, order, parameters and software environment; the report does not state the seed. For exact replay, retain the realised imputed matrix.

11.1 What the page answers

Imputation supplies model-based values, not measurements. It fills NA in eligible sample columns with a value drawn from a distribution placed where an unobserved measurement plausibly would be — low, narrow, below the bulk of the observed data — so that a method which cannot accept a gap can be run at all. Nothing about a drawn value is a measurement.

The page answers two questions: what the drawn values will look like beside the values that were measured, which is what the three figures show, and how much of the matrix they will

be, which is what the cards under **Applied imputation** report. A matrix that is two per cent imputed and one that is forty per cent imputed are different experiments, and the second is mostly the draw distribution. The figures are a preview — redrawn from the controls, touching nothing — and only **Impute Data Values** stores a matrix.

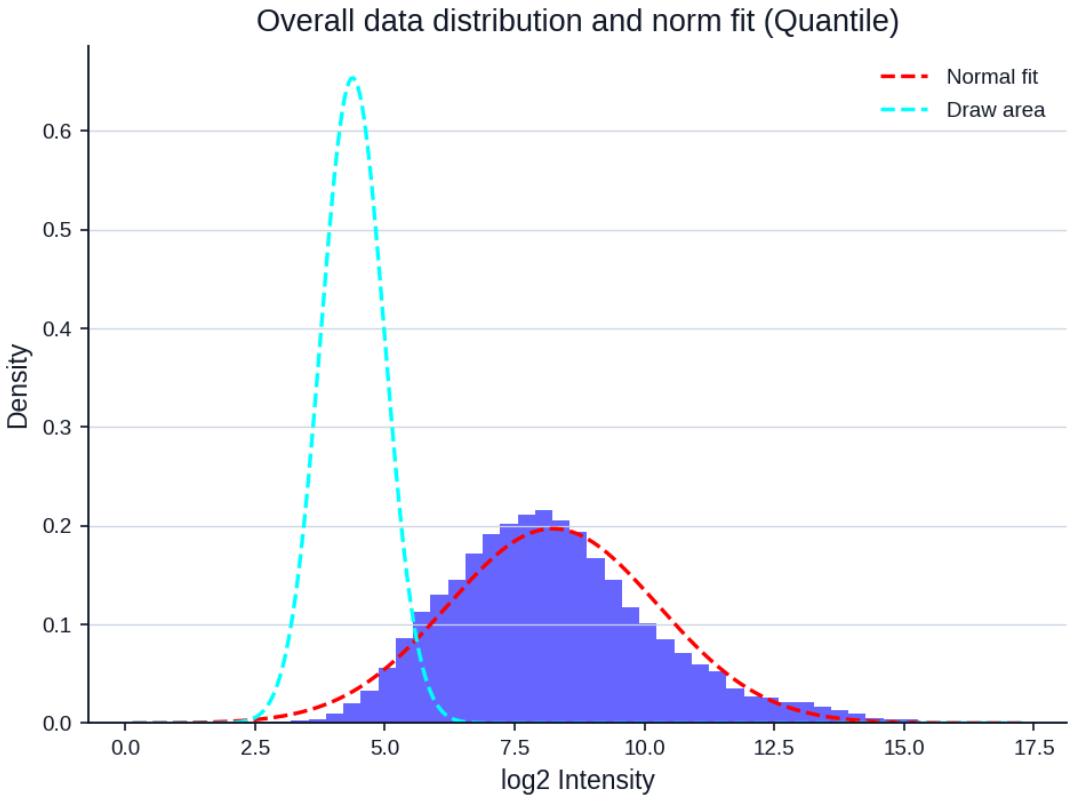


Figure 11.1. The observed distribution with its normal fit and the draw area, on the example data at protein level with the quantile method: every observed $\log_2$ protein intensity as a density, peaking near 8 and running from roughly 2.5 to 15, with the broad dashed normal fit over it and the narrow dashed draw area centred near 4.5, well below the bulk. The legend names the two curves and the title names the method.

11.1.1 Reading it

The figure above is the second of the three the page draws. The first is not reproduced here: it compares rows with no missing value against rows with at least one, and says whether missingness is concentrated in the low-intensity rows, which is the assumption the whole draw model rests on. The third is the after plot, which is the subject of the last bullet below.

- Every observed value in the sample columns, pooled over all samples, as a density.
- The red curve: a normal density with the mean and the standard deviation of those values. It is fitted to the whole distribution rather than its lower tail, and says how far from normal the data are before anything is drawn.
- The cyan curve: the draw area, the density the missing values are taken from. Sample-wise it is the mixture of the per-sample draw distributions, each weighted by the number of values that sample contributes; globally it is the one distribution.

- After the draw: measured values and invented values as two histograms on common bins. The imputed set should sit under the left shoulder of the observed one rather than displacing it.

> **EASY TO GET WRONG**
>
> Every plot drawn from Imputed Data will look better than the experiment was: no gaps are left in it, so a heatmap has no white, a correlation is computed on complete columns, and a coverage count would reach the number of rows. Impute only for the method that requires it and keep the non-imputed matrix for anything whose subject is completeness — which is why coverage and the missing-value pages never read the imputed matrix at all.

11.2 Controls

Controls on this page

Control	What it does	Default	Where
Dataset level	Protein or phospho. Phosphoprotein is not offered.	Protein	page
Method	Quantile or Numerical: where the draw is centred, and how its width is set.	Quantile	page
Sample-wise imputation	On: every sample gets its own draw distribution, from its own observed values. Off: one distribution for the whole matrix.	on	page
Quantile	Quantile method only. The quantile of the observed values the draw is centred on; 0.01 is the first percentile.	0.01	page
Adjust standard deviation	Quantile method only. Draw width as a multiple of the standard deviation of the observed values. Must be greater than 0; 0 is refused.	0.3	page
Downshift	Numerical method only. How many standard deviations below the mean the draw is centred.	1.8	page
Adjust standard deviation	Numerical method only, and a different parameter from the field of the same name above: draw width as a multiple of the standard deviation. Must be greater than 0; 0 is refused.	0.3	page
Random seed	Fixes the draw, so the same parameters reproduce the matrix value for value.	1337	page

Controls on this page (continued)

Control	What it does	Default	Where
Impute Data Values	Computes the imputation and stores it as this level's Imputed Data. Nothing else on the page stores anything.	—	page
Show imputation draw distribution	Draws the cyan draw-area curve over the middle figure.	on	page
Separately / Together	How the two histograms of the lower figure are scaled: each to its own density, or both by the combined count.	Separately	page

11.3 How it is computed

WHICH MATRIX IS IMPUTED

There is no **Data Source** selector here: the most processed matrix available is taken — the filtered matrix if filtering has been run, otherwise the annotated $\log_2$ matrix, otherwise the uploaded frame, which is reported as a warning. A filter that kept no rows stops the run with an error rather than quietly imputing the unfiltered matrix. Only NA counts as missing. Zero on the $\log_2$ scale is an observed value, because $2^0 = 1$, and is never drawn over; the convention that a raw intensity of zero means *not detected* is applied earlier.

THE DRAW

Let x be the observed values in scope — one sample's column when sample-wise, the whole matrix when global — with mean $\bar{x}$ and standard deviation s, taken with $n - 1$ in the denominator. Each missing value in that scope is replaced by an independent draw from

$$\mathcal{N}\left(Q_q(x),\, (a\,s)^2\right) \quad \text{or} \quad \mathcal{N}\left(\bar{x} - d\,s,\, (w\,s)^2\right),$$

the first for **Quantile**, the second for **Numerical**. Q_q is the q-quantile and q is **Quantile**; a, d and w are **Adjust standard deviation**, **Downshift** and the numerical **Adjust standard deviation**. At the defaults the quantile draw is centred on the $q = 0.01$ quantile of the observed values with a spread of $0.3\,s$ — a MinProb-like draw (Lazar et al. 2016) — and the numerical draw, the Perseus downshift (Tyanova et al. 2016),

1.8 s below their mean with the same spread. Both spreads must be greater than 0: a spread of 0 would give every imputed value in a sample the same number, and such constant replicates make a later t-test undefined, so the request is refused with Adjust standard deviation must be greater than 0.
Sample-wise, $\bar{x}$, s and $Q_q(x)$ are recomputed inside every sample column, so a sample that ran low is filled low and its imputed values stay in the company of its own measurements. Globally, one centre and one width are computed from all observed values pooled; this needs at least two observed values with non-zero spread and errors otherwise. The generator is created per request and seeded with **Random seed**.

11.3.1 Edge cases

- Sample-wise, a sample with fewer than two observed values or no spread among them has nothing to estimate a distribution from. It is skipped, its missing values survive the run, and a warning under the plots names it.
- Number imputed can therefore be smaller than the number of missing values the matrix started with: the cards report what was done, not what was asked.
- A phosphoprotein level is refused outright. Impute the phosphosites, then collapse from the imputed phosphosite source.

11.4 What it changes downstream

Imputation adds a matrix and replaces nothing: Data and Filtered Data remain as they were.

- Panels with a **Data Source** selector can be pointed at Imputed Data: PCA, the statistics pages, the comparison pipeline, Kinase Library, the single-protein page and the analysis canvas. One opened for the first time after a successful run opens on the most processed matrix that exists, which is now the imputed one; one that already holds a choice keeps it.
- Coverage, the missing-value plots and tables, the histograms, the boxplots, the correlation surfaces and the distribution page offer no such choice and do not follow the imputed matrix: they describe the data as annotated, which is the only way to go on answering *what was measured*.
- Every statistic computed on the imputed matrix counts drawn values as data: a test across a condition in which a feature was never measured compares two draw distributions. Check such a feature on the missing-value heatmap before believing it.

11.5 Normalization

Normalization is not on this page — it is the **Normalization** control on Data → Annotation, applied when the annotation is generated — but it is the other step that rewrites the values, and it runs before both filtering and imputation. Its values are None (the default), Median centering, Quantile normalization and VSN, and which one ran is stated in the annotation output. Left at None, sample loading and instrument drift are uncorrected unless the upload was already normalized upstream — MaxLFQ or PG.Normalised, say.

WHAT EACH ONE DOES

All three are global, over all sample columns at once; normalising within conditions would remove the differences the later tests look for.
Median centering works on the $\log_2$ scale, where a multiplicative loading difference is an additive offset: each sample's median is subtracted and the grand median of those medians added back. The sample-specific offsets can change fold changes; the shared constant added afterwards cancels in a comparison.
Quantile normalization sorts each sample's observed values, interpolates them onto a common grid with as many positions as the matrix has rows, and averages those curves into one reference distribution, off which every sample is read at its own quantile positions. Missing values neither are filled nor move a sample's ranks.
VSN is fitted on linear intensities and returns a generalised-log scale approaching $\log_2$ for large intensities, so it replaces the $\log_2$ transform rather than following it. NA is carried through.

VSN is delegated to the Bioconductor vsn package rather than reimplemented, so the option appears in the list only where Rscript is present *and* that package is installed in it; otherwise the list has three entries, and where R is there but the package is not, the message names the installation command. The annotation output records the package version that ran. Phosphoprotein data is never normalized at the phosphoprotein level, and the **Normalization** control is not shown there.

11.6 In the notebook

A notebook starting point, using the frames introduced in *The function reference*. These compact examples do not encode every UI option; use the plot's notebook ZIP, where available, for its saved settings and preprocessing recipe:

PYTHON

```
imputed = impute_values(data, meta)
fig = impute_values(data, meta, ret=2)
```

```r
imputed <- impute_values(data, meta)
fig <- impute_values(data, meta, ret = 2)
```

One function does both jobs, and ret chooses which: 0 returns the imputed matrix, 1 the missing-value comparison, 2 the figure above and 3 the after plot, so the page's three figures and its stored matrix all come out of the same call. Every other argument is a control on the page under the same name and at the same default — q, adj_std, sample_wise, seed, method, down_shift and width. A fixed seed supports repeatability within the same implementation, environment and input order; Python and R need not generate identical draws. The one thing missing from ret=2 is the cyan draw-area curve: the package plots the observed histogram and the normal fit, and the application adds the draw area over them.

11.7 Exporting

The diagnostic figures intentionally have no plot or notebook download. **Download Full Imputed CSV** writes the stored matrix as imputed_protein.csv or imputed_phospho.csv, using the parameters of the applied run rather than the ones currently in the fields, so what is downloaded is what the rest of the session is using. The table at the foot of the page is its first twenty rows.

11.8 Further reading

The vignette *Missing value handling* of the Bioconductor package DEP (Bioconductor 3.22) provides worked proteomics examples of filtering and different imputation assumptions. It is useful methodological background, not a claim that CoPYlot implements every DEP option or reproduces its random draws.

Behind this page

Function	What it produces
impute_values	The draw model in both packages: same parameters, aligned method and parameters; Python and R random draws need not be identical
impute_values_with_diagnostics	The same draw, returning also the before and after value sets, the draw components and any skipped samples
run_imputation	Runs it and stores the session's imputed matrix
imputed_dataframe	The imputed matrix alone, for the CSV download
normalize_data	Median, quantile and VSN normalization of a $\log_2$ matrix
vsn_normalize_linear	The call into Bioconductor vsn

12 Distributions

How the pooled observed intensities differ from a normal distribution.

AT A GLANCE

Where	Data → Distribution
Data levels	Protein, phosphosite, phosphoprotein
Draws	One Q-Q plot: every observed value of the annotated $\log_2$ matrix against the normal distribution.
Matrix	The annotated $\log_2$ matrix, always. There is no **Data Source** selector, so filtering and imputation do not change what is drawn; normalization does.
Sampling	Above 100,000 observed values a seeded random sample of 100,000 is plotted, and a warning says so.
Table	None.

AT PHOSPHOSITE LEVEL

The shared **Localization probability cutoff** selects sites across phosphosite views. At 0 it keeps all sites, including those with no probability; above 0 those unknowns are excluded. This is separate from completeness filtering and does not rerun normalization or imputation. An already collapsed phosphoprotein table has its own recorded collapse cutoff.

12.1 What the plot answers

The Q-Q plot compares ordered observed intensities with theoretical normal quantiles. An approximately straight pattern is consistent with a normal shape; curvature shows how the shapes differ. This is a visual diagnostic, not a pass/fail test of the analysis pipeline. An upward bend at the high end means the largest observations exceed the fitted normal reference. Check the scale, sample distributions and upstream missing-value handling before assigning a cause.

All sample columns are pooled. This mixes features with different abundances and samples from different conditions. It does not assess the residuals for each feature or the within-pair differences in a paired test, which are the relevant distributions for those parametric tests. Pearson correlation can be calculated without normal intensities; its inferential assumptions are a separate question.

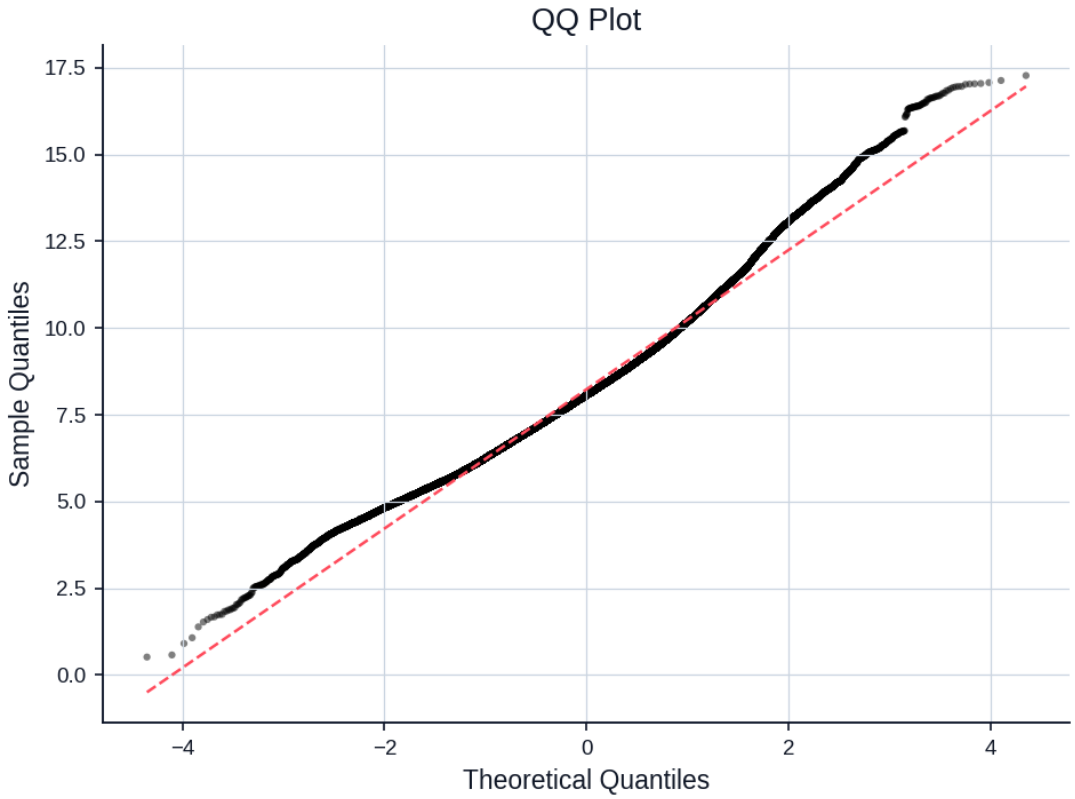


Figure 12.1. The page's own Q-Q plot on the example data, at protein level: sample quantiles of the $\log_2$ matrix, running from about 0 to 17.5, against the theoretical quantiles of the normal distribution, with the fit line through the plotted points. The cloud lies above the line at the left end and above it again at the right while the middle of the range sits on it, and there is no step or shelf anywhere along it.

12.1.1 Reading it

- One point, one observed value. Its horizontal position is the quantile a normal distribution would put it at, its vertical position the value itself.
- The red line: a least-squares straight line through the plotted points. It is the line the points would lie on if they were normal, fitted to them rather than assumed.
- The ends. Points above the line at the right and below it at the left are a distribution with heavier tails than normal; the reverse is lighter tails. A bend at one end only is a skew in that direction.
- A step or a shelf near the lower end is usually a floor in the data — a detection limit, a filled value, an imputation already carried in the upload — rather than a property of the biology.

The page carries a second panel, **QQ Norm Reference**, which is a fixed figure with its own explanatory text. It is part of the application, is the same whatever is loaded, and is there to be compared against the plot above.

> EASY TO GET WRONG
>
> A curve in the Q-Q plot is not automatically a problem. A proteomics matrix pools tens of thousands of features whose abundances span several orders of magnitude, and the mixture of them is not expected to be exactly normal; mild departure at both ends is the normal case. What is worth acting on is a departure in the middle of the range, a step, or two visibly separate straight segments — the last usually means the matrix contains two populations, for example samples normalized differently, or a level mixed with values that are ratios rather than abundances.

12.2 Controls

Controls on this page

Control	What it does	Default	Where
Dataset level	Protein, phospho or phosphoprotein. Only the levels actually uploaded are offered, and the choice is remembered for the session.	Protein	page
Random seed	Fixes which values are drawn when the matrix is larger than the sampling limit, so the same figure is reproduced exactly. Below the limit it changes nothing.	187	page
Refresh	Reloads the summary behind the page, including the sampling warning.	—	page

In the plot appearance panel

Control	What it does	Default	Where
Marker style	Size, colour, shape and opacity of the points.	small black dots, half opaque	appearance
Reference line	The fit line: whether it is drawn at all, and its colour, dash pattern and width.	shown, red, solid	appearance

WORTH KNOWING

The appearance panel is hidden until advanced mode is turned on (**Ctrl** + **Shift** + **E**), and is described in Part XI. This plot has no legend, no conditions and no sample labels, so the panel offers none of those cards here; the title, axis, font and background cards work as they do everywhere else.

12.3 How it is computed

HOW IT IS COMPUTED

The sample columns of the annotated $\log_2$ matrix are coerced to numbers and flattened into one vector, and every non-finite entry — NA and any infinity — is dropped. Nothing else is removed: a zero and a negative value are kept, which matters because a $\log_2$ intensity is negative wherever the linear intensity is below 1, and a phosphoprotein matrix collapsed with robust_site_median holds ratios centred on zero.
If more than 100,000 values remain, 100,000 of them are drawn without replacement, using a generator seeded with **Random seed**.
The remaining n values are sorted ascending and become the sample quantiles. The theoretical quantile of the i-th of them is

$$t_i = \Phi^{-1}\left(\frac{i - 0.5}{n}\right), \qquad i = 1, \ldots, n,$$

with Φ^{-1} the inverse standard normal distribution function. Each point is $(t_i, x_{(i)})$.

THE FIT LINE

The line is the ordinary least-squares fit of the sample quantiles on the theoretical quantiles — the straight line through *all* the plotted points, not a line through two chosen quantiles — drawn from the smallest theoretical quantile to the largest. Its slope is therefore an estimate of the standard deviation of the values and its intercept an estimate of their mean, and the ends of the cloud pull on it: a heavy lower tail tilts the line, which is why a departure is read as a shape rather than as a distance from a fixed reference.

12.3.1 Edge cases

- Conditions and replicates play no part. All samples are pooled, so a condition with one replicate is drawn exactly like any other.

- Missing values are not plotted and not replaced; they simply do not enter the vector.
- Fewer than two usable values gives one point and no fit line. No usable values at all is an error rather than an empty figure.
- Where the values are all equal the line is horizontal, which is the correct picture of a matrix with no spread.
- Because the page reads the annotated matrix, running filtering or imputation does not change this figure. Regenerating the annotation with a different **Normalization** does.

12.4 The three shapes

Beside the figure the page shows a reference panel: three quantile–quantile plots of data whose distributions are known, so that the shape on screen can be compared with a shape whose meaning is settled. They are the same three here.

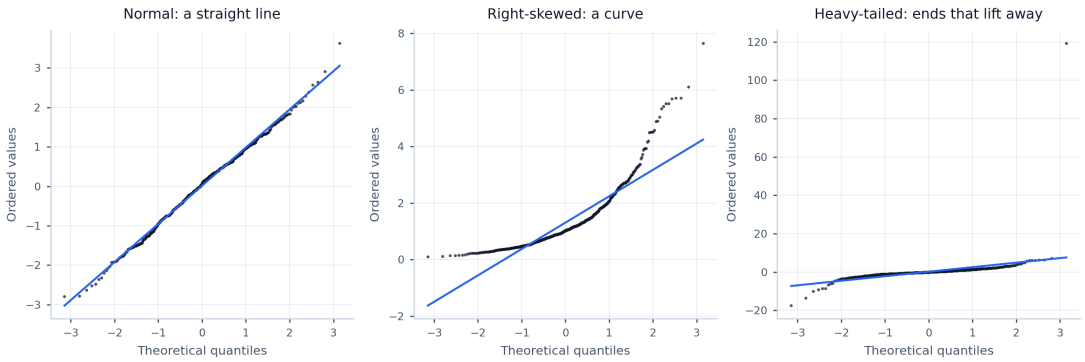


Figure 12.2. The three shapes a quantile–quantile plot takes. Left: values drawn from a normal distribution. Middle: values with a long right tail. Right: values from a distribution with heavy tails at both ends.

An approximately straight pattern is consistent with a normal shape for the pooled observations. It does not prove normality or validate every downstream test. Log transformation often reduces right skew, but neither transformation nor normalization guarantees a normal distribution.

Asymmetric curvature can indicate skewness. A long upper tail is one possible pattern. Verify the declared scale, but do not infer that it is wrong or that all p-values are optimistic from this plot alone. A mixture of valid feature distributions can be curved.

Ends that lift away from the line while the middle stays on it means heavy tails: more extreme values, in both directions, than a normal distribution would produce. In a proteomics matrix this is ordinary in moderation — the dynamic range is real, and a handful of very high and very low values is the expected consequence. It matters when it is not moderate: extreme values pull means and standard deviations, so a fold change or a t-statistic computed over a

feature whose values sit in such a tail rests on one or two measurements. The opposite, ends that bend *towards* the line, means the values are truncated — a quantitation floor, a filter applied upstream, or a report that writes a fixed value in place of a low measurement.

> **WORTH KNOWING**
>
> Read the middle of the plot first and the ends second. A departure in the central mass of the points says something about the whole matrix; a departure confined to the last few points on either side describes a handful of features and is usually not worth acting on.

12.5 In the notebook

A notebook starting point, using the frames introduced in *The function reference*. These compact examples do not encode every UI option; use the plot's notebook ZIP, where available, for its saved settings and preprocessing recipe:

PYTHON

```
fig = qqnorm_plot(data, meta)
```

R

```
fig <- qqnorm_plot(data, meta)
```

One call and one argument beyond the frames: `seed` defaults to 187, the same value the page's **Random seed** starts at, so the sampling that kicks in above 100,000 observed values draws the same subset in a notebook as on screen. What the packages do not carry is the rest of the page — there is no reference panel of the three shapes beside the figure, and no `distribution_summary` counts under it, so the comparison the page sets up has to be made against this chapter instead.

12.6 Exporting

This is an in-app diagnostic. It intentionally has no plot download or notebook-download button. The notebook function above remains available for an independently written analysis; it is not a code export from this page.

12.7 Further reading

NIST: probability plots explains comparison with theoretical quantiles. A pooled intensity plot is not a substitute for checking the assumptions of a particular statistical model.

Behind this page

Function	What it produces
qqnorm_plot	The Q-Q plot in both packages, including the 100,000-value sampling and its seed
qqnorm_plot_data	The points and the fit line as numbers
distribution_qqnorm_plot	The figure this page shows
distribution_summary	The counts behind the page: observed values, missing values, rows with and without a missing value, and the sampling warning

13 Translating identifiers

Converting the identifiers in one column of a loaded table into another identifier system, and writing the result back as a new column.

AT A GLANCE

Where	Data → ID Translator
Data levels	Protein and phosphosite
Works on	One column of the uploaded table, chosen by name
Systems	Eighteen, from HGNC and UniProt. Any of them can be the input and any of them the output.
Source	A lookup table bundled with the application, BioNamesDB.txt: a prepared merge of UniProt and HGNC content, about 107,000 mappings.
Writes	A new column on the active dataset, plus a downloadable copy of the whole table.
In viewer mode	Not available: the page is not shown, and running or downloading a translation is refused.

13.1 What it translates

An uploaded report names its features in whatever system the search engine used — a UniProt accession, most often — while the rest of the world asks for gene symbols. This page does that conversion on the table itself: one column in, one new column out, beside the original. Nothing is replaced and no row is dropped, so a translation that fails is visible as an NA in the new column rather than as a shorter table.

Eighteen systems are offered, and each of them can be either side of the translation:

- HGNC: HGNC ID, HGNC Symbol, HGNC Name, HGNC Alias Symbol.
- UniProt: UniProt Accession, UniProt Name, UniProt Gene, UniProt Description, UniProt Secondary Accession, and UniProt All Accessions, which is the primary accession and the

secondary ones treated as one set.

- Other cross-references carried in the same table: Ensembl, VEGA, ENA, RefSeq, CCDS, PubMed, Enzyme Commission (EC), RNAcentral.

The mapping is whatever the bundled table holds. It is a snapshot, prepared from UniProt and HGNC downloads and shipped with the application; it is not fetched from anywhere at run time, so translation works with no network and gives the same answer on every machine running the same release. A record absent from that table cannot be translated here however current it is elsewhere.

> **NOTE**
>
> The release the snapshot was built from is not recorded in this version, so the manual cannot state it. Record the CoPYlot version and retain the exported mapping with the analysis. For time-sensitive identifiers, check the current UniProt or HGNC record independently rather than assuming this bundled snapshot is current.

13.2 Translating a column

1. Choose **Dataset level**. The card beside it names the file currently loaded at that level and its size, so it is clear which table is about to be changed.
2. Choose **Column to translate**. Every column of the uploaded table is offered by name, including ones added by an earlier translation.
3. Choose **Translate to**: the system the new column will be written in.
4. Leave **Auto-detect input database from the selected column** on, or turn it off and set **Translate from** by hand.
5. Press **Translate and Preview**. The four cards report which input system was used, which output system, how many rows were translated out of how many, and the name of the appended column; the first fifty rows are shown below them.

Pressing that button also writes the translated table back as the active dataset for that level, which is why the appended column is available to everything downstream. It does not re-run the annotation: the filtered and $\log_2$ matrices already computed still have the columns they had. Generate the annotation again to carry the new column into them.

Download Translated Table runs the same translation and gives the whole table as a file without writing anything back.

Controls on this page

Control	What it does	Default	Where
Dataset level	Protein or phospho. Only the levels actually uploaded are offered.	Protein	page
Column to translate	Which column of the uploaded table holds the identifiers.	the first column	page
Translate to	The output system, one of the eighteen.	HGNC Symbol	page
Translate from	The input system. Disabled, and ignored, while auto-detection is on.	HGNC Symbol	page
Auto-detect input database from the selected column	Works the input system out from the values themselves.	on	page
Translate and Preview	Translates, shows the summary and the first fifty rows, and applies the result to the active dataset.	—	page
Download Translated Table	The whole translated table as a file. Does not change the active dataset.	—	page

13.3 How it is computed

THE LOOKUP

The bundled table is read from disk once and then held in memory. For the chosen pair of systems an index is built from it: in every row, each value of the input column is paired with every value of the output column, and rows where either side is empty contribute nothing. The index is therefore many-to-many by construction — one identifier may have several translations, and several identifiers may share one.

ONE CELL

A cell is split on semicolons, its parts trimmed, and repeats within the cell dropped, so P12345;P12345;Q9Y6K9 contributes two identifiers. Each is looked up, and the union of everything found is sorted and written back into one cell, joined by semicolons. A cell that is empty, or whose identifiers are all absent from the index, becomes the literal string NA. Translated Rows on the summary counts the rows that are *not* that

string, so it is the number of rows with at least one translation, not the number of identifiers matched.

AUTO-DETECTION

Up to 5000 non-empty values are taken from the chosen column and split into at most 20,000 identifiers. For each of the eighteen systems the number of those identifiers that appear anywhere in that system's column of the lookup table is counted, and the system with the highest count wins. A tie is resolved in favour of UniProt All Accessions if it is among the leaders, and otherwise by the order the systems are listed in.

If nothing matches at all and no input system was set by hand, the translation stops and says so. If nothing matches but **Translate from** had been set before auto-detection was switched on, that manual choice is used and a warning records it.

EASY TO GET WRONG

Auto-detection overrides **Translate from** whenever it succeeds, so a manual choice made with the checkbox left on is not what runs. Turn the checkbox off to hold the input system fixed — which is what to do when a column contains identifiers of two systems, since detection then sees only the more common one.

13.3.1 The appended column

The new column is named after the output system, not after the source column, so translating twice into the same system overwrites the first result rather than accumulating columns.

Column names written, by output system

Column	What it holds
GeneNames	HGNC Symbol — and also UniProt Gene, which writes the same column name
HGNCIDNames	HGNC ID
HGNCNames	HGNC Name
HGNCAliasSymbolNames	HGNC Alias Symbol
UniProtAccessionNames, UniProtNames, UniProtDescriptionNames, UniProtSecondaryAccessionNames, UniProtAllAccessionsNames	the five remaining UniProt systems

Column names written, by output system (continued)

Column	What it holds
EnsemblNames, VEGANames, ENANames, RefSeqNames, CCDSNames, PubMedNames, ECNames, RNAcentralNames	the cross-reference systems, in the same pattern

One further column can appear unasked. If the table has no GeneNames column and the output system is not one of the two that write it, the same identifiers are translated a second time into HGNC symbols and added as GeneNames — provided at least one of them maps — so that the analyses keyed on gene names can run. A warning on the summary says when this happened.

13.3.2 Edge cases

- An identifier that maps to several targets keeps all of them in one cell, separated by semicolons and sorted. Downstream pages that expect one name per row see the whole string.
- An identifier that maps to none leaves NA, and the row stays in the table.
- If the bundled lookup table cannot be found, the page reports that and the locations it searched are written to the server log.

13.4 Exporting

The download is the whole table with the new column, not just the mapping: a .tsv or .txt upload comes back tab-separated with the same extension, everything else comes back as CSV, and the file is named after the upload with _translated before the extension. It is generated from the current settings on the page, so it can be taken without applying anything to the session.

Behind this page

Function	What it produces
run_id_translation	Translates and applies the result to the active dataset
export_id_translation	The same translation as a file, applying nothing
translated_dataframe	The translated table, and the input system that was used

Behind this page

Function	What it produces
autodetect_input_db	Works the input system out from the values
build_index	The many-to-many index between two systems
translate_cell	One cell: split, look up, join, or NA

14
Condition colours

Setting the colour of each condition once, for every figure the session draws.

AT A GLANCE

Where	Data → Colors
Data levels	Protein, phosphosite, phosphoprotein — each with its own palette
Needs	An annotation. The conditions listed here are the conditions it defines.
Applies to	Every plot that groups by condition, at that level
Overridden by	The per-plot condition colours in the plot appearance panel
Persistence	Held for the session, saved as each colour is changed, written into a save state and restored with it.

14.1 Why a palette rather than per-plot colours

A figure set is read as a set. If the treated condition is orange on the coverage plot and blue on the PCA, every reader of the figure has to re-learn the key on each panel, and at least one of them will misread a result. Setting the colour here costs less than that: one colour per condition, used by every plot at that level for the rest of the session, including the ones drawn before the colour was chosen — they are redrawn when the palette changes.

Each dataset level keeps its own palette, because protein and phosphosite levels may have different conditions. Where they have the same number of conditions, **Copy Colors** carries one palette over to the other level rather than making them match by hand.

14.2 Setting the palette

1. Choose **Dataset level**. The conditions of the annotation at that level are listed, one row each, in the order the annotation gives.
2. Set a colour either from the swatch, which opens the operating system's colour picker, or by typing a hexadecimal value such as #1F77B4 into the field beside it.
3. There is no save button. Each colour is written as soon as it is complete, and the plots follow immediately.
4. **Reset Default Palette** puts every condition back on the built-in cycle and saves that.
5. With more than one level loaded, choose a level under **Copy colors to dataset level** and press **Copy Colors** to give it the same colours.

Controls on this page

Control	What it does	Default	Where
Dataset level	Which level's palette is being edited. Only the levels actually uploaded are offered.	Protein	page
Colour swatch	The condition's colour, from the system colour picker.	the built-in cycle	page
Hex field	The same colour as #RRGGBB. Anything that is not a complete six-digit value is held while it is being typed and reverts when the field is left.	the built-in cycle	page
Reset Default Palette	Writes the built-in cycle back for every condition at this level, and saves it.	—	page
Copy colors to dataset level	The level **Copy Colors** will write to. Shown only when more than one level is loaded.	the first other level	page
Copy Colors	Copies this level's colours to that level by position: first condition to first condition, and so on.	—	page

14.3 How a colour is chosen

RESOLUTION ORDER

When a plot needs a colour for a condition it asks three sources in turn, and takes the first answer:

- the condition colours set for *that plot* in the appearance panel, which are part of that plot's own settings;
- this page's palette for that level, if it holds an entry for that condition name;
- otherwise a generated colour, by the condition's position in the list.

The generated colours are a fixed sequence of twenty, beginning #1F77B4, #FF7F0E, #2CA02C, #D62728; beyond twenty conditions they continue by rotating the hue in steps of the golden ratio, alternating saturation and brightness so that neighbours stay distinguishable.

STORAGE

A palette is a map from condition name to a #RRGGBB string, one map per dataset level, held for the browser session. Values are normalised to upper case, and anything that is not six hexadecimal digits is discarded rather than stored. Writing a palette invalidates the cached figures at that level, which is what makes the plots redraw. The palettes of all three levels are written into a save state and come back with it. Reloading the application in the browser starts a new session and clears them, as it clears the uploaded data.

EASY TO GET WRONG

The list on this page shows a repeating cycle of ten colours, while a plot that has not been given a saved palette continues to twenty distinct ones. With eleven or more conditions the swatch shown here for condition eleven onwards is therefore not the colour the plot is using — until something is saved. Pressing **Reset Default Palette**, or setting any one colour, writes the whole list and the plots follow it from then on.

14.3.1 Edge cases

- The palette is keyed by condition name. Renaming a condition in the annotation leaves its old entry behind and the renamed condition falls back to a generated colour, so set it again after a rename.
- **Copy Colors** works by position and refuses when the two levels have different numbers of conditions; it says so rather than copying part of the palette.
- Renaming a condition in the appearance panel changes only a label on one figure. It does not change what this page is keyed on.
- With no annotation there are no conditions and the list says so. Generate the annotation first.

WORTH KNOWING

A colour that separates well on screen may not survive print or a colour-blind reader. The safest check is to look at the figure in greyscale — if two conditions become the same grey, they differ only in hue — and to keep the pairs red/green and blue/purple apart in lightness as well as hue. The built-in cycle is designed for this, which is a reason to change as few of its entries as the figure needs.

14.4 Exporting

There is nothing to download here: the palette is not a file, it travels in the save state and in the figures that use it. To hand the colours to somebody else, save a state; to reuse them at another level, use **Copy Colors**.

Behind this page

Function	What it produces
get_condition_palette	The palette stored for one level
set_condition_palette	Validates and stores it, and invalidates the cached figures
build_condition_color_map	The colour every plot actually uses: stored entry, else generated by position
reset_condition_palettes	Clears the palettes of a session

15
Saving and restoring a state

Writing the analysis to a file, and reading it back into a session.

AT A GLANCE

Where	Data → Restore
The file	A compressed .copylotstate file, named copylot_state_ and the date and time it was written.
Saves	Annotations and their metadata tables, filter and normalization settings, the imputation and its matrix, the condition palettes, the metadata profiles, which package and profile were active, and the state of the interface.
Does not save	The measured data. Datasets are recorded by filename and path and re-read from disk on restore.
Never writes	Your uploaded files, or the settings of the installation. A restore only reads them.
Two kinds of save	Ordinary, which records where the datasets are, and portable, which does not.

15.1 What a save state is

A save state records the analysis setup: decisions made about a dataset — how the samples were annotated, what was filtered, what was normalized, what was imputed, which colours the conditions have, which page was open — plus stored resources such as the realised imputed matrix. It can therefore be large and can contain scientific data. Original measurement files are still required on restore, either at their recorded locations or supplied as replacements. Keep them alongside the state.

15.2 Saving

Three buttons, under **Save State**:

Controls on this page

Control	What it does	Default	Where
Save State… / Update Save State	Writes the state. Once a save file has been chosen in this session, its name is shown above the buttons as the current save target and the button writes to it again.	—	page
Save State As…	Writes the state to a file chosen now, which then becomes the save target.	—	page
Save Portable State…	The same state with every local dataset path left out.	—	page

The difference between the last two matters. An ordinary state records the absolute path of each dataset file, so restoring it on the same machine finds the data without being asked. A portable state deliberately does not, so those local dataset paths are omitted — and every dataset has to be supplied by hand when it is restored. Save portable when the file is going to somebody else; save ordinary when it is for yourself tomorrow. Portable does not mean anonymous: filenames, labels, metadata and stored values may still identify the study. Review them before sharing.

15.3 Restoring

1. Select the file in the field under **Restore State**. Where the platform allows it, **Open Save File…** opens a proper file dialogue instead.
2. Press **Restore**.
3. If any dataset the file refers to cannot be found, nothing is changed: a **Missing Files** panel lists each one with its package, its level, the path that was recorded and the filename it had. Choose a replacement file for each, then press **Restore** again.
4. On success the page reports how many datasets and how many annotations were restored, and lists any warnings.

Controls on this page

Control	What it does	Default	Where
Save state file to restore	The .copylotstate file.	none until chosen	page

Controls on this page (continued)

Control	What it does	Default	Where
Open Save File...	A native file dialogue, where the platform provides one.	—	page
Restore	Runs the restore. Disabled until a file is chosen, and while a missing dataset has no replacement.	—	page
Replacement file	One per missing dataset, in the **Missing Files** panel.	none	page

15.3.1 What comes back

- The datasets, re-read from their recorded paths or from the replacement files supplied, for every package the state held.
- The metadata profiles and their tables, and which profile was active.
- For every annotation: the sample-to-condition table, the filter settings, the normalization, the $\log_2$ declaration, the imputation settings and the imputed matrix, and the phosphoprotein aggregation mode.
- The condition palettes of all three levels.
- The interface: the section and tab that were open, the right-hand panel, whether advanced plot mode was on and the appearance settings of each plot, and the per-page settings each panel keeps — selected levels, data sources, seeds and the rest.

15.3.2 What does not

- The measured values. Only filenames and, in an ordinary state, paths are recorded.
- The $\log_2$ and filtered matrices. They are recomputed from the re-read dataset and the restored filter and normalization settings, which is why a restore can take a moment on a large table. The imputed matrix, which could not be recomputed identically without its draw, is carried in the file.
- Anything belonging to the installation rather than to the analysis: the settings pages, access configuration and viewer configuration are not written into a save state and are not touched by restoring one.
- Your uploaded files. A restore reads them and never writes to them, so a state restored against the wrong data changes the analysis, not the data.

15.4 How it is applied

HOW IT IS COMPUTED

The file is a gzip-compressed JSON document carrying a format tag, the application version that wrote it, the time it was written and the portable flag. Restoring reads it, then checks every dataset it refers to before touching anything: if one is missing and no replacement was supplied, the whole restore is refused and the missing files are reported.
When everything needed is present, the session's datasets, annotations, uploaded metadata, condition palettes and peptide metadata are cleared, and then rebuilt from the file. The annotation for each level is recomputed from the restored dataset and its stored settings. If anything fails part-way, every one of those stores is put back exactly as it was before the restore began, and the error is reported: a failed restore never leaves half a session behind.

LIMITS

A state file larger than about 64 MB, or one that expands beyond about 256 MB, is refused, as is a replacement set beyond the configured number of files or total size. Those are the defaults; whoever runs the instance can raise them.

15.4.1 Failure modes

- *An empty file, or one that is not this format.* Reported as such: the state must carry the `copylot_state_v1` tag.
- *A missing dataset.* The **Missing Files** panel, as above. A portable state always lands here, for every dataset it holds.
- *A dataset that could not be restored.* Its annotation is skipped with a warning naming the package, profile and level, rather than being restored against nothing.
- *A stored imputed matrix whose sample columns no longer match the annotation.* Discarded with a warning, and the imputation is marked as not applied; run imputation again.
- *A state written by a different release.* Restored, with a warning giving both versions. Settings that release did not have take the current defaults, which is stated in a notice on every restore.

EASY TO GET WRONG

A state written before a breaking change restores the same settings, and those settings can now mean something different. The concrete case is fold-change orientation: the difference is now computed as condition 1 minus condition 2, so a volcano regenerated from an older state selects the same two conditions and comes out with every sign re-

versed against the figure that state was saved beside. Magnitudes, *p*-values and counts are unaffected — only the direction is. When a restore reports a version difference, regenerate the figures rather than trusting the ones already in the manuscript.

15.5 Exporting

The save state *is* the export: there is nothing else to download from this page. Keep it beside the data it refers to, since an ordinary state is only as good as the paths it records, and prefer a portable state plus the raw files when the analysis is being handed over.

Behind this page

Function	What it produces
export_state_blob	Builds the save state, with or without local paths
decode_restore_blob	Reads and validates the file
inspect_missing_files	The datasets a state refers to and cannot find
apply_restore_state	Applies the state, or rolls the session back unchanged
migrate_restore_ui_state	The disclosed migrations of older interface settings

PART III

Quality control



16
The coverage plot

How many features were quantified in each sample, and how far each sample falls short of the dataset as a whole.

AT A GLANCE

Where	Quality control → Coverage
Data levels	Protein, phosphosite, phosphoprotein, peptide, precursor
Needs	An uploaded dataset and its annotation. Conditions come from the annotation; sample order follows it.
Two modes	**Summary Type** off draws one bar per sample; **Summary Type** on draws one bar per condition, with the standard deviation and the individual samples over it.
Reference line	The number of features quantified in at least one sample — the ceiling the bars are measured against.
Table	Yes, below the plot, downloadable as CSV.

AT PHOSPHOSITE LEVEL

The shared **Localization probability cutoff** selects sites across phosphosite views. At 0 it keeps all sites, including those with no probability; above 0 those unknowns are excluded. This is separate from completeness filtering and does not rerun normalization or imputation. An already collapsed phosphoprotein table has its own recorded collapse cutoff.

16.1 What the plot answers

The coverage plot shows how many features were quantified in each sample. It is a quick way to find differences in detection depth before interpreting a comparison. Both the absolute counts and differences between comparable runs matter: equal bars can still represent consistently shallow coverage.

For example, 4,000 measured proteins in one run against about 6,000 in its replicates is worth investigating. It does not establish that the run failed or that the missing proteins are biologically absent. Check the missing-value pattern, intensity distributions and experimental design alongside it.

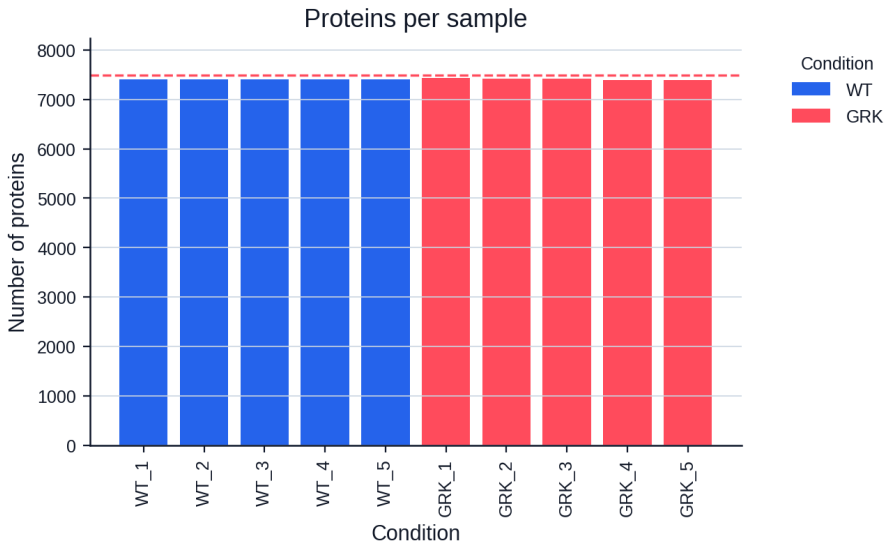


Figure 16.1. Per-sample mode, at protein level, on the example dataset: five WT and five GRK samples, and the reference line across the top.

16.1.1 Reading it

- **A** One bar, one sample. Its height is the count of non-missing values in that column.
- **B** The red dashed reference line: the number of features quantified in at least one sample. No bar can exceed it, and the distance to it is what that sample missed.
- **C** Bar colour: the condition the sample belongs to, taken from the annotation and shared with every other plot in the session.
- **D** The label under each bar: condition_n, numbered within the condition rather than across the experiment. With **Toggle IDs** on, the original sample identifier is added on a second line.

> **EASY TO GET WRONG**
>
> A low bar is not by itself a failed sample. The count depends on everything upstream of it — injection amount, gradient length, the search settings — so a bar that is low in every condition alike is usually a property of the run, while a single low bar inside a set of replicates is the one worth investigating. Check it against the missing-value heatmap before excluding anything: a sample that is low but complete in the features it does have behaves very differently from one whose missing values are scattered.

16.2 Controls

Controls on this page

Control	What it does	Default	Where
Dataset level	Which dataset to draw: protein, phosphosite, phosphoprotein, peptide or precursor. Only the levels actually uploaded are offered; phosphoprotein appears once a collapse has been run.	Protein	page
Summary Type	Off: one bar per sample. On: one bar per condition, showing the mean, the standard deviation and the individual samples.	off	page
Toggle IDs	Adds the original sample identifier under each label.	off	page
Toggle Text	Prints the count on each bar.	off	page
Random Seed	Only in summary mode. Fixes the horizontal jitter of the individual sample points so that the same figure is reproduced exactly.	187	page
Abundance Column	Peptide and precursor levels only. Which quantity column of the report to count.	none until chosen	page
Total Line	Draws the reference line at the number of features quantified in at least one sample. The y axis always leaves the same 10 % headroom: above the line when it is drawn, above the tallest bar when it is not.	on	page
Width, Height, DPI	The size of the *downloaded* figure. The browser scales the rendered figure to fit the page.	20 cm × 10 cm, 300	page
Download format	PNG, JPG, WEBP, SVG or PDF.	PNG	page

In the plot appearance panel

Control	What it does	Default	Where
Title	Shown or hidden, and its text, size and font. The title is written from the data level: Proteins per sample and its equivalents.	shown	appearance
Legend	Draws the condition legend beside the plot.	shown	appearance
Value labels	Size of the numbers printed on the bars by **Toggle Text**.	9	appearance

In the plot appearance panel (continued)

Control	What it does	Default	Where
Reference line	The line's colour, dash pattern and width. Offered only while **Total Line** is on; whether it is drawn is switched there.	red, dashed	appearance
Condition colours	One colour per condition, shared with every other plot in the session.	the condition palette	appearance

WORTH KNOWING

The second table is the plot appearance panel, which is hidden until advanced mode is turned on (**Ctrl** + **Shift** + **E**). Everything it contains is described in Part XI; only the entries particular to this plot are repeated here.

16.3 How it is computed

HOW IT IS COMPUTED

Every sample column is reduced to a 0/1 matrix: 1 where the value is present, 0 where it is NA. The height of a bar is the column sum of that matrix, so for sample j over m rows

$$\text{count}_j = \sum_{i=1}^{m} \mathbf{1}\left[x_{ij} \neq \text{NA}\right].$$

Only NA counts as missing. A value of zero is a measured zero and is counted as present, which matters for reports that write 0 rather than leaving a cell empty. Samples are renamed condition_n in the order the annotation gives, and the bars are drawn grouped by condition rather than in file order.

SUMMARY MODE

The per-sample counts are computed exactly as above and then aggregated within each condition: the bar is the mean of the counts of the samples of that condition, the error bar is their standard deviation, and each sample is additionally drawn as a point over the bar. With one replicate in a condition there is no spread and no error bar is drawn.

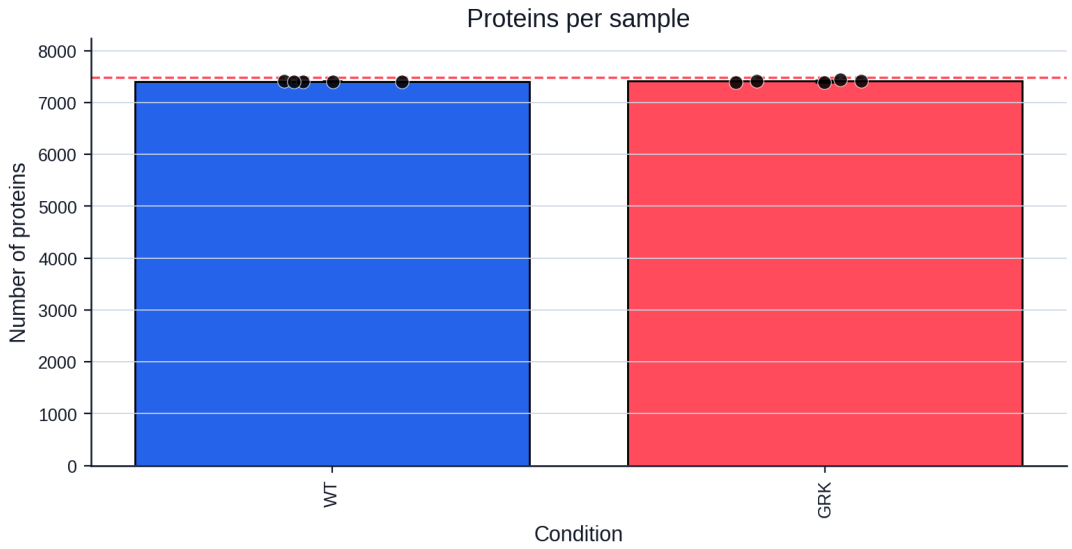


Figure 16.2. Summary mode on the same data: one bar per condition, the standard deviation as an error bar, and the individual samples over it.

THE REFERENCE LINE

The red dashed line is the number of features quantified in *at least one* sample: the size of the union of the bars, not the row count of the matrix. The two differ, because a feature can be identified — a real sequence, a localized site, an assigned protein — and still carry no quantity in any run. Such a row occupies the matrix but appears in no bar, and counting it would put the line above a ceiling no sample could reach. It is the same number as the All samples row of the table below.

NOTE

The same figure drawn in a notebook sits slightly differently: the package function coverage_plot still puts its reference line at the row count of the matrix it is given, so on a dataset with identified but unquantified rows the notebook line is a little higher than the one on screen. The two numbers differ only by the identified-but-unquantified rows, which the table under the plot states on its own row, so a figure from either source can be reconciled with the other.

16.3.1 Edge cases

- A sample with no values at all draws a bar of height zero rather than disappearing from the axis, so the gap is visible.
- A condition with a single replicate gets a bar and a point in summary mode, but no error bar.
- Peptide and precursor levels come from the long-format report rather than a wide matrix, and are counted after the chosen abundance column has been pivoted; until a column is

chosen, no plot is drawn.

- The page always counts the dataset as annotated, before filtering and before imputation. There is no data-source selector here, and that is deliberate: coverage describes what was measured, and counting an imputed matrix would raise every bar to the reference line by construction.

16.4 The table below the plot

Columns in this table

Column	What it holds
Sample	The sample, labelled condition_n as on the plot.
Number	Features quantified in that sample — the height of its bar.
Original_Name	The name the sample carried in the uploaded file. Present when original names are requested.

Two further rows close the table. All samples is the number of features quantified in *at least one* sample, which is the size of the union of the bars rather than their sum. Identified, not quantified appears only when the matrix contains rows that are missing in every sample, and gives their count.

In summary mode the table is replaced by one row per condition, with the mean, minimum, maximum and standard deviation of the per-sample counts of that condition, rounded to two decimals.

16.5 In the notebook

A notebook starting point, using the frames introduced in *The function reference*. These compact examples do not encode every UI option; use the plot's notebook ZIP, where available, for its saved settings and preprocessing recipe:

```python
fig = coverage_plot(data, meta, id=False, data_scale="log2")
fig = coverage_plot_summary(data, meta, data_scale="log2")
```

```r
fig <- coverage_plot(data, meta, id = FALSE, data_scale = "log2")
fig <- coverage_plot_summary(data, meta, data_scale = "log2")
```

id=True adds the original sample identifier under each label, as **Toggle IDs** does on the page. The reference line is drawn at the row count of the frame handed in rather than at the number of features quantified anywhere, which is the one difference from the figure on screen.

16.6 Exporting

The download reproduces the plot at the width, height and resolution set on the page. The browser scales the rendered figure to fit the page. Choose PDF for a figure that will be scaled or edited, PNG for a slide. The table underneath downloads separately as CSV.

Behind this page

Function	What it produces
coverage_plot	Per-sample coverage for wide-format data: protein, phosphosite and phosphoprotein
coverage_plot_pep	Per-sample coverage from the long-format report: peptide and precursor
coverage_plot_summary	The condition-level summary, with spread and individual samples
prepare_coverage_df	The per-sample table under the plot
prepare_coverage_summary_df	The condition-level table

17
The intensity histogram

The distribution of intensities in each condition, drawn as one curve per condition on a shared axis.

AT A GLANCE

Where	Quality control → Histogram
Data levels	Protein, phosphosite, phosphoprotein
Needs	An uploaded dataset and its annotation. The conditions, and which samples belong to each, come from the annotation.
Data source	The data as loaded, on the $\log_2$ scale. There is no selector: quality control describes what the instrument produced, before filtering or imputation.
One curve	One condition. The value drawn for a feature is its mean across the replicates of that condition.
Table	None.

AT PHOSPHOSITE LEVEL

The shared **Localization probability cutoff** selects sites across phosphosite views. At 0 it keeps all sites, including those with no probability; above 0 those unknowns are excluded. This is separate from completeness filtering and does not rerun normalization or imputation. An already collapsed phosphoprotein table has its own recorded collapse cutoff.

17.1 What the plot answers

Every quantitative comparison in the tool assumes the conditions sit on the same intensity scale. This page is where that assumption is checked: the curves are drawn on one axis, so a condition that is shifted, broader or bimodal is visible against the others rather than in isolation.

Compare both shape and position. A second mode, shoulder or tail can reflect sample com-

position, processing, technical effects or biology. There is no requirement that a valid proteomics distribution be a single symmetric hump, and this across-feature distribution is not the normality assumption of a feature-wise t-test.

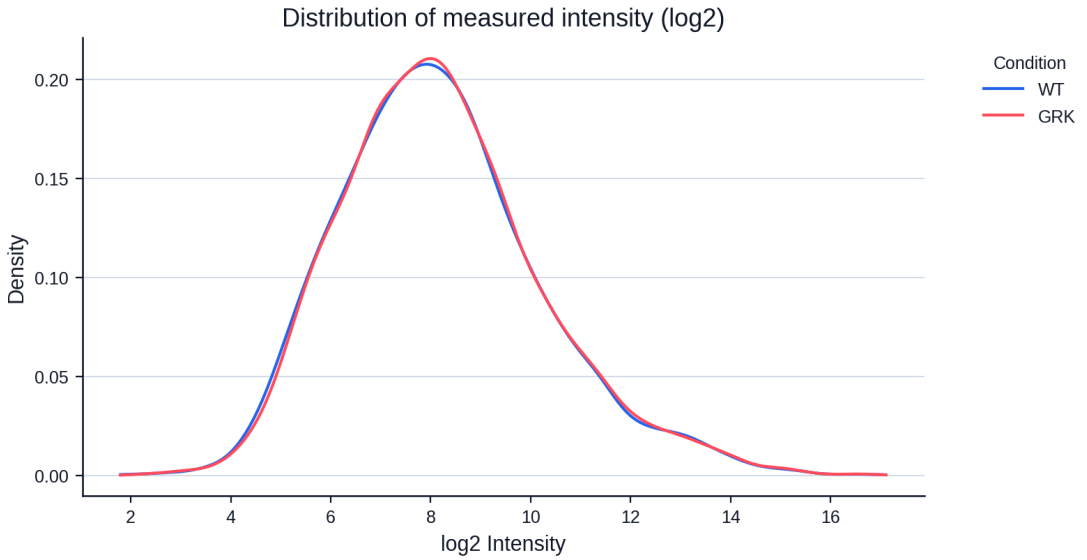


Figure 17.1. Protein level on the example dataset: one curve for the five WT samples and one for the five GRK samples, on the shared $\log_2$ intensity axis. The two lie almost exactly on top of one another — a single hump peaking near 8 with a long right shoulder — which is what two conditions on the same scale look like.

17.1.1 Reading it

- One curve, one condition. Its colour is the condition colour shared with every other plot in the session.
- The x axis: $\log_2$ intensity. At phosphoprotein level collapsed with `robust_site_median` it reads `log2 Ratio to Site Median` instead, because those values are ratios and not abundances.
- The y axis: density. Each curve integrates to one, so a condition built from fewer features is not drawn shorter — the heights are comparable, the counts behind them are not.
- The overlap. Curves that lie on top of one another share a scale; a curve displaced sideways from the rest does not.

> **NOTE**
>
> The curve is a kernel density estimate, not a bar histogram. It is smooth everywhere, including where no feature lies, so a thin tail is the estimator spreading a handful of points rather than evidence of features at those intensities. A condition with fewer than two values, or with no spread at all, cannot be smoothed and is drawn as a 20-bin step histogram instead.

EASY TO GET WRONG

A displaced curve is a difference in how much material was measured at least as often as it is a difference in biology. Injection amount, digestion efficiency and instrument response all move a whole condition sideways, and none of them is a result. Read the shift here against the boxplot, which shows the same distributions per sample, and against the coverage plot: a condition that is both lower and thinner has a loading problem, while one that is lower but equally wide has been scaled.

17.2 Controls

Controls on this page

Control	What it does	Default	Where
Dataset level	Which dataset to draw: protein, phosphosite or phosphoprotein. Only the levels actually uploaded are offered; phosphoprotein appears once a collapse has been run. The selector sits at the top of the page and is shared by every quality-control tab.	Protein	page
Width (cm), Height (cm), DPI	The size of the *downloaded* figure. The browser scales the rendered figure to fit the page. Width and height are held between 1 and 60 cm, DPI between 72 and 600, and a very large figure has its resolution reduced so the image stays under 24 megapixels.	20 cm × 10 cm, 300	page
Download format	PNG, JPG, WEBP, SVG or PDF.	PNG	page

In the plot appearance panel

Control	What it does	Default	Where
Title	Shown or hidden, and its text, size, weight and colour. The title is written by the plot: Distribution of measured intensity (log2).	shown	appearance
Legend	Shown or hidden, and its title, size, colour, frame and position. Its entries are the conditions.	shown	appearance

In the plot appearance panel (continued)

Control	What it does	Default	Where
Condition names, Condition colors	One display name and one colour per condition, for the figure only. The colours are shared with every other plot in the session.	the condition palette	appearance

WORTH KNOWING

The second table is the plot appearance panel, which is hidden until advanced mode is turned on (**Ctrl** + **Shift** + **E**). Everything it contains is described in Part XI; only the entries particular to this plot are repeated here.

17.3 How it is computed

HOW IT IS COMPUTED

The matrix is the data as loaded, on the $\log_2$ scale. Non-numeric and non-finite entries are treated as NA.

For every feature i and every condition c, the value drawn is the mean over the samples of that condition that carry a value:

$$\bar{x}_{i,c} = \frac{1}{|O_{i,c}|} \sum_{j \in O_{i,c}} x_{ij}, \qquad O_{i,c} = \{\, j \in S_c \,:\, x_{ij} \neq \mathrm{NA} \,\}.$$

A feature missing in every sample of a condition contributes nothing to that condition's curve and is dropped from it.

The resulting vector is smoothed with a Gaussian kernel density estimate and evaluated on 1000 equally spaced points spanning the smallest to the largest value over *all* conditions, so every curve is drawn on the same grid and the shapes can be compared directly.

WHEN THE ESTIMATE CANNOT BE FITTED

A condition with fewer than two finite means, or whose means are all equal, has no bandwidth to smooth with. That condition alone is drawn as a 20-bin outline histogram, normalised to a density so that it shares the axis with the curves beside it.

17.3.1 Edge cases

- A condition with one replicate is drawn from that sample's own values; the mean over one observation is the observation.
- A feature missing in some replicates of a condition is averaged over the ones it has, so curves can rest on different numbers of features. Nothing on the figure says how many; the coverage plot does.
- If every value in the dataset is identical, the shared axis would have zero width, so it is widened by one unit either side.
- With no numeric value anywhere in the matrix no figure is drawn and the page reports that instead.

17.4 In the notebook

A notebook starting point, using the frames introduced in *The function reference*. These compact examples do not encode every UI option; use the plot's notebook ZIP, where available, for its saved settings and preprocessing recipe:

```python
fig = histo_int(data, meta)
```

```r
fig <- histo_int(data, meta)
```

The function takes no data-source and no level argument: the level is whichever frame is handed in, and the frame is expected on the $\log_2$ scale, as the page's is. `legend=False` and `header=False` drop the legend and the title, and `plot_colors` takes one colour per condition in the order the annotation gives them.

17.5 Exporting

The download reproduces the plot at the width, height and resolution set on the page. The browser scales the rendered figure to fit the page. Choose PDF for a figure that will be scaled or edited, PNG for a slide.

Behind this page

Function	What it produces
histo_int	One kernel density curve per condition, from the condition-wise mean $\log_2$ intensities

18
The intensity boxplot

The same distributions as the histogram, summarised so that every condition — or every sample — fits on one axis.

AT A GLANCE

Where	Quality control → Boxplot
Data levels	Protein, phosphosite, phosphoprotein
Needs	An uploaded dataset and its annotation. The conditions, and which samples belong to each, come from the annotation.
Data source	The data as loaded, on the $\log_2$ scale. There is no selector: quality control describes what the instrument produced, before filtering or imputation.
Two modes	**Type** Mean draws one box per condition; **Type** Single draws one box per sample.
Table	Yes, below the plot, downloadable as CSV.

AT PHOSPHOSITE LEVEL

The shared **Localization probability cutoff** selects sites across phosphosite views. At 0 it keeps all sites, including those with no probability; above 0 those unknowns are excluded. This is separate from completeness filtering and does not rerun normalization or imputation. An already collapsed phosphoprotein table has its own recorded collapse cutoff.

18.1 What the plot answers

A boxplot answers the same question as the histogram — do the samples sit on one scale — for far more samples at once. Twenty overlaid density curves are unreadable; twenty boxes are not, and a single displaced median is found faster here than anywhere else in the tool. What it costs is shape: a box states five numbers, so two distributions that differ everywhere except

in their quartiles draw the same box. Use this page to find the sample worth looking at, then the histogram to see what is wrong with it.

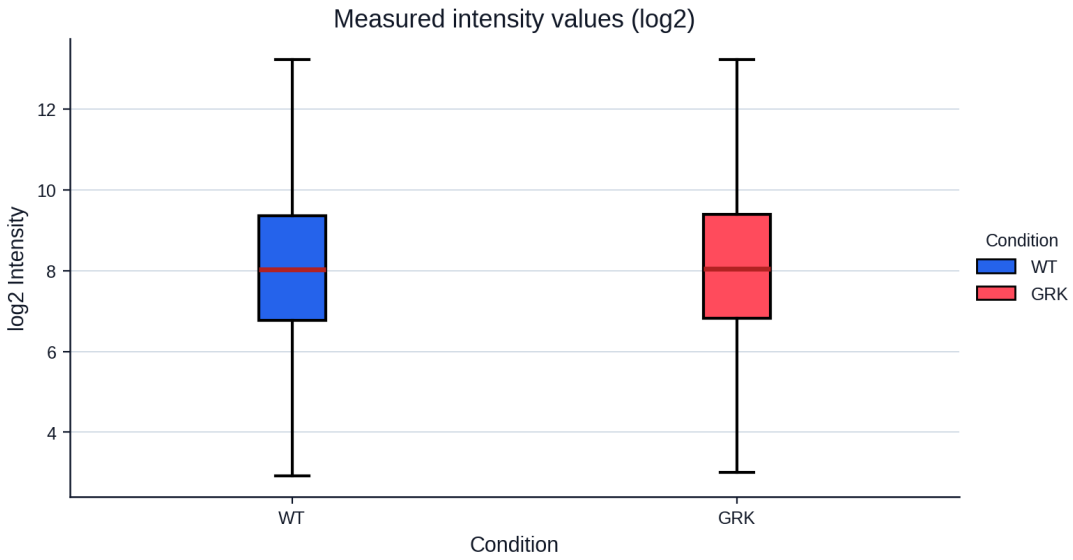


Figure 18.1. Type Mean at protein level on the example dataset: one box per condition, built from the condition means of the five WT and the five GRK samples. The two boxes sit at the same median and span the same range, which is the reading this page exists to give.

18.1.1 Reading it

- The box: the interquartile range, from the 25th to the 75th percentile. Its fill is the condition colour.
- The line across the box: the median. It is drawn heavier and in its own colour so it reads at a glance across many boxes.
- The whiskers: the most extreme values still inside 1.5 × IQR of the box. The caps are data points, not the limits themselves.
- The outlier markers: everything beyond the whiskers, drawn only when **Toggle Outliers** is on.
- The label under each box: the condition in Mean mode, and condition_n in Single mode — numbered within the condition rather than across the experiment, with the original sample identifier added in brackets when **Toggle IDs** is on.

EASY TO GET WRONG

Two very different distributions can draw the same box. A bimodal sample and a unimodal one with the same quartiles are indistinguishable here, and so are a symmetric distribution and a skewed one whose median happens to sit centrally. Whenever a box looks acceptable but a downstream result does not, open the histogram for the same level: it is the same data, drawn without the summary.

18.2 Controls

Controls on this page

Control	What it does	Default	Where
Dataset level	Which dataset to draw: protein, phosphosite or phosphoprotein. Only the levels actually uploaded are offered; phosphoprotein appears once a collapse has been run.	Protein	page
Type	Mean: one box per condition, built from the condition-wise mean of each feature. Single: one box per sample, built from every value in that column.	Mean	page
Toggle Outliers	Draws the points beyond the whiskers. With it off the whiskers are still computed the same way; the points outside them are simply not marked.	off	page
Toggle IDs	Adds the original sample identifier in brackets after each sample label. Single mode only — in Mean mode the labels are conditions and the switch does nothing. It does not change the table.	off	page
Toggle Text	Prints the median above each box, to two decimals.	off	page
Width (cm), Height (cm), DPI	The size of the *downloaded* figure. The browser scales the rendered figure to fit the page. Width and height are held between 1 and 60 cm, DPI between 72 and 600.	20 cm × 10 cm, 300	page
Download format	PNG, JPG, WEBP, SVG or PDF.	PNG	page

In the plot appearance panel

Control	What it does	Default	Where
Box border color	The outline of every box, and of its whiskers and caps.	black	appearance
Median color	The line across the box.	firebrick	appearance
Outlier dot color, Outlier outline color, Outlier size (pt)	The markers beyond the whiskers. The three rows appear only while **Toggle Outliers** is on.	red, black, 5	appearance

In the plot appearance panel (continued)

Control	What it does	Default	Where
Value labels	Size of the number printed on each box by **Toggle Text**. Until it is set the plot uses 9.	9	appearance
Condition names, Condition colors	One display name and one colour per condition; the colour is the box fill. In Single mode the sample names can be overridden as well.	the condition palette	appearance
Title	Shown or hidden, and its text, size, weight and colour. The title is written by the plot: Measured intensity values (log2).	shown	appearance
Legend	Shown or hidden, and its title, size, colour, frame and position. Its entries are the conditions.	shown	appearance

WORTH KNOWING

The second table is the plot appearance panel, which is hidden until advanced mode is turned on (**Ctrl** + **Shift** + **E**). Everything it contains is described in Part XI; only the entries particular to this plot are repeated here.

18.3 How it is computed

The box, the whiskers and what counts as an outlier follow the standard definition given in **Concepts**; it is not repeated here. What differs between the two modes is only what goes into the box.

MEAN MODE

The matrix is the data as loaded, on the $\log_2$ scale; non-numeric and non-finite entries are NA. For every feature i and condition c the mean over the replicates that carry a value is taken,

$$\bar{x}_{i,c} = \frac{1}{|O_{i,c}|} \sum_{j \in O_{i,c}} x_{ij}, \qquad O_{i,c} = \{\, j \in S_c \,:\, x_{ij} \neq \mathrm{NA} \,\},$$

and the box of condition c is built from the vector $\{\bar{x}_{i,c}\}$ — one value per feature. Averaging first removes the sample-to-sample scatter, so what remains is the shape of the condition.

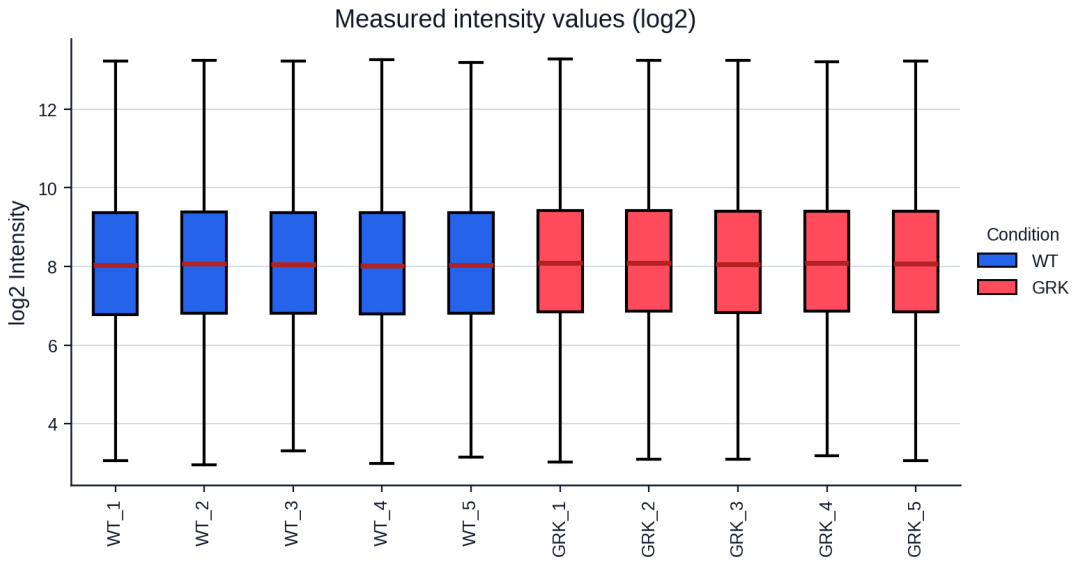


Figure 18.2. Single mode on the same data: ten boxes, one per sample, labelled WT_1 to GRK_5 and grouped by condition. Nothing is averaged away, and every box still sits at the same median — the evenness the summary in Mean mode could have hidden.

SINGLE MODE

No averaging. The box of a sample is built from every non-missing value in that column, one per feature. This is the distribution the instrument actually produced for that run, and it is the mode to use when looking for one bad sample inside a set of replicates.

Samples are relabelled condition_n, numbered within the condition in the order the annotation gives, and the boxes are drawn grouped by condition rather than in file order.

Compare samples with similar roles in the design. A low median or unusually wide box can reflect loading, instrument response, biological composition, missingness or processing; the plot alone does not decide which. Check Single mode before Mean, since averaging across replicates can hide an unusual sample. If normalisation aligned the distributions, their similarity is partly a result of that operation, not independent validation.

18.3.1 Edge cases

- In Mean mode a condition whose features are all missing yields no finite means and is left out of the axis entirely. In Single mode a sample with no values keeps its position on the axis, so the gap is visible.
- A feature missing in some replicates of a condition is averaged over the ones it has, so in Mean mode the boxes of different conditions can rest on different numbers of features.
- A condition with one replicate is legitimate here: in Mean mode its box is that sample's own distribution.
- **Toggle Text** draws the number at 1.02 times the median, so on a negative median — possi-

ble at phosphoprotein level after a robust_site_median collapse — the label sits just below the line rather than above it.

- At phosphoprotein level collapsed with robust_site_median the y axis reads log2 Ratio to Site Median: those values are ratios centred on each site's own median, and comparing their absolute height with an intensity is meaningless.

18.4 The table below the plot

The table reports the numbers that draw the boxes, each rounded to two decimals. Its first column follows the mode: one row per condition in Mean mode, one row per sample in Single mode.

Columns in this table

Column	What it holds
Condition	Mean mode. The condition, as the annotation names it.
Sample	Single mode. The sample, labelled condition_n as on the plot. **Toggle IDs** does not change this column.
mean	The arithmetic mean of the values in the box. It is not drawn on the plot, and it is not the median.
median	The line across the box.
Q1	The lower edge of the box, the 25th percentile.
Q3	The upper edge of the box, the 75th percentile.
lower_whisker	The smallest value at or above $Q1 - 1.5 \times \text{IQR}$ — where the lower whisker ends.
upper_whisker	The largest value at or below $Q3 + 1.5 \times \text{IQR}$ — where the upper whisker ends.

The two whisker columns are data values, not the cutoffs themselves, which is why they move in steps rather than continuously as a sample changes.

18.5 In the notebook

A notebook starting point, using the frames introduced in *The function reference*. These compact examples do not encode every UI option; use the plot's notebook ZIP, where available, for its saved settings and preprocessing recipe:

```python
fig = boxplot_int(data, meta)
fig = boxplot_int_single(data, meta, id=False)
```

```r
fig <- boxplot_int(data, meta)
fig <- boxplot_int_single(data, meta, id = FALSE)
```

The two modes are two functions rather than an argument: boxplot_int is **Type** Mean and boxplot_int_single is Single. outliers=True is **Toggle Outliers**, text=True is **Toggle Text**, and id — which only the single-sample function takes, and which defaults to True there rather than to off as on the page — adds the original identifier in brackets.

18.6 Exporting

The download reproduces the plot at the width, height and resolution set on the page. The browser scales the rendered figure to fit the page. Choose PDF for a figure that will be scaled or edited, PNG for a slide. The table underneath downloads separately as CSV.

18.7 Further reading

NIST: box plots introduces quartiles, whiskers and observations beyond the whiskers. Such observations are not automatically measurement errors.

Behind this page

Function	What it produces
boxplot_int	One box per condition, from the condition-wise mean $\log_2$ intensities
boxplot_int_single	One box per sample, from every value in the column
prepare_boxplot_df	The condition-level table
prepare_boxplot_single_df	The per-sample table

19
The coefficient of variation

How much the replicates of a condition disagree about the same feature, in per cent of their mean.

AT A GLANCE

Where	Quality control → Coefficient of variation
Data levels	Protein, phosphosite, phosphoprotein
Needs	An uploaded dataset and its annotation, and at least one condition with two or more samples.
Data source	The normalized, unfiltered, non-imputed QC matrix, converted back to the linear scale for this calculation; after VSN, the unnormalized data.
Varies across	Replicates of one condition, for one feature. Not across features, and not across conditions.
One box	One condition. Its values are the CVs of all features quantified in at least two of that condition's replicates.
Table	Yes, below the plot, downloadable as CSV.

AT PHOSPHOSITE LEVEL

The shared **Localization probability cutoff** selects sites across phosphosite views. At 0 it keeps all sites, including those with no probability; above 0 those unknowns are excluded. This is separate from completeness filtering and does not rerun normalization or imputation. An already collapsed phosphoprotein table has its own recorded collapse cutoff.

WHEN NORMALIZATION IS VSN

The calculation exponentiates the analysis values to obtain positive values for a CV. VSN output is log-like rather than an ordinary logarithm, so exponentiating it does not give a linear intensity. After VSN normalization this page therefore uses the unnormalized data of the same rows, and a note under the plot says so: VSN output is

not on a linear intensity scale, so this plot uses the unnormalized data. With any other normalization the normalized values are used.

19.1 What the plot answers

The coefficient of variation asks how far apart the replicates of a condition are when they measure the same thing, as a fraction of what they measured. Two conditions can have different mean intensities and similar relative variability. Read the median and the width of the box together. A low CV describes consistent values under the chosen processing; it does not establish accuracy, independent biological replication or suitability for every downstream test.

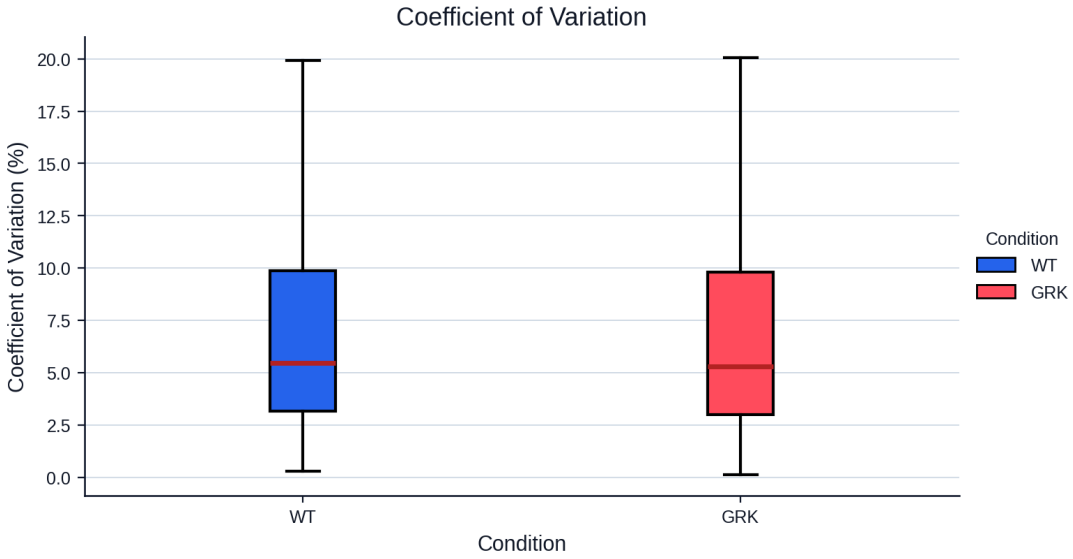


Figure 19.1. Protein level on the example dataset: one box for the five WT replicates and one for the five GRK replicates. Both medians sit near 5 % and both boxes close below 10 %. These are distributions of per-feature relative standard deviations, not confidence intervals or error bounds for each measurement.

19.1.1 Reading it

- One box, one condition, built from one CV per feature.
- The median line: the CV of the typical feature in that condition. With **Toggle Text** on its value is printed at the line, to two decimals.
- The upper whisker and the outliers above it: the features whose replicates disagreed most. A few are expected — low-abundance features have poor precision — but a long tail common to every condition is a property of the run.
- The y axis is per cent of the mean, so the boxes of different conditions are directly comparable even when their intensities are not.

EASY TO GET WRONG

Compute an intensity CV on a positive linear scale, not by dividing the standard deviation of log intensities by their mean. The problem is the origin: multiplying linear intensities by a unit-conversion factor adds a constant to their logarithms, changing that log-scale denominator. Changing only the logarithm base scales both numerator and denominator equally; it is not the reason the log-scale calculation is unsuitable. The notebook function requires an explicit linear-scale declaration.

19.2 Controls

Controls on this page

Control	What it does	Default	Where
Dataset level	Which dataset to draw: protein, phosphosite or phosphoprotein. Only the levels actually uploaded are offered; phosphoprotein appears once a collapse has been run.	Protein	page
Toggle Outliers	Draws the points beyond the whiskers. With it off the whiskers are still computed the same way; the points outside them are simply not marked.	off	page
Toggle Text	Prints the median CV at each box, to two decimals.	off	page
Width (cm), Height (cm), DPI	The size of the *downloaded* figure. The browser scales the rendered figure to fit the page. Width and height are held between 1 and 60 cm, DPI between 72 and 600.	20 cm × 10 cm, 300	page
Download format	PNG, JPG, WEBP, SVG or PDF.	PNG	page

In the plot appearance panel

Control	What it does	Default	Where
Box border color	The outline of every box, and of its whiskers and caps.	black	appearance
Median color	The line across the box.	firebrick	appearance

In the plot appearance panel (continued)

Control	What it does	Default	Where
Outlier dot color, Outlier outline color, Outlier size (pt)	The markers beyond the whiskers. The three rows appear only while **Toggle Outliers** is on.	red, black, 5	appearance
Value labels	Size of the number printed at each box by **Toggle Text**. Until it is set the plot uses 9.	9	appearance
Condition names, Condition colors	One display name and one colour per condition; the colour is the box fill.	the condition palette	appearance
Title	Shown or hidden, and its text, size, weight and colour. The title is written by the plot: Coefficient of Variation.	shown	appearance
Legend	Shown or hidden, and its title, size, colour, frame and position. Its entries are the conditions.	shown	appearance

WORTH KNOWING

The second table is the plot appearance panel, which is hidden until advanced mode is turned on (**Ctrl** + **Shift** + **E**). Everything it contains is described in Part XI; only the entries particular to this plot are repeated here.

19.3 How it is computed

The box, the whiskers and what counts as an outlier follow the standard definition given in **Concepts**; it is not repeated here. The box on this page contains one CV per feature, and the CVs of one condition only.

HOW IT IS COMPUTED

1. Use the normalized QC matrix, before completeness filtering and imputation — after VSN, the unnormalized values of the same rows — and convert its stored values to linear values. Missing entries remain missing. The result describes variation after the selected normalization, not raw instrument precision.
2. Conditions with fewer than two samples are dropped. A single replicate has nothing to vary across, and such a condition gets no box and no place on the axis.

3. Within each remaining condition c, and for each feature i, the mean and the sample standard deviation are taken across that condition's replicates, over the observed values only:

$$m_{i,c} = \frac{1}{n_{i,c}} \sum_{j \in O_{i,c}} x_{ij}, \qquad s_{i,c} = \sqrt{\frac{1}{n_{i,c} - 1} \sum_{j \in O_{i,c}} \left(x_{ij} - m_{i,c}\right)^2},$$

   where $O_{i,c}$ are the replicates of c in which feature i was quantified and $n_{i,c} = |O_{i,c}|$. The denominator is $n - 1$, so a feature seen in only one replicate of the condition has no standard deviation.

4. The coefficient of variation, in per cent:

$$\mathrm{CV}_{i,c} = \frac{s_{i,c}}{m_{i,c}} \times 100.$$

5. Every value that is not finite is removed — the features seen in one replicate, and any whose mean is zero. What remains is the vector the box of condition c is built from.

> **NOTE**
>
> Note what the CV is *not* taken over. It is not the spread of one sample's intensities across features, which is a property of the dynamic range and not of reproducibility; and it is not the spread of a feature across conditions, which is the biological effect the experiment is looking for. It varies across replicates within one condition, one feature at a time.

For values 80, 100 and 120, the mean is 100 and the sample standard deviation is 20, giving a CV of 20 %. Multiplying all three by ten leaves the CV unchanged. With only two observed replicates a CV can be calculated, but it is an uncertain estimate.

A higher box means greater relative variation among the observed values. Possible explanations include biological heterogeneity, low signal and technical variability; the box alone cannot separate them. Inspect sample coverage, intensity distributions and the individual features before making a quality decision. There is no universal acceptable CV threshold for all proteins, experimental designs and abundance ranges.

19.3.1 Edge cases

- A condition with one replicate has no box. It is not drawn empty and it is not drawn at zero: it is absent from the axis, and from the table.
- A feature quantified in only one replicate of a condition contributes no CV to that condition, but may contribute one to another condition where it was seen twice. Boxes therefore rest on different numbers of features.
- If no condition has two samples, no figure is drawn and the page reports that instead.

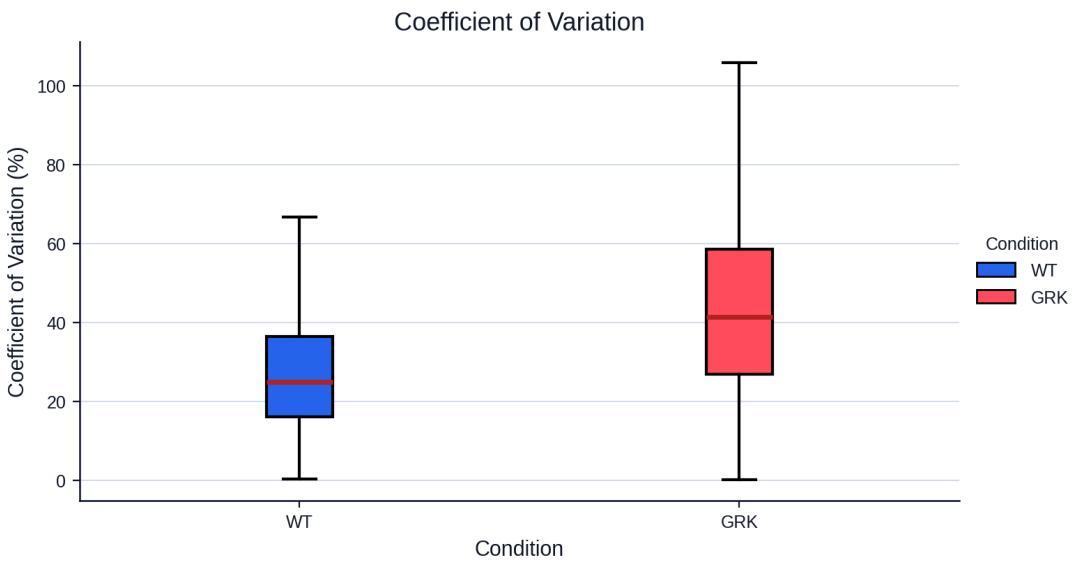


Figure 19.2. The same experiment at phosphosite level, and a different axis: the scale now runs past 100 %, the WT median sits near 25 % and the GRK box is both higher and much wider. Enrichment costs precision, and the comparison worth making is between the two boxes here rather than between either of them and the protein-level figure above.

- Features whose replicates agree exactly give a CV of zero, which is a legitimate value and is kept.
- At phosphoprotein level collapsed with `robust_site_median` the stored values are ratios to each site's own median, so on the linear scale they are centred on one rather than on an abundance. A ratio to the standard deviation of such values is arithmetically defined but is not a coefficient of variation of an abundance, and should not be compared with the protein-level numbers. The page draws them anyway, and prints a warning under the plot when it does, because the number is defined but is not the quantity the page is named after.

19.4 The table below the plot

One row per condition that has a box, holding the numbers that draw it, each rounded to two decimals.

Columns in this table

Column	What it holds
Condition	The condition, as the annotation names it. Conditions with fewer than two samples do not appear.

Columns in this table (continued)

Column	What it holds
mean	The arithmetic mean of that condition's CVs. It is not drawn on the plot, and it is not the median.
median	The line across the box — the CV of the typical feature.
Q1	The lower edge of the box, the 25th percentile of the CVs.
Q3	The upper edge of the box, the 75th percentile of the CVs.
lower_whisker	The smallest CV at or above $Q1 - 1.5 \times \text{IQR}$.
upper_whisker	The largest CV at or below $Q3 + 1.5 \times \text{IQR}$ — the most useful single number on the table for comparing conditions.

Every column is in per cent, and the two whisker columns are observed CVs rather than the cutoffs themselves.

19.5 In the notebook

A notebook starting point, using the frames introduced in *The function reference*. These compact examples do not encode every UI option; use the plot's notebook ZIP, where available, for its saved settings and preprocessing recipe:

PYTHON

```
fig = cov_plot(org_data, meta, data_scale="linear")
```

R

```
fig <- cov_plot(org_data, meta, data_scale = "linear")
```

The frame is org_data, the linear one obtained with inverse_log2_transform_data, because a CV on $\log_2$ values is not a CV; and data_scale is a declaration rather than a conversion — the function refuses to run without it instead of turning a $\log_2$ matrix into a plausible-looking number. outliers=True and text=True are **Toggle Outliers** and **Toggle Text**.

19.6 Exporting

The download reproduces the plot at the width, height and resolution set on the page. The browser scales the rendered figure to fit the page. Choose PDF for a figure that will be scaled or edited, PNG for a slide. The table underneath downloads separately as CSV.

19.7 Further reading

NIST: coefficient of variation explains the ratio of standard deviation to mean and the importance of a meaningful zero.

Behind this page

Function	What it produces
cov_plot	One box of CVs per condition. It requires the caller to state that the matrix is linear, and refuses a $\log_2$ one
prepare_cov_df	The table under the plot

20
Principal component analysis

How samples relate in a low-dimensional view of the chosen matrix, and what dominates the variation when they do not.

AT A GLANCE

Where	Quality control → PCA
Data levels	Protein, phosphosite, phosphoprotein
Needs	An uploaded dataset and its annotation, at least two samples, and enough features observed in *every* sample.
Data source	The only quality-control page with a choice — Data, Filtered Data or Imputed Data — because a projection cannot use a feature that is missing anywhere, so which matrix goes in is a scientific decision.
Methods	PCA, UMAP and t-SNE. Only PCA offers three dimensions and only PCA labels its axes with explained variance.
Two renderers	**Type** Normal draws a static figure; Interactive opens a zoomable, rotatable plot in a new tab.
Table	None. The feature count the projection rests on is reported with the controls instead.

AT PHOSPHOSITE LEVEL

The shared **Localization probability cutoff** selects sites across phosphosite views. At 0 it keeps all sites, including those with no probability; above 0 those unknowns are excluded. This is separate from completeness filtering and does not rerun normalization or imputation. An already collapsed phosphoprotein table has its own recorded collapse cutoff.

20.1 What the plot answers

PCA summarises the differences between samples using a few new axes. Each axis combines the retained features: PC1 captures the largest remaining variance, PC2 the next largest in a perpendicular direction, and so on. Nearby samples have similar scores on the displayed components. Samples far apart differ in the patterns represented by those components, but may still be similar in other respects.

Use the plot to ask whether variation follows condition, batch or an individual sample. Separation is not a significance test and does not identify its cause. A treatment effect need not dominate PC1, and overlapping conditions can still contain differentially abundant proteins.

The axes are quantitative component scores. Their interpretation depends on feature scaling; their signs are arbitrary, so a mirrored PCA can represent the same result. The variance percentages refer to the matrix actually analysed, not every row of the uploaded file.

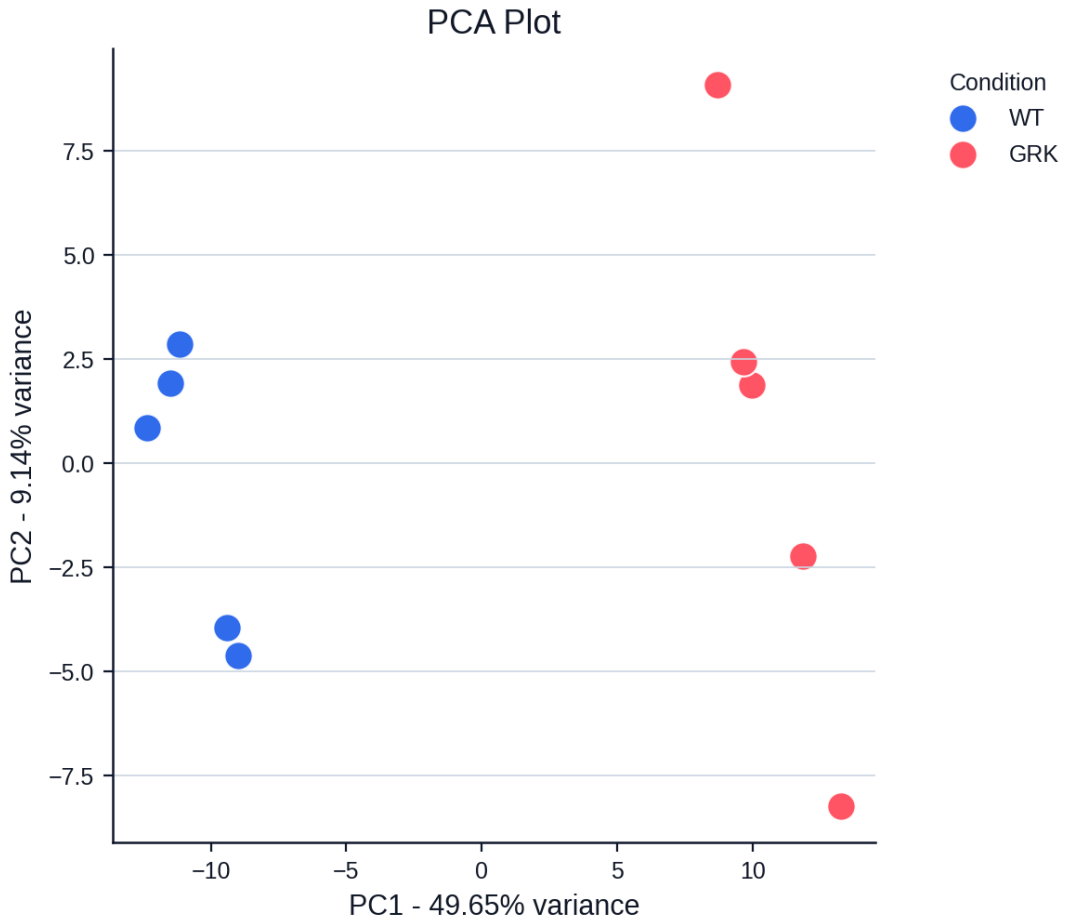


Figure 20.1. PCA of the protein-level matrix of the example dataset, in two dimensions and without ellipses: ten points, five WT on the left and five GRK on the right, separated along PC1 and mixed along PC2. PC1 carries 49.65 % of the variance in the reduced matrix and PC2 9.14 %. These values describe this saved example, not a target that another experiment should reproduce.

20.1.1 Reading it

- One point, one sample — never a feature. The features are what define the axes; the samples are what is plotted on them.
- The axis labels. For PCA they read PC1 - 49.65% variance: the share of the total variance this component accounts for, so the two percentages together say how much of the dataset the picture represents. UMAP and t-SNE axes carry no percentage, because neither preserves variance.
- The ellipse, with **Add Ellipses (2D)** on: the *smallest ellipse containing every point of that group*. It summarises where the group lies; it is not a confidence region, and it can only grow as replicates are added.
- The colour: the condition, from the annotation and shared with every other plot in the session, or the cluster when **Color By** says so.

> **EASY TO GET WRONG**
>
> This implementation uses complete, non-constant features. Features with missing values or no variation are excluded, and the page reports the usable count. This can favour consistently detected features. A small retained set is a reason to inspect missingness, not an automatic reason to impute. If an imputed matrix is used, compare it with the observed-data result: the positions may partly reflect the imputation assumptions.

20.2 Controls

Controls on this page

Control	What it does	Default	Where
Dataset level	Which dataset to project: protein, phosphosite or phosphoprotein.	Protein	page
Data Source	Which matrix to project: Data as uploaded, Filtered Data or Imputed Data. Only the ones that exist are offered.	the most processed matrix available	page
Method	PCA, UMAP or t-SNE. Changing it resets **Feature scaling** to that method's own default, and returns **Dimensions** to 2D and **Clustering** to None where those do not apply.	PCA	page

Controls on this page (continued)

Control	What it does	Default	Where
Feature scaling	Centred subtracts each feature's mean across the samples and stops there, so a feature contributes in proportion to its variance. Unit variance also divides by the standard deviation, giving every feature equal weight.	Centred for PCA, Unit variance for UMAP and t-SNE	page
Type	Normal draws a static figure that can be downloaded. Interactive builds a Plotly figure with zoom, rotation and per-sample hover, opened in a new tab.	Normal	page
Dimensions	2D plots PC1 against PC2, 3D adds PC3. **Method** PCA only.	2D	page
Add Ellipses (2D)	Draws the minimum enclosing ellipse of each group. Offered in two dimensions only.	off	page
Clustering	None, KMeans, Agglomerative or DBSCAN. It is run on the two embedding coordinates rather than on the original features, so it clusters what the picture shows rather than what the data holds. Offered for UMAP and t-SNE only.	None	page
Color By	Condition, or Cluster once a clustering method has been chosen.	Condition	page
Cluster Count	How many clusters to form. KMeans and Agglomerative only; held between 2 and 20.	3	page
DBSCAN Eps, DBSCAN Min Samples	The neighbourhood radius in embedding coordinates, and the number of points that makes one. DBSCAN only.	0.5, 2	page
UMAP Neighbors, UMAP Min Dist	How much neighbourhood structure UMAP preserves, and how tightly it may pack points. Held between 2 and 200, and between 0 and 1.	15, 0.1	page
t-SNE Perplexity, t-SNE Learning Rate	The effective neighbourhood size, capped at one less than the number of samples, and the step size of the optimisation.	30, 200	page
Random State	Seeds UMAP and t-SNE so that the same figure is reproduced exactly. Neither is deterministic without it.	42	page
Width (cm), Height (cm), DPI	The size of the *downloaded* figure, shown only while Normal is selected. The interactive plot has none: it fills its frame, Plotly's camera button saves a picture of it, and **Open Interactive Plot** opens it in a window of its own.	20 cm × 10 cm, 300	page
Download format	PNG, JPG, WEBP, SVG or PDF. Normal mode only.	PNG	page

In the plot appearance panel

Control	What it does	Default	Where
Points	Diameter in points, shape, opacity and outline — for all points at once, or for one condition at a time.	6 static, 9.6 interactive	appearance
Condition names, Condition colors	One display name and one colour per group. With **Color By** set to Cluster the rows address the clusters instead.	the condition palette	appearance
Title	Shown or hidden, and its text, size, weight and colour. The title is written from the method: PCA Plot, UMAP Plot, t-SNE Plot, prefixed 3D in three dimensions.	shown	appearance
Legend	Shown or hidden, and its title, size, colour, frame and position. Its heading follows **Color By**.	shown	appearance
X axis, Y axis	Overwriting an axis label removes the explained-variance percentage with it.	plot's own	appearance
Frame and grid	The two renderers start from different conventions: spines on and no grid for the static figure, grid on and no spines for the interactive one.	by renderer	appearance

WORTH KNOWING

The second table is the plot appearance panel, which is hidden until advanced mode is turned on (**Ctrl** + **Shift** + **E**). Everything it contains is described in Part XI; only the entries particular to this plot are repeated here.

20.3 How it is computed

HOW IT IS COMPUTED

1. The matrix is the one **Data Source** names. Non-numeric and non-finite entries are read as NA.
2. Every feature with an NA in any sample is dropped. If fewer complete features remain than the projection has dimensions, no figure is drawn and the notice says so.
3. The matrix is transposed — the samples become the rows and the features the

columns — and any feature that takes the same value in every sample is dropped, having no variation to contribute.

4. Each feature is centred on its mean across the samples. With **Feature scaling** set to Unit variance it is then divided by its standard deviation across the samples, taken with $n-1$ in the denominator as R's prcomp(scale. = TRUE) does; a feature whose standard deviation is zero is divided by one instead of being lost.

5. For PCA: the singular value decomposition of that matrix, $X = U\Sigma V^{\top}$. The sample coordinates are the first components of $U\Sigma$, and the label on component k is $\sigma_k^2 / \sum_l \sigma_l^2 \times 100\,\%$ — a share of the variance *in the reduced matrix*, after incomplete and constant features were dropped, not of the uploaded dataset.

6. For UMAP and t-SNE: the same prepared matrix goes to the respective algorithm with the parameters set on the page. Neither produces an ordered decomposition, so the axes are numbered and carry no percentage.

THE ELLIPSE

For each group, the smallest ellipse containing all of its points is fitted by Khachiyan's algorithm and drawn at 15 % opacity in the group's own colour. The fit needs at least three points that span the plane. Three points on a line, or two at the same position and a third, leave it undefined, and an ellipse whose shorter axis collapses would draw as a line — which reads as a region of zero area rather than as the absence of one. In either case that group gets no ellipse, its points are drawn as usual, the other groups are unaffected, and a note with the controls names the groups skipped. The figure itself never says so: a plot that leaves the tool should carry no caption about how the tool drew it.

For example, PC1 at 40 % and PC2 at 10 % display half the variance in the analysed matrix. They leave the other half out of view. Separation on a 4 % component can still be biologically relevant; the percentage measures variance, not importance or confidence. Compare sample positions with the experimental design, coverage and intensity distributions before excluding a sample. Biological replicates need not be as similar as technical repeats.

UMAP and t-SNE answer a related but different question about neighbourhood structure. Their axes are not fold changes or percentages of variance. Cluster area and the distance between separate islands should not be read as quantitative measures of biological difference. Check whether an apparent pattern persists across reasonable settings; clustering an embedding is exploratory, not an independent validation of the groups.

20.3.1 Edge cases

- Two samples are needed for a two-dimensional projection and three for a three-dimensional one; below that no figure is drawn.
- Selecting Imputed Data can increase the complete-feature count, but remaining non-finite

or constant features are still excluded. Imputation is not a guarantee that every row is usable.

- If every complete feature is constant, there is nothing to decompose and the page reports that instead.
- Ellipses exist in two dimensions only; the switch is hidden in 3D. A condition with one or two samples cannot bound one and is named among the skipped groups.
- UMAP requires the umap-learn package and t-SNE requires scikit-learn. If one is absent the page says which, and how to install it.

20.4 In the notebook

A notebook starting point, using the frames introduced in *The function reference*. These compact examples do not encode every UI option; use the plot's notebook ZIP, where available, for its saved settings and preprocessing recipe:

PYTHON

```
fig = pca_plot(data, meta)
```

R

```
fig <- pca_plot(data, meta)
```

add_ellipses=True and plot_3d=True are the page's **Add Ellipses (2D)** and **Dimensions** switches, and scaling takes the same "auto", "centred" and "unit_variance" the **Feature scaling** selector offers. There is no data-source argument: which matrix is projected is decided by which frame is handed in, so an imputed projection means passing the imputed frame.

20.5 Exporting

In Normal mode the download reproduces the plot at the width, height and resolution set on the page, which is why those three controls do not change the figure on screen. In Interactive mode there is no image to download: the plot opens in its own tab, where Plotly's camera button saves what is on screen at screen resolution. For a figure that will be scaled or edited, use Normal and PDF.

20.6 Further reading

For an accessible introduction, see Wikipedia: principal component analysis. For nonlinear projections, Wattenberg, Viégas and Johnson, How to Use t-SNE Effectively illustrates why island size and spacing need care. These explain the methods; the controls and row exclusions above describe CoPYlot.

Behind this page

Function	What it produces
pca_plot	The static projection, in two or three dimensions, with the optional ellipses
pca_plot_interactive	The interactive one. The package function takes no scaling argument and always projects the centred matrix; the page applies **Feature scaling** in both modes
resolve_feature_scaling	Which scaling a method runs on
fitted_group_ellipse	One group's ellipse, or nothing when its points do not span the plane

21
The abundance plot

The dynamic range of the experiment, drawn as every feature in rank order, and where any one of them sits in it.

AT A GLANCE

Where	Quality control → Abundance
Data levels	Protein, phosphosite, phosphoprotein
Needs	An uploaded dataset and its annotation. The conditions, and which samples belong to each, come from the annotation.
Data source	The normalized, unfiltered, non-imputed QC matrix, converted to linear values before averaging; after VSN, the unnormalized data.
One point	One feature in one condition: its mean intensity over that condition's replicates, on a $\log_{10}$ axis.
Two renderers	**Type** Normal draws a static figure; Interactive opens a hoverable plot in a new tab.
Table	None.

AT PHOSPHOSITE LEVEL

The shared **Localization probability cutoff** selects sites across phosphosite views. At 0 it keeps all sites, including those with no probability; above 0 those unknowns are excluded. This is separate from completeness filtering and does not rerun normalization or imputation. An already collapsed phosphoprotein table has its own recorded collapse cutoff.

A PROCESSED ABUNDANCE SCALE

Normalization changes the values ranked here. VSN is the exception: its output is only log-like, and exponentiating it does not give a linear intensity, so after VSN normalization this page uses the unnormalized data of the same rows, and a note under

> the plot says so: VSN output is not on a linear intensity scale, so this plot uses the unnormalized data. Record the normalization and treat this as a relative abundance overview, not an absolute concentration measurement.

21.1 What the plot answers

The abundance plot shows how far the experiment reaches. Sorting every feature by its intensity and plotting it against its own rank turns the dataset into one curve, and the shape of that curve says how many orders of magnitude were covered, how quickly the signal falls away, and where the measurement runs into the floor.

It also places a selected feature within the measured intensity range. A feature near the right-hand end has a relatively low recorded intensity. That does not establish an instrument detection limit or an absolute concentration: peptide response, ionisation and data processing also affect intensity. Use missingness and replicate measurements to assess how well that feature was quantified.

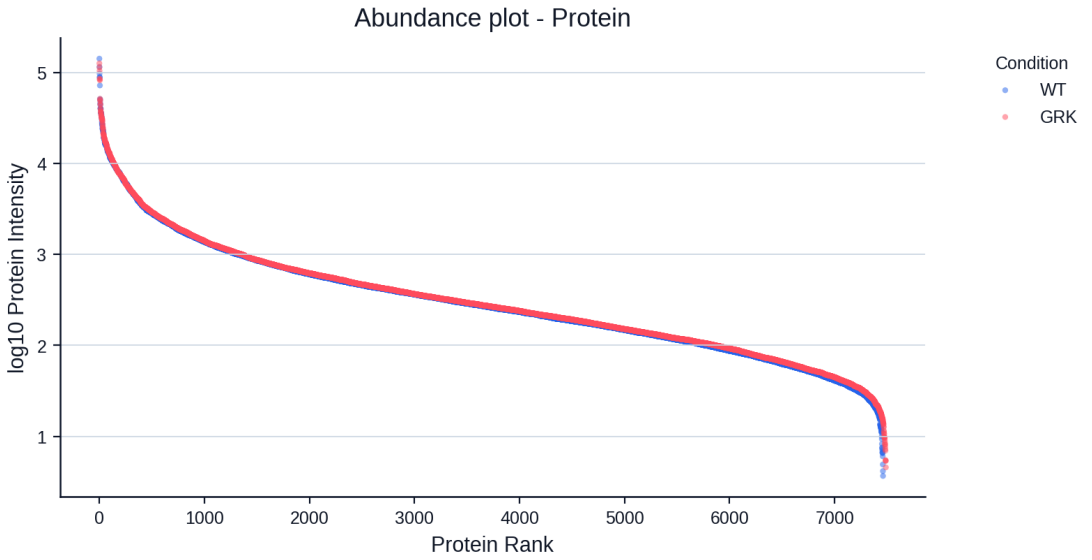


Figure 21.1. Protein level on the example dataset, All Conditions: the WT and the GRK curve drawn over one another, each running from rank 1 to about 7 500 and falling from above 10^5 to below 10^1 — four and a half orders of magnitude, a short steep shoulder, a long body and a tail that turns down at the right-hand end. The two curves are nearly indistinguishable, which is what two conditions of equal depth look like.

21.1.1 Reading it

- One point, one feature in one condition. Its height is the $\log_{10}$ of the mean of that feature over the condition's replicates.
- The x axis: the feature's rank within its own condition, 1 being the most abundant. It is an ordering, not a quantity — the distance between rank 10 and rank 20 means nothing beyond "ten features".

- The y axis: $\log_{10}$ intensity, so one unit is a factor of ten. At phosphoprotein level collapsed with robust_site_median it reads log10 Ratio to Site Median instead, because those values are ratios, not abundances.
- The colour: the condition. With one condition chosen there is a single curve; with All Conditions the curves are drawn over one another, and where they separate is where the conditions differ in depth.

EASY TO GET WRONG

The rank is computed separately for each condition, so a vertical line through the figure does not pick out one feature. Rank 500 in one condition and rank 500 in another are different proteins, and reading a gap between two curves at the same x as a fold change is wrong. To compare one feature between conditions, use the single-protein pages; to compare depth, compare the *lengths* of the curves and the height at which each ends.

21.2 Controls

Controls on this page

Control	What it does	Default	Where
Dataset level	Which dataset to draw: protein, phosphosite or phosphoprotein. Only the levels actually uploaded are offered; phosphoprotein appears once a collapse has been run.	Protein	page
Type	Normal draws a static figure that can be downloaded. Interactive builds a Plotly figure with zoom and per-feature hover — name, condition, rank and intensity — opened in a new tab; there is no image download in that mode.	Normal	page
Condition	All Conditions draws every condition, each ranked within itself. Choosing one draws that condition alone, ranked 1 to *n*. The list is the conditions in the annotation, and a saved choice that no longer exists falls back to All Conditions.	All Conditions	page
Width (cm), Height (cm), DPI	The size of the *downloaded* figure, shown only while Normal is selected. The interactive plot has none: it fills its frame, Plotly's camera button saves a picture of it, and **Open Interactive Plot** opens it in a window of its own.	20 cm × 10 cm, 300	page

Controls on this page (continued)

Control	What it does	Default	Where
Download format	PNG, JPG, WEBP, SVG or PDF. Normal mode only.	PNG	page

In the plot appearance panel

Control	What it does	Default	Where
Points	Diameter in points, shape, opacity and outline — for all points at once, or for one condition at a time. The plot's own default is deliberately small: a dataset of eight thousand features drawn with larger marks is a solid band.	2.3 static All Conditions, 2.7 static single condition, 6 interactive	appearance
Condition names, Condition colors	One display name and one colour per condition, shared with every other plot in the session.	the condition palette	appearance
Title	Shown or hidden, and its text, size, weight and colour. The title is written from the data level: Abundance plot - Protein and its equivalents.	shown	appearance
Legend	Shown or hidden, and its title, size, colour, frame and position. Its entries are the conditions.	shown	appearance
Frame and grid	The two renderers start from different conventions: spines on and no grid for the static figure, grid on and no spines for the interactive one.	by renderer	appearance

> **WORTH KNOWING**
>
> The second table is the plot appearance panel, which is hidden until advanced mode is turned on (**Ctrl** + **Shift** + **E**). Everything it contains is described in Part XI; only the entries particular to this plot are repeated here.

21.3 How it is computed

HOW IT IS COMPUTED

1. Take the normalized, unfiltered, non-imputed QC matrix — after VSN, the unnormalized values of the same rows — and convert its stored values to linear values. Missing observations remain missing; they are not replaced by zero before averaging.
2. For every feature i and condition c, the arithmetic mean over the replicates in which it was quantified:

$$m_{i,c} = \frac{1}{|O_{i,c}|} \sum_{j \in O_{i,c}} x_{ij}, \qquad O_{i,c} = \{\, j \in S_c \,:\, x_{ij} \neq \mathrm{NA} \,\}.$$

3. The height of the point is $\log_{10} m_{i,c}$, with no pseudocount. The axis names the quantity, so the number has to be the logarithm of the quantity: adding one before taking the logarithm would change low values disproportionately: a mean of 1 would move from 0 to about 0.301, while a mean of 0.1 would move from -1 to about 0.041. Means that are not positive have no logarithm and are dropped rather than drawn at a value they do not have.
4. Within each condition the remaining features are ranked by $\log_{10} m_{i,c}$, descending, and the rank is the x coordinate. Ties are broken by the order the features appear in the matrix, so equal intensities take consecutive ranks rather than a shared average one.

WITH ONE CONDITION CHOSEN

The arithmetic is identical; only that condition's samples are averaged, and the ranks run 1 to n over the features that condition quantified. The static figure draws the same points in one colour; the interactive one adds hover text carrying the feature name, the condition, the rank and the intensity to four decimals.

Compare curve lengths and shapes, then inspect the features behind a difference. A shorter curve contains fewer quantified features; a vertical shift can reflect sample loading, normalization, measurement response or biology. Overlapping curves do not imply identical protein regulation, because each condition is ranked independently. Do not force curves to overlap merely to make the figure look consistent.

21.3.1 Edge cases

- A feature missing in every replicate of a condition has no mean there and is left out of that condition's curve. Curves therefore differ in length, and that difference is the point of the figure rather than a defect in it.

- A feature missing in some replicates is averaged over the ones it has, so a point can rest on fewer measurements than its neighbours.
- Features quantified nowhere at all are dropped from the figure entirely; the coverage table counts them under Identified, not quantified.
- With a single replicate in a condition the mean is that sample's own value, and the curve is that run's dynamic range.
- Interactive produces an HTML plot rather than an image: it opens in its own tab, and the download format selector is replaced by a link to it.

21.4 In the notebook

A notebook starting point, using the frames introduced in *The function reference*. These compact examples do not encode every UI option; use the plot's notebook ZIP, where available, for its saved settings and preprocessing recipe:

PYTHON

```
fig = abundance_plot(org_data, meta, workflow="Protein", data_scale="linear")
```

R

```
fig <- abundance_plot(org_data, meta, workflow = "Protein", data_scale = "linear")
```

workflow names the level — "Protein", "Phosphosite" or "Phosphoprotein" — and decides which column identifies a feature; data_scale states the scale of the frame handed in rather than converting it, and a $\log_2$ frame with data_scale="log2" works just as well — the function inverts the transform itself. There is no **Condition** argument: abundance_plot always draws every condition, and the single-condition curve is interactive_abundance_plot, which takes the condition as its third argument.

21.5 Exporting

In Normal mode the download reproduces the plot at the width, height and resolution set on the page, which is why those three controls do not change the figure on screen. In Interactive mode there is no image to download: the plot opens in its own tab, where Plotly's camera button saves what is on screen at screen resolution. Choose PDF for a figure that will be scaled or edited — at this many points a PDF is large but stays sharp — and PNG for a slide.

Behind this page

Function	What it produces
abundance_plot	The static figure: every condition, or one, ranked within itself
interactive_abundance_plot	The hoverable one, for a single condition. The package function additionally takes a list of features to highlight in red, and a limit on how many modification sites a phosphoprotein's hover label spells out; the page exposes neither
abundance_axis_label	Which y-axis label the level and the collapse method call for

22
The correlation matrix

How similar every sample is to every other, in one square, with the similar ones brought next to each other.

AT A GLANCE

Where	Quality control → Correlation
Data levels	Protein, phosphosite, phosphoprotein
Needs	An uploaded dataset and its annotation, and at least two samples that are not entirely empty.
Data source	The data as loaded, on the $\log_2$ scale. There is no selector: quality control describes what the instrument produced, before filtering or imputation.
Coefficient	Pearson, over the features both samples of a pair quantified.
Ordering	Hierarchical, by complete linkage on $1 - r$ — not the order of the annotation, unless a correlation could not be computed.
Table	None.

AT PHOSPHOSITE LEVEL

The shared **Localization probability cutoff** selects sites across phosphosite views. At 0 it keeps all sites, including those with no probability; above 0 those unknowns are excluded. This is separate from completeness filtering and does not rerun normalization or imputation. An already collapsed phosphoprotein table has its own recorded collapse cutoff.

22.1 What the plot answers

Every other quality-control page looks at samples one at a time. This one looks at pairs, and it answers the question those pages cannot: which samples resemble which. A set of replicates that belong together forms a block on the diagonal; a sample that resembles the wrong condition, or nothing at all, breaks the block, and that is visible here long before it

becomes an unexplained result.

The rows are not in the order of the annotation. They are reordered so that similar samples sit next to one another, which is what turns the figure from a table of numbers into a picture of the experiment's structure.

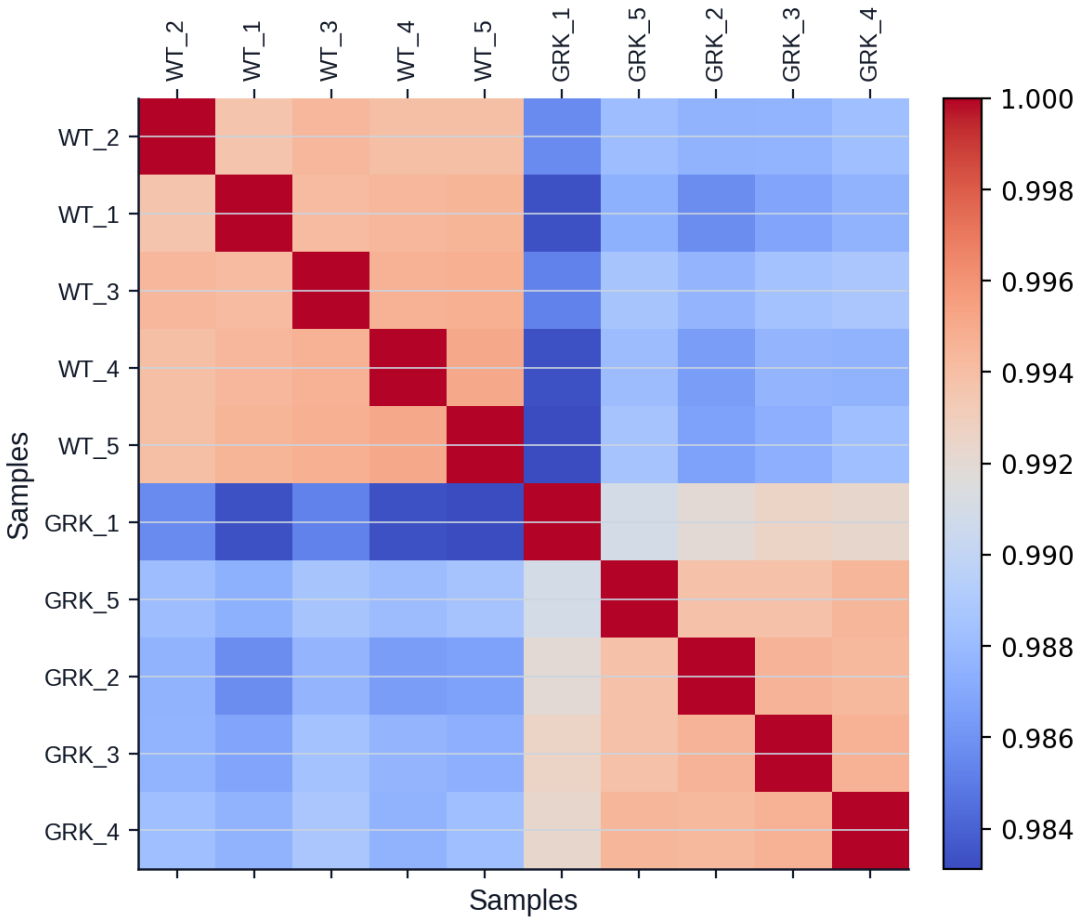


Figure 22.1. The ten samples of the example dataset at protein level, in Matrix mode. The clustering has put the five WT samples together and the five GRK samples together, so the two conditions appear as two blocks on the diagonal — and the colour bar, which runs from about 0.983 to 1, is the whole of the story: every pair correlates above 0.98, and the scale has been stretched across a range of two hundredths.

22.1.1 Reading it

- One cell, one ordered pair of samples — but the matrix is symmetric, so the cell above the diagonal and its mirror below carry the same number.
- The diagonal: every sample against itself, so exactly 1. It is the reference for the colour scale, not a result.
- The colour bar. Read its two ends before reading any cell: unless **Use Full Range** is on, they are the smallest and largest correlation actually observed, not −1 and 1.
- The order of the rows and columns, which is the dendrogram's, not the annotation's. Replicates that belong together end up adjacent; a sample that lands among another condition's

replicates is the finding.

- A cell reading NA, in flat grey: that pair shares no feature that both quantified, so no correlation exists. When any such cell is present the clustering is skipped for the whole matrix and the samples stay in annotation order; the x-axis label says so.

> **EASY TO GET WRONG**
>
> With **Use Full Range** off — the default — the colour scale is stretched to the observed range, so a dataset in which every pair lies between 0.97 and 0.99 is painted from one end of the gradient to the other. That is useful for finding the least similar sample and disastrous for judging whether the samples are similar in absolute terms: the figure looks as dramatic at a range of 0.02 as at a range of 0.8. Turn **Use Full Range** on to fix the scale at −1 to 1 before concluding anything about the size of a difference. In the other direction, uniformly high correlations are not by themselves good news: proteomics samples correlate well because most of the proteome does not change, and a matrix with no structure at all can mean the conditions do not differ rather than that the data are clean.

22.2 Controls

Controls on this page

Control	What it does	Default	Where
Dataset level	Which dataset to correlate: protein, phosphosite or phosphoprotein. Only the levels actually uploaded are offered; phosphoprotein appears once a collapse has been run.	Protein	page
Type	Matrix paints each cell with the colour of its correlation. Ellipse draws an ellipse per cell whose shape carries the magnitude and whose tilt carries the sign.	Matrix	page
Toggle IDs	Adds the original sample identifier in brackets after each label on both axes. It changes the labels only, never the ordering or the values.	off	page
Use Full Range	Fixes the colour scale at −1 to 1. With it off the scale spans the smallest and largest correlation in the matrix, which magnifies small differences.	off	page

Controls on this page (continued)

Control	What it does	Default	Where
Width (cm), Height (cm), DPI	The size of the *downloaded* figure. The browser scales the rendered figure to fit the page. This plot starts taller and at a higher resolution than the others on the page, because the labels are small and numerous.	20 cm × 16 cm, 400	page
Download format	PNG, JPG, WEBP, SVG or PDF.	PNG	page

In the plot appearance panel

Control	What it does	Default	Where
Correlation gradient	Which colour scale the cells — or the ellipses — are painted from: Coolwarm, Red/blue, Spectral, Pink/green, Purple/green, Brown/teal, Viridis or Cividis. A diverging scale is the right choice while the range crosses zero; a sequential one is easier to read when every correlation is positive.	Coolwarm	appearance
Sample names	A display name per sample, for the figure only. Applied to the composite condition_n label, so a renamed condition follows through to these ticks.	as drawn	appearance
Title	This plot writes no title of its own. Entering one in the panel adds it, with the usual control over its size, weight and colour.	none	appearance
X axis, Y axis	Both labels read Samples and can be overwritten. The tick label size is set here too, which is the control that matters most on a wide matrix.	Samples, 7	appearance

WORTH KNOWING

The second table is the plot appearance panel, which is hidden until advanced mode is turned on (**Ctrl** + **Shift** + **E**). Everything it contains is described in Part XI; only the entries particular to this plot are repeated here. This page offers no legend card and no condition colours: the colour bar is the legend, and the cells are not coloured by condition.

22.3 How it is computed

HOW IT IS COMPUTED

1. The matrix is the data as loaded, on the $\log_2$ scale. Non-numeric and non-finite entries are read as NA.
2. Samples are relabelled condition_n, numbered within the condition in the order the annotation gives, with the original identifier appended in brackets when **Toggle IDs** is on. A sample column that is entirely missing is dropped; at least two must remain.
3. For each pair of samples (a, b), the Pearson correlation over the features that both quantified:

$$r_{ab} = \frac{\sum_{i \in O_{ab}} (x_{ia} - \bar{x}_a)(x_{ib} - \bar{x}_b)}{\sqrt{\sum_{i \in O_{ab}} (x_{ia} - \bar{x}_a)^2} \sqrt{\sum_{i \in O_{ab}} (x_{ib} - \bar{x}_b)^2}},$$

   where O_{ab} is the set of features present in both. Pairs are handled independently, so different cells can rest on different numbers of features, and a pair sharing nothing gives NA.
4. The samples are reordered by hierarchical clustering: the distance between two samples is $d_{ab} = 1 - r_{ab}$, clipped to $[0, 2]$, the linkage is complete, and the rows and columns are permuted into the dendrogram's leaf order. If any correlation is NA the whole reordering is skipped and the annotation order is kept, because a distance matrix with a hole in it has no dendrogram.
5. The colour limits are the smallest and largest finite correlation, or -1 and 1 when **Use Full Range** is on. If every correlation is the same value, the fixed limits are used instead, since a scale of zero width paints nothing.

ELLIPSE MODE

The same matrix, drawn differently. Each cell holds an ellipse of fixed width whose height is $0.9\,(1 - |r|)$, so it collapses towards a line as $|r|$ approaches 1 and opens into a circle as r approaches 0. It is tilted $+45^\circ$ for a positive correlation and -45° for a negative one, and filled from the same colour scale as Matrix mode at 60 % opacity — so the shape and the colour say the same thing twice, which is what makes a small difference in a uniformly high matrix visible.

Look for samples whose relationships differ from comparable replicates, then inspect coverage and sample-level distributions. A high correlation does not establish agreement: adding a constant to every log intensity can leave Pearson's correlation unchanged. Shared differences between abundant and low-abundance proteins can also make every sample pair highly correlated. Distinct condition blocks are therefore neither required for a valid experiment nor proof of a treatment effect. Read the colour scale and the number of shared observations before comparing cells.

22.3.1 Edge cases

- Fewer than two non-empty sample columns leaves nothing to correlate, and the page reports that instead of drawing.
- A sample whose values are all identical has no variance, so every correlation with it is NA — one grey row and one grey column, and no clustering for the whole figure.
- Cells that are NA are painted flat grey and labelled NA in both modes, so an absent correlation is never mistaken for a low one.
- Pairwise handling means a cell can rest on very few shared features. A high correlation over two hundred features is not the same evidence as a high correlation over eight thousand, and the figure does not distinguish them; the missing-value heatmap does.
- The pairwise scatter of two conditions — the plot behind a correlation rather than its summary — is a separate page: Comparison → Pearson correlation.

22.4 In the notebook

A notebook starting point, using the frames introduced in *The function reference*. These compact examples do not encode every UI option; use the plot's notebook ZIP, where available, for its saved settings and preprocessing recipe:

```python
fig = corr_plot(org_data, meta, id=False, data_scale="linear")
```

```r
fig <- corr_plot(org_data, meta, id = FALSE, data_scale = "linear")
```

data_scale declares the scale of the frame rather than converting it, and it decides what counts as missing, so the call above correlates linear intensities while the page correlates $\log_2$ ones — corr_plot(data, meta, id=False, data_scale="log2") is the call that reproduces the figure on this page. method="Ellipse" is **Type** Ellipse, and full_range=True is **Use Full Range**.

22.5 Exporting

The download reproduces the plot at the width, height and resolution set on the page. The browser scales the rendered figure to fit the page. This is the one plot in the section where

the defaults are worth raising rather than lowering — with twenty or more samples the tick labels need the height. The Matrix cells are drawn as one vector shape each, so an SVG or PDF download contains no embedded image and each cell can be selected in a drawing program. Choose PDF for a figure that will be scaled or edited, PNG for a slide.

Behind this page

Function	What it produces
corr_plot	The correlation matrix, in either mode, with the hierarchical reordering

PART IV

Completeness



23
The missing value plot

How many features are missing in no sample, in one sample, in two, and so on to the end of the experiment.

AT A GLANCE

Where	Completeness → Missing Value Plot
Data levels	Protein, phosphosite, phosphoprotein, peptide, precursor
Needs	An uploaded dataset and its annotation. Peptide and precursor additionally need a chosen abundance column.
Data source	The matrix as uploaded. Completeness has no Filtered or Imputed choice: it describes what arrived, before any decision taken about it.
One bar	The number of features missing in exactly that many samples.
Table	No. The same counts, with percentages, are on the Tables page.

AT PHOSPHOSITE LEVEL

The shared **Localization probability cutoff** selects sites across phosphosite views. At 0 it keeps all sites, including those with no probability; above 0 those unknowns are excluded. This is separate from completeness filtering and does not rerun normalization or imputation. An already collapsed phosphoprotein table has its own recorded collapse cutoff.

23.1 What the plot answers

The bar over 0 counts features quantified in every included sample. Bars to its right count increasingly incomplete features. Completeness alone does not decide whether to filter or impute: the experimental design, location of the gaps and plausible reasons for missingness also matter.

The plot says how *much* is missing per feature, not *where*. A feature with one missing value

is one bar to the right whether the gap is in the first sample or the last; the heatmap on the next page is what distinguishes them.

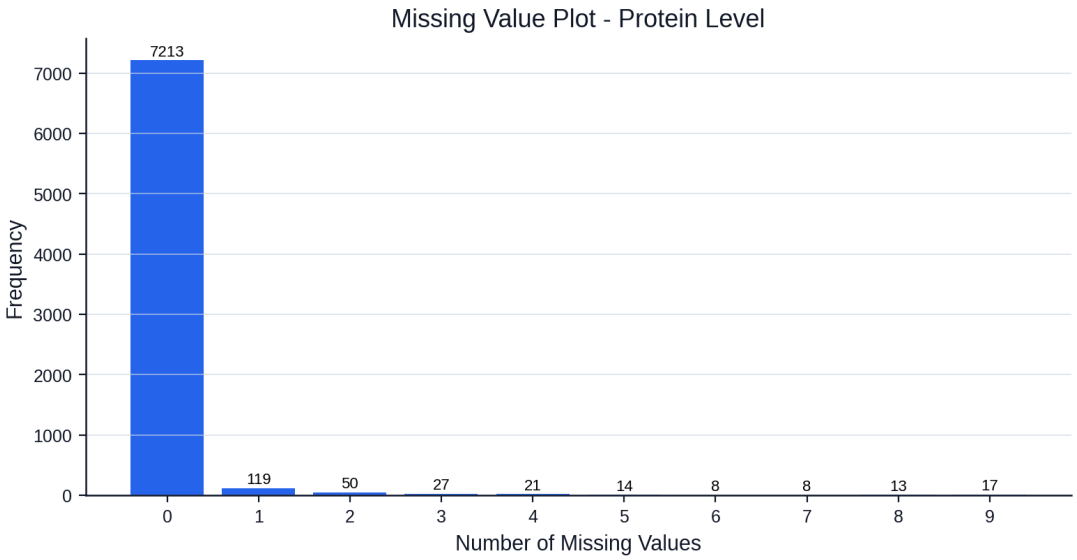


Figure 23.1. Protein level on the example project, ten samples: 7213 proteins are missing in none of them, and the tail from 1 to 9 holds the 277 that are missing somewhere. The axis stops at 9, one short of the sample count.

23.1.1 Reading it

The bullets below read a figure from a larger experiment — 96 samples rather than the ten above — so that the tail has something in it to read.

- 3663 proteins sit over 0: they carry a value in all 96 samples.
- 448 proteins sit over 1: 95 samples measured them and one did not. Which one is not shown here.
- 27 proteins sit over 95: each was reported in a single sample.
- There is no bar over 96, and there cannot be one. A feature missing everywhere is dropped before the counting, so the axis stops one short of the sample count.

> **EASY TO GET WRONG**
>
> The x axis is a list of the counts that actually occur, not a number line. If no feature is missing in exactly 2 samples, the axis runs 0, 1, 3 with no gap where 2 would be, and the bars either side of the absent value sit next to each other. Read the tick labels before reading the shape, and read the Tables page when the exact numbers matter.

23.2 Controls

Controls on this page

Control	What it does	Default	Where
Dataset level	Which dataset to describe: Protein, Phospho, Phosphoprotein, Peptide or Precursor. Only the levels actually uploaded are offered; Phosphoprotein appears once a collapse has been run, and Peptide and Precursor only on this page, from the full report.	Protein	page
Toggle Text	Prints the count over each bar.	on	page
Bin missing values	Folds the long tail into one bar. With the value *b* above zero, every feature missing in more than *b* samples is counted into a single >b bar and the axis runs 0 to b and then >b. Zero means no binning.	0	page
Abundance Column	Peptide and precursor only. Which quantity column of the report is pivoted before the missing values are counted. Precursor.Quantity is chosen automatically when the report has it; otherwise, with more than one candidate column, no plot is drawn until one is picked.	the report's own	page
Width, Height, DPI	The size of the *downloaded* figure. The browser scales the rendered figure to fit the page. Width and height are held between 1 and 60 cm, the resolution between 72 and 600.	20 cm × 10 cm, 300	page
Download format	PNG, JPG, WEBP, SVG or PDF.	PNG	page

In the plot appearance panel

Control	What it does	Default	Where
Title	Shown or hidden, and its text, size, weight, colour and position. The title is written from the data level: Missing Value Plot - Protein Level and its equivalents for phosphosite, phosphoprotein, peptide and precursor.	shown	appearance
X axis, Y axis	The text and styling of Number of Missing Values and Frequency, and of the tick labels under the bars.	plot's own	appearance

In the plot appearance panel (continued)

Control	What it does	Default	Where
Frame and grid	Which of the four spines are drawn, and the grid behind the bars. This plot opens with all four spines and a horizontal dashed grid.	four spines; horizontal dashed grid, #b0b0b0, width 0.8	appearance
Font	One face for the whole figure, or one per text.	plot default	appearance
Background	Theme, paper colour and transparency.	Default, white, opaque	appearance

NOTE

Bar Color and **Text Size**, which the earlier manual listed beside the plot, are gone from the page. The renderer still reads a bar colour and a label size, but neither the **Bars** card nor the **Value labels** card is offered on this page, so the bars are drawn in the default blue and the printed counts at size 8. The **Bars** and **Value labels** cards are offered here: this renderer reads both the bar colour and the label size.

WORTH KNOWING

The second table is the plot appearance panel, which is hidden until advanced mode is turned on (**Ctrl** + **Shift** + **E**). Everything it contains is described in Part XI; only the entries particular to this plot are repeated here.

23.3 How it is computed

HOW IT IS COMPUTED

The matrix is restricted to the sample columns named in the annotation. A value is missing when it is NA, when it is not a number, or when it is infinite; a zero counts as missing too, because a zero in a linear intensity means not detected. The one exception is a dataset uploaded already on the $\log_2$ scale, where zero is a real measurement — an intensity of 1 — and is kept.

For feature i over n sample columns,

$$m_i \;=\; \sum_{j=1}^{n} \mathbf{1}\big[x_{ij} = \mathrm{NA}\big], \qquad h_k \;=\; \#\{\, i : m_i = k \,\}.$$

Features with $m_i = n$ are removed before h_k is formed, so k runs from 0 to $n - 1$ and the height of a bar is the number of features missing in exactly that many samples. Only the values of k that occur are drawn.

WITH BINNING

With **Bin missing values** set to $b > 0$, every m_i above b is relabelled $>$b before counting, and the axis is fixed to 0, …, b, $>$b. The exclusion of the missing-everywhere features happens first, on the numbers, so the $>$b bar never quietly absorbs them.

PEPTIDE AND PRECURSOR

The full report is long, so it is pivoted first: rows are the stripped sequence at peptide level and the precursor identifier at precursor level, columns are the run file names mapped onto the annotated samples, and the values are the chosen abundance column. Duplicate entries for the same feature and run are reduced with the maximum — not the first row, which would depend on the order the file happened to be written in. Zeros then become NA, and the counting is the same.

23.3.1 The example worked

Three proteins over three samples:

THE MATRIX, AND THE TWO BARS IT GIVES

	A1	A2	B1	m_i
Protein 1	3.34	3.54	NA	1
Protein 2	3.34	3.54	2.67	0
Protein 3	3.34	NA	2.67	1

Number of missing values	Frequency
0	1
1	2

Two bars, not three. There is no bar over 2 because no protein is missing in exactly two samples, and there could be none over 3 in any case: that is the sample count. The Tables page prints the same two rows with their percentages, 33.33 % and 66.67 %.

23.3.2 Edge cases

- A feature missing in every sample never appears. It occupies a row of the matrix, but it is left out of this plot, the heatmap and the Tables page alike, and the Tables page says how many such features were excluded.
- A missing-value count that no feature has gets no bar and no tick.
- A dataset with nothing missing draws a single bar over 0.
- At peptide and precursor level, nothing is drawn until an abundance column is chosen, and the plot then describes that column alone: a second abundance column in the same report can give a different figure.
- Binning changes the picture but not the underlying counts: the same dataset summed over the bars gives the same total at every setting of *b*.

23.3.3 Interpreting the pattern

A tall bar over 0 means many features were observed in every selected sample. A rise near the other end means many were observed only sparsely. Neither shape is universally right or wrong. A peak around half the sample count could come from features restricted to one of two equal-sized conditions, or from missing values scattered across both. This histogram cannot distinguish them; the missing-value heatmap supplies the sample and condition pattern.

23.4 In the notebook

A notebook starting point, using the frames introduced in *The function reference*. These compact examples do not encode every UI option; use the plot's notebook ZIP, where available, for its saved settings and preprocessing recipe:

PYTHON

```
fig = missing_value_plot(data, meta, data_scale="log2")
```

R

```
fig <- missing_value_plot(data, meta, data_scale = "log2")
```

There is no data level to choose: the level is the frame handed in, so the phosphosite and phosphoprotein figures are the same call on those frames, and the title is written from the frame's own identifier column. data_scale carries the convention the page applies silently — "log2" keeps a zero as a measurement, "linear" reads it as a non-detection — and bin is the binning control, at 0 unless it is given.

23.5 Exporting

The download reproduces the plot at the width, height and resolution set on the page, which is why those three controls do not change the figure on screen: the displayed plot is fitted to the window, the downloaded one to the numbers. Choose PDF for a figure that will be scaled or edited, PNG for a slide. There is no table under this plot; the same counts, with percentages, download as CSV from the Tables page.

Behind this page

Function	What it produces
missing_value_plot	The plot for wide-format data: protein, phosphosite and phosphoprotein
missing_value_plot_pep	The peptide-level version, from the long-format report
missing_value_plot_prec	The precursor-level version, from the long-format report

- Missing values

24
The missing value heatmap

Where in the matrix the gaps are: which samples, which features, and whether the two are related.

AT A GLANCE

Where	Completeness → Missing Value Heatmap
Data levels	Protein, phosphosite, phosphoprotein. Peptide and precursor are not offered here.
Needs	An uploaded dataset and its annotation.
Shows	One cell per feature and sample: measured, or not.
Colours	The viridis scale over a yes/no matrix, so two colours only: pale yellow for a missing value, dark purple for a measured one. Fixed; there is no colour control on this page.
Table	No.

AT PHOSPHOSITE LEVEL

The shared **Localization probability cutoff** selects sites across phosphosite views. At 0 it keeps all sites, including those with no probability; above 0 those unknowns are excluded. This is separate from completeness filtering and does not rerun normalization or imputation. An already collapsed phosphoprotein table has its own recorded collapse cutoff.

24.1 What the plot answers

The missing value plot counts the gaps; this one places them. Its whole content is the pattern: a yellow column has missing measurements throughout, a yellow row is absent across samples, and a yellow block can show condition-associated missingness. These are observations, not diagnoses of sample failure or of a missing-data mechanism. A feature observed in only one condition may survive an at-least-one-condition filter; filling the other condition with low values can then strongly influence its fold change.

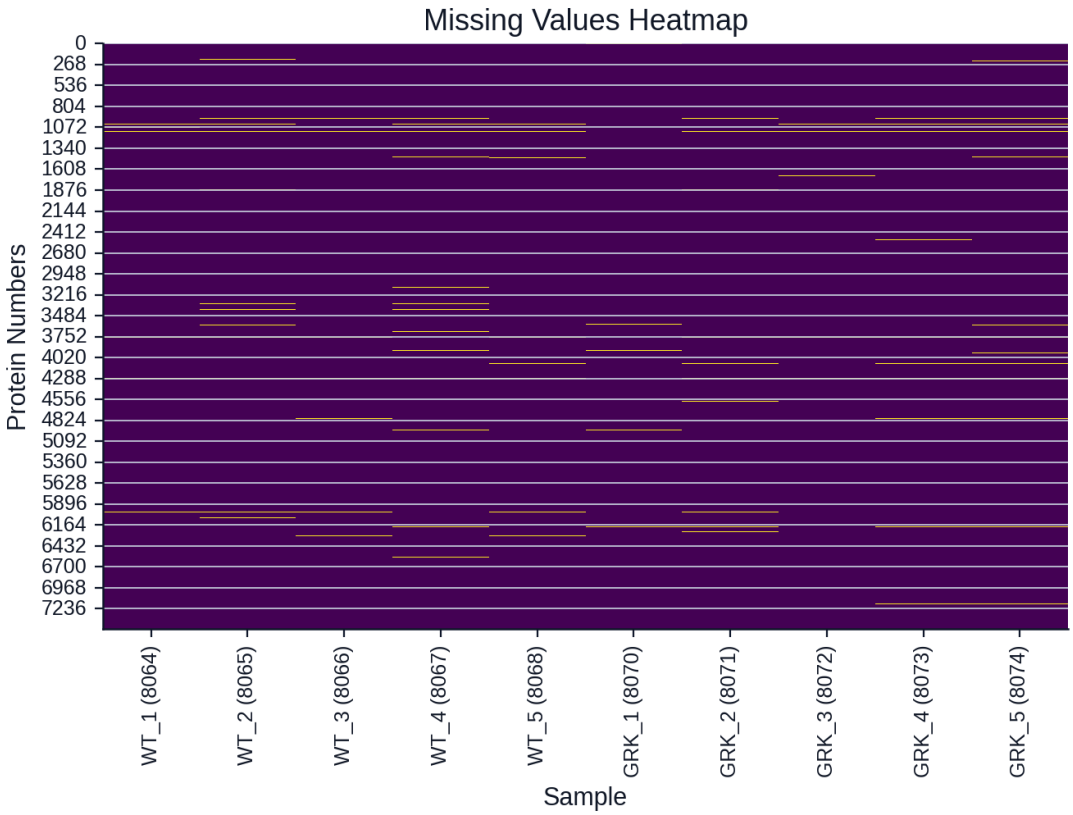


Figure 24.1. Protein level on the example project: ten columns, five WT then five GRK, against every protein of the matrix, with **Toggle IDs** on. Pale yellow is a missing value, dark purple a measured one; there is no colour bar, because the matrix has only two states. This dataset is almost complete, so the figure is nearly solid purple and the few gaps read as isolated horizontal lines — no column and no condition is paler than the rest.

24.1.1 Reading it

- One cell: one feature in one sample, yellow if that value is missing.
- A vertical yellow stripe: a sample missing a great many features. Check it against the coverage plot, where it is the short bar.
- A horizontal yellow band: features missing across most of the experiment, usually the low-abundance end of the dataset.
- A yellow block confined to the samples of one condition: missingness that is not random with respect to the comparison to be made.
- The label under each column: condition_n, numbered within the condition. With **Toggle IDs** on it becomes condition_n (id), where the id is the last number in the original sample name, or the whole name when it has no number.

> **EASY TO GET WRONG**
>
> Neither axis is sorted. The rows are in the order the matrix has them — the order of the uploaded file — and the columns are in the order of the annotation, grouped by condition. Nothing is clustered and nothing is reordered to make a pattern, which is what makes a visible block worth something; but it also means that a dataset whose file order happens to carry structure, such as a report sorted by intensity, will show horizontal banding that is a property of the sort and not of the experiment.

24.2 Controls

Controls on this page

Control	What it does	Default	Where
Dataset level	Which dataset to draw: Protein, Phospho or Phosphoprotein. Only the levels actually uploaded are offered; Phosphoprotein appears once a collapse has been run.	Protein	page
Toggle IDs	Adds the original sample identifier to each column label, as condition_n (id).	on	page
Width, Height, DPI	The size of the *downloaded* figure. The browser scales the rendered figure to fit the page. Width and height are held between 1 and 60 cm, the resolution between 72 and 600.	10 cm × 10 cm, 300	page
Download format	PNG, JPG, WEBP, SVG or PDF.	PNG	page

In the plot appearance panel

Control	What it does	Default	Where
Title	Shown or hidden, and its text, size, weight, colour and position. The title is Missing Values Heatmap at every data level.	shown	appearance
X axis, Y axis	The text and styling of Sample and Protein Numbers, and of the column labels, which are printed vertically at size 7.	plot's own	appearance
Sample names	One row per sample, to replace the condition_n label with a name of your own. Empty means the generated label.	generated	appearance
Frame and grid	Which of the four spines are drawn, and the grid behind the cells.	four spines; no grid	appearance
Font	One face for the whole figure, or one per text.	plot default	appearance
Background	Theme, paper colour and transparency. The paper is only visible in the margins: the matrix itself covers the axes completely.	Default, white, opaque	appearance

WORTH KNOWING

The second table is the plot appearance panel, which is hidden until advanced mode is turned on (**Ctrl** + **Shift** + **E**). Everything it contains is described in Part XI; only the entries particular to this plot are repeated here. There is no colour card here: the palette of this figure is fixed.

24.3 How it is computed

HOW IT IS COMPUTED

The matrix is restricted to the sample columns named in the annotation and renamed condition_n in annotation order, so the columns arrive grouped by condition. A value is missing when it is NA, when it is not a number, or when it is infinite; a zero counts as missing as well, unless the dataset was uploaded already on the $\log_2$ scale, where zero is an intensity of 1 and is a real measurement.

Each cell is then the truth value of that one test: the whole figure is the matrix $\mathbf{1}[x_{ij} = \text{NA}]$, painted with the viridis scale. A two-valued matrix takes only the two ends of

that scale, which is why the figure has exactly two colours and no colour bar is drawn. No row or column is sorted, clustered or aggregated. Every feature with a value in at least one selected sample is one row of the image; a feature with no value in any of them is left out, as on the other completeness pages.

MORE FEATURES THAN PIXELS

Every quantified feature is a row, and the image has a finite number of pixel rows: height in centimetres, divided by 2.54, times the DPI. At the default 10 cm and 300 dpi that is about 1180 rows of pixels, so a dataset of 8000 proteins cannot give each feature its own row and neighbouring features are merged into one line of the image. Isolated single missing values therefore thin out or vanish at large feature counts, while stripes and blocks survive. Raise the height and the DPI before concluding that a region is clean.

24.3.1 Edge cases

- A sample with no values at all is a solid yellow column; a feature measured everywhere is a solid dark row.
- A dataset with nothing missing is a solid dark rectangle, and one with nothing measured a solid yellow one. Neither draws a legend to say which it is: the two states are distinguished only by colour, so a uniform figure has to be read against the coverage plot or the Tables page.
- The y axis is labelled Protein Numbers at every data level, including phosphosite and phosphoprotein. The Y axis card in the appearance panel replaces the text.
- Filtering and imputation do not reach this page. It always draws the matrix as uploaded, which is the only matrix on which the question makes sense — an imputed matrix has no gaps left to show.

24.3.2 Interpreting the pattern

A column with many missing entries identifies a sample to investigate; a condition-aligned block identifies structured missingness worth checking against the design. Neither pattern establishes its cause or justifies automatic exclusion. Horizontal bands can reflect low detectability or genuinely restricted occurrence. Check feature identities, coverage and the original export before choosing a filter or imputation assumption.

24.4 In the notebook

A notebook starting point, using the frames introduced in *The function reference*. These compact examples do not encode every UI option; use the plot's notebook ZIP, where available, for its saved settings and preprocessing recipe:

PYTHON

```
fig = missing_value_heatmap(data, meta, id=False, data_scale="log2")
```

R

```
fig <- missing_value_heatmap(data, meta, id = FALSE, data_scale = "log2")
```

id=False drops the bracketed sample identifier from the column labels, leaving condition_n; the page has **Toggle IDs** on by default, so the figure above is the id=True labelling. The one trap is the figure size: width and height are inches here, not the centimetres of the download controls, and they default to 20 × 10 — so the number of feature rows the image can resolve, the thing this page depends on most, is set by a different unit in the notebook than on the page.

24.5 Exporting

The download reproduces the plot at the width, height and resolution set on the page, which is why those three controls do not change the figure on screen. Height and DPI matter more here than on any other page, because they decide how many features the image can resolve; a heatmap meant to be read at the level of individual cells needs both raised well above their defaults. PDF keeps the figure scalable, but the matrix inside it is still drawn at the chosen resolution.

Behind this page

Function	What it produces
missing_value_heatmap	The heatmap, with the sample renaming and the two-colour matrix
missing_sample_table	The per-sample counts behind a yellow column, as numbers

25
The completeness tables

The missing values as numbers: one line for the whole matrix, one row per sample, and one row per missing-value count.

AT A GLANCE

Where	Completeness → Tables
Data levels	Protein, phosphosite, phosphoprotein. Peptide and precursor are not offered here.
Needs	An uploaded dataset and its annotation.
Data source	The matrix as uploaded. There is no Filtered or Imputed choice on this page.
Shows	Three figures across the top, then three tables: per sample, the outliers among them, and per missing-value count.
Download	Each table separately, as CSV.

AT PHOSPHOSITE LEVEL

The shared **Localization probability cutoff** selects sites across phosphosite views. At 0 it keeps all sites, including those with no probability; above 0 those unknowns are excluded. This is separate from completeness filtering and does not rerun normalization or imputation. An already collapsed phosphoprotein table has its own recorded collapse cutoff.

25.1 What the page answers

Two questions that a figure answers only approximately. *How bad is it* — one percentage for the whole matrix, and one per sample, so that a run can be named rather than pointed at. And *which samples are past the line* — a threshold the user sets, applied to the per-sample percentage, with the samples that exceed it listed by name.

The three figures across the top are the whole-matrix summary:

- Overall Missing Values: the percentage of *cells* of the matrix that are missing, over every feature with at least one value and every sample at once.
- Outlier Threshold: the value currently set below, repeated so that the list beneath it can be read without scrolling back.
- Outlier Samples: how many samples exceed it.

The tables are recomputed when the level or the threshold changes, and the **Refresh** button re-runs them against the dataset as it now stands — after a collapse, a re-annotation or a new upload.

25.2 Controls

Controls on this page

Control	What it does	Default	Where
Dataset level	Which dataset to describe: Protein, Phospho or Phosphoprotein. Only the levels actually uploaded are offered; Phosphoprotein appears once a collapse has been run.	Protein	page
Toggle IDs	Labels the samples condition__n (id) rather than condition__n, where the id is the last number in the original sample name, or the whole name when it has no number.	on	page
Outlier Threshold (%)	The percentage of missing values above which a sample is listed as an outlier. Held between 0 and 100. The test is strict: a sample at exactly the threshold is not an outlier.	50	page

> **NOTE**
>
> This page draws no figure, so the plot appearance panel is not shown on it, and neither are the width, height, resolution and format controls that belong to a download of an image.

25.3 How the numbers are computed

HOW IT IS COMPUTED

The matrix is restricted to the sample columns named in the annotation and renamed as **Toggle IDs** requires. A value is missing when it is NA, when it is not a number, or when it is infinite; a zero counts as missing as well, unless the dataset was uploaded already on the $\log_2$ scale, where zero is an intensity of 1 and is a real measurement. Features with no value in any selected sample are removed first. Write m for the number of remaining features (rows), n for the number of samples (columns).
The line at the top. Every cell of the matrix, counted once:

$$\text{overall } \% \;=\; \frac{\#\{\text{ missing cells }\}}{m \times n} \times 100.$$

The sample table. Down each column, the count of missing values in that sample, over the number of features:

$$\%\text{ for sample } j \;=\; \frac{\#\{\, i \,:\, x_{ij} = \text{NA} \,\}}{m} \times 100.$$

The feature table. Across each row, the count of missing values for that feature; the features are then grouped by that count, and each group is given as a share of all features:

$$\%\text{ for count } k \;=\; \frac{\#\{\, i \,:\, m_i = k \,\}}{m} \times 100.$$

All three are rounded to two decimals. The percentages of the feature table sum to 100; those of the sample table do not, and are not meant to — each is a share of the same denominator, not a share of the missing values.

25.4 The sample table

Headed Sample-level Missing Values.

Columns in this table

Column	What it holds
Sample	The sample, labelled condition_n or condition_n (id), in the order of the annotation.
Counts	How many features are missing in that sample.
%_missing_values	That count as a percentage of all features, to two decimals.

25.5 The outlier table

Headed Outliers by Missing Value %. One column, Sample, and one row per sample whose %_missing_values is strictly greater than **Outlier Threshold (%)**. When no sample exceeds it the table is replaced by a line saying so, and the count at the top of the page reads zero.

Being past the threshold is not a verdict. It is the point at which a sample is worth looking at in the heatmap and the coverage plot: a sample that is missing 60 % of the features but is otherwise ordinary in the ones it has may still support some questions, while a lower overall percentage can still hide missingness concentrated in a scientifically important subset. Neither percentage alone decides exclusion.

25.6 The feature table

Headed Feature-level Missing Value Distribution.

Columns in this table

Column	What it holds
n_MissingProteinValues	A number of missing values, from 0 upwards. The column carries this name at every data level, not only for proteins.
Counts	How many features are missing in exactly that many samples.
%_missing_values	That count as a percentage of all features, to two decimals.

This table is the missing value plot, as numbers. Counts is the height of a bar and n_MissingProteinValues is its position on the x axis, with the percentage added. Both leave out the features that are missing in *every* sample, so no row here has n_MissingProteinValues equal to the number of samples.

25.6.1 The example worked

Three proteins over three samples, A1, A2 and B1:

THE MATRIX, AND THE THREE OUTPUTS

	A1	A2	B1
Protein 1	3.34	3.54	NA
Protein 2	3.34	3.54	2.67
Protein 3	3.34	NA	2.67

Two of the nine cells are missing, so the line at the top reads 22.22%.

Sample	Counts	%_missing_values
A1	0	0.0
A2	1	33.33
B1	1	33.33

n_MissingProteinValues	Counts	%_missing_values
0	1	33.33
1	2	66.67

At the default threshold of 50 the outlier table is empty: 33.33 is not greater than 50. Set the threshold to 30 and both A2 and B1 appear in it.

25.6.2 Edge cases

- A feature with no value in any selected sample is left out of this page — of the line at the top, of every sample's count and of the feature table — as it is left out of the missing value plot and the heatmap. A line under the cards says how many features were excluded. The notebook functions do the same in Python and R.
- A sample with no values at all reads 100.0 and is an outlier at any threshold below 100. At a threshold of exactly 100 nothing can ever be an outlier, because the test is strict.
- Both percentage columns carry the same name, %_missing_values, but they have different denominators — features per sample in one table, features per count in the other. The two downloads are separate files, so the names do not collide.

25.7 In the notebook

A notebook starting point, using the frames introduced in *The function reference*. These compact examples do not encode every UI option; use the plot's notebook ZIP, where available, for its saved settings and preprocessing recipe:

PYTHON

```
missing_protein_table(data, meta, id=False, data_scale="log2")
missing_sample_table(data, meta, id=False, data_scale="log2")
```

R

```
print(missing_protein_table(data, meta, id = FALSE, data_scale = "log2"))
print(missing_sample_table(data, meta, id = FALSE, data_scale = "log2"))
```

These two return frames rather than figures, which is why there is no fig = in front of them: what comes back is the feature table and the sample table, with the same three columns and the same names as the page, but unrounded — the two decimals are the page's, not the function's. The other two numbers on the page have their own functions: missing_value_outlier takes the frame that missing_sample_table returned and gives the names past the threshold, and all_miss_val gives the one percentage for the whole matrix.

25.8 Exporting

Each of the three tables has its own **Download CSV** button and downloads exactly what is on screen, with the same column names and the same rounding. The outlier table downloads its one Sample column; when it is empty there is no button, because there is nothing to write.

Behind this page

Function	What it produces
missing_sample_table	The per-sample table
missing_protein_table	The per-count table
missing_value_outlier	The samples past the threshold
all_miss_val	The one percentage for the whole matrix

PART V

Statistics



26
The volcano plot

Which features differ between two conditions, by how much, and how certain that is.

AT A GLANCE

Where	Statistics → Volcano
Data levels	Protein, phosphosite, phosphoprotein
Needs	Two conditions with at least two samples each; the paired test also needs a pair_id per sample.
Two sources	Volcano compares two conditions; Volcano Control compares two differences, each against its own control.
Two renderers	**Type** Interactive for exploring, Normal for a figure for print.
Fold change	$\log_2$ FC = mean(Condition 1) − mean(Condition 2). Positive means higher in Condition 1.
Table	Yes, below the plot, downloadable as CSV.

AT PHOSPHOSITE LEVEL

The shared **Localization probability cutoff** selects sites across phosphosite views. At 0 it keeps all sites, including those with no probability; above 0 those unknowns are excluded. This is separate from completeness filtering and does not rerun normalization or imputation. An already collapsed phosphoprotein table has its own recorded collapse cutoff.

CHANGED IN version 1.0.0

The fold change is computed as **Condition 1 minus Condition 2**, so a positive $\log_2$ FC means the feature is higher in the condition selected as **Condition 1**, and the axis is labelled with that subtraction in full. Earlier releases, and the volcano manual that described them, defined it the other way round — mean(Condition 2) − mean(Condition 1) — while the title of the plot read the same. Every direction in a result produced be-

> fore this change is therefore reversed: what was reported as upregulated is reported here as downregulated, at the same magnitude and the same p-value.

26.1 What the plot answers

Deciding that a feature has moved needs two numbers, and neither is sufficient alone: the effect size says how large the difference is and in which direction, the p-value says how well the replicates support it. The volcano puts one on each axis so that both are read at once, and the two thresholds divide the result into the classes the table reports.

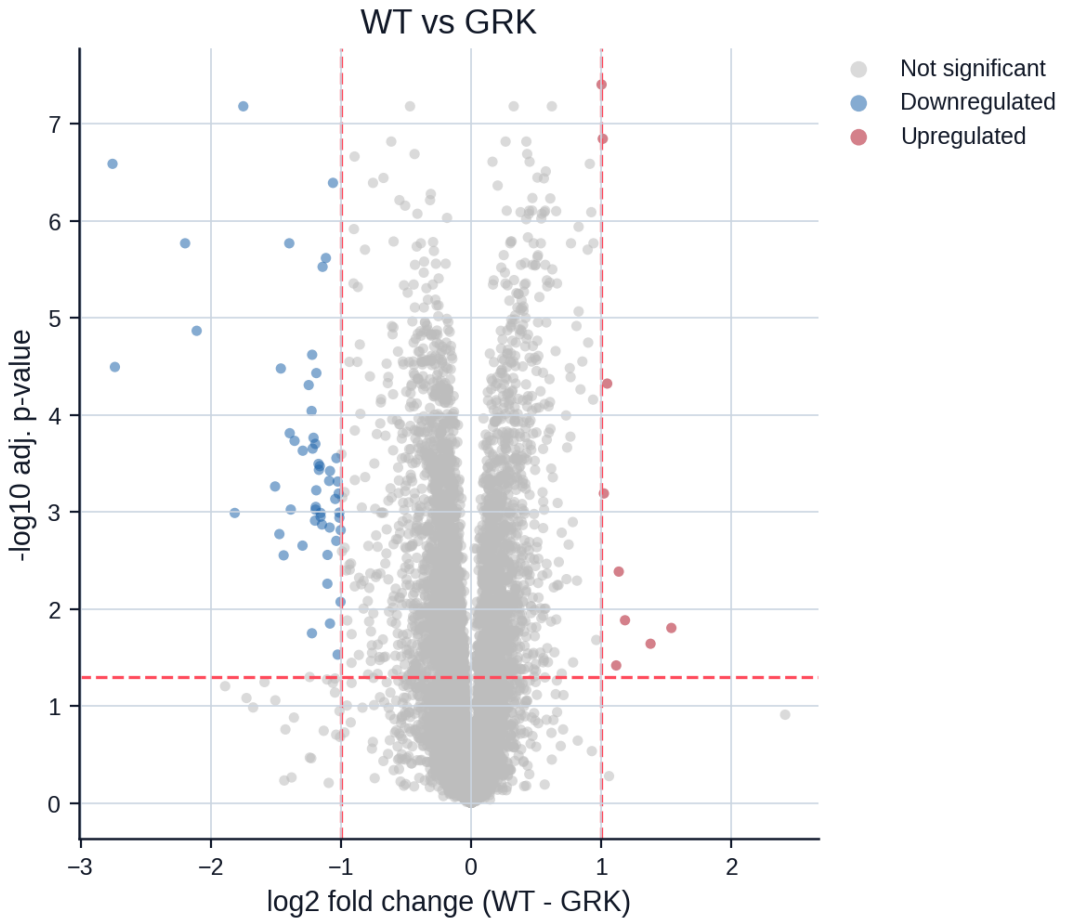


Figure 26.1. The static volcano on the example data: WT against GRK at protein level, at the default thresholds. The axis is labelled log2 fold change (WT - GRK), so the figure states its own orientation — the red points on the right are higher in WT, the blue ones on the left higher in GRK. The three dashed lines are the two $\log_2$ FC thresholds at ±1 and the p-value threshold, drawn at $-\log_{10}(0.05)$; far more features clear them on the left than on the right.

26.1.1 Reading it

- One point, one feature, named by **Label column**.

- The x axis, $\log_2$ FC, labelled log2 fold change (Condition 1 - Condition 2) with the real condition names in it. Right is higher in Condition 1, left is higher in Condition 2.
- The y axis, $-\log_{10}$ of the adjusted p-value, or of the raw p-value when **Use uncorrected p-values** is on; the axis title says which. Higher is stronger evidence.
- Three threshold lines: horizontal at $-\log_{10}(\alpha)$, vertical at $\pm\tau$, for the two threshold controls.
- Colour is the class: red up, blue down, grey not significant — saturated in the interactive plot, muted in the static one.
- A feature named in **Highlight terms** keeps its class colour and gains an outline, and in the interactive plot is drawn larger and at full opacity.

A volcano need not be symmetric. An asymmetric pattern may reflect real biology, normalization assumptions, sample composition or technical effects. Check those possibilities against the design rather than adjusting the data to produce a familiar shape. Few significant points may reflect small effects, high variability, limited replication or multiple-testing correction.

For example, a log2 fold change of 1 means a twofold ratio of geometric mean intensities in the ordinary two-condition comparison; -1 means one half. An adjusted p-value of 0.01 appears at height 2. The p-value is not the probability that this protein is unchanged, and a fold-change threshold does not by itself establish biological importance. Inspect the replicate values of a candidate in the single-protein plots before interpreting it.

> EASY TO GET WRONG
>
> Do not take the direction of the x axis from memory, or from any manual written before version 1.0.0. Check it on your own data in one step: type the label of a feature you know is higher in one of the two conditions into **Highlight terms** and see which side it lands on. A feature higher in the condition you put in **Condition 1** belongs on the right; if it is on the left, the two selectors are the other way round.

26.2 Controls

Controls on this page

Control	What it does	Default	Where
Dataset level	Protein, phosphosite or phosphoprotein. Only the levels uploaded are offered; phosphoprotein appears once a collapse has been run.	Protein	page

Controls on this page (continued)

Control	What it does	Default	Where
Source	Volcano Control subtracts a control condition from each side first and compares the two differences; it needs four distinct conditions.	Volcano	page
Data Source	The matrix the test runs on: Data, Filtered Data or Imputed Data, whichever exist.	the most processed matrix available	page
Condition 1, Condition 2	The two conditions compared. Condition 1 is the minuend: a positive $\log_2$ FC means higher there.	first condition in the annotation	page
Condition 1 Control, Condition 2 Control	Volcano Control only. The control each side is measured against; all four conditions must differ.	first condition in the annotation	page
Type	Interactive draws the plot with hover text on every point; Normal renders a static figure for print.	Interactive	page
Width (cm), Height (cm), DPI	The size and resolution of the static figure, only while Normal is selected. The interactive plot has none: it fills its frame, Plotly's camera button saves a picture of it, and **Open Interactive Plot** opens it in a window of its own.	18 cm × 14 cm, 300	page
Download format	PNG, JPG, WEBP, SVG or PDF, beside **Download Plot** in the plot's header. Normal only.	PNG	page
Label column	What names a feature and is matched against **Highlight terms**: the workflow name or the gene names.	the workflow name	page
Test type	Unpaired (Student's t, equal variance), Unpaired (Welch, unequal variance) or Paired t-test (matched samples). In Volcano Control the pooled unpaired test is not offered, because the control means carry their own variance; the entries there are Unpaired (propagated variance) and Paired (propagated control variance).	Unpaired (Student's t, equal variance)	page
P-value threshold, log2 FC threshold	α and τ: a feature must be below the first and beyond the second in absolute value to be called up or down.	0.05, 1	page
Use uncorrected p-values	Applies the threshold to the raw p-value instead of the adjusted one and plots that on the y axis. It does not change the test.	off	page
Label top up-regulated, Label top down-regulated	Normal only. How many hits in each direction to name, from the top of the y axis. A label that cannot be placed clear of another is left off.	0	page

Controls on this page (continued)

Control	What it does	Default	Where
Highlight terms	Labels to emphasise, space-separated. Matching ignores case and a trailing _HUMAN; a semicolon-separated annotation matches on any token.	empty	page
Add Volcano To Report, Remove Volcano From Report	Stores the current configuration in the report list, or removes a matching entry. Plain Volcano only.	—	page

In the plot appearance panel

Control	What it does	Default	Where
Volcano points	Size, opacity, shape and colour, for all three classes at once and then per class.	6 and 0.85 significant, 5 and 0.55 not significant	appearance
Highlight size, Highlight outline	Marker size and outline colour of a highlighted feature.	10, #22c55e	appearance
Hit label size	The on-figure labels written by the **Label top** controls. Static plot only.	7	appearance
Reference line	The three threshold lines: shown or hidden, colour, dash and width. Both renderers read the same settings.	shown, dashed	appearance
Title, axis labels, font, frame, grid, background	As for every other plot; described in Part XI.	the plot's own	appearance

> **WORTH KNOWING**
>
> The appearance panel is hidden until advanced mode is on (**Ctrl** + **Shift** + **E**). Its condition-renaming card is offered here, but both renderers build the title and the x-axis label from the annotation's own condition names, so a rename reaches neither.

26.3 How it is computed

Choose the test from the experiment, not from which option produces more significant points. Independent biological samples and genuinely matched pairs are different designs. Welch's

test permits unequal variances; the pooled test assumes equal variances. With few replicates, distributional assumptions and outlying measurements matter especially. Neither test automatically models batches, technical-replicate dependence or uncertainty introduced by imputation.

THE FOLD CHANGE

For the two selected conditions the $\log_2$-transformed values of every feature are taken from the chosen matrix, and a mean is formed within each condition over the replicates present:

$$\bar{x} = \text{mean(Condition 1)}, \quad \bar{y} = \text{mean(Condition 2)}, \quad \log_2 \text{FC} = \bar{x} - \bar{y}.$$

Positive means higher in Condition 1, negative higher in Condition 2. For ordinary $\log_2$ intensities, the difference of means is the $\log_2$ ratio of geometric means; +1 is a doubling on that processed scale. VSN is only log-like, particularly at low intensity, so its differences should not be presented as exact raw-intensity ratios.
In Volcano Control each side is centred on its own control first — every value of Condition 1 loses the mean of Condition 1 Control, and likewise for Condition 2 — before the means are taken. The axis then reads (C1 - C1 control) - (C2 - C2 control).

THE T-TEST

Unpaired, equal variance — independent samples, pooled variance. With n_1, n_2 the numbers of values actually observed for that feature and s_1^2, s_2^2 their within-condition variances:

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}, \qquad SE = \sqrt{s_p^2 \left(\frac{1}{n_1} + \frac{1}{n_2}\right)},$$

$$t = \frac{\log_2 \text{FC}}{SE}, \quad df = n_1 + n_2 - 2, \quad p = 2\,P(T_{df} > |t|).$$

Unpaired, unequal variance pools nothing: it uses $SE = \sqrt{s_1^2/n_1 + s_2^2/n_2}$ with Welch–Satterthwaite degrees of freedom. An unpaired control-corrected comparison also propagates the two control means' variances into SE and uses Satterthwaite degrees of freedom, rather than treating estimated control means as known constants.
Paired — matched samples, read from the pair_id column of the annotation; each identifier may appear at most once per condition, and samples are ordered by it rather than by file order. The test is on the within-pair differences $d_i = x_i - y_i$, in the same orientation as everything else:

$$t = \frac{\bar{d}}{s_d/\sqrt{n}}, \qquad df = n - 1,$$

with $\bar{d}$ and s_d their mean and standard deviation and n the number of pairs observed on both sides, reported as n_pairs. The $\log_2$ FC on the plot is $\bar{d}$ itself, which is not the difference of the two condition means when some pairs are incomplete. With control correction, the two control-mean variance terms are added to the variance of the paired

mean difference and the degrees of freedom are recomputed. This models those control means as independent components; it is not a general repeated-measures model over all four conditions.

MULTIPLE TESTING CORRECTION

One test runs per feature, so several thousand run per comparison and at $\alpha = 0.05$ several hundred features would clear the threshold on noise alone if all null hypotheses were true and tests calibrated. Under independence or appropriate positive dependence, the Benjamini–Hochberg procedure controls the false discovery rate instead — the expected share of false positives among the features called significant.

1. Sort the m p-values into ascending order, $p_{(1)} \leq p_{(2)} \leq \ldots \leq p_{(m)}$.
2. Scale each by its rank:
$$\text{adj_p}_{(i)} = \frac{m}{i} p_{(i)}.$$
3. Enforce monotonicity, walking from the largest rank down, so that an adjusted value never exceeds the one above it:
$$\text{adj_p}_{(i)} = \min\left(\text{adj_p}_{(i)}, \text{adj_p}_{(i+1)}\right),$$
and cap the result at 1.

Features whose test produced no finite p-value take no part: they are excluded from m, keep an adjusted value of NA and are classed Unavailable. m is the number of features actually tested in this comparison, not the number of rows in the dataset. adj_pval always retains the Benjamini–Hochberg adjusted value. **Use uncorrected p-values** selects the raw value for selected_pval, the y axis and classification. The correction column then says none (uncorrected) rather than Benjamini-Hochberg — so a downloaded table read without the interface still states what was done.

Classification follows from the two thresholds and the value q in selected_pval — adjusted, or raw when the uncorrected switch is on. **Upregulated**: $q < \alpha$ and $\log_2 \text{FC} > \tau$, so higher in Condition 1. **Downregulated**: $q < \alpha$ and $\log_2 \text{FC} < -\tau$, so higher in Condition 2. **Unavailable**: the row has no p-value, either because the test produced no finite one or because the row has no variation (see *Edge cases*). **Not significant**: everything else. Both conditions have to hold, which is what the two sets of threshold lines mean.

26.3.1 Edge cases

- A feature needs at least two observed values in each condition (for a paired test, also at least two complete pairs), or it is not tested. It is not drawn on the plot and is not in the result table, so a feature measured in only one condition is never shown at an extreme fold change. It is not lost silently either: a warning above the plot reads *u* of *T* rows were not tested: fewer than 2 values in a compared condition or, for paired tests, fewer than 2 complete pairs., and the rows are listed in their own table, described below.

- A condition with a single replicate makes every feature fail that rule and the comparison is refused. The paired test additionally needs two complete pairs.
- A constant condition is flagged in zero_variance. One constant group does not necessarily make the whole standard error zero: variation in the other group still contributes. A row without any variation — every compared group constant; for a paired test, identical differences; for Volcano Control, the controls constant too — has a standard error of zero and an undefined test. It gets no p-value, is classed Unavailable and counted under No p-value, and a warning on the page says how many rows this applies to. Inspect rounding, imputation and detection floors before interpreting these cases.
- A p-value at or below 10^{-300} is clipped there before the logarithm, so the y axis has a ceiling near 300.
- Imputed Data tests imputed values as though measured.

26.4 The table below the plot

The table is the plot as numbers: one row per feature tested, in the order of the matrix, carrying the label column, every sample column of the two conditions as it went into the test, and the following.

Columns in this table

Column	What it holds
the label column	The feature, as **Label column** selects it. Phosphoprotein data also carries site_num and PTM_Collapse_keys where they exist.
log2FC	mean(Condition 1) − mean(Condition 2), or the mean of the paired differences.
pval	The raw two-sided p-value of the t-test.
adj_pval	The Benjamini–Hochberg adjusted p-value, retained even when the display uses raw p-values.
selected_pval, correction	The p-value used for classification and plotting, and whether the selected basis is adjusted or raw.
neg_log10_selected_pval	$-\log_{10}$ of the value on the y axis, clipped at 10^{-300}.
significance	Upregulated, Downregulated, Not significant or Unavailable.
zero_variance	True where every observed replicate of one condition holds the same value (for a paired test, where the differences are identical). Where every compared group is constant the row has no p-value; otherwise the p-value is unaffected and the flag says where it came from.
n_pairs	Paired test only: complete pairs behind the row.

The counts above the table — Rows, Upregulated, Downregulated, Not Significant and No p-value — are over the same set of tested features.

When some rows could not be tested, a second table, Rows not tested, follows the result table with its own **Download CSV** (the file name ends in _not_tested.csv). It has one row per untested feature: its identifier columns, n_<condition> and mean_<condition> for each compared condition — the observed count and mean before any control correction — n_pairs for a paired test, and reason, for example fewer than 2 values in GRK. These are often the on/off cases worth inspecting by hand.

26.5 In the notebook

A notebook starting point, using the frames introduced in *The function reference*. These compact examples do not encode every UI option; use the plot's notebook ZIP, where available, for its saved settings and preprocessing recipe:

PYTHON

```
volcano_data = volcano_data_f(data, meta, "WT", "GRK", workflow="Protein")
fig = volcano_plot(volcano_data, "WT", "GRK")
```

R

```
volcano_data <- volcano_data_f(data, meta, "WT", "GRK", workflow = "Protein")
fig <- volcano_plot(volcano_data, "WT", "GRK")
```

The volcano is two calls and not one: volcano_data_f runs the tests and returns the table — fold changes, p-values, adjusted p-values and classes — and volcano_plot draws that table, so the numbers can be inspected, corrected or exported before anything is plotted. Give both calls the same in_pval and in_log2fc: the plot colours its points from the significance column the table already carries, but draws its three threshold lines from its own arguments, so a mismatch leaves the lines somewhere the colours do not agree with. The package draws the figure with Plotly, so what comes back is the *interactive* volcano with hover text on every point; the static figure above is the application's own second renderer, which the packages do not carry.

26.6 Exporting

Type chooses between two renderings of identical numbers. Interactive keeps its gridlines and its full frame and names a point on hover, which is how a feature is identified while exploring; Open Plot opens it in its own tab. Normal is the version for a paper: no grid, two spines, the muted palette, the legend outside the plotting area, the points rasterised, and nothing labelled

unless the **Label top** controls ask for it; Open Image gives the PNG. The table downloads separately as CSV, named after the two conditions.

The static figure is rendered at a fixed 18 by 14 cm and 300 dpi. Unlike the quality-control plots this page offers no width, height or resolution controls, which is deliberate: a volcano is read at one size, and the two threshold lines and the label placement are laid out for it.

26.7 Further reading

SciPy: independent-sample t-tests describes the equal-variance and Welch tests. SciPy: false discovery control explains adjusted p-values and the assumptions behind Benjamini–Hochberg control. The application may pin a different library version; these links are conceptual references.

Behind this page

Function	What it produces
volcano_data_f	The table: fold changes, p-values, adjusted p-values and classes
volcano_data_f_control	The same, with each side centred on its own control condition
volcano_plot	The interactive volcano
volcano_plot_h	The interactive volcano with named features highlighted

27 ANOVA

Which features differ somewhere across three or more conditions, and which conditions those are.

AT A GLANCE

Where	Statistics → ANOVA
Data levels	Protein, phosphosite, phosphoprotein
Needs	At least two conditions with at least two samples each. The post-hoc needs at least three conditions.
The test	One-way ANOVA per feature, with the condition as the single factor and every condition in the annotation as a level.
Two sources	ANOVA draws the significant features; ANOVA Cluster additionally cuts them into groups that move together.
Post-hoc	Tukey HSD, optional, run only on the features the ANOVA called significant.
Tables	The result table, the selected cluster's rows, and the post-hoc comparisons — each downloadable as CSV.

AT PHOSPHOSITE LEVEL

The shared **Localization probability cutoff** selects sites across phosphosite views. At 0 it keeps all sites, including those with no probability; above 0 those unknowns are excluded. This is separate from completeness filtering and does not rerun normalization or imputation. An already collapsed phosphoprotein table has its own recorded collapse cutoff.

27.1 What the plot answers

A volcano compares two conditions. With four there are six pairs, and testing all six separately answers six questions when the question asked was one — does this feature differ anywhere across the design — and gives six chances to clear α by accident.

The ANOVA asks that question directly. For each feature it compares the variation *between* the condition means with the variation *within* them, and the p-value is the probability of a ratio at least that large if all the condition means were equal. One test per feature, one correction across features, and a result that says *something moved* without saying where. The figure is a heatmap of the features that passed, so the shape of the movement is visible even though the omnibus test does not name it.

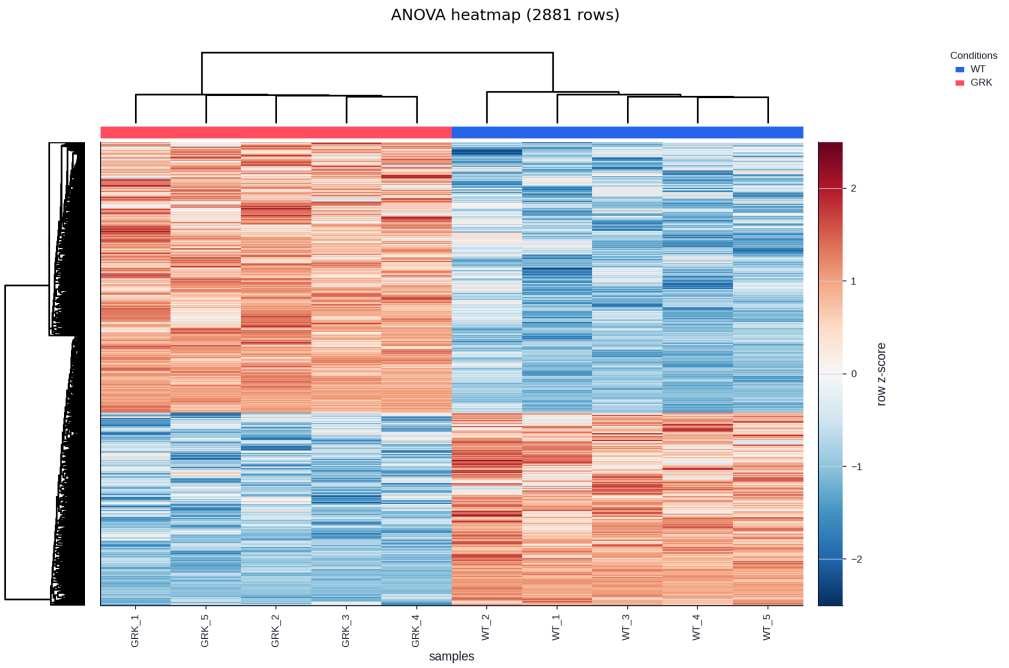


Figure 27.1. The ANOVA heatmap on the example data, at protein level with its two conditions: the 2,881 rows its title counts, against the ten sample columns, each cell a row z-score on the fixed −2.5 to +2.5 scale. The strip above the columns is the condition, and the column dendrogram separates the five GRK samples from the five WT ones without having been told which is which.

27.1.1 Reading it

- One cell: one feature in one sample as a row z-score — how far that sample sits from the feature's own mean, in the feature's own standard deviations across samples. Blue is below, red above; a missing value is grey.
- One row: a feature the ANOVA called significant. Rows are not labelled — there are usually too many — so a feature is found in the table rather than on the figure.
- One column: one sample, labelled condition_n.
- The strip above the columns: each sample's condition, in the colours every other plot in the session uses.
- The dendrograms, when **Show dendrograms** is on: average linkage on euclidean distances, over the z-scored rows and over the samples. In ANOVA Cluster the row tree is replaced by a strip of cluster colours.
- The colourbar, row z-score, fixed from −2.5 to +2.5 so two heatmaps can be compared.

The heatmap visualises features selected by the statistical analysis. Condition separation here is not independent confirmation of that analysis: the rows were selected because their means differ. A row z-score describes relative position within one feature, not an absolute intensity or a fold change shared across rows. For values 8, 9 and 10, centring produces -1, 0 and 1 before division by the row standard deviation; a much more abundant feature with the same relative pattern can have the same colours. Inspect original replicate values and post-hoc results when following up a cluster. Lack of a neat band is not, by itself, a reason to change the significance threshold.

EASY TO GET WRONG

A significant ANOVA says the condition means are not all equal. It does not say which condition differs, or in which direction, and the heatmap is not evidence for a particular pair either — the eye will find one. Set **Post-hoc test** to Tukey HSD to get the pairwise comparisons with their own p-values, and read the direction from the log2FC column of the post-hoc table rather than from the colours.

27.2 Controls

Controls on this page

Control	What it does	Default	Where
Dataset level	Protein, phosphosite or phosphoprotein. Only the levels uploaded are offered.	Protein	page
Source	ANOVA draws every feature below the threshold. ANOVA Cluster cuts those features into groups and colours them.	ANOVA	page
Data Source	The matrix the test runs on: Data, Filtered Data or Imputed Data, whichever exist.	the most processed matrix available	page
Label column	Which column names a feature in the tables: the workflow name or the gene names.	the workflow name	page
P-value threshold	The cutoff for significance, and the gate on both the heatmap and the post-hoc.	0.05	page
Use uncorrected p-values	Uses the raw p-value in place of the adjusted one everywhere it is read, including the two gates.	off	page
Show dendrograms	Draws the row and column trees beside the heatmap. Turning them off does not change the ordering, which is computed either way.	on	page

Controls on this page (continued)

Control	What it does	Default	Where
Clusters	ANOVA Cluster only. How many row clusters to cut the tree into.	4	page
Cluster method	ANOVA Cluster only. Average (UPGMA), Ward, Complete, Weighted (WPGMA) or Single, always on euclidean distances. Average chains and can return two large clusters and two of a handful of features; Ward is the usual remedy.	Average (UPGMA)	page
Cluster	ANOVA Cluster only. Draws one cluster on its own below the full heatmap and lists its rows. The menu shows each cluster's size.	None	page
Show feature names	ANOVA Cluster only. Writes the **Label column** beside each row of the cluster heatmaps, to the right of the heatmap. The size fits the number of rows, so a single small cluster is readable and a cluster of a thousand rows is not; the plain ANOVA heatmap never names its rows.	off	page
Post-hoc test	None or Tukey HSD.	None	page
Condition 1, Condition 2	In the **Post-hoc Comparison** card. Restricts which pair is *reported*; the numbers are unchanged, because the studentized range still takes k from the whole design and the pooled error from all of it.	the first two conditions in metadata order	page

In the plot appearance panel

Control	What it does	Default	Where
Condition colours, condition labels	The strip above the columns and the names in its legend, shared with every other plot in the session.	the condition palette	appearance
Legend	Hides the cluster and condition legends beside the heatmap.	shown	appearance
Row labels	Size and colour of the feature names. Offered only while **Show feature names** is on. An empty size fits the names to the rows.	fitted, black	appearance
Title, axis labels, font, frame, grid, background	As for every other plot; described in Part XI.	the plot's own	appearance

WORTH KNOWING

The appearance panel opens with **Ctrl** + **Shift** + **E**. It offers no colour-scale card here: the heatmap gradient is fixed at RdBu reversed between −2.5 and +2.5 so that two ANOVA figures can be compared. The figure is also rendered at a fixed size, with no width, height or resolution controls on this page.

27.3 How it is computed

The ordinary one-way ANOVA assumes independent observations and a common within-condition variance; its small-sample reference distribution also relies on approximately normal errors. It is not a repeated-measures or Welch ANOVA. Check the experimental design before choosing it. A pooled intensity Q–Q plot does not test these assumptions for every feature. Imputed values enter the calculation as values, without propagating their imputation uncertainty.

THE TEST

The factor is the condition, and its levels are *every* condition in the annotation — there is no condition selector on this page, so a condition is excluded from the ANOVA by excluding its samples from the annotation. Each condition must contribute at least two sample columns, and each feature must have at least two observed values in every condition or its row is dropped before the test.

For a feature with k conditions and N observed values in total, the statistic is the ratio of the variation between the condition means to the variation within them:

$$F = \frac{\text{between-condition mean square}}{\text{within-condition mean square}}, \qquad df = (k-1,\ N-k),$$

and the p-value is the upper tail of that F distribution. Missing values are omitted rather than imputed, so N and the within-condition counts are per feature.

A feature whose values are identical in every condition has no defined F. Such rows return no finite p-value and are removed *before* the correction, so they do not sit in the Benjamini–Hochberg denominator. A feature that is constant within every condition but differs between them is different: its F is infinite (or extremely large) and its p-value 0; it is kept, marked in the zero_variance column, and a warning says how many such rows there are, because they rank first for absent variance rather than a large difference.

The page states how many rows were not tested, in two warnings: *u* of *T* rows were not tested: fewer than 2 values in at least one condition., and *v* rows were not tested: their values are identical in every condition, so no F statistic exists.

CORRECTION

The p-values of the surviving features are adjusted by the Benjamini–Hochberg procedure, exactly as on the volcano page: sorted ascending, scaled by m/i, made monotonic from the largest rank downwards and capped at 1, with m the number of features actually tested. adj_pval retains that adjusted result. **Use uncorrected p-values** puts the raw value in selected_pval; the correction column records the selection. Downstream selection — the significance label, which rows reach the heatmap, which rows are clustered, which rows reach the post-hoc — reads that one column.

TUKEY HSD

Run only when **Post-hoc test** is Tukey HSD, only on features whose adjusted p-value is below the threshold, and only when the design has at least three conditions; with two, the omnibus F and the single pairwise comparison test the same hypothesis and the post-hoc is skipped with a notice.

Within a feature the error is pooled across all k conditions as the ANOVA itself pools it, and for a pair a, b:

$$q = \frac{|\bar{x}_a - \bar{x}_b|}{\sqrt{\frac{1}{2}\,\mathrm{MSE}\left(\frac{1}{n_a} + \frac{1}{n_b}\right)}},$$

with the p-value from the studentized range distribution on k and $N - k$. This is the Tukey–Kramer form, so unequal group sizes arising from missing values are handled correctly. The correction is two-layered and deliberately so: Benjamini–Hochberg controls the false discovery rate across features, Tukey the family-wise rate across the condition pairs within each feature.

The reported difference is *signed* and follows the same convention as everywhere else in CoPYlot: the condition named first in the comparison is the minuend, so log2FC is mean(group1) − mean(group2) and a positive value means higher in group1. The q statistic uses the absolute difference, so the p-value itself is symmetric.

CLUSTERING, IN ANOVA CLUSTER

The significant rows are z-scored across samples — each row minus its own mean, divided by its own standard deviation, with a zero standard deviation replaced by 1. Rows not observed in every sample cannot be clustered and are dropped; the count is reported under the plot. The rest are clustered by the chosen linkage on euclidean distances, cut into the requested number of groups, and the groups renumbered down the dendrogram, so cluster 1 is the topmost band of the figure rather than an arbitrary label.

27.3.1 Edge cases

- A condition with one sample column stops the whole ANOVA with a message naming it; it is not run on the remaining conditions.
- When nothing passes the threshold, no heatmap and no clusters are produced and the panel says so.
- Requesting more clusters than there are significant rows gives as many as the tree can supply.
- The heatmap tolerates missing values — drawn grey — but the dendrograms and the clustering do not, so switching either on can drop rows the plain heatmap showed.
- Tukey costs one studentized-range evaluation per feature per pair. Naming a pair is the cheap way to ask, and returns identical numbers.

27.4 The table below the plot

The result table has one row per feature tested, sorted by adjusted p-value, then raw p-value, then label. It carries the label column, every sample column that went into the test, and:

Columns in this table

Column	What it holds
f_statistic	The ANOVA F for that feature.
pval	The raw p-value from the F distribution.
adj_pval	The Benjamini–Hochberg adjusted p-value.
selected_pval, correction	The raw or adjusted p-value selected for classification, and its stated basis.
significance	Significant or Not significant, against the **P-value threshold**.
neg_log10_selected_pval	$-\log_{10}$ of the value used, clipped at 10^{-300}.
mean_<condition>	One column per condition: the mean of that condition's observed values for the feature.
cluster	ANOVA Cluster only, in the cluster table: which group the row was cut into.

The post-hoc table is separate: one row per feature and condition pair, sorted by Tukey p-value, with a header stating which comparison was run and how many were significant.

Columns in the post-hoc table

Column	What it holds
comparison, group1, group2	The pair, as group1 vs group2 and as its two names, in the order compared.
log2FC	mean(group1) − mean(group2).
n_group1, n_group2	Observed values behind each mean.
q_statistic, tukey_pval	The studentized range statistic and its p-value for this pair. The p-value is not adjusted again across features.
significance	Against the same **P-value threshold**.
anova_adj_pval	The feature's ANOVA adjusted p-value, carried through so the row reads on its own.

27.5 In the notebook

A notebook starting point, using the frames introduced in *The function reference*. These compact examples do not encode every UI option; use the plot's notebook ZIP, where available, for its saved settings and preprocessing recipe:

PYTHON

```
anova_df = anova_data_f(data, meta, workflow="Protein")
fig = anova_plot(anova_df, meta)
```

R

```
anova_df <- anova_data_f(data, meta, workflow = "Protein")
fig <- anova_plot(anova_df, meta)
```

As on the volcano page the analysis is two calls: anova_data_f returns the result table and anova_plot draws the rows of it that pass, so the *F* statistics and the adjusted p-values are in hand before any heatmap exists. Leaving conditions unset uses every condition in the annotation, which is what the page does, since it has no condition selector; the two other panels are anova_cluster_data into anova_plot_cluster, and anova_posthoc_data for Tukey. The one visible difference is the key to the condition strip: the strip is drawn either way, but the legend naming its colours — which the page always shows — is behind show_condition_legend, and that is off by default.

27.6 Exporting

Each figure has its own Open Image link and each table its own Download CSV: the full result table, the selected cluster's rows, and the post-hoc comparisons. The heatmap is a PNG at a fixed size, so a figure for a paper is usually better assembled from the cluster table than cropped out of the full square.

Behind this page

Function	What it produces
anova_data_f	The result table: *F*, p-values, adjusted p-values, per-condition means
anova_cluster_data	The z-scored clustering of the significant rows
anova_posthoc_data	Tukey HSD for every condition pair, or one named pair
anova_plot	The heatmap of the significant features
anova_plot_cluster	The same heatmap with the cluster strip

28
Over-representation analysis

Which biological terms appear in a list of features more often than chance would put them there.

AT A GLANCE

Where	Statistics → ORA, titled GO Enrichment (ORA)
Data levels	Protein and phosphoprotein. Phosphosite level is refused: a site list is not a gene list.
Needs	Gene names in the dataset, and an upstream result — a volcano comparison or a selected ANOVA cluster.
Organism	Human (*Homo sapiens*), the only bundled Gene Ontology database. Nothing falls back to it: another organism is refused rather than silently answered.
The test	One-sided hypergeometric test per term, Benjamini–Hochberg corrected across the terms that overlap the query.
Output	One dot plot and one table per gene list — up and down from a volcano, one from an ANOVA cluster — plus the gene lists themselves as text.

AT PHOSPHOSITE LEVEL

The shared **Localization probability cutoff** selects sites across phosphosite views. At 0 it keeps all sites, including those with no probability; above 0 those unknowns are excluded. This is separate from completeness filtering and does not rerun normalization or imputation. An already collapsed phosphoprotein table has its own recorded collapse cutoff.

28.1 What the page answers

A comparison returns a few hundred genes, and the question is whether that list is about anything: whether its genes share an annotation more often than a list of the same length

drawn at random would.

Over-representation answers it by counting. For one Gene Ontology term, four numbers suffice: how many genes carry the term, how many are in the query list, how many are in both, and how large the universe all three are drawn from is. The hypergeometric distribution gives the probability of an overlap at least that large by chance — the term's p-value — and the procedure is repeated for every term of a usable size and corrected across them. The fourth number decides the answer, and it is the one the page makes you choose.

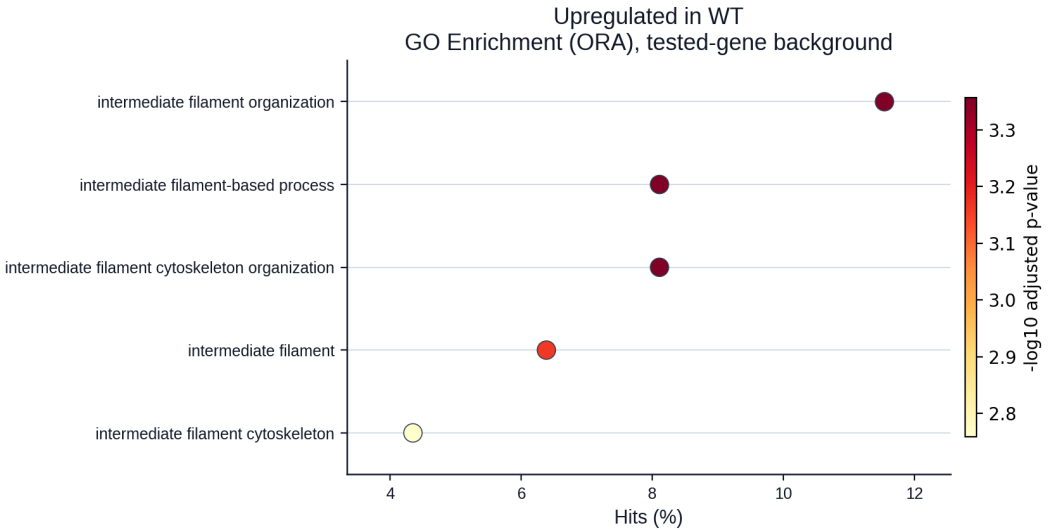


Figure 28.1. The dot plot for the upregulated gene list of WT against GRK, at protein level and at the default settings. Five terms survived the cutoff, all of them about intermediate filaments; the x axis is Hits (%), the colour is $-\log_{10}$ of the adjusted p-value with the most significant term at the top, and the dot area is the number of query genes the term contains.

28.1.1 Reading it

- One dot, one term.
- The x axis, Hits (%): the share of the term's genes that are in the query list — the overlap divided by the term's size within the background.
- The dot's area: the size of the overlap, in genes. It is relative rather than absolute — the figure carries no size key, and the exact count is the intersectionSize column of the table below — so read it as a comparison between terms in the same figure and nowhere else. A large dot far left is a large term barely covered; a small dot far right is a small term covered almost entirely.
- The colour and colourbar: $-\log_{10}$ of the adjusted p-value. Terms are ordered by it, most significant at the top.
- The title names the direction, the condition it belongs to and the background, so the figure carries all three with it.

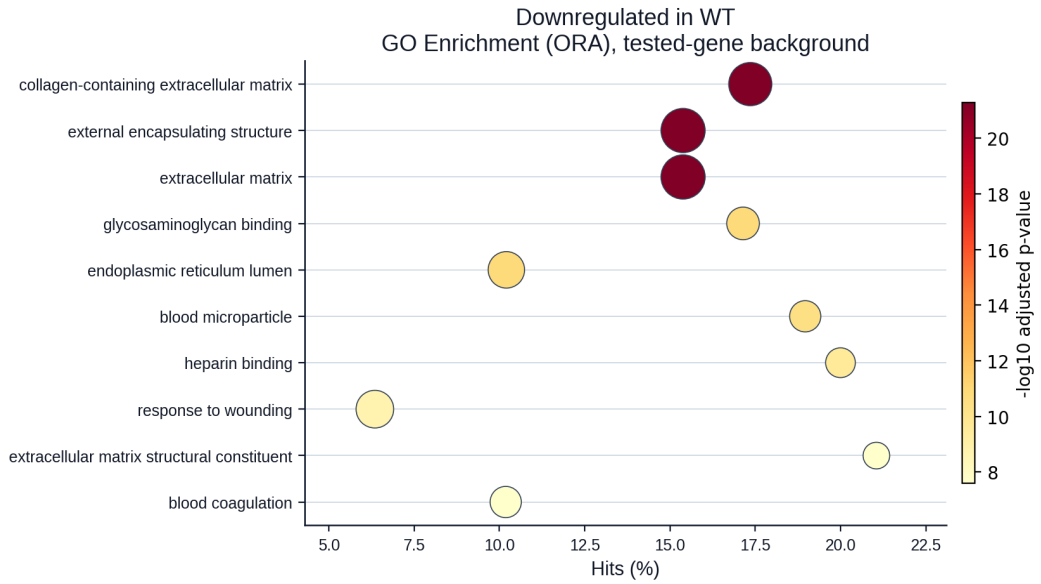


Figure 28.2. The downregulated list of the same comparison, which the page draws directly under the one above. Ten terms, all extracellular matrix and wound response. The colour scale is this term set's own — the two figures are separate analyses of separate gene lists, and neither their colours nor their dot sizes are comparable across the pair. What is comparable is what they are about.

THE DIRECTION IS ONE CONDITION'S

The title of the upper figure reads Upregulated in WT, and names no other condition. That is the whole statement: the upstream volcano computes $\log_2$ FC as mean(Condition 1) − mean(Condition 2), so a feature called Upregulated is higher in Condition 1, and the terms enriched among those features are the terms that are over-represented among the genes higher in Condition 1. Saying *and lower in GRK* as well would be the same fact a second time, and the second telling is what makes a reader stop and work out which way round the figure is.

EASY TO GET WRONG

The background is the set of genes the query could have been drawn from, and the two settings answer different questions. Tested genes in this analysis — the default — uses every gene that received the upstream test, so the result is about what moved among the things this experiment could see. All annotated human genes uses the whole Gene Ontology universe, and will report terms enriched only because their genes are detectable by mass spectrometry at all. Whichever is chosen is intersected with the annotated universe before anything is counted, and the mode and both sizes are printed above the table and written into the exported CSV.

28.2 Controls

Controls on this page

Control	What it does	Default	Where
Dataset level	Protein or phosphoprotein.	Protein	page
Analysis source	Where the gene list comes from: Volcano or ANOVA. The card below changes with it.	Volcano	page
Source	Volcano or Volcano Control under the volcano source; fixed at ANOVA Cluster under the ANOVA source.	Volcano	page
Data Source	The matrix the upstream test runs on.	the most processed matrix available	page
Condition 1, Condition 2	Volcano source. The comparison, in the same orientation as the volcano page: up means higher in Condition 1.	first condition in the annotation	page
Condition 1 Control, Condition 2 Control	Volcano Control only. Four distinct conditions are required.	first condition in the annotation	page
Test type, P-value threshold, log2 FC threshold, Use uncorrected p-values	The upstream volcano settings, as on the volcano page. They decide which features are called up or down, and so what the gene lists contain.	Unpaired (Student's t, equal variance), 0.05, 1, off	page
Clusters, Cluster method, Protein cluster	ANOVA source. The clustering to rebuild and which of its clusters supplies the gene list. All three must match the ANOVA page the cluster was chosen on, or the numbering differs.	4, Average (UPGMA), none	page
Organism	Which Gene Ontology database to match against. Human (Homo sapiens) is the only entry.	Human (Homo sapiens)	page
Adj p-value cutoff	Terms with an adjusted p-value above this are dropped from the result.	0.05	page
Top terms	How many surviving terms to keep, taken from the most significant end.	10	page
Min term size, Max term size	The size a term must fall between to be tested, counted *after* intersection with the background rather than in the database. Very small terms give unstable percentages, very large ones are too general to interpret.	20, 300	page
Background	Tested genes in this analysis (recommended) or All annotated human genes (global).	Tested genes in this analysis	page

Controls on this page (continued)

Control	What it does	Default	Where
Width, Height, DPI	The size of both figures, on screen and in the download alike — the two are the same picture here, unlike the pages where the download is rendered separately. They are a *minimum*: see the note below.	26 cm × 12 cm, 300	page
Download format	PNG, JPG, WEBP, SVG or PDF, beside each figure. One setting serves both figures, so a pair downloaded for the same document arrives in the same format.	PNG	page

> **WHY THE SIZE IS A MINIMUM**
>
> Gene Ontology descriptions can be too long for a compact figure. The page does three things about it, in this order: the term labels are wrapped to the strip of width beside the plot, the canvas is enlarged — beyond **Width** and **Height** — until every label is inside it, and only if the largest permitted canvas still cannot hold them is the label text set smaller than asked for. This improves label fitting but cannot guarantee a figure for invalid or empty input. Set the two numbers to the smallest figure you want and let it grow; setting them large does not make the terms any easier to read, it only spreads five dots over a page.

In the plot appearance panel

Control	What it does	Default	Where
Title, axis labels, ticks, font, frame, grid, background	As for every other plot; described in Part XI. The dot colours are the significance scale and are not a setting.	the plot's own	appearance

28.3 How it is computed

1. **The gene list.** The upstream analysis runs through the same code path the Volcano and ANOVA pages use. From a volcano the Upregulated features form one query list and the Downregulated features another, analysed separately; from an ANOVA the selected cluster forms a single list. Gene labels are split on semicolons where an entry names a group, trimmed and deduplicated.
2. **The background.** Tested genes takes every gene that received the upstream test, not

only the significant ones; Global takes every gene carrying at least one term. Either is intersected with the annotated universe, and that intersection is the population the test draws from.

3. **The terms.** Every term is intersected with the background, and those whose remaining size falls between **Min term size** and **Max term size** are eligible. The database covers the three Gene Ontology branches — biological process, molecular function, cellular component — tested together in one family.

4. **The test and the cut.** Each eligible term sharing at least one gene with the query is tested, the p-values are corrected, and the terms surviving **Adj p-value cutoff** are sorted and cut to **Top terms**.

THE HYPERGEOMETRIC TEST

For one term, with N the number of genes in the background, K the number of them that carry the term, n the number of query genes that are in the background and k the overlap, the p-value is the probability of drawing at least k of the term's genes in n draws without replacement:

$$p = \sum_{j=k}^{\min(K,n)} \frac{\binom{K}{j}\binom{N-K}{n-j}}{\binom{N}{n}}.$$

The test is one-sided: only over-representation is reported, never depletion. Hits (%) on the figure is $100\,k/K$.

CORRECTION

Benjamini–Hochberg, run over the terms that share at least one gene with the query. Zero-overlap terms are excluded from this implementation's correction family. This is a specific analysis convention, not a universal requirement of Benjamini–Hochberg: correcting over all size-eligible terms, including zero-overlap terms with $p = 1$, can give different adjusted values. Report the convention when comparing with another enrichment tool. The two counts are in the table: testedTermCount is how many terms passed the size filter, overlapTested the smaller number the correction actually used.

The two corrections on this page are separate and are reported separately. correctionMethod is always Benjamini-Hochberg and describes the term p-values; upstreamCorrectionMethod describes the volcano or ANOVA that chose the genes, and reads none (uncorrected) when **Use uncorrected p-values** was on. Each query list — up, down or the cluster — is corrected independently of the others.

28.3.1 Edge cases

- A query gene absent from the background is ignored; in Global mode a symbol absent from the database is dropped.

- An empty query list, or one with no eligible overlapping term, produces no figure and a message saying which.
- Where at least 50 distinct query symbols are supplied and under 20 % of them are in the human database, a warning is shown — the practical sign that the data is from another organism, for which the result would be arithmetically valid and biologically meaningless.
- **Min term size** above **Max term size** is refused.

28.4 The result table

One table per gene list, holding the terms that survived the cutoff. Each row carries the term and, repeated on every row so a downloaded file can be read alone, the settings it was produced under.

Columns in this table

Column	What it holds
source	The Gene Ontology branch: BP, MF or CC.
termId	The GO identifier.
name	The term description, as shown on the figure.
termSize	Genes carrying the term *within the background*.
intersectionSize	Genes in both the term and the query.
hitPercent	100× intersectionSize / termSize — the x axis of the figure.
pValue	The raw hypergeometric p-value.
adjPValue	Benjamini–Hochberg adjusted across the overlapping terms of this query.
intersectingGenes	The overlap itself, named.
backgroundMode, backgroundSize, annotatedBackgroundSize	Which background was chosen, how many genes were submitted as it, and how many of those carry any annotation.
correctionMethod, upstreamCorrectionMethod	The correction on the term p-values, and the one used upstream to choose the genes.
testedTermCount, searchSpace, overlapTested, passingCutoff, displayed	How the search space narrowed: terms passing the size filter, terms actually tested, terms under the cutoff, terms kept by **Top terms**.
method, organism, analysisSource, datasetLevel, dataSource	What was run, against which database, from which upstream analysis, at which level and on which matrix. The upstream conditions, test type, thresholds and cluster settings follow on the same row.

28.5 In the notebook

A notebook starting point, using the frames introduced in *The function reference*. These compact examples do not encode every UI option; use the plot's notebook ZIP, where available, for its saved settings and preprocessing recipe:

PYTHON

```
genes = different_genes(data, meta, "WT", "GRK", workflow="Protein")
fig = enrichment_analysis(genes["Upregulated"], background=genes["Background"])
```

R

```
genes <- different_genes(data, meta, "WT", "GRK", workflow = "Protein")
fig <- enrichment_analysis(genes$Upregulated, background = genes$Background)
```

The page hides the join between the two analyses; in a notebook it is explicit. different_genes runs the same volcano and hands back three gene lists — Upregulated, Downregulated and Background, the last being every gene that was tested — and each query list is then enriched on its own, so the figure above is one of two the page produces from the same call. The background is passed in by hand rather than chosen from a menu: background_mode stays at "experiment", which is the page's Tested genes in this analysis, and enrichment_data returns the same table the figure is drawn from when the numbers are wanted instead of the picture.

28.6 Exporting

Each figure has its own format selector and Download Plot button beside it, and its own Open Image link; each table has its own Download CSV. The downloaded figure is the one on screen, at the width, height and resolution set on the page — there is no separate render, so what is downloaded cannot differ from what was looked at. Choose PDF or SVG for a figure that will be scaled or edited, PNG for a slide. The file is named for its direction and its comparison, so ora_up_protein_WT_vs_GRK.png and its downregulated counterpart do not have to be told apart afterwards.

The gene lists download separately as plain text: the up list, the down list, and — in Tested genes mode — the background itself, which is what makes the result reproducible by somebody else.

The bundled Gene Ontology table is fixed for a release, so the result does not print a database identity: the release of the application is the provenance. Quote the version on this manual's title page when a method section needs one.

28.7 Further reading

Gene Ontology: enrichment analysis explains why the reference population matters. SciPy: false discovery control discusses multiple testing. CoPYlot's tested-term family is the specific convention documented above.

Behind this page

Function	What it produces
enrichment_data	The over-representation table for one gene list
enrichment_analysis	The dot plot of that table
load_go_db	The bundled Gene Ontology term-to-gene table

29
The pathway heatmap

The measured values behind one Gene Ontology term, gene by gene and sample by sample, in one square.

AT A GLANCE

Where	Statistics → Pathway heatmap
Data levels	Any uploaded level that carries gene names
Needs	Gene names in the dataset. The term list comes from the bundled human Gene Ontology database and is available without running anything first.
One row	One gene of the term, as the median of every dataset row whose annotation names it.
Two value types	Z-score across the displayed samples, or Intensity as the matrix holds it.
Table	None. The page produces the figure only.

AT PHOSPHOSITE LEVEL

The shared **Localization probability cutoff** selects sites across phosphosite views. At 0 it keeps all sites, including those with no probability; above 0 those unknowns are excluded. This is separate from completeness filtering and does not rerun normalization or imputation. An already collapsed phosphoprotein table has its own recorded collapse cutoff.

29.1 What the plot answers

An enrichment result says that a term is over-represented in a list. It says nothing about whether the genes behind it actually move together, or in which samples — a term can be called enriched on a set of genes that go up, down and nowhere. This page puts those genes back on the data: one row per gene of the term, one column per sample, so that a coherent block and a scattering of unrelated rows look different.

Because the term list is the whole bundled database rather than the output of a run, the page also works the other way round: pick a pathway you already care about and look at it, without an enrichment result at all.

29.1.1 Reading it

- One cell: one gene of the term in one sample. Missing values are left uncoloured rather than filled.
- One row: one gene named by the term. Its value is the *median* across every row of the dataset whose gene annotation contains that gene — one row for most genes, several where a protein group or a phosphoprotein names it.
- One column: one sample of a selected condition, labelled condition_n, with the sample identifier on a second line when **Toggle ID** is on.
- The colourbar. Under Z-score it is centred on each row's own mean and measured in that row's own standard deviations; under Intensity it is the matrix's own unit, named in the label.

A GO term can contain genes with different responses; coherent colour bands are not a requirement for a valid enrichment result. The term may describe a cellular component or molecular function rather than a signalling pathway. Check how many annotated genes were measured, which contributed to the query, and which move in different directions. Turn **Remove empty genes** off to inspect coverage. The heatmap itself is not an enrichment test and cannot establish pathway activation or inhibition.

> **EASY TO GET WRONG**
>
> Z-score is computed per row, so every row is stretched to fill the same colour range whatever its real spread. A gene that varies by a factor of ten and one that varies in the third decimal come out looking equally variable, and a row with no spread at all is blanked entirely rather than drawn flat. Use Z-score to compare the *shape* of the rows and Intensity to compare their level; the title says which is on screen.

29.2 Controls

Controls on this page

Control	What it does	Default	Where
Dataset level	Which dataset to draw. Only levels carrying a gene-name column can be used.	Protein	page

Controls on this page (continued)

Control	What it does	Default	Where
Pathway	The term to draw, chosen from every term description in the bundled Gene Ontology database — 22,074 of them, in alphabetical order.	the first term in the list	page
Value type	Z-score standardises each row across the displayed samples; Intensity draws the values as the matrix holds them.	Z-score	page
Data source	Which matrix to read: Data, Filtered Data or Imputed Data, whichever exist.	the most processed matrix available	page
Conditions	Which conditions supply the columns. Samples are drawn in annotation order within the selection.	all conditions	page
Width (cm), Height (cm), DPI	The size of the *downloaded* figure.	20, 12, 300	page
Show gene labels	Prints the gene name beside each row. Turn it off for a term of several hundred genes.	on	page
Toggle ID	Adds the sample identifier under each column label.	off	page
Remove empty genes	Drops rows for genes of the term that no dataset row names — the usual case for most of a large term. Off, they are drawn as empty rows, which is the honest picture of how much of the term was measured.	on	page
Show collapsed row count (n)	Appends (n=k) to each gene label, where *k* is how many dataset rows were collapsed onto it. Off by default: for most genes the answer is 1.	off	page
Cluster rows, Cluster columns	Reorders rows or columns by average linkage on euclidean distances. Both require complete values.	off	page

In the plot appearance panel

Control	What it does	Default	Where
Colour scale	The gradient the cells are drawn in.	plasma	appearance
Title, axis labels, ticks, font, frame, grid, background	As for every other plot; described in Part XI.	the plot's own	appearance

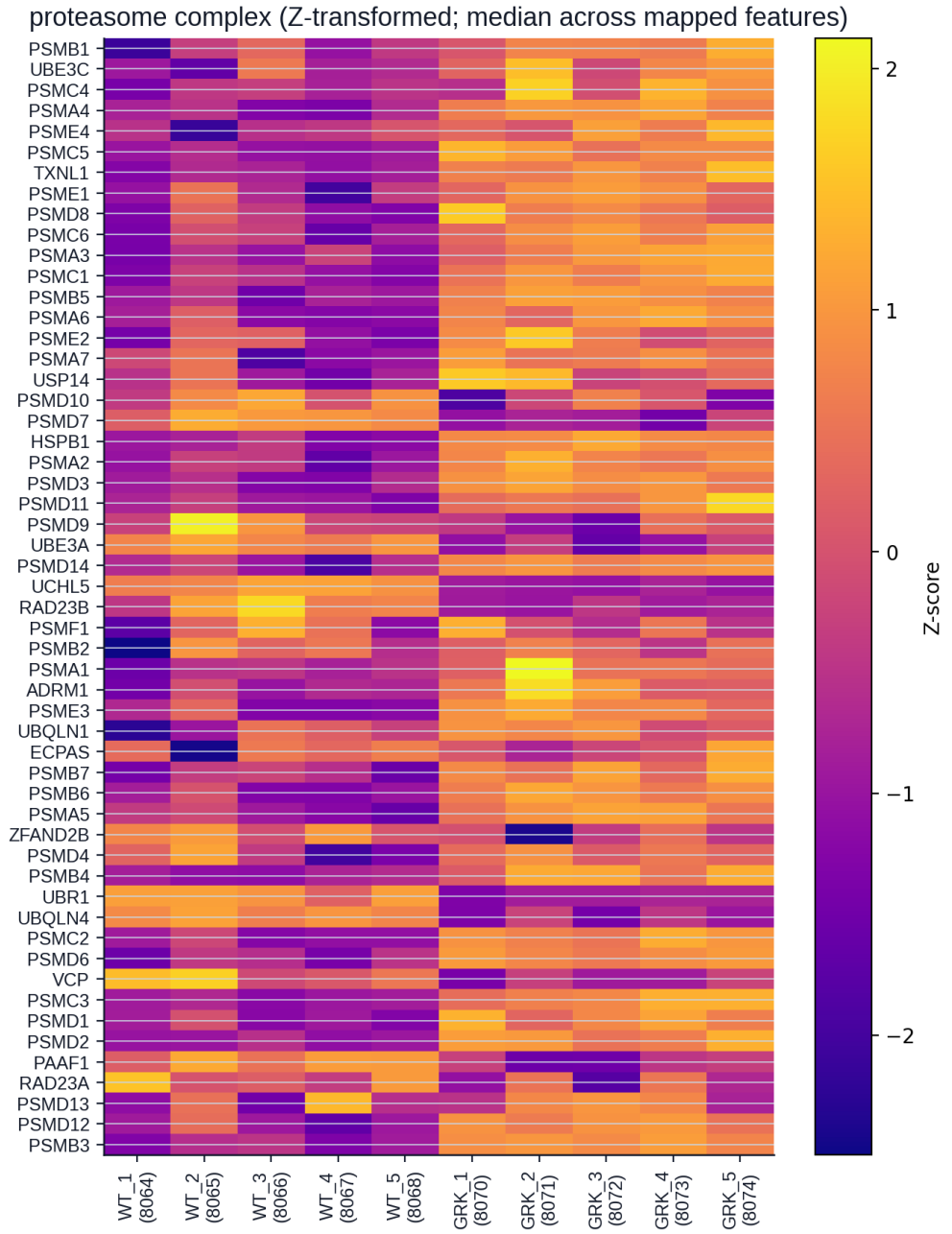


Figure 29.1. proteasome complex at protein level on the example project, Z-scored: one row per gene of the term that the dataset quantified, one column per sample, the five WT samples then the five GRK ones. The block of colder colours on the left against warmer ones on the right is the whole term moving together between the two conditions — which is what this plot exists to show, and what no single volcano point says.

WORTH KNOWING

The appearance panel opens with **Ctrl** + **Shift** + **E**.

29.3 How it is computed

FROM TERM TO ROWS

The chosen term is matched in the database by its description, ignoring case, and its gene list is read. Those genes, in the term's own order and with repeats removed case-insensitively, are the rows of the figure.

Each dataset row contributes to a gene when its gene annotation contains that gene as one of its semicolon-separated tokens, matched without regard to case. A token repeated inside one annotation still counts as one contribution. Where several dataset rows map to the same gene — a protein group, or several phosphoproteins of one gene — the row's value in each sample is the **median** of theirs, over the values that are present. A gene of the term that no dataset row names becomes a row of NA, and is removed when **Remove empty genes** is on.

THE COLOUR

Under Intensity the numbers are drawn as they stand, and the colourbar takes the matrix's unit — log2 Intensity, or log2 Ratio to Site Median for phosphoprotein data collapsed by the robust site median.

Under Z-score each row i is standardised across the displayed samples only:

$$z_{ij} = \frac{x_{ij} - \bar{x}_i}{s_i},$$

with $\bar{x}_i$ and s_i the mean and standard deviation of that row over the selected samples, both taken over the values that are present. A row whose standard deviation is zero or undefined has no meaningful z-score and is blanked and then dropped rather than divided by zero. The scores therefore change when the condition selection changes: two heatmaps of the same term over different conditions are not comparable cell for cell.

ORDERING

Without clustering the rows are in the term's own gene order and the columns in

annotation order. **Cluster rows** and **Cluster columns** each replace that with the leaf order of an average-linkage tree on euclidean distances, computed on the matrix as drawn — so on z-scores under Z-score and on the values themselves under Intensity. Clustering cannot be computed through a missing value, so when either switch is on the rows that are not observed in every displayed sample are dropped first. On a matrix with scattered missing values that can remove most of the term, which is why a notice appears beside the two switches whenever they are used on anything other than an imputed matrix. Nothing is imputed for you.

29.3.1 Edge cases

- A term none of whose genes appear in the dataset produces no figure and an error saying so.
- With **Remove empty genes** off and few genes matched, the figure is mostly blank: that is the measured coverage of the term, not a fault.
- Under Z-score, a gene measured in only one of the selected samples has no standard deviation and disappears from the figure.
- A single replicate per condition is not a problem here — nothing on this page is a test — but the z-score is then taken across conditions rather than within them.
- Selecting Imputed Data draws imputed values in the same colours as measured ones, with nothing to distinguish them.

29.4 In the notebook

A notebook starting point, using the frames introduced in *The function reference*. These compact examples do not encode every UI option; use the plot's notebook ZIP, where available, for its saved settings and preprocessing recipe:

PYTHON

```
db = load_go_db()
fig = proteins_in_pathway(data, meta, db, "proteasome complex")
```

R

```
db <- load_go_db()
fig <- proteins_in_pathway(data, meta, db, "proteasome complex")
```

The term list is not a menu here: load_go_db returns the bundled table and the term is named to proteins_in_pathway as a string, matched against its description, so any of the 22,074 terms

can be drawn without an enrichment result in front of it. The defaults line up with the page — value_type="z", remove_empty=True, both clustering switches off, every condition drawn when conditions is left unset — with one exception: id is on by default and puts the sample identifier under each column label, which is **Toggle ID**, and that is off on the page.

29.5 Exporting

The figure is rendered at the width, height and resolution set on the page, which is why those three controls do not change what is on screen: the displayed image is fitted to the window, the downloaded one to the numbers. Raise the height for a term of many genes before turning **Show gene labels** off — a taller figure keeps the labels readable where a wider one does not. Open Image gives the PNG.

29.6 Further reading

Gene Ontology: enrichment analysis distinguishes an enrichment result from a list of annotated genes. A heatmap visualizes values for such a list; it does not itself test activation of the term.

Behind this page

Function	What it produces
proteins_in_pathway	The heatmap of one term's genes across samples
load_go_db	The bundled Gene Ontology term-to-gene table the term list is read from

30
The replicate and variance projection

How the calculated significance changes when an assumed replicate count or variance is substituted while the observed effects are held fixed.

AT A GLANCE

Where	Statistics → Replicate & Variance Projection, titled Projected volcano plot
What it is	A what-if projection, not a power calculation.
Data levels	Protein, phosphosite, phosphoprotein
Needs	The same two conditions a volcano needs, each with at least two samples
What changes	The replicate count and the variance that go into the standard error. Nothing else.
What does not	The fold changes. They are the measured ones, and no setting on this page moves a point sideways.
Not random	Nothing is resampled, drawn or generated. There is no seed, and the same settings always give the same figure.
Renderer	Interactive only. There is no static figure and no appearance panel on this tab.

AT PHOSPHOSITE LEVEL

The shared **Localization probability cutoff** selects sites across phosphosite views. At 0 it keeps all sites, including those with no probability; above 0 those unknowns are excluded. This is separate from completeness filtering and does not rerun normalization or imputation. An already collapsed phosphoprotein table has its own recorded collapse cutoff.

30.1 What it simulates

The panel states what the page is, under **Projection settings**:

> **IN THE PANEL'S WORDS**
>
> A what-if projection, not a power calculation: the observed mean fold changes are held fixed and the t-test is recomputed for the variance factor and replicates per group set here. Nothing is drawn at random, so the same settings always give the same plot. The observed fold changes include noise, so the projection is optimistic for effects that look large, and it does not estimate the probability of detecting a true effect.

The recomputation is arithmetic, not an experiment. For every feature the observed condition means and the observed within-condition variances are taken from the selected matrix, exactly as the volcano takes them. The variances are then multiplied by **Variance × factor**, the replicate counts are replaced by **Replicates per group** if a fixed count is set, and the same two-sample t-test is recomputed with those substituted numbers. The fold change is untouched, so every point stays where the real volcano put it; only its height changes.

That is the whole of it. No values are drawn at random, no dataset is generated, and no feature is invented: the page is a what-if on the *precision* of the effects already in the data.

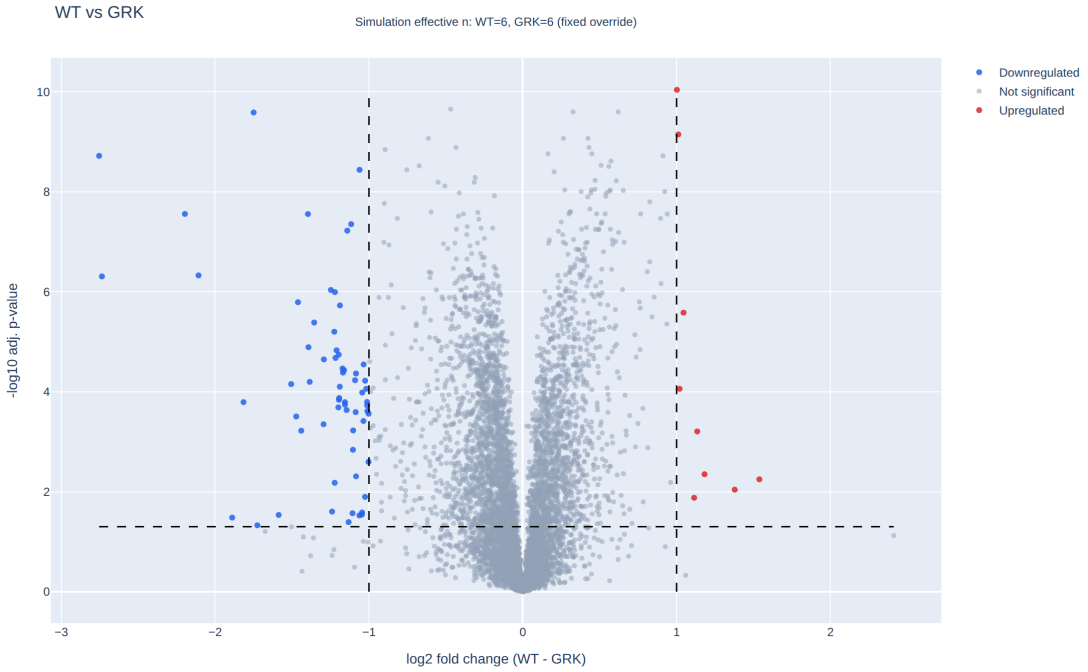


Figure 30.1. The projected volcano for WT against GRK at protein level, with the replicate count fixed at 6. The line above the plot states the effective *n* the simulation used and where it came from; the axes are the measured $\log_2$ fold change and the p-value the test would return under that substituted count. The three dashed lines are the two fold change thresholds and the p-value threshold.

30.1.1 Reading it

- The line above the plot: the effective *n* actually used for each condition, as a single number or a range, and whether it came from the data (observed) or from the control (fixed override).
- The x axis: the measured $\log_2$ fold change, Condition 1 − Condition 2, identical to the

volcano page.

- The y axis: $-\log_{10}$ of the p-value the test *would* return under the substitution. This is the only thing the two simulation controls move.
- The colours and the three threshold lines, read exactly as on the volcano.
- The summary above the plot: the method, the sample-size mode, the effective n of each condition, the number of rows, and how many features are called up, down, not significant, or have no p-value under these settings.

Holding an observed effect fixed while raising the assumed replicate count lowers its standard error. The figure shows the arithmetic consequence of that assumption and nothing more: a candidate crossing the line at a fixed count of six is significant in the substituted calculation under those fixed inputs, not a prediction of what an experiment with six replicates would find.

> **EASY TO GET WRONG**
>
> A what-if projection, not a power calculation. Every feature keeps its own measured effect and its own measured variance, so a variance estimated from three replicates is no more reliable at a fixed count of 20 than it was at three; the page says nothing about effects this dataset did not detect at all. The observed fold changes include noise, so the projection is optimistic for effects that look large, and it does not estimate the probability of detecting a true effect.

30.2 Controls

Controls on this page

Control	What it does	Default	Where
Dataset level	Protein, phosphosite or phosphoprotein.	Protein	page
Source	Fixed at Volcano: the projection always starts from a two-condition comparison.	Volcano	page
Data Source	Which matrix the observed means and variances come from.	the most processed matrix available	page
Condition 1, Condition 2	The comparison. Condition 1 is the minuend, as on the volcano page.	first condition in the annotation	page
Label column	Which column names a feature in the hover text: the workflow name or the gene names.	the workflow name	page

Controls on this page (continued)

Control	What it does	Default	Where
Test type	Unpaired (Student's t, equal variance) or Unpaired (Welch, unequal variance). The paired test is not offered here.	Unpaired (Student's t, equal variance)	page
P-value threshold, log2 FC threshold	The two thresholds the simulated result is classified against.	0.05, 1	page
Use uncorrected p-values	Classifies on the raw simulated p-value instead of the adjusted one.	off	page
Variance × factor	Multiplies both within-condition variances before the standard error is formed. Below 1 is a cleaner experiment, above 1 a noisier one; the field clamps to a minimum of 0.01.	1	page
Replicates per group (0 = observed, 2-100 fixed)	0 keeps each feature's own observed replicate counts; a value from 2 to 100 replaces both counts with that number for every feature. 1 is refused, because a t-test cannot estimate variance from one sample per condition.	0	page

WORTH KNOWING

This page has no plot appearance panel: the figure it draws is not one of the styled plots, so **Ctrl** + **Shift** + **E** adds nothing here.

30.3 How it is computed

THE SUBSTITUTION

A feature needs at least two observed values in each condition, as on the volcano; everything else is dropped before anything is computed. For a surviving feature, with $\bar{x}, \bar{y}$ the observed condition means, s_1^2, s_2^2 the observed within-condition variances and $n_1^{\text{obs}}, n_2^{\text{obs}}$ the observed counts:

$$\log_2 \text{FC} \;=\; \bar{x} - \bar{y} \qquad \text{(unchanged by any setting on this page),}$$

$$s_k'^2 = m \cdot s_k^2, \qquad n_k' = \begin{cases} n_k^{\text{obs}} & \text{override} = 0, \\ n^{\text{fixed}} & \text{otherwise,} \end{cases}$$

where m is **Variance × factor** and n^{fixed} is **Replicates per group**. Note that n_k' replaces the count in the standard error and in the degrees of freedom only; the variance it is divided by is still the one estimated from the replicates that were actually measured.

THE TEST

With the substituted numbers, the pooled option forms

$$s_p'^2 = \frac{(n_1' - 1)s_1'^2 + (n_2' - 1)s_2'^2}{n_1' + n_2' - 2}, \quad SE = \sqrt{s_p'^2 \left(\frac{1}{n_1'} + \frac{1}{n_2'}\right)}, \quad df = n_1' + n_2' - 2,$$

and the Welch option forms $SE = \sqrt{s_1'^2/n_1' + s_2'^2/n_2'}$ with Welch–Satterthwaite degrees of freedom. Either way

$$t = \frac{\log_2 \text{FC}}{SE}, \qquad p = 2\,P(T_{df} > |t|).$$

Because SE falls as $1/\sqrt{n'}$ while $\log_2$ FC is fixed, raising the replicate count moves every point straight up the figure and never sideways.

CORRECTION AND CLASSIFICATION

The simulated p-values are adjusted by Benjamini–Hochberg over the features tested, unless **Use uncorrected p-values** is on, and classified against the two thresholds exactly as on the volcano page: Upregulated above $+\tau$ and below α, Downregulated below $-\tau$ and below α, Unavailable where no finite p-value resulted, Not significant otherwise. The correction is therefore run on a set of p-values that were never observed, which is the point — but it means the adjusted values are as hypothetical as the raw ones.

30.3.1 Edge cases

- An override applies the same n to every feature, including one observed in only two of five samples. The summary reports both the mode and the effective n so the substitution is never silent.
- With the override at 0, the effective n is a range rather than a number whenever coverage differs between features, and the annotation above the plot shows it as one.
- Zero variance in both conditions gives zero standard error. Distinct constant means produce an extreme test statistic; identical constant means give an undefined test, not evidence

of a difference. Scaling zero cannot supply missing variability.

- A saved panel state holding the old value 1, or any other value the field cannot hold, falls back to 0 on load, because a t-test cannot estimate variance from one sample per condition. Typing 1 is refused with a message.
- This tab is not available in viewer mode.

30.4 In the notebook

A notebook starting point, using the frames introduced in *The function reference*. These compact examples do not encode every UI option; use the plot's notebook ZIP, where available, for its saved settings and preprocessing recipe:

PYTHON

```
fig = volcano_plot_sim(data, meta, "WT", "GRK", workflow="Protein", mod_var=1,
    ↪ mod_n=6)
```

R

```
fig <- volcano_plot_sim(data, meta, "WT", "GRK", workflow = "Protein", mod_var = 1,
    ↪ mod_n = 6)
```

This is one call rather than the page's two stages: mod_var and mod_n are **Variance × factor** and **Replicates per group**, and mod_n=0 keeps each feature's own observed counts exactly as 0 does on the page. Two things the page offers are not here — there is no Welch option, because the function forms the pooled variance and nothing else, and there is no summary line, so the effective *n* has to be read from the override you set; what comes back is the Plotly figure, which is the only renderer either side has for this plot.

30.5 Exporting

There is one renderer and no download button: Open Plot opens the interactive figure in its own tab, where the browser's own save is the only export. There is no static version of this plot and no result table under it, so a simulated comparison that has to appear in a document is best described by its settings — the two conditions, the test, the variance factor and the effective *n*; the summary above the plot states the test and the effective *n*.

The page reports counts and a figure, not a per-feature export. Do not report these hypothetical p-values as experimental findings: the page is a what-if projection, not a power calculation,

and it does not estimate the probability of detecting a true effect.

Behind this page

Function	What it produces
volcano_plot_sim	The projected volcano. Its mod_var and mod_n arguments are **Variance × factor** and **Replicates per group**; it offers the pooled variance model only.

PART VI

Peptide level



31
The retention time plot

Observed retention time against the retention time written into the report as predicted, for every row of the peptide report.

AT A GLANCE

Where	Peptide Level → RT Plot
Needs	A peptide report, loaded as the Peptide level in Data → Upload. The peptide and precursor levels are read from the long-format report on disk rather than from a wide matrix; every other page of this part needs an abundance column in that report, and this one does not. It reads two columns, Predicted.RT and RT, and needs no annotation either.
One point	One row of the report. Nothing is grouped or de-duplicated first, so a peptide identified in several runs, or at several charges, contributes one point for each.
Two renderers	**Type** Normal draws a static figure in one of three modes. Interactive draws a scatter that can be hovered and zoomed, and opens in its own tab.
Table	None.

31.1 What the plot answers

Both numbers on this figure come out of the report: RT is when a precursor was actually seen, Predicted.RT is when whatever produced the report expected to see it. The tool computes neither. It draws one against the other, so the question the page answers is how far the chromatography of this experiment departed from the run the prediction was made for.

A well-aligned experiment puts the whole cloud on the diagonal, narrow enough that its width is the scatter of single identifications rather than a property of the gradient. Departures come in two shapes, and they mean different things: a *bend* is systematic and affects every precursor in a retention window, while *individual points far off the line* are single identifications whose prediction and observation disagree.

> **A SMALL EXAMPLE**
>
> If both axes are in minutes, a precursor predicted at 20 and observed at 23 lies above the identity line: it eluted three minutes later than predicted. A whole band with that displacement suggests a systematic mismatch worth checking, not that every identification is wrong. An indexed prediction and an observed time cannot be compared this way without calibration.

31.1.1 Reading it

- One point, one row of the report. In hexbin and density mode the points are binned and it is the colour of a cell, not a point, that is drawn — the colour bar gives how many rows fell into it.
- The identity line, $y = x$. It is drawn only with **Add Line** on. A point above it eluted later than predicted, a point below it earlier.
- A point far from the line: one identification whose observed and predicted times disagree. In a report of a few hundred thousand rows a scatter of these is expected.
- A systematic curve or step in the band. Every identifications in that retention window may be affected together. Check calibration, the prediction model, gradient and column conditions before interpreting individual departures.

> **EASY TO GET WRONG**
>
> The axes do not carry a unit, because the tool takes both columns as the report wrote them and converts neither. If the two columns were produced on different scales — one in minutes, one in seconds, or one indexed rather than timed — the cloud will lie along a straight line that is not the diagonal, and the identity line will look wrong when nothing about the chromatography is. Read the slope of the band before reading its width.

Confirm that both columns use the same units and calibration before treating the identity line as agreement. The application does not establish that from the column names alone.

31.2 Controls

Controls on this page

Control	What it does	Default	Where
Mode	How the points are drawn: Scatter Plot draws them individually, Hexbin Plot bins them into hexagons coloured by count, Density Plot bins them into a square grid coloured by count on a logarithmic scale. Both binned modes add a colour bar. Ignored when **Type** is Interactive.	Hexbin Plot	page
Type	Normal renders a static image. Interactive renders a scatter with hover text, zoom and pan, shown in a frame with its own **Open Interactive Plot** link instead of a download button.	Normal	page
Bins	The resolution of the binned modes: the hexagon grid size in Hexbin Plot, the number of bins per axis in Density Plot. It is held between 10 and 2000, and has no effect in Scatter Plot or in the interactive renderer.	1000	page
Random Seed	Fixes the sub-sample drawn when the report is larger than the renderer draws in full, so that the same figure is reproduced exactly. It has no effect below that size, and none in the binned modes.	69	page
Add Line	Draws the identity line $y = x$ across the data.	off	page
Width (cm), Height (cm), DPI	The size of the *downloaded* figure. The browser scales the rendered figure to fit the page. Width and height are held between 1 and 60 cm and DPI between 72 and 600, and a very large figure has its resolution reduced so that the image stays under 24 megapixels. Shown only while Normal is selected. The interactive plot has none: it fills its frame, Plotly's camera button saves a picture of it, and **Open Interactive Plot** opens it in a window of its own.	20 cm × 15 cm, 100	page
Download format	PNG, JPG, WEBP, SVG or PDF. Offered for the static renderer only.	PNG	page

In the plot appearance panel

Control	What it does	Default	Where
Reference line	The identity line: shown or hidden, and its colour, dash pattern and width. Hiding it here removes the line even with **Add Line** on; **Add Line** decides whether there is a line at all, this decides what it looks like.	shown, red, dashed, width 1.5	appearance
Points	Colour, shape, size, opacity and outline of the dots. The plot asks for a diameter of 1 pt at opacity 0.2 in the static scatter and 4 pt at 0.35 in the interactive one, and the panel overrides both. The outline is off unless it is asked for.	the plot's own	appearance
Title, X axis, Y axis	Text, size, weight, colour and position of each label, and the size, rotation and colour of the tick labels. The plot writes Retention Time Plot, Predicted RT and Actual RT.	the plot's own	appearance
Font	One face for the whole figure, or one per text.	plot default	appearance
Frame and grid	Which of the four spines are drawn, and the grid behind the data.	all four spines	appearance
Background	Paper colour, transparency and the preset themes.	default	appearance

WORTH KNOWING

The second table is the plot appearance panel, which is hidden until advanced mode is turned on (**Ctrl** + **Shift** + **E**). Everything it contains is described in Part XI; only the entries particular to this plot are repeated here. The panel keeps one set of settings per plot, and this page has two renderers, so its defaults follow whichever of them **Type** has selected.

31.3 How it is computed

HOW IT IS COMPUTED

The columns Predicted.RT and RT are required by those exact names; a report without either draws nothing and says which is missing. Both are read as numbers, anything that cannot be read as one becomes NA, and a row is kept only if *both* of its values

survive. The surviving rows are the points: x is Predicted.RT, y is RT.
No grouping happens at any stage. One row of the report is one point, so a peptide seen in twelve runs at two charges contributes twenty-four points, and all runs are pooled into a single cloud. Nothing on this page splits by sample or by condition.

THE THREE STATIC MODES

Scatter Plot draws every point. Above 100 000 rows it first draws a random sample of 100 000 of them, chosen with **Random Seed**.
Hexbin Plot tiles the plane with hexagons whose grid size is **Bins**, clamped to [10, 2000], and colours each by how many rows fell into it on a linear scale.
Density Plot bins the plane into a square grid of **Bins** cells per axis, again clamped to [10, 2000], and colours each cell by its count on a logarithmic scale, so that a sparse tail is still visible beside a dense diagonal.
Both binned modes draw a colour bar; the scatter does not, because each of its marks is one row.

THE IDENTITY LINE

With **Add Line** on, a straight line is drawn from $(x_{\min}, x_{\min})$ to $(x_{\max}, x_{\max})$, where $x_{\min}$ and $x_{\max}$ are the smallest and largest Predicted.RT among the points drawn. It is the locus of perfect agreement: a precursor on it was observed exactly when it was predicted. The interactive renderer spans the same line over the smaller of the two minima and the larger of the two maxima, so its line reaches the corners of the frame.

THE INTERACTIVE RENDERER

Interactive always draws a scatter, whatever **Mode** says. Above 250 000 rows it draws a random sample of 250 000, again fixed by **Random Seed**. Each point carries hover text with the feature, the predicted and actual times to four decimals, their difference as Actual − Predicted, and the sample.
The feature is the first of Stripped.Sequence, Modified.Sequence, Precursor.Id, PrecursorID, Precursor or ProteinNames that the report has, failing that any column whose name contains *sequence*, and failing that a running number. The sample is the filename column reduced to the file's own name without its folders, truncated to 14 characters.

Mode and **Bins** apply to the static renderer only; with **Type** set to Interactive they are greyed out, because the interactive figure is always a scatter.

31.3.1 Edge cases

- Replicates are not a concept on this page: every run in the report is pooled into one cloud, and a single-run report is drawn exactly as a twelve-run one is.
- A row whose Predicted.RT or RT is missing, blank or non-numeric is dropped. It is dropped silently, and nothing on the figure says how many were.
- If no row has both values the page draws nothing and reports that no numeric values are available.
- If every point has the same predicted time, the identity line collapses to a single point and the axis to a single value; the figure is drawn, but there is nothing on it to read.
- The interactive renderer needs the plotly package. Where it is not installed the page says so and names the package.

31.4 Exporting

The static figure downloads at the width, height and resolution set on the page, which is why those three controls do not change what is on screen: the displayed plot is fitted to the window, the downloaded one to the numbers. Choose PDF for a figure that will be scaled or edited, PNG for a slide. A scatter of several hundred thousand points is large in PDF and slow to open; the binned modes are the better choice for a vector figure.

The interactive renderer has no download. It opens in its own tab, where its own toolbar saves the current view as a PNG. Its settings are also written into the summary report, which renders this page as a static figure whatever **Type** is set to here.

Behind this page

Function	What it produces
rt_vs_pred_rt_plot	Actual against predicted retention time, in scatter, hexbin or density mode. The package fixes the density mode at 200 bins per axis where the page passes **Bins**.
peptide_rt_plot	The static figure as the page renders it, including the appearance-panel settings. Its interactive twin is peptide_rt_interactive_html, which produces the scatter and its hover text.

32 Modified peptides

How many distinct modified sequences carrying each of three tracked modifications were detected in each sample.

AT A GLANCE

Where	Peptide Level → Modification Plot
Needs	A peptide report, loaded as the Peptide level in Data → Upload, and an annotation. The peptide and precursor levels are read from the long-format report on disk rather than from a wide matrix, and this page needs an abundance in it: the report must carry the column Precursor.Quantity, which is fixed in the code and is not the **Abundance Column** offered on the quality-control pages. It also needs Modified.Sequence.
Annotation	With a protein dataset loaded, the protein annotation is used. Without one, the peptide annotation made in Data → Annotation is. Until one of them exists the page says so and draws nothing.
Recognises	Three classes, matched by name or by UniMod identifier: Carbamidomethylation (UniMod:4), Oxidation (UniMod:35) and Acetylation (UniMod:1). Nothing else is counted.
One bar	One sample and one modification. Its height is a count of distinct *modified sequences*, not of precursors and not of occurrences.
Table	None.

32.1 What the plot answers

Use these counts to compare the tracked modification classes between samples. Differences can reflect preparation, storage, search settings, identification depth or biology. Acetylation and oxidation are not automatically preparation artefacts. These are counts of detected sequences, not modification occupancy: a high bar does not say what fraction of all molecules carries a modification.

The bars are grouped by sample, so the comparison the figure invites is across samples within one modification, and that is the comparison it supports. Comparing the three modifications with each other on one sample compares three different things: how often a residue was chemically derivatised on purpose, how often one was oxidised, and how often one was acetylated.

> **READ COUNTS AS COUNTS**
>
> Suppose one sample has 100 distinct sequences carrying oxidation and another has 200. The second bar is twice as high, but this is not evidence of twice the oxidation rate: the second sample may also have twice as many identified sequences. Read modification counts alongside coverage and the search settings that determined which modifications were reported.

32.1.1 Reading it

- One bar: one modification in one sample. Its height is how many distinct modified sequences carrying that modification were detected there.
- The group: the three bars over one sample, drawn side by side at a fixed width and centred on the sample's tick.
- The label under each group: `condition_n`, numbered within the condition rather than across the experiment, in annotation order. With **Toggle IDs** on, the sample's own identifier is added on a second line.
- The legend, titled Modification. Only the classes with at least one positive count anywhere in the dataset appear in it; a class detected nowhere is drawn nowhere and named nowhere.

> **EASY TO GET WRONG**
>
> A sequence contributes at most once to a class per sample, however many times it carries that modification, and a sequence carrying two of the three tracked classes contributes once to each. So the bar is a count of sequences, not of modification sites and not of spectra — a peptide with two oxidised methionines is one, and a peptide seen at three charges in one run is also one.

32.2 Controls

Controls on this page

Control	What it does	Default	Where
Toggle IDs	Adds the sample's own identifier under its condition_n label, on a second line.	off	page
Width (cm), Height (cm), DPI	The size of the *downloaded* figure. The browser scales the rendered figure to fit the page. Width and height are held between 1 and 60 cm and DPI between 72 and 600, and a very large figure has its resolution reduced so that the image stays under 24 megapixels.	25 cm × 15 cm, 100	page
Download format	PNG, JPG, WEBP, SVG or PDF.	PNG	page

In the plot appearance panel

Control	What it does	Default	Where
Legend	Shown or hidden, and its title, size, colour, frame and position. Its entries are the modifications.	shown	appearance
Title, X axis, Y axis	Text, size, weight, colour and position of each label, and the size, rotation and colour of the tick labels. The plot writes Modifications per sample, Sample and Number of modified peptides, and rotates the sample labels by 90°.	the plot's own	appearance
Font	One face for the whole figure, or one per text.	plot default	appearance
Frame and grid	Which of the four spines are drawn, and the grid behind the bars.	all four spines	appearance
Background	Paper colour, transparency and the preset themes.	default	appearance
Value labels	Offered on this plot, but the figure prints no numbers on its bars for it to size.	8	appearance

> NOTE
>
> There is no bar-colour control here. The three colours are fixed in the code — carbamidomethylation blue, oxidation red, acetylation green — and the appearance panel offers no colour card for this plot, so the same class is the same colour in every figure and in the summary report.

The panel offers no **Value labels** card on this page, because these bars carry no printed counts.

WORTH KNOWING

The second table is the plot appearance panel, which is hidden until advanced mode is turned on (**Ctrl** + **Shift** + **E**). Everything it contains is described in Part XI; only the entries particular to this plot are repeated here.

32.3 How it is computed

HOW IT IS COMPUTED

The long report is pivoted into a wide matrix: one row per distinct Modified.Sequence, one column per value of the file-name column, and in each cell the *largest* Precursor.Quantity of the rows that share that sequence and that run. Several precursors of one sequence in one run therefore collapse into one number before anything is counted.

The columns are then matched to the annotation and renamed condition_n. An exact run name is matched first; only a metadata sample that names no column exactly falls back to matching on the trailing number in the name. Columns the annotation does not account for are dropped, and the samples are drawn in annotation order.

A sequence counts as *detected* in a sample when its cell holds a finite number other than zero. Zero is read as not detected here, not as a measured zero.

For each tracked class m and each sample j the bar is then

$$\text{count}_{j,m} \;=\; \sum_{s \in S} \mathbf{1}[\, m \text{ matches } s \,] \cdot \mathbf{1}[\, s \text{ detected in } j \,],$$

where S is the set of distinct modified sequences in the report. The indicator on the left is a property of the sequence string alone, so it is the same in every sample; the indicator on the right is what changes between bars.

HOW A MODIFICATION IS RECOGNISED

The match is made against the Modified.Sequence string, ignoring case, and a class matches if either its UniMod identifier or its name appears in it:

- Carbamidomethylation: UniMod:4 or Carbamidomethyl, optionally followed by ation.
- Oxidation: UniMod:35 or Oxidation.
- Acetylation: UniMod:1 or Acetyl, optionally followed by ation.

The identifiers are matched so that a longer number cannot satisfy a shorter one:

UniMod:4 does not match UniMod:44, and UniMod:1 does not match UniMod:121. The names are matched so that they cannot be a fragment of a longer word.

The three classes are fixed. A modification outside them — a phosphorylation, an isobaric label, a pyroglutamate — is not counted here and does not appear in the figure, so this page describes the three it tracks and says nothing about the rest of the modifications in the report.

32.3.1 Edge cases

- A class with no positive count anywhere is dropped entirely: no bar, no legend entry, and the remaining bars widen to share the group.
- An annotated sample that matches no column in the report keeps its tick and its label and is drawn with no bars at all, so the gap is visible rather than silent.
- If no annotated sample matches any column, nothing is drawn and the page reports that the annotation does not match the report.
- A single replicate in a condition is drawn like any other sample; nothing on this page aggregates within a condition.
- A sequence whose cell is missing, non-numeric or exactly zero in a sample is not counted in that sample. It may still be counted in every other.
- A report without Modified.Sequence or without Precursor.Quantity draws nothing and names the missing column.

32.4 Exporting

The download reproduces the plot at the width, height and resolution set on the page. The browser scales the rendered figure to fit the page. Choose PDF for a figure that will be scaled or edited, PNG for a slide. The plot also appears in the summary report, which renders it from the settings left on this page.

Behind this page

Function	What it produces
modification_plot	Grouped bars of modified-sequence counts per sample, for the three tracked classes
peptide_modification_plot	The figure as the page renders it, including the appearance-panel settings. The counts behind the bars — one row per sample, one column per class, plus UniqueTrackedModifiedPeptides — are built by peptide_modification_counts.

33
Missed cleavages

How completely the digest cut, read off the sequences of the peptides each sample actually produced.

AT A GLANCE

Where	Peptide Level → Missed Cleavage Plot
Needs	A peptide report, loaded as the Peptide level in Data → Upload, and an annotation. The peptide and precursor levels are read from the long-format report on disk rather than from a wide matrix, and this page needs an abundance in it: the report must carry the column Precursor.Quantity, which is fixed in the code and is not the **Abundance Column** offered on the quality-control pages. It also needs a stripped-sequence column and a file-name column.
Annotation	With a protein dataset loaded, the protein annotation is used. Without one, the peptide annotation made in Data → Annotation is. Until one of them exists the page says so and draws nothing.
Each bar	One sample, stacked to 100 %. Each segment is the share of the distinct *stripped* sequences — sequences without their modifications — detected in that sample that carry that many missed cleavages. Not precursors, not spectra, and not sites.
The rule	Cleavage after K and R, with proline blocking taken from a table in the code. Which of the two rules applies is chosen on the page with **Cleavage Rule**; it is not read from the search parameters.
Table	None.

33.1 What the plot answers

A digest that ran to completion leaves almost every peptide cut at every available site. This page counts the sites that survived, per peptide, and shows how the surviving counts are distributed in each sample. It is a figure about sample preparation: the enzyme, how much of

it, how long, at what temperature, and how well the protein was denatured before it arrived.

Because each bar is normalised within its own sample, the comparison the figure supports is between the *shapes* of the bars, not their heights. A sample with a smaller zero-missed-cleavage segment has a larger fraction of detected sequences containing internal cleavage sites under the selected rule. This is consistent with less complete digestion, but search settings, sequence composition and detectability also affect the observed proportion.

> **A WORKED BAR**
>
> Of 100 detected sequences, suppose 80 have no missed cleavage, 15 have one and 5 have two. The segments are 80 %, 15 % and 5 %; the share with at least one missed cleavage is 20 %. These describe the identified sequences under the chosen cleavage rule, not all molecules in the original digest.

33.1.1 Reading it

- One segment: the share of the sequences detected in that sample that carry that many missed cleavages. The segments of one bar sum to 100 % by construction.
- The legend, titled Number: the missed-cleavage counts present anywhere in the dataset, in ascending order from the foot of the bar upwards. A count nobody has gets no segment and no entry.
- With **Toggle Text** on, one percentage is printed near the foot of each bar: the share of that sample's sequences with *at least one* missed cleavage — the number usually quoted when a digest is assessed, and the sum of every segment above the zero segment rather than the segment it is printed over.
- The title names the rule the counts were made under, because the same peptides give different counts under the two rules and the figure would otherwise be ambiguous.

> **EASY TO GET WRONG**
>
> The percentages are shares of what was *identified*, not of what was digested. A long peptide with three missed cleavages that the search never identified is absent from the report and therefore absent from the bar, so this is not an unbiased estimate of digestion completeness. It also says nothing about how many peptides are behind each bar: a sample with a tenth of the identifications of its neighbours still reaches 100 %. Read this page beside the coverage plot, which gives the counts.

33.2 Controls

Controls on this page

Control	What it does	Default	Where
Cleavage Rule	Which enzyme rule the count assumes: Trypsin or Trypsin / Lys-C. It changes the counts, not the data, and must match the digest that was actually run.	Trypsin	page
Toggle IDs	Adds the sample's own identifier under its condition_n label, on a second line.	off	page
Toggle Text	Prints, near the foot of each bar, the percentage of that sample's sequences with at least one missed cleavage, to one decimal.	on	page
Width (cm), Height (cm), DPI	The size of the *downloaded* figure. The browser scales the rendered figure to fit the page. Width and height are held between 1 and 60 cm and DPI between 72 and 600, and a very large figure has its resolution reduced so that the image stays under 24 megapixels.	25 cm × 15 cm, 100	page
Download format	PNG, JPG, WEBP, SVG or PDF.	PNG	page

In the plot appearance panel

Control	What it does	Default	Where
Value labels	The size of the percentage printed by **Toggle Text**. It is never set below 4.	8	appearance
Legend	Shown or hidden, and its title, size, colour, frame and position. Its entries are the missed-cleavage counts.	shown	appearance
Title, X axis, Y axis	Text, size, weight, colour and position of each label, and the size, rotation and colour of the tick labels. The plot writes Missed cleavages per sample followed by the rule in brackets, and Percentage of peptides with missed cleavage (%), and rotates the sample labels by 90°.	the plot's own	appearance
Font	One face for the whole figure, or one per text.	plot default	appearance
Frame and grid	Which of the four spines are drawn, and the grid behind the bars.	all four spines	appearance
Background	Paper colour, transparency and the preset themes.	default	appearance

NOTE

There is no bar-colour control here. The segments take the drawing library's default colour cycle in the order of the missed-cleavage counts, so the colour of a segment depends on which counts are present in the dataset and is not fixed across datasets.

WORTH KNOWING

The second table is the plot appearance panel, which is hidden until advanced mode is turned on (**Ctrl** + **Shift** + **E**). Everything it contains is described in Part XI; only the entries particular to this plot are repeated here.

33.3 How it is computed

THE CLEAVAGE RULE

A missed cleavage is a cleavable residue *inside* a peptide. The rule is not read from the search parameters and is not inferred from the data: it is a table in the code, selected by **Cleavage Rule**.
Both rules cleave after lysine (K) and after arginine (R). They differ only in whether a following proline blocks the cut:

- Trypsin: K–P and R–P are both blocked, so neither counts.
- Trypsin / Lys-C: the Lys-C component is not inhibited by proline, so K–P counts; R is cut by trypsin alone, so R–P stays blocked.

For a sequence of residues $r_1 r_2 \ldots r_L$, read in upper case, the count is

$$\mathrm{mc} \;=\; \sum_{i=1}^{L-1} \mathbf{1}[\, r_i \in \{\mathrm{K}, \mathrm{R}\}\,] \cdot \mathbf{1}[\, r_i \text{ is not blocked by } r_{i+1}\,] \,.$$

The sum stops at $L - 1$: the last residue of a peptide is the expected cut site, not a missed one, and is never counted whatever it is. So SAMPLER has no missed cleavage, SAMKPLER has none under Trypsin and one under Trypsin / Lys-C, and SAMKPLEKR has one and two respectively.

FROM THE REPORT TO THE BARS

The long report is pivoted into a wide matrix: one row per distinct stripped sequence, one column per value of the file-name column, and in each cell the *largest* Precur-

sor.Quantity of the rows that share that sequence and that run, so modified and multiply charged forms of one sequence collapse into one number.
The columns are matched to the annotation and renamed condition_n, exact run names first and the trailing number in the name only as a fallback. A sequence counts as *detected* in a sample when its cell holds a finite number other than zero; zero is read as not detected.
Each distinct sequence is then given its missed-cleavage count once, from its string alone, and for every sample j and count k

$$\mathrm{pct}_{j,k} \;=\; 100 \times \frac{\#\{\, s \,:\, \mathrm{mc}(s) = k \text{ and } s \text{ detected in } j \,\}}{\#\{\, s \,:\, s \text{ detected in } j \,\}}.$$

The denominator is the sample's own number of detected sequences, which is why every bar reaches 100 % and why the bars carry no information about depth.

There is no fixed share above which a digest is called failed, and this manual does not name one. The number that matters is the one your own laboratory reaches on a digest that worked: compare a new run against that, and against the other samples in the same experiment, rather than against a value from elsewhere.

Those two are the rules the tool supports, and no other protease is planned. The rule is a setting of this page rather than a property of the dataset, so it has to be set to match the digest that was actually run — which is why the figure carries it in its title.

33.3.1 Edge cases

- Rows with an empty sequence are dropped before anything is counted.
- A sequence missing, non-numeric or exactly zero in a sample does not enter that sample's numerator or its denominator.
- A sample in which nothing is detected, and an annotated sample that matches no column in the report, are both drawn as an empty bar of height zero rather than being left off the axis.
- A single replicate in a condition is drawn like any other sample; nothing on this page aggregates within a condition.
- If nothing at all is detected anywhere, no figure is drawn and the page reports that no peptide intensities are available.

33.4 Exporting

The download reproduces the plot at the width, height and resolution set on the page, which is why those three controls do not change the figure on screen. Choose PDF for a figure that will be scaled or edited, PNG for a slide.

> **IN THE SUMMARY REPORT**
>
> The plot also appears in the summary report, under the same **Cleavage Rule** as the page, and its title states which rule that was.

Behind this page

Function	What it produces
missed_cleavage_plot	The stacked percentage bars, one per sample, under the selected cleavage rule
count_missed_cleavages	The count for one sequence, or for a column of them. The per-sample table behind the bars — Sample, Count, Occurrences, Total, Percentage — is built by peptide_missed_cleavage_summary.
normalize_protease	Resolves the written rule to trypsin or trypsin_lysc and refuses anything else

34
Sequence coverage

Which residues of one protein were covered by an identified peptide, and what fraction of its sequence that is.

AT A GLANCE

Where	Peptide Level → Sequence Coverage
Needs	A peptide report, loaded as the Peptide level in Data → Upload, carrying ProteinNames and a stripped-sequence column. The peptide and precursor levels are read from the long-format report on disk rather than from a wide matrix, but this is the one page of this part that needs no abundance in it at all: coverage is computed from identifications, so no abundance column is chosen and no annotation is required.
Output	A monospace text map and three summary figures, not a plot. There is no plot appearance panel on this tab, because there is nothing drawn for it to style.
Denominator	The full protein sequence as it stands in a reference proteome that ships with the application — human or mouse, chosen with **Database** — and not the part of it the report knows about.
Scope	Every peptide assigned to the protein anywhere in the report. Nothing is filtered by sample, by condition or by abundance, so the map is the union over the whole experiment.
In the DataViewer	Absent. The tab is not offered in viewer mode.

34.1 What the map shows

Coverage answers a question the identification counts cannot: not how many peptides a protein produced, but which parts of it they came from. A protein covered at 40 % by peptides spread over its length has been sampled evenly; one covered at 40 % by peptides packed into two tryptic hot spots has three quarters of its sequence unobserved, and any modification or variant in that stretch was never in a position to be found.

The map is the protein sequence with the observed peptides written over it, so the unobserved stretches are the runs of dashes. The percentage under the heading is the same information reduced to one number.

> **OVERLAPPING PEPTIDES DO NOT DOUBLE THE COVERAGE**
>
> On a 100-residue reference sequence, peptides covering positions 1–10 and 6–15 cover 15 distinct residues: 15 %, not 20 %. They may need separate overlay rows to remain readable. An uncovered stretch means no reported peptide matched there; it does not prove that stretch was absent from the sample.

34.1.1 Reading it

- A run of letters in an overlay row: one observed peptide, written at the position where it occurs in the protein sequence. A run of dashes is sequence no identified peptide covered.
- A second or third overlay row. A peptide is written into the first row whose residues at its position are still all dashes; where two peptides overlap, the second is pushed onto a new row. The rows are printed with the first one nearest the sequence line.
- The line above each block: positions, printed right-aligned at every tenth residue, so the number 50 sits over the fiftieth residue.
- The line below: the protein sequence itself, from the selected database, in blocks of ten residues separated by spaces.
- The Coverage card above the map: covered residues as a percentage of the length of that sequence, to two decimals. The Peptides card beside it is the number of distinct peptide sequences the report assigns to this protein.

> **EASY TO GET WRONG**
>
> The percentage is a fraction of length, not a score of quantification quality. The same eight peptides can give a high percentage on a 150-residue protein and a low one on a 1500-residue protein, and neither number says anything about how well either protein was quantified. Percentages can be compared as fractions, but interpret them with protein length, peptide detectability and the reference sequence. Shared peptides and repeated motifs do not uniquely locate their origin: marking every sequence match is coverage of possible positions, not proof that each copy was separately observed.

34.2 Controls

Controls on this page

Control	What it does	Default	Where
Database	Which reference proteome the sequence and its length come from: Human or Mouse. Both ship with the application; nothing is downloaded.	Human	page
Protein	Which protein to map. The list is built from the report: the first entry of every ProteinNames value, de-duplicated and sorted. The first of them is selected when the page opens, and the selection is reset if the report no longer offers it.	the first protein in the list	page
Chunk Size	Roughly how many residues are written per line of the map. It is never taken below 10. The map is cut every $\lfloor 1.1 \times \text{Chunk Size} \rfloor$ characters of the spaced sequence, and a ten-residue block with its separating space costs eleven characters, so 100 gives exactly ten blocks — a hundred residues — on a line.	100	page

34.3 How it is computed

WHICH PEPTIDES BELONG TO THE PROTEIN

ProteinNames is split on semicolons and the selected protein is looked for among the parts, so a peptide assigned to several proteins counts towards each of them. The stripped — unmodified — sequences of the matching rows are taken, de-duplicated, and used as they are.

Note that this is a wider test than the one that built the **Protein** list: the list offers only the *first* name of each ProteinNames value, while the matching accepts the name in any position.

THE SEQUENCE AND THE DENOMINATOR

The reference proteome is read as FASTA. For each entry the header is cut at its first space and split on |: the last field is taken as the entry name, the middle field as the accession. The selected protein is looked up first against the entry names and then against the accessions, as an exact string in both cases.

The sequence found is the denominator. Coverage is a fraction of the *whole* protein as that file records it, not of the part the report mentions and not of a tryptic subset of it.

COVERAGE

A mask of zeros, one per residue of the protein sequence, is set to one everywhere a peptide occurs. Each peptide is searched for as a literal substring, and *every* occurrence is marked, not only the first, so a repeated motif is covered wherever it appears. Overlapping peptides set the same positions to one and are therefore counted once. Then

$$\text{coverage} \;=\; 100 \times \frac{\#\{\, i \,:\, \text{mask}_i = 1 \,\}}{L},$$

with L the length of the sequence, rounded to two decimals.

THE MAP

The map is built separately from the percentage, by writing the peptides into overlay strings of dashes rather than into a mask. Each peptide is placed, at each position where it occurs, into the first overlay row whose residues at that position are still all dashes; if none is free a new row is opened. Rows therefore accumulate only where peptides overlap, and their number is the deepest overlap in the protein.
The sequence, the overlays and the position line are then split into blocks of ten residues joined by single spaces, and cut into lines of $\lfloor 1.1 \times \text{Chunk Size} \rfloor$ characters. Each block of the output prints the position line, then the overlay rows in reverse order so that the first-placed peptides sit closest to the sequence, then the sequence line, then a blank line.

Coverage is deliberately an overall figure: every peptide identified anywhere in the report counts, with no sample, condition or abundance filter, so the number describes the experiment rather than a run.

Human and mouse are the reference proteomes the tool provides, and a protein absent from them cannot be drawn here.

34.3.1 Edge cases

- A peptide that does not occur verbatim in the database sequence contributes nothing and is not reported. This is what happens when the report's assignment names an isoform or a variant that the bundled proteome does not contain.
- If the protein carries no peptides in the report, nothing is drawn and the page says so by name.
- If the selected protein is in the report but in neither the entry names nor the accessions

of the selected database, the page reports that the variant does not exist in the loaded database. Switching **Database** between human and mouse is the usual cause.

- If the database file itself cannot be found the page says which species it was looking for; the locations it searched go to the server log rather than to the screen.
- If the report has no ProteinNames column the protein list is empty, the page says that sequence coverage is unavailable for this report, and the other tabs of this part are unaffected.
- Coverage does not distinguish a protein seen once from one seen in every run: a peptide identified in a single sample covers its residues exactly as one identified in all of them does.

34.4 The summary above the map

The three cards

Column	What it holds
Protein	The protein as selected, spelled as the report spells it.
Coverage	Covered residues as a percentage of the length of the database sequence, to two decimals.
Peptides	The number of distinct stripped sequences the report assigns to this protein — the peptides that went into the map, whether or not each was found in the sequence.

34.5 Exporting

The notebook download beside **Download TXT** reconstructs the calculation and its text output with the required reference resources. It does not invent a separate image plot. Keep the reference sequence identity with the result: changing an isoform or database sequence can change both positions and the coverage denominator.

There is no figure to download and no image format to choose. **Download TXT** saves the map exactly as it is shown, as a plain text file named for the protein and the database, in the form PROTEIN_species_coverage.txt. It is monospaced by construction, so it only lines up in a monospaced font; pasted into a proportional one the overlay rows no longer sit over the residues they refer to. Neither the map nor the percentage is carried into the summary report,

which covers the other three tabs of this part only.

34.6 Further reading

The UniProt help page *Sequence processing* explains canonical precursor and mature sequences. Coverage depends on the actual reference sequence selected; an accession, isoform and sequence version should accompany a reproducible map.

Behind this page

Function	What it produces
calculate_coverage	The percentage: covered residues over the length of the database sequence
vis_coverage	The text map, with its overlay rows and position lines
fasta_to_dataframe	Reads a FASTA database into headers, sequences, entry names and accessions
peptide_sequence_coverage	The page's own call: the percentage, the peptide count and the map together

PART VII

Single protein



35
One protein, by condition

The intensities of a single feature, condition by condition.

AT A GLANCE

Where	Single protein → Boxplot
Data levels	Protein, phosphosite, phosphoprotein
Needs	An uploaded dataset and its annotation, and one feature selected. Conditions come from the annotation.
Data source	Data, Filtered Data or Imputed Data; the page opens on the most processed one that exists.
One box	The replicates of the selected feature within one condition — one value per sample, not one per feature.
Table	Yes, below the plot, downloadable as CSV.

AT PHOSPHOSITE LEVEL

The shared **Localization probability cutoff** selects sites across phosphosite views. At 0 it keeps all sites, including those with no probability; above 0 those unknowns are excluded. This is separate from completeness filtering and does not rerun normalization or imputation. An already collapsed phosphoprotein table has its own recorded collapse cutoff.

35.1 What the plot answers

This is the plot drawn after the volcano: not where a feature's point landed, but what the feature actually did. Every box is one condition and what fills it are that condition's replicates, so the question it answers is whether the difference a test reported is carried by all of them or by one.

A box here holds as many values as the condition has samples, which makes it a far smaller summary than the boxes in quality control, where a box holds one value per feature. Read the spread, not the exact whisker.

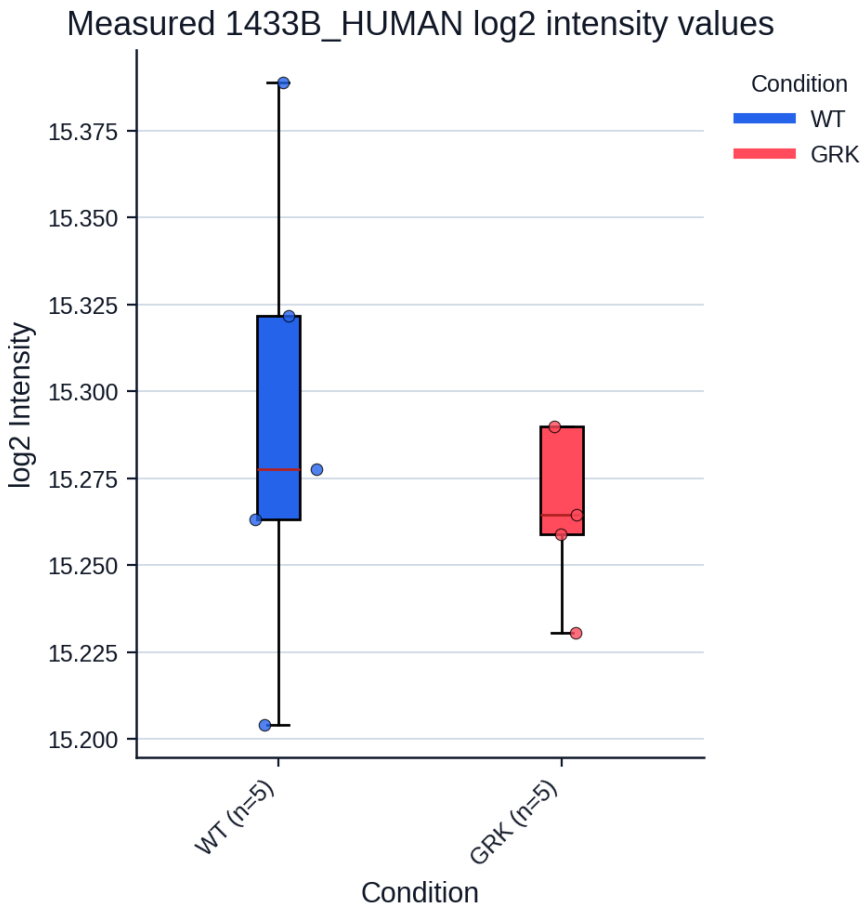


Figure 35.1. The protein 1433B_HUMAN on the example dataset, with **Toggle Dots** on: one box per condition, each built from the five replicates of that condition and labelled (n=5) underneath. The whole y axis spans under two tenths of a $\log_2$ unit, so the two boxes overlap almost completely — this is a feature that did not move, and the figure says so at a glance.

35.1.1 Reading it

- One box, one condition: the distribution of that feature's values over the samples of the condition.
- The line across it: the median of those values.
- The dots over the box, when **Toggle Dots** is on: the replicates that fall inside the whiskers, spread sideways so that equal values do not hide one another.
- The markers beyond the whiskers, when **Toggle Outliers** is on: the replicates the dots leave out. One of these in a set of five is a sample to look at, not a result.
- The label under each box: the condition, followed by (n=...) — the number of values the box was built from — as long as **Toggle Sample n** is on.

> EASY TO GET WRONG
>
> Three points are not a distribution. With three or four replicates the median is one value or the mean of two and a quartile is an interpolation between neighbours, so the box draws those few numbers rather than estimating a population. Compare the dots across conditions before the boxes, and treat a difference that rests on one replicate as one that rests on one replicate.

35.2 Controls

Controls on this page

Control	What it does	Default	Where
Dataset level	Which dataset the feature is taken from: Protein, Phospho or Phosphoprotein. Only the levels actually uploaded are offered; phosphoprotein appears once a collapse has been run.	Protein	page
Data source	Which matrix the values are read from: Data, Filtered Data or Imputed Data. Only the ones that exist are offered.	the most processed matrix available	page
Identifier	Which name the feature is chosen by: Protein Names, Phosphosite Names or Phosphoprotein Names according to the level, and Gene Names where the dataset carries a gene column. The phosphosite level offers no gene option.	Protein Names	page

Controls on this page (continued)

Control	What it does	Default	Where
Protein / Site	The one feature to draw; at protein level only the first accession of a group is offered. Labelled **Gene** under Gene Names.	the first entry	page
Conditions	Which conditions to draw. Their order is the annotation's.	all	page
Toggle Dots	Draws the individual replicates over each box, the outliers excluded.	off	page
Toggle Sample n	Appends (n=...) to each condition label — how many values that box was built from.	on	page
Toggle Outliers	Draws the values beyond the whiskers as individual markers.	off	page
Random Seed	Appears only with **Toggle Dots** on. Fixes the sideways jitter of the replicate dots, so that the same figure is reproduced exactly.	69	page
Width (cm), Height (cm), DPI	The size of the *downloaded* figure. The browser scales the rendered figure to fit the page. Width and height are held between 1 and 60 cm, DPI between 72 and 600.	20, 10, 300	page
Download format	PNG, JPG, WEBP, SVG or PDF.	PNG	page

In the plot appearance panel

Control	What it does	Default	Where
Show title	The plot's own title is Measured <feature> log2 intensity values, and Imputed ... on Imputed Data. Its text, size, weight, colour and position are set here too.	shown	appearance
Show legend	The legend of condition colours, and its title, size, colour and frame.	shown	appearance
Box border color, Median color	The outline of every box, and the line across it.	black, #b22222	appearance
Data dot size, opacity, outline, outline width	The replicate dots drawn by **Toggle Dots**.	4.5, 0.8, black, 0.4	appearance
Outlier dot color, outline color, size	The markers drawn by **Toggle Outliers**, styled separately because they are the values the replicate dots exclude.	red, black, 5	appearance
Condition markers	The same four dot settings, one condition at a time, over the general ones above.	the general settings	appearance

In the plot appearance panel (continued)

Control	What it does	Default	Where
Condition names and colours	Renames a condition on the axis and in the legend, and recolours it. The colour is shared with every other plot in the session.	the condition palette	appearance
Frame, grid, font, background	Which of the four spines are drawn and the grid behind the boxes; one face for the whole figure or one per text; paper colour, transparency and the preset themes.	all four spines, no grid	appearance

WORTH KNOWING

The second table is the plot appearance panel, which is hidden until advanced mode is turned on (**Ctrl** + **Shift** + **E**). Everything it contains is described in Part XI; only the entries particular to this plot are repeated here. **Toggle Header** and **Toggle Legend** are no longer beside the plot: they are **Show title** and **Show legend** in the panel.

35.3 How it is computed

The box, the whiskers and what counts as an outlier follow the standard definition given in **Concepts**; it is not repeated here. What is particular to this page is only what goes into the box: the replicates of one feature within one condition, one value per sample.

HOW IT IS COMPUTED

The rows of the chosen matrix whose identifier column matches the selection are taken: at protein level on the first accession of the group, by gene on any of a row's semicolon-separated gene tokens, so more than one row can match.
For every sample column j the value plotted is the median over the matching rows,

$$v_j \;=\; \text{median}\{\, x_{ij} \;:\; i \in M \,\}, \qquad M = \text{the matching rows},$$

computed over the rows that carry a value in that sample. With one matching row — the ordinary case — v_j is that row's value. Non-numeric entries, $\pm\infty$ and NA are removed first, and a sample left with nothing is dropped rather than counted as zero. The box of condition c is built from $\{\, v_j \;:\; j \in S_c \,\}$, the values of the samples of that condition, so its height is the spread *between replicates* and n is at most the number of samples in the condition.

WHICH CONDITIONS GET A BOX

A box is drawn for a condition with **three or more** values. A condition below that is not dropped: it keeps its place on the axis, its label, its colour in the legend and its row in the table, and is drawn as its individual points instead — which appear whether or not **Toggle Dots** is on. Only a condition in which the feature has *no* finite value leaves the axis entirely.

THE REPLICATE DOTS

With **Toggle Dots** on, the values within $Q_1 - 1.5\times\text{IQR}$ and $Q_3 + 1.5\times\text{IQR}$ are drawn over the box, each displaced sideways by a draw from a normal distribution of standard deviation 0.05 around the box's position. The generator is seeded with **Random Seed**, so one seed reproduces one scatter exactly.

35.3.1 Edge cases

- A missing value removes that sample from the box; it is not imputed here and does not become a zero. The *n* in the label is what survived, which is why it can differ between conditions of equal size. A feature missing throughout a condition leaves that condition off the plot and out of the table, and if that is true everywhere the figure reads No measured values in the selected conditions.
- When the selection matches several rows — duplicates, or a gene mapping to several proteins — every plotted value is their per-sample median, and an amber note above the table says how many rows were combined.
- On Imputed Data some values in the box are draws from the imputation model rather than measurements, which is why the title then reads Imputed instead of Measured: the exported figure carries no data-source selector.
- At phosphoprotein level collapsed with robust_site_median the values are ratios to each site's own across-sample median, and the axis says log2 Ratio to Site Median instead.

The threshold of three is fixed and there is no control for it: below three values a box would describe nothing, so the replicates are shown as points instead.

35.4 The table below the plot

One row per condition that reached the plot, in the order the conditions were drawn.

Columns in this table

Column	What it holds
Condition	The condition, as labelled on the axis.
N, SampleCount	How many values the box was built from — the same number as (n=...) in the label, twice, under the two names the single-protein tables use.
Mean, Median	Mean and median of those values, to four decimals. The median is the line across the box.
Q1, Q3	The 25th and 75th percentiles: the bottom and top of the box.
Min, Max	The smallest and largest value, outliers included — the data, not the whisker caps.
SourceFeatureCount, SourceFeatures	How many rows the selection matched and which — 1 unless a gene maps to several features or the matrix holds duplicates.
Aggregation, Warning	Per-sample median, and a note filled in only when more than one row was combined. The note is shown above the table as well.

35.5 In the notebook

A notebook starting point, using the frames introduced in *The function reference*. These compact examples do not encode every UI option; use the plot's notebook ZIP, where available, for its saved settings and preprocessing recipe:

The function looks the feature up in the frame's index, so the frame has to be indexed by the identifier first:

PYTHON

```
data_by_protein = data.set_index("ProteinNames", drop=False)
fig = boxplot_int_single_prot(data_by_protein, meta, "1433B_HUMAN", dots=True)
```

R

```
data_by_protein <- data
rownames(data_by_protein) <- data_by_protein$ProteinNames
fig <- boxplot_int_single_prot(data_by_protein, meta, "1433B_HUMAN", dots = TRUE)
```

At phosphosite level the same call takes data3 indexed by PTM_Collapse_key and annotated with meta2. dots and outliers are **Toggle Dots** and **Toggle Outliers**, and the per-sample median over matching rows is the same aggregation the page performs — the function writes it into the title rather than into a warning above a table. Two things differ: the (n=...) in each label is always printed, there being no **Toggle Sample n**, and the boxes take their colours from Matplotlib's default cycle instead of the session's condition palette unless plot_colors is given.

35.6 Exporting

Width and height control the rendered figure; the browser then scales its preview to the available space. Resolution controls raster detail. Choose PDF for a figure that will be scaled or edited, PNG for a slide. The table downloads separately as CSV, and it is the better thing to keep: it holds the numbers the box is drawn from.

35.7 Further reading

NIST: box plots explains the summary geometry. CoPYlot's minimum-count rule and optional individual points are described above.

Behind this page

Function	What it produces
single_protein_boxplot_plot	The figure, with the condition palette and the appearance panel applied
single_protein_boxplot_table	The per-condition summary under it
boxplot_int_single_prot	The same plot in the package, for a notebook

36
One protein, across samples

The same values, joined, sample by sample with the conditions in metadata order.

AT A GLANCE

Where	Single protein → Lineplot
Data levels	Protein, phosphosite, phosphoprotein
Needs	An uploaded dataset and its annotation, and at least one feature selected.
Data source	Data, Filtered Data or Imputed Data; the page opens on the most processed one that exists.
One point	One feature in one *sample*. The axis is the samples, not the conditions: nothing is averaged.
Several features	One line each, in the colours of the tab10 cycle.
Table	Yes, below the plot, downloadable as CSV — one row per feature and sample.

AT PHOSPHOSITE LEVEL

The shared **Localization probability cutoff** selects sites across phosphosite views. At 0 it keeps all sites, including those with no probability; above 0 those unknowns are excluded. This is separate from completeness filtering and does not rerun normalization or imputation. An already collapsed phosphoprotein table has its own recorded collapse cutoff.

36.1 What the plot answers

The boxplot summarises a condition; this page refuses to. Every marker is one sample, and the line joins them with the selected conditions in metadata order, so what the figure shows is the individual runs in sequence: which replicate sits high, whether a condition moves as a block, and where a feature simply was not measured. It is also the only single-feature plot that draws more than one feature at once, which is how two phosphosites on the same protein

are compared.

The x axis is an ordering, not a scale. The samples are equally spaced because they are drawn in a list, and the gaps between conditions carry no quantity — a doubling of dose and the next time point look alike here. The line between two markers is drawn to make the sequence readable, not because anything was measured between them.



Figure 36.1. Two proteins, 1433B_HUMAN and 1433E_HUMAN, across the ten samples of the example dataset: the five WT runs first, then the five GRK ones, in metadata order. Each marker is one sample and each line one feature, coloured from the fixed set rather than from the condition palette. Both features are measured in every sample, so neither line has a gap. They run roughly a $\log_2$ unit apart; 1433B_HUMAN is flat across all ten runs, while 1433E_HUMAN sits a little higher in GRK than in WT — a shift of about a tenth of a $\log_2$ unit, which the y range makes visible and which is exactly the kind of step this plot is read for.

36.1.1 Reading it

- One line, one feature. Its colour comes from the fixed tab10 set, in the order the features were selected; it is not the condition palette, because a line crosses every condition.
- One marker, one sample: the value of that feature in that run.
- The spread within a condition is the scatter of the markers standing over consecutive labels of the same condition. There is no error bar — nothing here is a mean.
- The label under each marker: condition_n, numbered within the condition rather than across the experiment, with the sample's own identifier on a second line when **Toggle ID** is on.

> **EASY TO GET WRONG**
>
> A missing value is not drawn, but the line is: the markers on either side of it are joined directly, so a feature missing in the middle of a condition leaves a segment that looks like a measurement and is not. The gap is visible only as a missing marker over a

label that is still on the axis. When a trajectory matters, count the markers against the labels, or read the table, which retains missing feature–sample combinations as NA.

36.2 Controls

Controls on this page

Control	What it does	Default	Where
Dataset level	Which dataset the features are taken from: Protein, Phospho or Phosphoprotein. Only the levels actually uploaded are offered.	Protein	page
Data source	Which matrix the values are read from: Data, Filtered Data or Imputed Data. Only the ones that exist are offered.	the most processed matrix available	page
Identifier	Which name the features are chosen by: Protein Names, Phosphosite Names or Phosphoprotein Names according to the level, and Gene Names where the dataset carries a gene column. The phosphosite level offers no gene option.	Protein Names	page
Proteins / Sites	The features to draw, one line each. Labelled **Genes** under Gene Names.	the first three in the list	page
Conditions	Which conditions contribute samples. They are listed, and laid out on the x axis, in metadata order; selecting them in another order does not reorder the axis. Within a condition the samples follow the annotation.	all	page
Toggle ID	Adds the sample's own identifier under each condition_n label, on a second line.	off	page
Show NA as low	Draws the line down to 0 on the log2 axis (equivalent to linear 1) at a missing sample, without a marker. This changes only the drawing, not data or statistics.	off	page
Width (cm), Height (cm), DPI	The size of the *downloaded* figure. The browser scales the rendered figure to fit the page. Width and height are held between 1 and 60 cm, DPI between 72 and 600.	20, 10, 300	page
Download format	PNG, JPG, WEBP, SVG or PDF.	PNG	page

Controls on this page (continued)

Control	What it does	Default	Where

In the plot appearance panel

Control	What it does	Default	Where
Show title	The plot's own title is Protein expression across samples at protein level, Gene … under Gene Names and Phosphosite … at both phospho levels. Its text, size, weight, colour and position are set here too.	shown	appearance
Show legend	The legend of features beside the plot, headed Feature, and its title, size, colour and frame.	shown	appearance
Frame, grid, font, background	Which of the four spines are drawn and the grid behind the lines; one face for the whole figure or one per text; paper colour, transparency and the preset themes.	all four spines, grid on both axes	appearance
Condition names	Offered, but this figure builds its sample labels from the annotation's own condition names and does not read the overrides. There is no condition-colour card at all: the lines are coloured by feature.	the annotation's names	appearance

> **WORTH KNOWING**
>
> The second table is the plot appearance panel, which is hidden until advanced mode is turned on (**Ctrl** + **Shift** + **E**). Everything it contains is described in Part XI; only the entries particular to this plot are repeated here. **Toggle Header** and **Toggle Legend** are no longer beside the plot: they are **Show title** and **Show legend** in the panel.

The panel shows neither a **Points** card nor the condition-renaming rows on this page: the line plot draws a fixed circle marker, and the sample labels are built from the annotation, so neither would have any effect.

36.3 How it is computed

HOW IT IS COMPUTED

The rows of the chosen matrix whose identifier column matches any selected feature are taken: at protein level the match is on the first accession of the group, by gene on any of a row's semicolon-separated gene tokens.
The matched rows are reshaped to one row per feature and sample. Where several rows carry the same feature — duplicates, or several proteins under one gene — their values are reduced to one per sample by the median,

$$v_{fj} \;=\; \text{median}\{\, x_{ij} \;:\; i \in M_f \,\},$$

over the rows M_f of feature f that have a value in sample j.
Non-numeric and non-finite values become NA. Missing combinations stay in the table and on the sample axis but have no marker. By default the line connects successive observed points. With **Show NA as low**, it visits the display baseline at missing samples without adding a point. No missing value is replaced in the matrix or used as a measurement.

THE ORDER OF THE AXIS

The samples of the selected conditions are relabelled condition_n, numbered within the condition in the order the annotation gives, and laid out condition by condition in the order **Conditions** lists them, which is metadata order. Each sample takes the next integer position; the markers of one feature are then joined in that order.

Use the markers to inspect the individual measurements behind a summary. A lone extreme value deserves a check of that sample and feature; a condition-wide shift may have biological or technical explanations. Missing points show unavailable measurements, not a measured low abundance. The connecting line is a visual guide in the chosen sample order, not a trajectory, interpolation model or evidence of a treatment effect. Use a suitable statistical comparison to assess a difference.

36.3.1 Edge cases

- A condition with one replicate contributes one marker and needs no special handling; there is no minimum number of samples on this page.
- Missing samples remain on the axis. With no observed values there are no measured markers; the optional baseline line must not be mistaken for evidence of a low measured abundance. Zero and negative log2 measurements are valid values and retain their markers.
- When one selection matches several rows, the note above the table says how many were combined and that each value is their per-sample median.

- A selected name that matches no row of the dataset is not drawn. It is named under the figure, in an amber notice box, as Not found in this dataset, so not plotted: …. The notice is shown in the application only.
- The line colours are drawn from tab10, which holds ten, so beyond ten features some lines share a colour.
- At phosphoprotein level collapsed with robust_site_median the values are ratios to each site's own across-sample median, and the axis says log2 Ratio to Site Median instead.

36.4 The table below the plot

One row per feature and sample, including rows whose Value is NA. The drawing baseline is never written into this table.

Columns in this table

Column	What it holds
Feature	The feature, as labelled in the legend.
Sample	The sample, labelled condition_n as on the axis.
Condition	The condition that sample belongs to.
Value	The plotted value, to four decimals.
SampleCount	How many samples this feature has a value in.
SourceFeatureCount, SourceFeatures	How many rows the selection matched and which — 1 unless a gene maps to several features or the matrix holds duplicates.
Aggregation, Warning	Per-sample median, and a note filled in only when more than one row was combined.

36.5 In the notebook

A notebook starting point, using the frames introduced in *The function reference*. These compact examples do not encode every UI option; use the plot's notebook ZIP, where available, for its saved settings and preprocessing recipe:

```python
conditions = ["WT", "GRK"]
fig = compare_prot_line(data, meta, conditions, ["1433B_HUMAN", "1433E_HUMAN"],
                        workflow="Protein", id=False)
```

```r
conditions <- c("WT", "GRK")
fig <- compare_prot_line(data, meta, conditions, c("1433B_HUMAN", "1433E_HUMAN"),
                         workflow = "Protein", id = FALSE)
```

This call needs no indexed frame: the features are matched against a column, and workflow says which one — ProteinNames, PTM_Collapse_key or Phosphoprotein — in place of the data level chosen on the page. At phosphosite level the same call takes data3 and meta2 with workflow="Phosphosite", and the features are then PTM_Collapse_key values. The conditions list does the work the page's condition selector does, choosing both which conditions appear and the order they run in along the x axis, and id is **Toggle ID**. Matching rows are reduced to a per-sample median exactly as on the page, but the legend is headed with the workflow name rather than Feature.

36.6 Exporting

The download reproduces the plot at the width, height and resolution set on the page; the browser scales that render to fit the page. With many samples the labels along the bottom are the first thing to become unreadable, so widen the figure before reducing the font. The table downloads separately as CSV.

Behind this page

Function	What it produces
single_protein_lineplot_plot	The figure
single_protein_lineplot_table	The long table under it
compare_prot_line	The same plot in the package, for a notebook

37
The phosphosites of one protein

Every phosphosite measured on one protein, against every sample.

AT A GLANCE

Where	Single protein ⟶ Heatmap
Data levels	Phosphosite only: the dataset-level selector is fixed on this tab and the page says so.
Needs	A phospho dataset carrying PTM_Collapse_key and a protein-group column, and one protein selected.
Data source	Data, Filtered Data or Imputed Data; the page opens on the most processed one that exists.
Rows, columns	One row per phosphosite of the selected protein; one column per sample of the selected conditions.
Two scales	**Value Type** log2 colours the stored values; z colours each row's deviation from its own mean.
Table	Yes, below the plot, downloadable as CSV — the matrix as drawn.

AT PHOSPHOSITE LEVEL

The shared **Localization probability cutoff** selects sites across phosphosite views. At 0 it keeps all sites, including those with no probability; above 0 those unknowns are excluded. This is separate from completeness filtering and does not rerun normalization or imputation. An already collapsed phosphoprotein table has its own recorded collapse cutoff.

37.1 What the plot answers

One protein can carry twenty phosphosites and the boxplot draws one at a time. This page draws them all at once, against every sample, so the question it answers is which measured sites have similar patterns. It cannot separate a change in total protein abundance from a

change in phosphorylation occupancy without additional measurements.

The cost is that a cell states only a colour. Nothing here is a test and nothing is aggregated by condition — a column is one run — so this page is for finding the pattern, and the boxplot or the volcano for the number that goes in the figure legend.

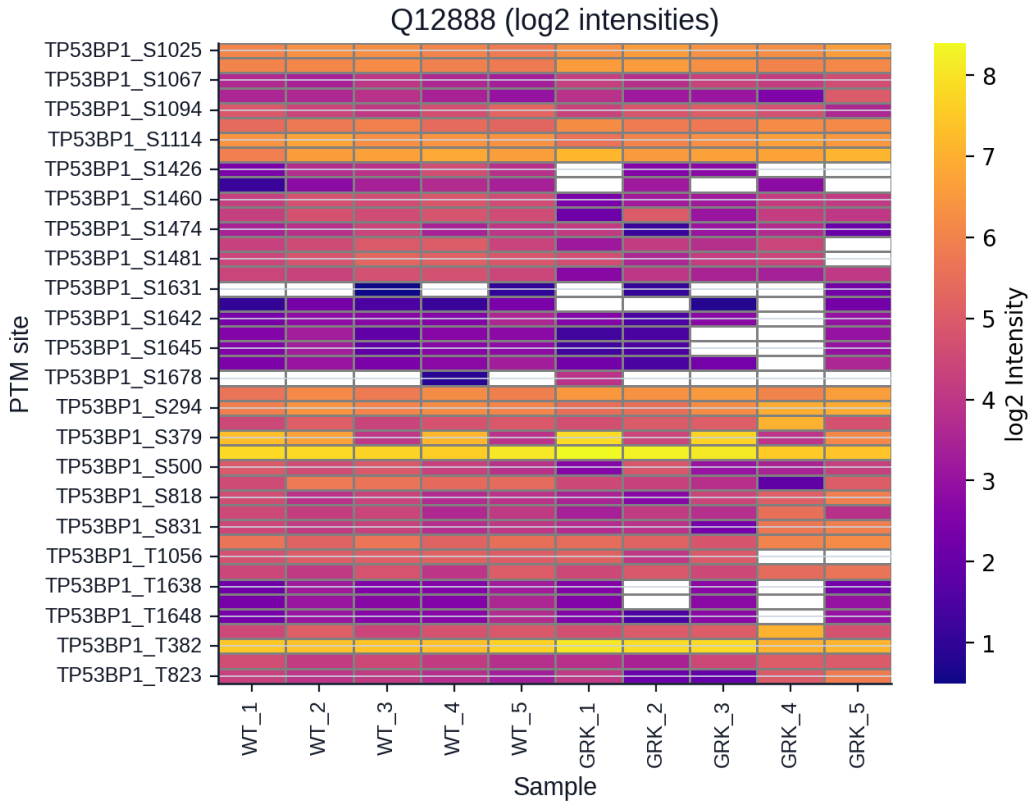


Figure 37.1. The phosphosites of protein group Q12888 — TP53BP1 — against the ten samples of the example dataset, five WT and five GRK, in **Value Type** log2 and with **Filter** *_**M1** at its default. That leaves 43 rows, which is why the row labels carry no multiplicity suffix. Colour is the stored value on the plasma scale at the right, and the white cells are the 38 values of this 43 × 10 matrix that are missing, several of them in the last two GRK runs. There are more rows than the figure has room to label, so every other site name is printed.

37.1.1 Reading it

- A cell: one phosphosite in one sample, coloured by that value on the scale **Value Type** chose. A cell left in the background colour is a missing value.
- A row: one phosphosite, labelled with the short site name — the accession is dropped because every row belongs to the same protein, and the multiplicity suffix too when **Filter** *_**M1** has already restricted the view to it. The whole identifier stays in the table.
- A column: one sample, labelled condition_n, numbered within its condition, with the sample's own identifier added on a second line when **Toggle ID** is on.
- The colourbar: log2 Intensity or Z-score, according to **Value Type**. It is the only thing that tells the two modes apart in an exported figure.

EASY TO GET WRONG

The two value types answer different questions. log2 colours the values as they stand, so an abundant site is bright in every sample and one a hundredfold below it is dark in every sample: the figure shows which sites are abundant, and a real regulation of the faint ones is invisible. z centres each row on its own mean, which makes the rows comparable and destroys the difference in level between them — a site whose values vary by a hundredth of a $\log_2$ unit gets the same full range of colour as one that moves fourfold. Read z for the pattern and log2 for the amount, never either for both.

37.2 Controls

Controls on this page

Control	What it does	Default	Where
Data source	Which matrix the values are read from: Data, Filtered Data or Imputed Data. Here the choice can change which *rows* are drawn, because clustering uses complete rows only.	the most processed matrix available	page
Identifier	Which name the protein is chosen by: Protein Names — the protein-group column of the phospho dataset — or Gene Names where the dataset carries a gene column.	Protein Names	page
Protein / Site	The one protein whose sites are drawn. The match is exact, so a stale selection draws nothing and says so rather than drawing a neighbouring protein group. Labelled **Gene** under Gene Names.	the first entry	page
Conditions	Which conditions contribute columns, and in which order.	all	page
Value Type	log2 colours the stored values; z colours each row's z-score across the selected samples.	log2	page
Color Map	The gradient the cells are painted with: plasma, viridis, magma, inferno or coolwarm. The appearance panel's own gradient overrides this one when it is set.	plasma	page
Toggle ID	Adds the sample's own identifier under each column label, on a second line.	off	page
Filter *_M1	Keeps only the rows whose PTM_Collapse_key ends in _M1, that is the singly modified form of each site, and shortens the row labels accordingly.	on	page

Controls on this page (continued)

Control	What it does	Default	Where
Cluster Rows	Orders the rows by hierarchical clustering instead of by the order they occur in the matrix.	off	page
Cluster Columns	Orders the columns by hierarchical clustering instead of by condition, so the grouping then has to be read off the labels.	off	page
Width (cm), Height (cm), DPI	The size of the *downloaded* figure. The browser scales the rendered figure to fit the page. Width and height are held between 1 and 60 cm, DPI between 72 and 600.	20, 12, 300	page
Download format	PNG, JPG, WEBP, SVG or PDF.	PNG	page

In the plot appearance panel

Control	What it does	Default	Where
Show title	The plot's own title is the selected protein with (log2 intensities) or (Z-transformed) after it; text, size, weight, colour and position are set here too.	shown	appearance
Colour scale	The gradient the matrix is painted with. Setting it overrides **Color Map** on the page; an unrecognised name falls back to the page's choice rather than failing.	unset, so **Color Map** applies	appearance
Axis labels, ticks, font, background	The Sample and PTM site labels, the row and column ticks, the font, and the paper colour — which is what a masked cell is drawn in.	plot's own	appearance
Condition and sample names	Offered, but this figure builds its column labels from the annotation and does not read the overrides. There is no legend card and no condition-colour card: the figure has a colourbar, not a legend.	the annotation's names	appearance

WORTH KNOWING

The second table is the plot appearance panel, which is hidden until advanced mode is turned on (**Ctrl** + **Shift** + **E**). Everything it contains is described in Part XI; only the entries particular to this plot are repeated here.

37.3 How it is computed

THE MATRIX

The rows of the phospho matrix whose protein-group column equals the selection are taken — by gene, those whose gene column contains the selected token. With **Filter** *_M1 on, only the rows whose PTM_Collapse_key ends in _M1 are kept. The columns are the samples of the selected conditions, relabelled condition_n in the order the annotation gives.

Every entry is coerced to a number; anything that is not one, and ±∞, becomes NA, and a row missing in every selected sample is dropped before anything is drawn.

SCALING

Value Type log2 colours the values as they are stored, which on this page is the $\log_2$ intensity, and the colourbar is captioned log2 Intensity.

Value Type z replaces every row by its z-score across the selected samples: for site i and sample j,

$$z_{ij} = \frac{x_{ij} - \bar{x}_i}{s_i},$$

where $\bar{x}_i$ and s_i are the mean and the standard deviation of that row over the samples in which it has a value, s_i with the $n - 1$ denominator. Both are computed *after* the conditions have been selected, so the centre of a row is the centre of what is on screen and adding a condition moves every colour in it. A row whose values are all equal has $s_i = 0$, cannot be scaled, and is removed rather than drawn flat.

CLUSTERING

With **Cluster Rows** or **Cluster Columns** on, the order is taken from hierarchical clustering with **average linkage** on **Euclidean** distances, computed on the matrix as scaled — distances between rows for **Cluster Rows**, between columns for **Cluster Columns** — and the plot is reordered to the leaves of the resulting tree. No dendrogram is drawn and no cluster boundaries are marked: the result is an ordering, not a partition, so nothing here states where one group of sites ends and the next begins.

A distance is undefined where a value is missing, so before either clustering runs *every row with a missing value in any drawn column is removed* — from the clustering, and from the figure and table with it. A notice beside the controls says so whenever clustering is on and the data source is not the imputed matrix. If no complete row survives, the error names the count and the two ways out: impute and select Imputed Data, or turn clustering off. Nothing switches the data source by itself.

37.3.1 Edge cases

- With one row, or one column, that clustering is skipped; the switch can be left on without effect.
- In z with a single sample selected a row's standard deviation is undefined, every row is removed, and the page reports an empty matrix. Use log2 to look at one sample.
- A missing value is masked: the cell is drawn in the background colour, with no colour from the scale — unless clustering is on, in which case its whole row has gone.
- **Filter *_M1** on a dataset whose collapse keys carry no multiplicity suffix removes every row, and the page reports that none remain after filtering. Turn the switch off for such datasets.
- On Imputed Data the cells that were missing carry imputed values instead and none are masked; the title does not say so, so label the figure yourself when it leaves the tool.

The filter stays on by default. On a dataset whose collapse keys carry no multiplicity suffix it therefore empties the heatmap and raises an error, and turning it off is the fix.

37.4 The table below the plot

The matrix exactly as drawn: the rows in the figure's order, scaled as the figure is scaled, rounded to four decimals.

Columns in this table

Column	What it holds
Site	The short row label the figure draws.
PTM_Collapse_key	The whole identifier the values were read under, accession included. Short labels are not guaranteed to be unique; these are.
one column per sample	Headed condition_n, holding the value the cell is coloured by.

37.5 In the notebook

A notebook starting point, using the frames introduced in *The function reference*. These compact examples do not encode every UI option; use the plot's notebook ZIP, where available,

for its saved settings and preprocessing recipe:

```python
conditions = ["WT", "GRK"]
fig = phossite_on_prot_regulation(data3, meta2, "Q12888",
                                  conditions=conditions, id=False)
```

```r
conditions <- c("WT", "GRK")
fig <- phossite_on_prot_regulation(data3, meta2, "Q12888",
                                   conditions = conditions, id = FALSE)
```

The name is misleading: nothing in this function is a regulation or a test. It draws exactly the heatmap on this page — one row per phosphosite of the protein, one column per sample of the chosen conditions, missing values masked — and value_type, cluster_rows, cluster_cols, filter_M2 and id are **Value Type**, **Cluster Rows**, **Cluster Columns**, **Filter *_M1** and **Toggle ID** under other names. Three of those defaults differ from the page. filter_M2 — which despite its name is the _M1 filter — starts *off* where the page has it on, so the call above draws all 45 rows of Q12888 with their multiplicity suffixes rather than the 43 in the figure. id starts on, so the sample identifiers are added unless the call turns them off. And the protein is matched as a *substring* of Protein_group, where the page's selector matches exactly, so a short accession can pull in more than one protein group.

37.6 Exporting

The download reproduces the plot at the width, height and resolution set on the page; the browser scales that render to fit the page. The default is taller than on the other single-protein pages because the number of rows is the number of sites. The table downloads separately as CSV: it holds the values behind the colours.

37.7 Further reading

Wu et al., absolute phosphorylation stoichiometries (2015) explains the distinction between phosphorylation stoichiometry and protein abundance. Row colours in this plot are not measurements of occupancy.

Behind this page

Function	What it produces
single_protein_heatmap_plot	The figure: scaling, clustering and colour scale

Behind this page

Function	What it produces
single_protein_heatmap_table	The matrix under it
drop_incomplete_features	The completeness filter clustering runs behind
phossite_on_prot_regulation	The same plot in the package, for a notebook

PART VIII

Phosphoproteomics



38
The phosphosite plot

How many phosphoproteins, phosphopeptides and phosphosites survive a given localisation cutoff.

AT A GLANCE

Where	Phospho → Phosphosite Plot
Data level	Phosphosite. The page reads the phospho matrix as it was uploaded, so filtering and imputation performed elsewhere do not change these counts.
Needs	A phospho dataset. PTM_localization, Protein_group, UPD_seq and PTM_Collapse_key are the columns it reads.
Three bars	Phosphoproteins, phosphopeptides, phosphosites, drawn in descending order with the count printed at the end of each bar.
Shared control	**Localization probability cutoff**, the minimum PTM localisation probability. Everything else is size, format and appearance.
Table	Yes, below the plot, downloadable as CSV.

AT PHOSPHOSITE LEVEL

The shared **Localization probability cutoff** selects sites across phosphosite views. At 0 it keeps all sites, including those with no probability; above 0 those unknowns are excluded. This is separate from completeness filtering and does not rerun normalization or imputation. An already collapsed phosphoprotein table has its own recorded collapse cutoff.

38.1 What the plot answers

The plot answers how deep the phosphoproteome is, and how much of that depth survives a demand for confident site localisation. Raising the shared localisation cutoff shows how many reported identities meet a stricter requirement. Counts alone do not validate the calibration

of the search software's localisation probabilities.

The three numbers are not three measurements of the same thing. They sit in a hierarchy — a protein carries several peptides, a peptide can carry several sites. Their ratios describe the reported identifications, not the fraction of a protein's molecules carrying a phosphate.

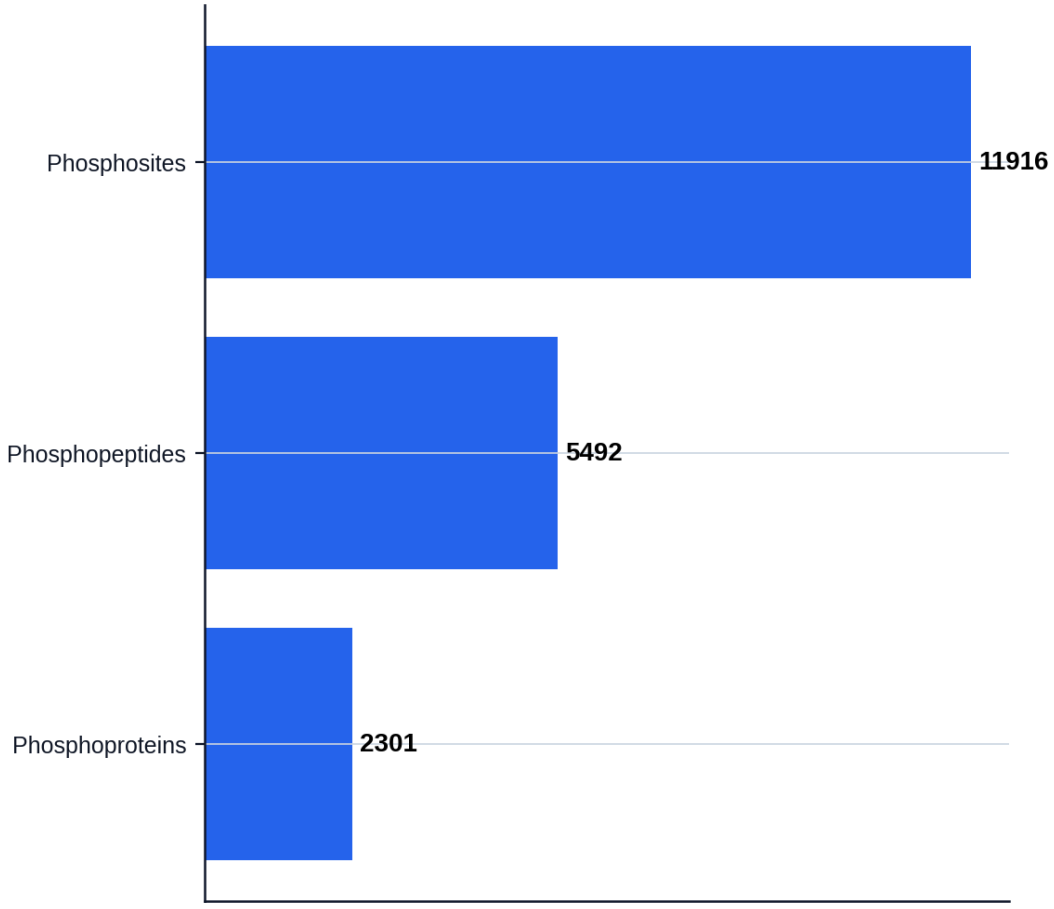


Figure 38.1. The phosphosite plot at **Cutoff** 0 on the example dataset: 11 916 phosphosites, 5 492 phosphopeptides and 2 301 phosphoproteins, drawn in descending order with the count printed at the end of each bar and no ticks on the x axis. Fold change and conditions play no part here — this page counts identifications, not differences.

38.1.1 Reading it

- Phosphoproteins — distinct values of Protein_group among the rows that pass the cutoff. A protein group counts once however many of its sites survive.
- Phosphopeptides — distinct peptide sequences, taken from UPD_seq with every non-word character removed and the rest upper-cased. The modification marker is one of the characters removed, so s*LSPILPGR and SLSPILPGR are the same peptide here.
- Phosphosites — distinct values of PTM_Collapse_key, the site identifier the search output carries.
- The number at the end of each bar. It is always printed; there is no switch for it. The x

axis carries no ticks, because the label is the number.

> **EASY TO GET WRONG**
>
> The shared cutoff selects which sites enter phosphosite analyses. Class I is a separate, fixed classification at 0.75. Setting the shared cutoff to 0.75 excludes lower-probability and unknown sites from other phosphosite views too; the coverage plot's remaining sites will therefore be Class I. It does not change the Class I definition.

38.2 Controls

Controls on this page

Control	What it does	Default	Where
Localization probability cutoff	Shared across phosphosite analyses, between 0 and 1. At zero all sites are retained; above zero a usable probability at least this large is required.	0	module
Width (cm), Height (cm), DPI	The size of the *downloaded* figure. The browser scales the rendered figure to fit the page.	15 × 10, 100	page
Download Plot	Exports the figure in the format chosen in the dropdown beside it: PNG, JPG, WEBP, SVG or PDF.	PNG	page

In the plot appearance panel

Control	What it does	Default	Where
Bar colour	The colour of all three bars. Until one is chosen the plot draws its own sky blue, #87CEEB; the picker itself opens on #2563eb.	#87CEEB	appearance
Title, axis labels	Text, size, weight, colour and position of every label, and whether the title is drawn at all. This plot sets no title and no axis labels of its own, so the fields start empty.	none	appearance
Font	One face for the whole figure, or one per text.	plot default	appearance

In the plot appearance panel (continued)

Control	What it does	Default	Where
Axis frame, grid	Which spines are drawn, and the grid over them. The top and right spines are off and the grid is off before the panel touches them. Tick styling is not offered here: the plot clears its x ticks, so there is nothing to style.	left and bottom spines, no grid	appearance
Background, theme	Paper colour, transparency, and the preset themes.	default	appearance

WORTH KNOWING

The second table is the plot appearance panel, which is hidden until advanced mode is turned on (**Ctrl** + **Shift** + **E**). Everything it contains is described in Part XI; only the entries particular to this plot are repeated here.

38.3 How it is computed

HOW IT IS COMPUTED

Above a shared cutoff of zero, PTM_localization is read as a number and only rows meeting the cutoff are kept. At zero the selection is disabled. Over the selected rows three counts are taken:

- phosphoproteins: distinct non-missing Protein_group;
- phosphopeptides: distinct non-empty sequences after UPD_seq has had every character outside [A-Za-z0-9_] removed and been upper-cased;
- phosphosites: distinct non-missing PTM_Collapse_key.

The three are sorted by count, largest first, and drawn as horizontal bars with the value written at the end of each. Nothing is divided by anything and no sample column is read: a site counts once whether it was quantified in one run or in all of them.

38.3.1 Edge cases

- At cutoff 0, missing probabilities do not remove rows. Above 0, they cannot meet the requirement and are excluded. A positive cutoff with no localisation column is refused; set it to 0 or supply a suitable probability column.

- If the cutoff removes every row the plot is replaced by the words No phospho rows available and the table comes back empty.
- Without UPD_seq the phosphopeptide bar falls back to the number of distinct PTM_Collapse_key values, which makes it equal to the phosphosite bar. Two bars of exactly the same height are the symptom.
- Sample values, replicates and missing values never enter this page; it counts identifiers.

38.4 The table below the plot

Columns in this table

Column	What it holds
Term	Phosphoproteins, Phosphopeptides or Phosphosites.
Count	The number behind that bar, at the current **Cutoff**.

The table is sorted the same way as the bars and downloads as CSV.

38.5 In the notebook

A notebook starting point, using the frames introduced in *The function reference*. These compact examples do not encode every UI option; use the plot's notebook ZIP, where available, for its saved settings and preprocessing recipe:

```python
fig = simple_phos_site_plot(data3)
```

```r
fig <- simple_phos_site_plot(data3)
```

The function takes the phosphosite frame alone — no annotation, because no sample column is read. **Cutoff** is the filter_value argument and defaults to 0, and color, width_cm, height_cm and dpi stand in for the appearance and size controls of the page; unlike the page, the function has no fallback for a frame without UPD_seq and will fail on one rather than repeating the phosphosite count.

38.6 Exporting

The download reproduces the plot at the width, height and resolution set on the page, which is why those three controls do not change the figure on screen: the displayed plot is fitted to the window, the downloaded one to the numbers. PDF for a figure that will be scaled or edited, PNG for a slide. The table underneath downloads separately as CSV.

In well-populated datasets the phosphopeptide and phosphosite bars stand well above the phosphoprotein bar, which is the hierarchy showing through. Counts that barely move as **Cutoff** is raised from 0 to 0.75 indicate confidently localised identifications; a sharp fall over the same range means a large share of the sites are assigned to a residue the evidence does not fix, and any site-level conclusion drawn from that dataset should be checked against the Class I count on the coverage page first.

Behind this page

Function	What it produces
simple_phos_site_plot	The bar chart, in the Python package
phosphosite_plot_table	The two-column table of counts
phosphosite_plot_png	The figure the page draws

39
Phosphosite coverage

How many sites each sample quantified, split by how confidently they are localised.

AT A GLANCE

Where	Phospho → Phosphosite Coverage Plot
Data level	Phosphosite. The page reads the phospho matrix as it was uploaded, so filtering and imputation performed elsewhere do not change these bars.
Needs	A phospho dataset and its annotation. Conditions and sample order come from the annotation.
Classes	Class I is PTM_localization ≥ 0.75, Not Class I is below it, and a third class, Unclassified, holds the rows whose localisation is missing or outside $[0, 1]$. The 0.75 is fixed in the code and follows no control.
Two modes	**Type** Normal draws one stacked bar per sample; Summary draws one bar per class per condition, with the standard deviation and the individual samples over it.
Table	Yes, below the plot, downloadable as CSV. Its columns change with **Type**.

AT PHOSPHOSITE LEVEL

The shared **Localization probability cutoff** selects sites across phosphosite views. At 0 it keeps all sites, including those with no probability; above 0 those unknowns are excluded. This is separate from completeness filtering and does not rerun normalization or imputation. An already collapsed phosphoprotein table has its own recorded collapse cutoff.

39.1 What the plot answers

The ordinary coverage plot asks whether every sample worked. This one asks the same question and one more: of the sites a sample quantified, how many are pinned to a particular residue. Two datasets can produce identical bar heights and differ entirely in how much of that height can be attributed to a site rather than to a peptide.

Read both the total height and the localisation categories. Comparable totals describe similar detection depth, not proof of equally good runs. The Class I fraction concerns confidence in the assigned residue; it can also depend on the peptides detected, their spectra and the search setup. A different fraction is a reason to inspect those factors, not a diagnosis of fragmentation failure or a requirement to make all fractions equal.

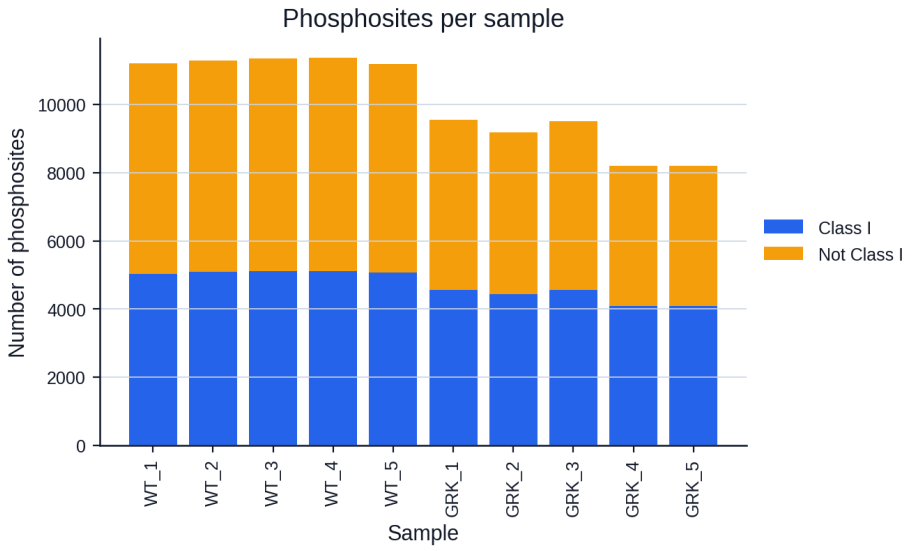


Figure 39.1. Per-sample mode on the example dataset: five WT and five GRK samples, Class I sites forming the lower blue segment of each bar and Not Class I the orange one above it, with the total height the number of phosphosites quantified in that sample. The five GRK bars are noticeably shorter than the five WT bars, in the Class I segment as well as in the total. No Unclassified segment is drawn, because every row of this dataset carries a usable localisation value.

39.1.1 Reading it

- The lower segment: sites in that sample whose PTM_localization is at least 0.75 and that carry a value in that column.
- The segment above it: sites with a localisation below 0.75, counted the same way.
- The full height of the bar: every site quantified in that sample, whatever its localisation.
- A third segment, slate grey, appears above the other two only if the dataset contains rows with no usable localisation value. It is labelled Unclassified in the legend, and its absence means every row carried a localisation between 0 and 1.
- The label under each bar: condition_n, numbered from 1 within each condition rather than across the experiment. With **Toggle ID** on, the sample's number from the original file name is added in brackets on a second line.

EASY TO GET WRONG

Class I is a statement about *where* the phosphate sits, not about how well the site was measured. A Class I site can have a poor quantitative signal, and a Not Class I site can be measured precisely; what is uncertain is which of two or three candidate residues on the peptide carries the modification. Treat the split as a guide to how far a site-level conclusion can be pushed — a regulated Not Class I site is a regulated *peptide* — and not as a quality filter on the intensities.

39.2 Controls

Controls on this page

Control	What it does	Default	Where
Type	Normal: one stacked bar per sample. Summary: one group of bars per condition, each class its own bar, with mean, standard deviation and the individual samples.	Normal	page
Conditions	Which conditions to draw. Samples of the conditions not selected are left out of the figure and the table altogether. Selecting none is the same as selecting all.	all conditions	page
Toggle ID	Adds the sample's number, taken from the original file name, under each label.	off	page
Random Seed	Summary only. Fixes the horizontal jitter of the individual sample points so the same figure is reproduced exactly.	69	page
Width (cm), Height (cm), DPI	The size of the *downloaded* figure. The browser scales the rendered figure to fit the page.	20 × 10, 300	page
Download Plot	Exports the figure in the format chosen in the dropdown beside it: PNG, JPG, WEBP, SVG or PDF.	PNG	page

In the plot appearance panel

Control	What it does	Default	Where
Class I	The colour of the Class I bar or segment. Until one is chosen the plot draws its own blue, #2563eb.	#2563eb	appearance

In the plot appearance panel (continued)

Control	What it does	Default	Where
Not Class I	The colour of the Not Class I bar or segment. The plot's own default is amber, #f59e0b; the picker opens on #94a3b8. The Unclassified segment is always slate grey and has no picker.	#f59e0b	appearance
Value labels	Size of numbers printed on bars. This plot prints none, so the field has nothing to act on — see the open question below.	8	appearance
Title	Shown or hidden, and its text, size, font and colour. The plot writes Phosphosites per sample, or Phosphosites per condition in summary mode.	shown	appearance
Legend	The class legend, drawn to the right of the plot. Turning it off also returns the width it was occupying.	shown	appearance
Condition labels	Renames a condition on the axis in summary mode without renaming it anywhere else.	the annotation's names	appearance
Axis frame, grid, background	Which spines are drawn, the grid over them, paper colour and the preset themes.	default	appearance

The panel shows no **Value labels** card here, because these bars carry no printed counts for it to size.

> **WORTH KNOWING**
>
> The second table is the plot appearance panel, which is hidden until advanced mode is turned on (**Ctrl** + **Shift** + **E**). Everything it contains is described in Part XI; only the entries particular to this plot are repeated here.

39.3 How it is computed

> **HOW IT IS COMPUTED**
>
> PTM_localization is read as a number and any value outside [0, 1] is discarded, which puts that row with the ones that had no value at all. Each row then belongs to exactly one of three classes: Class I at ≥ 0.75, Not Class I below it, Unclassified for the rest.

Every sample column is read as a number, and a value counts as present when it is finite. If the annotation is not log2-transformed, a zero is additionally treated as missing, because a report that writes 0 for an unmeasured phosphosite would otherwise inflate every bar. For sample j and class c over the m rows of the matrix,

$$\text{count}_{jc} \;=\; \sum_{i=1}^{m} \mathbf{1}\big[\, i \in c \,\big]\; \mathbf{1}\big[\, x_{ij} \text{ present} \,\big] \,.$$

The three counts are stacked in the order Class I, Not Class I, Unclassified, so the height of the bar is their sum: the number of phosphosites quantified in that sample.

SUMMARY MODE

The per-sample counts are computed exactly as above and then aggregated within each condition. Each class gets its own bar, the height being the mean of the counts of the samples of that condition and the error bar their standard deviation over $n - 1$ degrees of freedom. Every sample is additionally drawn as a black point over its bar, displaced sideways by a small random amount that **Random Seed** fixes. A condition with one replicate has no spread; its error bar is reported as zero rather than omitted.

39.3.1 Edge cases

- A sample that quantified nothing draws a bar of height zero rather than disappearing from the axis, so the gap is visible.
- The Unclassified segment is drawn only when at least one such row exists. Its absence from the legend is therefore informative.
- Sites that carry a localisation but no value in a given sample are counted in no bar of that sample; they are still counted in the samples that did quantify them.
- Deselecting a condition in **Conditions** removes its samples from the figure and from the table, and renumbers nothing: the remaining samples keep the condition_n labels they had.
- This page has no data-source selector. It always reads the matrix as uploaded, so a filter or an imputation applied on the Data page leaves these bars where they were.

39.4 The table below the plot

In Normal mode:

Columns in this table

Column	What it holds
Sample	The sample, labelled condition_n as on the plot.
Class I	Sites with localisation ≥ 0.75 quantified in that sample.
Not Class I	Sites with localisation below 0.75 quantified in that sample.
Unclassified	Sites with no usable localisation quantified in that sample.
Condition	The condition the sample belongs to.
Total	The three counts added — the height of the bar.

In Summary mode the table is replaced by one row per condition, holding Condition and then a mean and a standard deviation for each of the four quantities above: classIMean and classISd, notClassIMean and notClassISd, unclassifiedMean and unclassifiedSd, totalMean and totalSd. A standard deviation that cannot be computed is written as 0.0.

39.5 In the notebook

A notebook starting point, using the frames introduced in *The function reference*. These compact examples do not encode every UI option; use the plot's notebook ZIP, where available, for its saved settings and preprocessing recipe:

PYTHON

```
fig = phossite_coverage_plot(data3, meta2, id=False)
```

R

```
fig <- phossite_coverage_plot(data3, meta2, id = FALSE)
```

The stacking, the Class I boundary at 0.75 and the Unclassified segment are the same as on the page, and phossite_coverage_plot_summary draws the condition-level version. One difference is in the labels: the notebook function numbers the samples of a condition from zero, so its first bar is WT_0 where the page writes WT_1.

39.6 Exporting

The download reproduces the plot at the width, height and resolution set on the page; the figure on screen is fitted to the window instead. PDF for a figure that will be scaled or edited, PNG for a slide. The table underneath downloads separately as CSV, in whichever of its two shapes **Type** is currently producing.

Investigate a noticeably different run against its experimental peers. Lower coverage can have biological, acquisition or processing explanations; a lower Class I fraction can reflect a different peptide mixture as well as localisation evidence. Unclassified means a usable probability was not supplied for those observations. Check the source column and export format, but do not invent a localisation class for missing probabilities.

Behind this page

Function	What it produces
phossite_coverage_plot	The per-sample stacked plot, in the Python package
phossite_coverage_plot_summary	The condition-level summary, with spread and individual samples
phospho_coverage_table	The table under the plot, in either shape
phospho_coverage_png	The figure the page draws

- Phosphosite coverage

40
The phosphosite distribution

How many sites each phosphoprotein contributed, and how that is spread over the dataset.

AT A GLANCE

Where	Phospho → Phosphosite Distribution
Data level	Phosphosite. The page reads the phospho matrix as it was uploaded; filtering and imputation performed elsewhere do not change it.
Needs	A phospho dataset carrying either a site_num column, or a protein column (Phosphoprotein, else Protein_group) together with PTM_Collapse_key.
One bar	One bar per site count: its height is the number of proteins carrying exactly that many sites.
Cutoff	An upper quantile trim, not a site count. See the pitfall.
Localisation	Ignored. Class I and Not Class I sites count alike.
Table	Yes, below the plot, downloadable as CSV.

AT PHOSPHOSITE LEVEL

The shared **Localization probability cutoff** selects sites across phosphosite views. At 0 it keeps all sites, including those with no probability; above 0 those unknowns are excluded. This is separate from completeness filtering and does not rerun normalization or imputation. An already collapsed phosphoprotein table has its own recorded collapse cutoff.

40.1 What the plot answers

The plot answers whether the phosphorylation in the dataset is spread thinly over many proteins or concentrated in a few. That is partly biology and partly method: a protein with forty sites is either a genuine signalling hub or a large, well-covered protein whose peptides

the enrichment happened to like, and the shape of the distribution is what distinguishes a dataset where such proteins are the tail from one where they are the body.

It is a frequency distribution, so both axes count proteins in different ways. Along the x axis is the number of sites a protein carries; up the y axis is how many proteins carry that number. The bar at $x = 1$ is therefore the number of proteins seen with exactly one site, not the number of sites seen once.

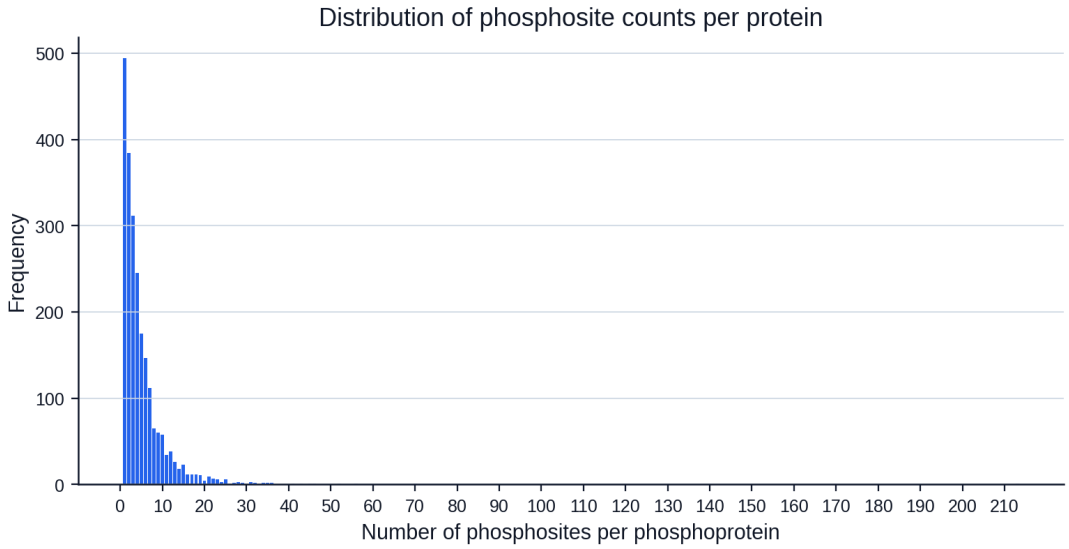


Figure 40.1. The distribution at **Cutoff** 0, the whole example dataset. The x axis is the number of phosphosites on a protein, the y axis how many proteins carry that many: 494 of the 2 301 phosphoproteins carry exactly one site, 384 carry two, and the frequency falls away steeply from there. The axis runs out to 213, the site count of the single most heavily modified protein — the tail to the right is a handful of proteins, not a large share of the data.

40.1.1 Reading it

- One bar: the number of phosphoproteins with exactly that many sites. The bars are drawn only at site counts that actually occur, so a gap on the axis means no protein in the dataset carries that number.
- The tail on the right: the heavily modified proteins. It is usually long and low, and the eye overstates it — read the y values before concluding that the dataset is dominated by them.
- The x ticks, placed every 10 sites once the largest count reaches 10, and every 1 below that. They are label positions and do not group the bars.

EASY TO GET WRONG

Cutoff is a quantile, not a number of sites. At 0.05 the tool takes the 95th percentile of the per-protein site counts and drops every protein above it — that is, the most heavily modified 5 % of proteins, whatever site count that turns out to be. It does *not* remove proteins with more than five sites. Ties at the quantile mean the removed fraction need not be exactly 5 %. This display trim is separate from the shared localisation-

probability cutoff. 0 disables the trim altogether; the field is clamped to the range 0 to 1, and 1 would keep only the proteins at or below the minimum.

40.2 Controls

Controls on this page

Control	What it does	Default	Where
Cutoff	The fraction of the most heavily modified proteins to remove, as an upper quantile. 0 keeps everything; the field is clamped to [0, 1].	0	page
Width (cm), Height (cm), DPI	The size of the *downloaded* figure. The browser scales the rendered figure to fit the page.	20 × 15, 300	page
Download Plot	Exports the figure in the format chosen in the dropdown beside it: PNG, JPG, WEBP, SVG or PDF.	PNG	page

In the plot appearance panel

Control	What it does	Default	Where
Bar colour	The colour of every bar. Until one is chosen the plot draws its own sky blue, #87CEEB; the picker opens on #2563eb.	#87CEEB	appearance
Title, axis labels	Text, size, weight, colour and position, and whether the title is drawn. The plot writes Distribution of phosphosite counts per protein, Number of phosphosites per phosphoprotein and Frequency.	shown	appearance
Font	One face for the whole figure, or one per text.	plot default	appearance
Axis frame, grid, ticks	Which spines are drawn, the grid over them, and the size, rotation and colour of the tick labels.	default	appearance
Background, theme	Paper colour, transparency, and the preset themes.	default	appearance

WORTH KNOWING

The second table is the plot appearance panel, which is hidden until advanced mode is turned on (**Ctrl** + **Shift** + **E**). Everything it contains is described in Part XI; only the entries particular to this plot are repeated here.

40.3 How it is computed

HOW IT IS COMPUTED

The per-protein site count is obtained in one of two ways, and which one applies depends on the dataset rather than on any control:

- if the frame carries a site_num column — as a phosphoprotein-level frame does, one row per protein — that column is read as a number and rows where it is not one are dropped;
- otherwise the rows are grouped by Phosphoprotein, or by Protein_group where the first is absent, and the count is the number of *distinct* PTM_Collapse_key values in the group. This is the path an ordinary phosphosite upload takes.

With **Cutoff** above zero, the threshold is the $(1 - \text{cutoff})$ quantile of those counts and proteins whose count exceeds it are removed. The remaining counts are tabulated: for every site count k that occurs,

$$\text{frequency}(k) \;=\; \#\{\,\text{proteins with exactly } k \text{ sites}\,\},$$

and the pairs are drawn as bars, ordered by k.

40.3.1 Edge cases

- A protein whose sites were all removed upstream does not appear at $x = 0$; it is absent from the frame and so from the distribution entirely. There is no zero bar.
- The quantile is taken over whatever the previous step produced. On the site_num path that is one value per row, so if a frame ever carries site_num with one row per *site*, each protein enters the quantile once per site and the trim is weighted towards the heavily modified proteins it is meant to remove.
- Neither a site_num column nor a protein column with PTM_Collapse_key means no distribution can be built, and the page reports that rather than drawing an empty axis.
- If the trim leaves nothing, the figure is replaced by the words No phosphosite distribution data available.

- Localisation probability plays no part here, and neither do the sample values: a site counts towards its protein whether or not it was quantified anywhere.

40.4 The table below the plot

Columns in this table

Column	What it holds
SiteCount	A number of phosphosites per protein — one x position.
Frequency	How many phosphoproteins carry exactly that many sites — the height of that bar.

Only the site counts that occur have a row; the table has no zero rows to skip over, and it downloads as CSV.

40.5 In the notebook

A notebook starting point, using the frames introduced in *The function reference*. These compact examples do not encode every UI option; use the plot's notebook ZIP, where available, for its saved settings and preprocessing recipe:

PYTHON

```
fig = phossite_dist(data4, cutoff=0.01)
```

R

```
fig <- phossite_dist(data4, cutoff = 0.01)
```

cutoff is the same quantile trim as **Cutoff** on the page, here dropping the most heavily modified 1 % of proteins so that the tail does not stretch the axis. The function only takes the site_num path, so it wants the phosphoprotein frame data4 rather than the phosphosite frame; handed a frame without site_num it raises an error instead of grouping by protein as the page does.

40.6 Exporting

The download reproduces the plot at the width, height and resolution set on the page; the figure on screen is fitted to the window instead. PDF for a figure that will be scaled or edited, PNG for a slide. The table downloads separately as CSV.

Many datasets have a long right tail, but its shape is not a pass/fail criterion. More reported sites can reflect protein length, abundance, sequence coverage, enrichment or biology. A second mode does not identify its cause, and a short tail does not prove that enrichment failed. Keep the untrimmed table when discussing the proteins omitted from the display.

Behind this page

Function	What it produces
phossite_dist	The distribution, in the Python package
phospho_distribution_table	The two-column table under the plot, and the counting behind it
phospho_distribution_png	The figure the page draws

41
Serine, threonine, tyrosine

Which residues the phosphorylations in the dataset sit on.

AT A GLANCE

Where	Phospho → STY Plot
Data level	Phosphosite, as uploaded. Filtering and imputation performed elsewhere do not change it.
Needs	A phospho dataset with a PTM_Collapse_key column. Nothing else is read — not the sample values, not the annotation, not PTM_localization.
What is counted	One count per row of the phosphosite matrix, so one per site identifier. Not per peptide, not per protein, and not per sample.
Controls	Only size, format and appearance. There is no cutoff and no condition selector on this page.
Table	Yes, below the plot, downloadable as CSV.
New	Not covered by any previous manual.

AT PHOSPHOSITE LEVEL

The shared **Localization probability cutoff** selects sites across phosphosite views. At 0 it keeps all sites, including those with no probability; above 0 those unknowns are excluded. This is separate from completeness filtering and does not rerun normalization or imputation. An already collapsed phosphoprotein table has its own recorded collapse cutoff.

41.1 What the plot answers

The plot answers what kind of phosphoproteome the enrichment produced. Serine, threonine and tyrosine are phosphorylated at very different rates in a cell and enriched at very different efficiencies by different chemistries, so the split between the three bars says more about the sample preparation than about the biology — which is exactly what makes it a useful check.

A dataset that was meant to be a phosphotyrosine experiment and comes back looking like a generic one has failed somewhere upstream of any analysis on this page.

The second bar of information is the fourth bar: Unparsed. It appears only when the site identifier could not be read, and it is a statement about the file format rather than about the chemistry.

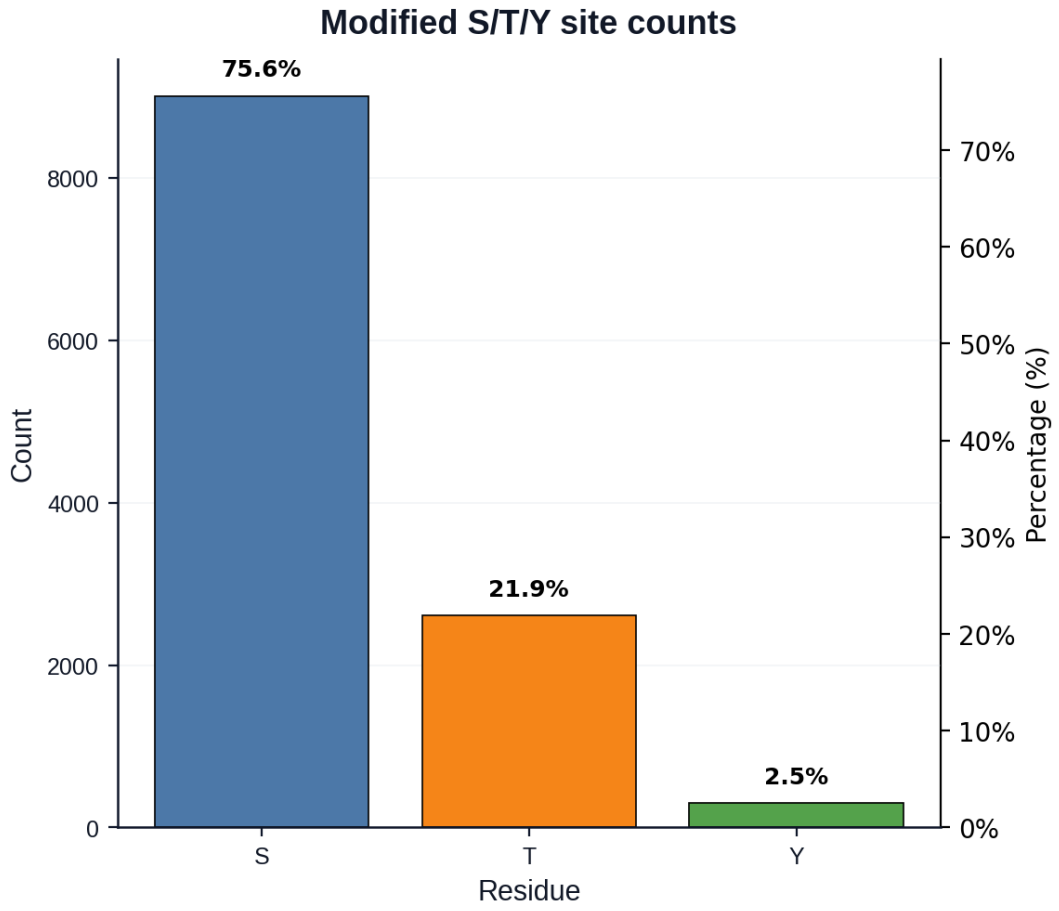


Figure 41.1. The S/T/Y plot on the example dataset, a standard non-selective enrichment: serine 75.6 %, threonine 21.9 % and tyrosine 2.5 % of the rows of the phosphosite matrix. Each bar is a count of site identifiers on the left-hand axis, with its percentage of the total printed above it and the same scale repeated on the right-hand axis. No Unparsed bar appears, so every identifier was read.

41.1.1 Reading it

- One bar per residue, in the fixed order S, T, Y, each in its own colour: blue for serine, orange for threonine, green for tyrosine. A residue that occurs nowhere still gets its bar, at height zero.
- The percentage printed above each bar, and the right-hand axis carrying the same scale. Both are shares of the total number of rows counted, including Unparsed.
- A fourth bar, slate grey and labelled Unparsed, drawn only when at least one identifier could not be read. Its presence means the counts below it are shares of an incompletely parsed dataset.

EASY TO GET WRONG

A high tyrosine share almost always means a phosphotyrosine enrichment worked, not that the biology is unusual. In a cell, phosphotyrosine is a small minority of all phosphorylation; a dataset in which it is not a small minority has been selected for it, by an antibody or by a fractionation, and its serine and threonine bars are then a biased sample of the phosphoproteome rather than a description of it. Read the three bars as a description of the method first and of the sample second.

41.2 Controls

Controls on this page

Control	What it does	Default	Where
Width (cm), Height (cm), DPI	The size of the *downloaded* figure. The browser scales the rendered figure to fit the page.	17.78 × 11.43, 140	page
Download Plot	Exports the figure in the format chosen in the dropdown beside it: PNG, JPG, WEBP, SVG or PDF.	PNG	page

In the plot appearance panel

Control	What it does	Default	Where
Bar colour	Offered on this page, but the renderer gives each residue its own fixed colour and does not read the setting — see the open question below.	—	appearance
Title, axis labels	Text, size, weight, colour and position, and whether the title is drawn. The plot writes Modified S/T/Y site counts, Residue and Count.	shown	appearance
Font	One face for the whole figure, or one per text.	plot default	appearance
Axis frame, grid, ticks	Which spines are drawn, the grid over them, and the tick labels. The top and right spines are off and a faint horizontal grid is on before the panel touches them.	left and bottom spines, y grid	appearance
Background, theme	Paper colour, transparency, and the preset themes.	default	appearance

The panel shows no **Bars** card on this page: each residue is drawn in its own fixed colour, so a single bar colour would have nothing to apply to.

> **WORTH KNOWING**
>
> The second table is the plot appearance panel, which is hidden until advanced mode is turned on (**Ctrl** + **Shift** + **E**). Everything it contains is described in Part XI; only the entries particular to this plot are repeated here.

41.3 How it is computed

HOW IT IS COMPUTED

Every row of the phosphosite matrix contributes exactly one count, and the residue is read off its PTM_Collapse_key by two patterns tried in order:

- first ~gene_X###_, which is the shape of an identifier such as A0A590UK10~ARHGEF18_S1035_M1 — a tilde, a gene name, then the residue letter and its position;
- then, for rows the first did not match, a looser _X### bounded by an underscore or the end of the string, which catches identifiers written without the tilde section.

X is S, T or Y in both patterns. A row matched by neither is counted as Unparsed. The counts are then reindexed to the fixed order S, T, Y — so a residue absent from the data still appears at zero — and Unparsed is appended when it is not empty. With N the total of all bars, each bar's label is $100\,c/N$ printed to one decimal, and the right-hand axis is the same conversion applied to the whole y axis.

41.3.1 Edge cases

- Rows are counted, not distinct identifiers. If the phosphosite matrix ever holds the same PTM_Collapse_key twice, that site counts twice.
- Replicates, missing values and intensities never enter this page: it reads one text column. A site counts the same whether it was quantified in every sample or in none.
- PTM_localization is not consulted either. Class I and Not Class I sites are counted alike, so this plot cannot be restricted to confidently localised sites — which matters, because the residue is exactly what a low localisation score is uncertain about.
- A dataset without a PTM_Collapse_key column produces no plot and says so.
- With no rows at all the bars are zero, and the right-hand percentage axis is omitted rather than dividing by zero.

41.4 The table below the plot

Columns in this table

Column	What it holds
Residue	S, T, Y, and Unparsed when that bar exists.
Count	The number of rows assigned to that residue.
Percentage	Its share of the total of the column above, to two decimals.

The counts always sum to the number of rows in the phosphosite matrix, which is the quickest way to confirm that nothing has been dropped silently.

41.5 In the notebook

A notebook starting point, using the frames introduced in *The function reference*. These compact examples do not encode every UI option; use the plot's notebook ZIP, where available, for its saved settings and preprocessing recipe:

```python
fig = sty_plot(data3)
```

```r
fig <- sty_plot(data3)
```

The function takes the phosphosite frame and nothing else: the two patterns, the fixed S, T, Y order, the Unparsed bar and the right-hand percentage axis are all as on the page, but there are no size, colour or title arguments, so the figure comes out at the function's own fixed size. Its x axis is labelled Peptide rather than Residue, which is the only wording that differs from the screen.

41.6 Exporting

The download reproduces the plot at the width, height and resolution set on the page; the figure on screen is fitted to the window instead. PDF for a figure that will be scaled or edited, PNG for a slide. The table downloads separately as CSV.

Serine commonly forms the largest share in broad phosphoproteomic datasets, but there is no required S–T–Y ratio for a valid experiment. Enrichment, search settings, biological material and detection all affect the observed composition. A large tyrosine fraction is compatible with a tyrosine-focused experiment; the bars alone do not establish how it was enriched or which kinase was active. These are counts of reported sites, not phosphorylation occupancies. Inspect Unparsed rows before interpreting percentages, because their residue category is unknown.

Behind this page

Function	What it produces
sty_plot	The bar chart, in the Python package
phospho_sty_table	The three-column table, and the parsing behind it
phospho_sty_png	The figure the page draws

42
Kinase Library motif enrichment

Which kinases the measured changes are consistent with — prepared here, computed elsewhere, and drawn here again.

AT A GLANCE

Where	Phospho → Kinase Library, headed Kinase Library Motif Enrichment. The tab is not shown in viewer mode.
Data level	Phosphosite.
Needs	An annotated phospho dataset with UPD_seq, two different conditions (four under Volcano Control), and — for the figure — two result files from the external tool.
Substrate sets	None are stored here: the motif and substrate definitions belong to the external kinase library the page sends you to.
Score	That tool's frequency factor, plotted as its base-2 logarithm.
Significance	The adjusted p-value in the uploaded file; no correction is applied here.
New	Not covered by any previous manual.

AT PHOSPHOSITE LEVEL

The shared **Localization probability cutoff** selects sites across phosphosite views. At 0 it keeps all sites, including those with no probability; above 0 those unknowns are excluded. This is separate from completeness filtering and does not rerun normalization or imputation. An already collapsed phosphoprotein table has its own recorded collapse cutoff.

42.1 What the enrichment infers, and what this page does

This page prepares input for the Kinase Library's motif-based Fisher enrichment workflow and visualises its returned results. The plot is not a direct measurement of kinase activity. Motif enrichment generates hypotheses about signalling, which also depends on substrate

availability, phosphatases, protein abundance and experimental coverage. This is not kinase-substrate enrichment analysis (KSEA, Casado et al. 2013) or another substrate-based z-score approach; their scores and input requirements are not interchangeable with this frequency-factor plot. CoPYlot does not run the external enrichment. The workflow has three steps:

1. **Prepare the input.** A phosphosite differential-abundance table is computed from your data and formatted for the external tool, each modified sequence re-centred on its modified residue, and downloads as TSV.
2. **Run the enrichment elsewhere.** The page links to the external kinase library's Fisher enrichment analysis, where the table is uploaded twice — once for serine/threonine kinases, once for tyrosine and non-canonical — with UPD_seq as the motif column, log2FC as the effect column and pval or adj_pval as the optional p-value column.
3. **Draw the results.** The two result files are uploaded back here and drawn as one volcano of kinases. Matching column names alone do not make results from another method comparable: verify its score definition, contrast and correction before importing it.

> **READ THE SCORE, NOT A KINASE FOLD CHANGE**
>
> A frequency factor of 4 has $\log_2$ value 2. In the down-regulated-site result it is placed on the negative side, so the side identifies the submitted site set. This is enrichment among that set, not a fourfold change in kinase abundance or a direct measurement of its activity. An adjusted p-value of 0.01 has height 2 on the vertical axis.

42.1.1 Reading it

- One point, one kinase. The two files are concatenated, so a kinase reported by both appears twice.
- The x axis, $\log_2$(frequency factor). Positive means the kinase's substrate set is enriched among the sites that rose in **Condition 1**; negative, among those that fell.
- The y axis, $-\log_{10}$ of the adjusted p-value the result file gives. A p-value of zero is drawn at $-\log_{10}(10^{-300})$ rather than at infinity.
- The dashed green line at $-\log_{10}$ of **Kinase Library Volcano P-value Threshold**. Points above it are red or blue by the sign of x; the rest are light grey, Not significant.
- How many substrates a point rests on is *not* on the figure. A substrate-count column is read and validated but neither plotted nor shown in the hover.

> **EASY TO GET WRONG**
>
> The marker size does not express the strength of biological evidence. Inspect the contributing motifs, counts where supplied, effect size, multiple-testing procedure and experimental context together. Neither a large substrate set nor a small p-value alone establishes kinase activity.

42.2 Controls

Controls on this page: the input table

Control	What it does	Default	Where
Source	Volcano for a plain two-condition comparison, Volcano Control to subtract a control from each side first; the latter requires four different conditions.	Volcano	page
Data Source	Which available matrix the comparison runs on: Data, Filtered Data or Imputed Data. Check the selection before exporting.	current selection	page
Condition 1, Condition 2	The two conditions compared; $\log_2$ FC = **Condition 1** − **Condition 2**. The same one twice produces no table.	first condition in the annotation	page
Condition 1 Control, Condition 2 Control	Volcano Control only; each is subtracted from its own side first.	first condition in the annotation	page
P-value Threshold, log2 FC Threshold	Together they set the significance label in the exported table: below the one in adjusted p and above the other in magnitude. Neither removes a row. The p-value is clamped to $[10^{-12}, 1]$.	0.05, 1	page
Test Type	Unpaired (Student's t, equal variance), Unpaired (Welch, unequal variance) or Paired t-test (matched samples). Volcano Control offers instead Unpaired (propagated variance) and Paired (propagated control variance), promoting an equal-variance choice to the Welch form: the control means add uncertainty of their own.	Unpaired (Student's t)	page
Motif Window Proteome	Human, Mouse, or no proteome. Protein sequence supplies flanking residues where the site can be matched; fallback peptide windows are reported.	Human	page
Use Uncorrected P-values	Selects raw pval for selected_pval and classification. adj_pval remains Benjamini–Hochberg adjusted.	off	page
Download Kinase Library Table (TSV)	Saves the table, named for the comparison.	—	page

Controls on this page: the volcano from uploaded results

Control	What it does	Default	Where
Serine/Threonine Result File, Tyrosine/Non-canonical Result File	The two files that come back from the external tool. .csv, .tsv or .txt. Both are required.	none	page
Kinase Library Volcano P-value Threshold	The adjusted p-value below which a kinase is coloured, and where the green line is drawn. Separate from **P-value Threshold** above.	0.1	page
Annotate Hits	Prints the kinase name beside every point that is not Not significant.	off	page
Generate Kinase Library Volcano	Builds the figure, once both files are present and the conditions validly chosen.	—	page

In the plot appearance panel

Control	What it does	Default	Where
Reference line	The significance line: shown or hidden, and its colour, dash pattern and width.	shown, green, dashed, 1.5	appearance
Title, axis titles, legend	Text, size, colour, and whether each is drawn. The plot writes Kinase Library Motif Enrichment: cond1 vs. cond2 and names the contrast in the x title.	shown	appearance
Points	Marker size, opacity, outline and shape of the interactive volcano. The colour stays the regulation class's, so the card has no colour field here.	size 10, opaque, black outline 0.5	appearance
Background, theme	Paper colour, preset themes.	default	appearance

An imputed phosphosite matrix is allowed. It changes the evidence entering the comparison: inspect the reported imputation counts and retain the settings rather than treating filled values as additional observations.

> **WORTH KNOWING**
>
> The third table is the plot appearance panel, hidden until advanced mode is on (**Ctrl** + **Shift** + **E**) and described in Part XI.

42.3 How it is computed

THE INPUT TABLE

Every phosphosite is tested between the two conditions exactly as the volcano plot tests it, and with the direction the volcano plot uses:

$$\log_2 \mathrm{FC} \;=\; \overline{x}_{\text{cond 1}} - \overline{x}_{\text{cond 2}},$$

or, under Volcano Control, $(\overline{x}_{\text{cond 1}} - \overline{x}_{\text{ctrl 1}}) - (\overline{x}_{\text{cond 2}} - \overline{x}_{\text{ctrl 2}})$: the same difference, taken after each side has had the mean of its own control subtracted from it.
UPD_seq contains a fifteen-residue window: the modified residue and seven positions on each side. With a selected proteome, the site position and peptide residues are checked against the protein sequence. If this succeeds, only positions beyond the protein terminus are padded with underscores. Otherwise a peptide-derived window is used and window_source says peptide. Inspect these fallbacks: a peptide end is not necessarily a protein end, and missing flanks can affect motif scoring. Every row of the comparison is exported; the thresholds only decide what each row's significance column says.

THE VOLCANO FROM THE UPLOADED RESULTS

Both result files are read and stacked. Two quantities are required per row, found by column name, ignoring case, punctuation and a subscript 2:

- the effect, from Log2(Frequency Factor) taken as it stands, or from Frequency Factor, which must be positive and whose base-2 logarithm is then taken;
- the significance, from Adjusted p-value, which must lie in $[0, 1]$.

A kinase name is taken from Kinase, Gene Name or Gene, a substrate count from Substrate Count, Number of Substrates or n substrates when one is present. Two columns that mean the same thing and disagree are an error rather than a silent choice, and so is a row missing either required quantity.
The figure is then arithmetic and nothing else: $y = -\log_{10}(p)$ with p clipped up to 10^{-300}; a point is Enriched in up-regulated sites if $p <$ **Kinase Library Volcano P-value Threshold** and the effect is positive, Enriched in down-regulated sites if $p <$ threshold and the effect is negative, Not significant otherwise; and a line is drawn at $-\log_{10}$ of the threshold. No p-value is adjusted here, no score normalised, no substrate set consulted.

Under Volcano Control the contrast computed, and printed on the x axis, is (cond 1 − ctrl 1) − (cond 2 − ctrl 2). Keep that orientation when interpreting the external results.

42.3.1 Edge cases

- A paired test needs every sample paired before either the table or the figure appears, and Volcano Control rejects any selection in which the four conditions are not distinct.
- Uploaded files with no rows, duplicate column names, a non-positive frequency factor or an adjusted p-value outside $[0, 1]$ are refused with a message naming the problem.
- A kinase in both result files is drawn twice; the two runs cover disjoint families, so an overlap is worth investigating.

Nothing records which release of the external library a set of result files came from, and the session does not keep the files themselves. Keep the downloaded result files with the rest of the analysis: they are the only record of what the enrichment was run against.

42.4 The input table

Columns in this table

Column	What it holds
UPD_seq	The re-centred motif — the column the external tool reads as the motif.
log2FC	The fold change of that site, in the direction stated above.
pval, adj_pval	The raw p-value and its Benjamini–Hochberg adjusted form, stored separately.
selected_pval	The raw or adjusted value chosen for classification.
significance	Upregulated, Downregulated, Not significant or Unavailable, from the two thresholds. Carried for reference only.
n_pairs, neg_log10_selected_pval	The usable pairs behind that site, on paired tests only, and $-\log_{10}$ of the selected p-value.
Protein, Gene, Site, PTM_Collapse_key, window_source	Site identity and whether the motif window came from the protein sequence or from the peptide fallback.

The table downloads as tab-separated text. The figure has no image download: it is interactive, and saves from its own toolbar.

There is no required number of significant kinases. No hits can reflect limited evidence, coverage, thresholds or genuine lack of enrichment; many hits can reflect related motifs as well as broad changes. The plot alone does not distinguish these explanations.

Upload the full eligible site comparison, not only its significant rows: the external method needs the comparison population to define its groups. Choose an optional p-value column if you intend the external classification to require both effect and statistical support. Omitting it

makes that classification depend on effect alone; it does not remove the external enrichment test. State whether the supplied values were raw or adjusted and record the external thresholds. The notebook download beside this table reproduces preparation of the export only, not the external analysis or its returned kinase plot.

42.5 In the notebook

A notebook starting point, using the frames introduced in *The function reference*. These compact examples do not encode every UI option; use the plot's notebook ZIP, where available, for its saved settings and preprocessing recipe:

The two steps of the page are two functions, with the trip to the external tool in between. kinase_library_data_f builds the input table, and kinase_library_volcano draws the result files once they come back:

PYTHON

```
kl_input = kinase_library_data_f(data3, meta2, "WT", "GRK")
fig = kinase_library_volcano(read_data("kl_st_results.csv"),
                             read_data("kl_tnc_results.csv"),
                             condition1="WT", condition2="GRK")
```

R

```
kl_input <- kinase_library_data_f(data3, meta2, "WT", "GRK")
fig <- kinase_library_volcano(read_data("kl_st_results.csv"),
                              read_data("kl_tnc_results.csv"),
                              condition1 = "WT", condition2 = "GRK")
```

The arguments of the first are the controls of the input-table card: in_pval and in_log2fc are the two thresholds, uncorrected is **Use Uncorrected P-values**, and paired and test_type carry **Test Type**; kinase_library_data_f_control is the Volcano Control source, taking the two control conditions as well. In kinase_library_volcano, pval_thresh is **Kinase Library Volcano P-value Threshold** and defaults to 0.1 as on the page. Neither function performs the enrichment: the two frames handed to kinase_library_volcano are the files the external tool produced, and the notebook does no more than the page does with them. The older names ksea_data_f, ksea_data_f_control and ksea_volcano still exist in both languages as aliases of these three, so existing notebooks keep working.

42.6 Further reading

The Kinase Library's official repository documents its motif-based kinase prediction and enrichment workflow. This is the relevant external implementation for the imported frequency-factor results, not an interchangeable label for every method called KSEA.

Behind this page

Function	What it produces
ksea_table	The prepared input table, standard or control-corrected. The application's function keeps the older name
kinase_library_data_f, kinase_library_data_f_control	The same table in the notebook packages, plain and control-corrected; ksea_data_f and ksea_data_f_control are aliases
ksea_uploaded_volcano_html	The interactive volcano built from the two result files. The application's function keeps the older name
kinase_library_volcano	The equivalent figure in the notebook packages; ksea_volcano is an alias

43
Phosphoprotein regulation

Which proteins carry the regulated phosphosites, and whether their sites move together or against each other.

AT A GLANCE

Where	Phospho → Phosprot Regulation
Data level	Phosphosite throughout. The page aggregates site-level results per protein; it does not read a protein matrix and performs no protein normalisation.
Needs	An annotated phospho dataset with Protein_group and PTM_Collapse_key, and two different conditions with at least two replicates each.
Direction	$\log_2$ FC = **Condition 1** − **Condition 2**. "Up" means higher in **Condition 1**.
Both axes	Non-negative. The figure occupies one quadrant, not four.
Figure	Interactive, drawn in a frame on the page. There is no image download; **Open Plot** opens it on its own.
Table	Yes, below the plot, downloadable as CSV.

AT PHOSPHOSITE LEVEL

The shared **Localization probability cutoff** selects sites across phosphosite views. At 0 it keeps all sites, including those with no probability; above 0 those unknowns are excluded. This is separate from completeness filtering and does not rerun normalization or imputation. An already collapsed phosphoprotein table has its own recorded collapse cutoff.

43.1 The question this page exists for

A volcano plot of phosphosites answers which sites changed; it does not answer which *proteins* the changes landed on, and that is usually what a signalling experiment is asking. This

page collapses the site-level result onto proteins and keeps the two directions apart instead of letting them cancel: for every protein it adds up the fold changes of its significant up sites, adds up the magnitudes of its significant down sites, and plots one against the other. A protein whose sites all move the same way lands on one of the axes; a protein whose sites disagree lands out in the field, and that disagreement is a result rather than noise.

> **NOTE**
>
> This is not protein-abundance correction. A site whose protein was itself induced appears here exactly like a site whose occupancy rose at constant protein level, because no protein matrix is consulted. To investigate the two, compare site-level changes with independently measured protein abundance. Collapsing sites alone does not provide that protein measurement and does not estimate absolute site occupancy.

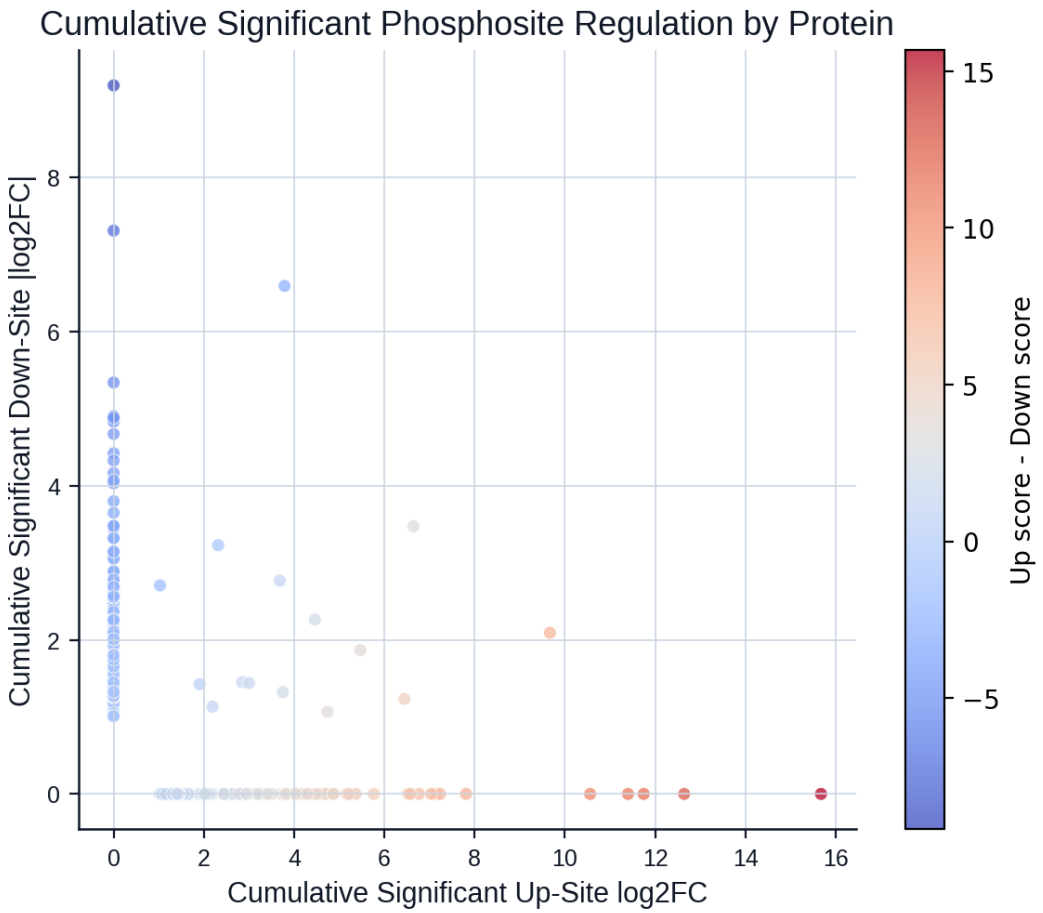


Figure 43.1. WT against GRK on the example dataset, at the default thresholds. Each point is one protein with at least one significant site: x is the sum of the $\log_2$ fold changes of its significant up sites, y the sum of the magnitudes of its significant down sites, and colour is $x - y$ on the blue-to-red scale at the right. Most points sit on one of the two axes — a dense row along $y = 0$ and a column along $x = 0$ — and only a scattering lies away from both. That describes this example; mixed directions at different sites can also be biologically meaningful.

43.1.1 Reading it

- One point, one protein. Only proteins with at least one significant site are drawn; the rest are absent from the figure and from the table.
- The x axis, Cumulative Significant Up-Site log2FC: the sum of $\log_2$ FC over the sites of that protein classified Upregulated. Zero when it has none, never negative.
- The y axis, Cumulative Significant Down-Site |log2FC|: the sum of the *magnitudes* of $\log_2$ FC over its Downregulated sites. Zero when it has none, never negative.
- Where a point sits. On the x axis: every significant site moved up. On the y axis: every one moved down. Away from both: significant sites in both directions, the distance from the origin being the total regulated signal on that protein, and the diagonal $x = y$ the balance point.
- Colour, from the scale Up score - Down score $= x - y$: red for net upward, blue for net downward, grey for balanced. It adds nothing the coordinates do not, but it makes the diagonal readable.
- Hover: the protein group, the number of significant sites in each direction, and those sites with their fold changes, up to **Max Sites in Table**.

> **CHANGED IN 1.0.0**
>
> Earlier documentation of this page stated that the function computes $\log_2$ fold changes and p-values for all phosphosites but applies no significance cutoff. That is no longer the case. **P-value Threshold** and **log2 FC Threshold** now decide which sites enter the sums, and a protein with no significant site is not drawn at all. A figure made with the old behaviour is not comparable to one made now.

43.2 Controls

Controls on this page

Control	What it does	Default	Where
Condition 1	The first condition of the comparison, and the minuend: $\log_2$ FC = **Condition 1** − **Condition 2**.	first condition in the annotation	page
Condition 2	The second condition. Choosing the same condition twice draws nothing and says so.	first condition in the annotation	page

Controls on this page (continued)

Control	What it does	Default	Where
P-value Threshold	A site is significant when its p-value is strictly below this. The value is the adjusted p-value unless **Use Uncorrected P-values** is on. Clamped to $[10^{-12}, 1]$.	0.05	page
log2 FC Threshold	A site must also exceed this in magnitude: above $+t$ to count as up, below $-t$ to count as down. 0 keeps the p-value criterion alone.	1	page
Test Type	Unpaired (Student's t, equal variance), Unpaired (Welch, unequal variance) or Paired t-test (matched samples); the last opens the pairing editor and needs every sample paired before a figure is drawn.	Unpaired (Student's t)	page
Use Uncorrected P-values	Compares raw p-values against **P-value Threshold** instead of Benjamini–Hochberg adjusted ones. This admits more sites, not fewer.	off	page
Show Phosphosites in Table	Whether the site identifiers and their fold changes are listed. Off leaves the four site columns empty and keeps the counts.	on	page
Max Sites in Table	How many site identifiers to list per protein and direction before the rest are summarised as … (+n more).	20	page
Width (cm), Height (cm), DPI	Present on the page, but the interactive figure this tab renders ignores them: it is sized to the frame it is drawn in.	20 × 12, 220	page

In the plot appearance panel

Control	What it does	Default	Where
Points	Size, opacity and outline of the markers. The fill is not offered: the points are coloured from $x - y$ through the colour scale.	size 8, opacity 0.7, outline 0.5	appearance
Title, axis labels	Text, size, weight, colour and position, and whether the title is drawn. The plot writes Cumulative Significant Phosphosite Regulation by Protein over the two axis titles quoted above.	shown	appearance
Font	One face for the whole figure, or one per text.	plot default	appearance
Axis frame, grid, background	Which axes are drawn, the grid over them, paper colour and the preset themes.	default	appearance

WORTH KNOWING

The second table is the plot appearance panel, which is hidden until advanced mode is turned on (**Ctrl** + **Shift** + **E**). Everything it contains is described in Part XI; only the entries particular to this plot are repeated here.

43.3 How it is computed

STEP ONE: THE SITE-LEVEL TEST

Every phosphosite is tested between the two conditions on the annotated log2 matrix, by exactly the procedure the volcano plot uses and described there: a row needs two finite values in each condition, the fold change is $\overline{x}_{\text{cond 1}} - \overline{x}_{\text{cond 2}}$, the p-value comes from the two-sided t-test **Test Type** selects, and Benjamini–Hochberg adjusts it unless **Use Uncorrected P-values** is on. A site is Upregulated when $p_{\text{adj}} <$ **P-value Threshold** and $\log_2 \text{FC} >$ **log2 FC Threshold**, Downregulated when $p_{\text{adj}} <$ **P-value Threshold** and $\log_2 \text{FC} <$ **−log2 FC Threshold**, and Not significant otherwise.

STEP TWO: THE COLLAPSE ONTO PROTEINS

The labelled sites are grouped by Protein_group. A group with no Upregulated and no Downregulated site is discarded. For the rest, with U the set of up sites and D the set of down sites of that protein,

$$x = \sum_{i \in U} \log_2 \text{FC}_i, \qquad y = -\sum_{i \in D} \log_2 \text{FC}_i, \qquad \text{colour} = x - y.$$

Both sums are over significant sites only, so a protein with twenty unchanged sites and one large change is placed by that one change. Because the members of U are all above $+t$ and those of D all below $-t$, both coordinates are non-negative, and a protein with sites in only one direction sits exactly on an axis.

43.3.1 Edge cases

- If no site anywhere passes both thresholds, the figure is drawn empty and carries the sentence No phosphosites pass the selected significance and fold-change thresholds. This is the answer to a legitimate question and not an error; earlier releases failed here instead, and the removed explanation box on the page is replaced by this section.
- Selecting the same condition in both boxes draws nothing and says so; so does a paired test whose samples are not fully paired.

- A site the test could not evaluate — fewer than two values in a condition — is labelled Unavailable and enters neither sum. Duplicate site identifiers are kept apart during the test, so a protein carrying the same PTM_Collapse_key twice contributes both rows.
- The page reads the annotated log2 matrix, not the filtered or imputed one.

> **EASY TO GET WRONG**
>
> Both axes are *sums*, so they grow with the number of significant sites as well as with the size of the changes. A protein far out along x may carry one site that moved eight-fold or eight sites that each moved a little, and the figure does not distinguish them: the hover and the Upregulated_count column do. Nor is either coordinate comparable to a fold change — adding $\log_2$ ratios multiplies the ratios, and the result is a score for ranking proteins within one figure, not a quantity to report.

43.4 The table below the plot

Columns in this table

Column	What it holds
Protein_group	The protein the point stands for.
Upregulated_count	Its significant up sites.
Downregulated_count	Its significant down sites.
Upregulated_log2FC_sum	The x coordinate.
Downregulated_log2FC_sum	The y coordinate, already made positive.
Upregulated_sites, Downregulated_sites	The site identifiers in each direction, separated by semicolons and truncated at **Max Sites in Table**. Empty when **Show Phosphosites in Table** is off.
Upregulated_site_effects, Downregulated_site_effects	The same identifiers with each site's $\log_2$ FC in brackets, signed.
x, y, color_val	The plotted coordinates and $x - y$, repeated as their own columns so the figure and the table cannot drift apart.

Rows are ordered by protein group and download as CSV. The figure has no image download on this tab: **Open Plot** opens the interactive version on its own, where it can be zoomed and saved from the plot's own toolbar.

A protein can genuinely carry sites changing in opposite directions. Do not adjust thresholds merely to move points onto the axes or increase their number. These coordinates are sums: a protein with many quantified sites has more opportunities to accumulate a large score than

one with few. They are neither a protein fold change nor a measure of kinase activity. Inspect the contributing sites, their replicate values and the multiple-testing setting before drawing a biological conclusion.

43.5 In the notebook

A notebook starting point, using the frames introduced in *The function reference*. These compact examples do not encode every UI option; use the plot's notebook ZIP, where available, for its saved settings and preprocessing recipe:

```python
fig = phosprot_regulation(data3, meta2, "WT", "GRK", in_pval=0.05, in_log2fc=1)
```

```r
fig <- phosprot_regulation(data3, meta2, "WT", "GRK", in_pval = 0.05, in_log2fc = 1)
```

The arguments are the page's controls under other names: `in_pval` and `in_log2fc` are the two thresholds, `uncorrected` is **Use Uncorrected P-values**, `paired` and `test_type` carry **Test Type**, and `max_hover_sites` and `show_phosphosites` govern the site lists in the hover and the table. What comes back is the interactive figure itself — the one the page hides behind **Open Plot** — and `phosprot_regulation_data` returns the same per-protein table without drawing anything.

43.6 Further reading

Wu et al., absolute phosphorylation stoichiometries (Nature Communications, 2015) discusses why phosphosite abundance alone does not distinguish occupancy from protein abundance. The plot here summarizes measured site changes; it does not perform that separation.

Behind this page

Function	What it produces
phosprot_regulation	The scatter, in the Python package
phosprot_regulation_data	The per-protein table the scatter plots
phosprot_regulation_table	The same table in the web application
phosprot_regulation_html	The interactive figure the page renders
phosprot_regulation_png	The static version of the same figure

PART IX

Comparison



44
Two samples against each other

Every feature of one sample plotted against another, with the correlation between them and the number of features it rests on.

AT A GLANCE

Where	Comparison → Pearson Correlation
Data levels	Protein, phosphosite, phosphoprotein
Needs	An uploaded dataset and its annotation, and two different samples or two different conditions.
Data source	Annotated data on the analysis scale, normalised if requested, but not completeness-filtered or imputed. Comparison has no Filtered or Imputed choice: filtering or imputing first would remove the disagreement being inspected.
Direction	The first selection is the x axis, the second the y axis. In the interactive plot the hovered difference is second minus first.
Two renderings	**Type** Normal draws a static figure; Interactive draws the same points in a pannable plot with hover and a fitted line.
Table	Yes, below the plot, downloadable as CSV.

AT PHOSPHOSITE LEVEL

The shared **Localization probability cutoff** selects sites across phosphosite views. At 0 it keeps all sites, including those with no probability; above 0 those unknowns are excluded. This is separate from completeness filtering and does not rerun normalization or imputation. An already collapsed phosphoprotein table has its own recorded collapse cutoff.

44.1 What the plot answers

A correlation matrix gives one number per pair of samples; this page gives the cloud behind one of those numbers. That is where the useful information is: whether the disagreement

between two samples is spread over the whole range or concentrated at the low end, whether it is a handful of features or all of them, and whether a high r is earned or merely a consequence of measuring intensities over four orders of magnitude.

Only features measured in *both* members of the pair are drawn. A feature present in one and missing in the other does not appear as a point on an axis; it is not on the figure at all, and the count N in the corner says how many features survived that test.

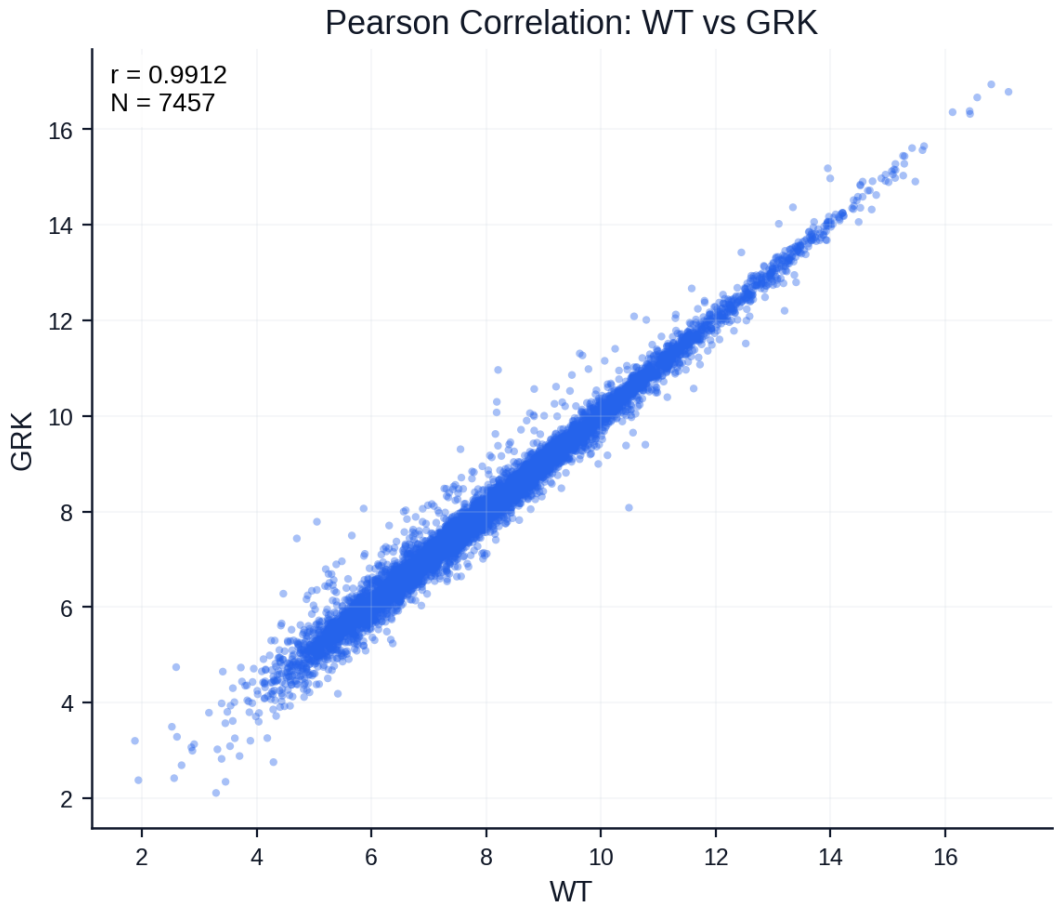


Figure 44.1. **Mode** Condition, **Type** Normal, protein level on the example project: the mean of the five WT samples on the x axis against the mean of the five GRK samples on the y. One point per protein, and the box in the corner gives the Pearson r and the number of features N it was computed on — here 0.9912 over the 7457 proteins the two conditions share. The cloud fans out below about $\log_2 6$ and is tight above it.

44.1.1 Reading it

- One point: one feature, at its $\log_2$ intensity in the first selection (x) and in the second (y).
- The box at the top left: r and N, the correlation and the number of features drawn.
- Points below the diagonal are lower in the second selection than in the first; points above it are higher.
- The fan at the low end: features near the detection limit, where the measurement is least reproducible. A widening of the cloud there is ordinary; a widening at the top is not.

- The axis labels are the sample or condition names, or the aliases if any were typed.

> **EASY TO GET WRONG**
>
> *r* is governed by the range of the data. Two samples that disagree about every regulated protein will still correlate above 0.98 if they agree about the four orders of magnitude separating the abundant proteins from the rare ones, so a high *r* is evidence that the measurement worked and almost no evidence about the biology. Read it together with *N*, which is not constant: each pair of samples is scored on the features that both of them measured, so a worse-covered sample is compared on fewer features and often scores *higher* for it.

44.2 Controls

Controls on this page

Control	What it does	Default	Where
Dataset level	Which dataset to compare: Protein, Phospho or Phosphoprotein. Only the levels actually uploaded are offered.	Protein	page
Mode	Single compares two samples. Condition compares two condition means, each the mean over that condition's replicates.	Single	page
Type	Normal renders a static image, which is what the download button writes. Interactive renders a plot with hover, zoom and a fitted line, opened in its own window.	Normal	page
Sample 1 / Condition 1	The first member of the pair, drawn on the x axis. The label changes with **Mode**.	the first sample or condition of the annotation	page
Sample 2 / Condition 2	The second member, drawn on the y axis. It may not be the same as the first.	the first one that differs	page
Alias 1 (optional)	Replaces the first name on the x axis and in the title. The data are not affected.	empty	page
Alias 2 (optional)	The same for the second name and the y axis.	empty	page

Controls on this page (continued)

Control	What it does	Default	Where
Width, Height, DPI	The size of the *downloaded* figure. Width and height are held between 1 and 60 cm, the resolution between 72 and 600. Shown only while Normal is selected. The interactive plot has none: it fills its frame, Plotly's camera button saves a picture of it, and **Open Interactive Plot** opens it in a window of its own.	20 cm × 12 cm, 300	page
Download format	PNG, JPG, WEBP, SVG or PDF. Shown for Normal only.	PNG	page

In the plot appearance panel

Control	What it does	Default	Where
Points	Colour, size, opacity, outline and shape of the dots. The figure uses its own values until a control here is touched: blue #1f77b4, about 7.7 pt across, 0.65 opaque, circular, with no outline.	as drawn	appearance
Reference line	The fitted line: shown or hidden, and its colour, style and width. Interactive only — the static figure draws no line, so this card has nothing to act on there.	shown, slate, solid, width 2	appearance
Title	Shown or hidden, and its text, size, weight, colour and position. The title is written as Pearson Correlation: first vs second.	shown	appearance
X axis, Y axis	The text and styling of the two axis labels and their tick labels.	the names or aliases	appearance
Frame and grid	Which of the four spines are drawn, and the grid behind the points. Normal opens with four spines and a solid grid on both axes; Interactive with no spines and its own pale grid.	by rendering	appearance
Font	One face for the whole figure, or one per text.	plot default	appearance
Background	Theme, paper colour and transparency.	Default, white, opaque	appearance

WORTH KNOWING

The second table is the plot appearance panel, which is hidden until advanced mode is turned on (**Ctrl** + **Shift** + **E**). Everything it contains is described in Part XI; only the entries particular to this plot are repeated here. **Dot Color** and **Dot Size**, which the

earlier manual listed beside the plot, are the **Points** card here.

44.3 How it is computed

HOW IT IS COMPUTED

Two vectors are taken from the $\log_2$ matrix, one per selection. In Single mode each is a sample column. In Condition mode each is the mean of that condition's replicates, taken over the replicates that have a value; a feature measured in one replicate of four gets a mean from that one replicate.
The pair is then reduced to the features present in both: a feature missing in either vector is dropped from both, and N is what remains. On those N pairs,

$$r = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_i (x_i - \bar{x})^2}\,\sqrt{\sum_i (y_i - \bar{y})^2}},$$

the product-moment correlation: the covariance of the two vectors over the product of their standard deviations, bounded by -1 and 1. It measures the straightness of the relationship, not its slope — a sample twice as intense as its replicate throughout correlates at 1.
r and N are printed in the corner of the figure. No p-value is computed here.

THE INTERACTIVE RENDERING

The same points, drawn with hover: each gives the feature identifier, its value in each selection, and the difference *second minus first*. A least-squares line $y = ax + b$ is fitted to the drawn points and extended over their range; the **Reference line** card hides it or restyles it. The feature identifier is taken from the dataset's own identifier column — protein names at protein level, the collapse key at phosphosite and phosphoprotein level — and is shortened in the hover box when it is long.

44.3.1 Edge cases

- The two selections must differ; the page refuses a pair of the same sample or the same condition rather than drawing a perfect diagonal.
- If no feature is present in both, no plot is drawn and the reason is stated.
- With a single surviving pair, r is undefined and the box reads r = n/a while the point is still drawn.

- A condition with one replicate is allowed: its mean is that replicate, and the axis label still says the condition name.
- N is specific to the pair on screen. Two comparisons from the same dataset are not scored on the same features, so their r values are not strictly comparable.
- Aliases are labels only. They change the axes, the title and the table, and nothing else.

44.3.2 Following up a difference

A narrow cloud indicates a strong linear relationship among the shared features, not necessarily equal measurements. For example, $y = x+1$ on a log2 plot can have $r = 1$ despite a twofold intensity offset. Inspect the cloud relative to equality as well as its correlation. Curvature, isolated points or changing spread are reasons to inspect those features and the preprocessing; they do not uniquely diagnose normalization failure. There is no universal pass mark such as $r > 0.95$. The overlap count and the experimental design matter alongside the coefficient.

44.4 The table below the plot

Two columns, Metric and Value, and four rows.

Rows in this table

Column	What it holds
Left	The first selection, as an alias if one was typed, otherwise the sample or condition name.
Right	The second selection, the same way.
Points	N: the features present in both, which is the number of dots on the figure.
PearsonR	r, to six decimals. Empty when it is undefined.

44.5 In the notebook

A notebook starting point, using the frames introduced in *The function reference*. These compact examples do not encode every UI option; use the plot's notebook ZIP, where available, for its saved settings and preprocessing recipe:

```python
fig = correlation_xy_intensity(data, meta, "WT", "GRK")
```

```r
fig <- correlation_xy_intensity(data, meta, "WT", "GRK")
```

That is **Mode** Condition: the two names are conditions, and each axis is the mean over that condition's replicates. Single mode is a different function, correlation_xy_intensity_single, which takes two sample column names and no annotation at all. Both draw more than the static page does — a least-squares line is always fitted, and the header carries *r and* a p-value where the page prints *r* and *N* in a corner box and leaves the line to the interactive rendering. min_present has no control on the page: raise it above 1 to drop condition means that rest on fewer than that many replicates.

44.6 Exporting

With **Type** at Normal the download reproduces the figure at the width, height and resolution set on the page, which is why those three controls do not change what is on screen. With Interactive there is no image to download: the button becomes **Open Interactive Plot**, which opens the plot in its own window, and the plot's own toolbar saves a snapshot of the current view. The table underneath downloads separately as CSV in either case.

Behind this page

Function	What it produces
correlation_xy_intensity_single	Two samples, in a notebook. It also fits a regression line and prints a p-value, which the page does not.
correlation_xy_intensity	The same for two condition means
pearson_correlation_png	The static figure behind the page
pearson_correlation_html	The interactive figure, with the fitted line and the hover
pearson_correlation_table	The four-row table under the plot

Pearson correlation

45
The Venn diagram

Which features two or three samples, or two or three conditions, have in common — and which are unique to one of them.

AT A GLANCE

Where	Comparison → Venn Diagram
Data levels	Protein, phosphosite, phosphoprotein
Needs	An uploaded dataset and its annotation, and two different samples or conditions; a third is optional.
Data source	Annotated, optionally normalised data. Comparison has no Filtered or Imputed choice — an imputed matrix would describe filled values rather than observed detection.
Membership	Quantified, not regulated: a feature is in a set when it has a value there.
Sets	Two or three, of samples (Single) or conditions (Condition).
Table	Yes, below the plot: every region and every total, downloadable as CSV.

AT PHOSPHOSITE LEVEL

The shared **Localization probability cutoff** selects sites across phosphosite views. At 0 it keeps all sites, including those with no probability; above 0 those unknowns are excluded. This is separate from completeness filtering and does not rerun normalization or imputation. An already collapsed phosphoprotein table has its own recorded collapse cutoff.

45.1 What the plot answers

Whether two runs, or two conditions, saw the same features. Nothing more than that: the diagram is built from presence and absence, so a feature sits in the intersection if both selections measured it, however differently. It is a statement about coverage, and it answers the

question a volcano plot cannot — what was never in the comparison at all.

In Condition mode a feature belongs to a condition when at least one of its replicates measured it. The sets are therefore unions, and a condition with six replicates has more chance of including a marginal feature than one with three; part of any unique wedge is that difference rather than biology.

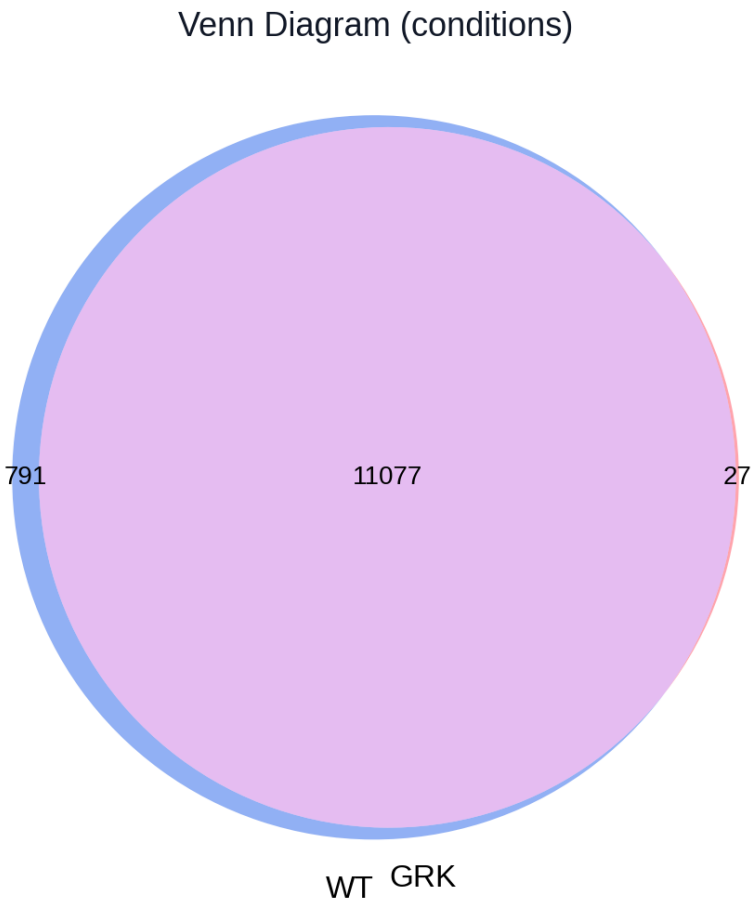


Figure 45.1. Mode Condition, phosphosite level on the example project: the sites quantified in at least one of the five WT samples against those quantified in at least one of the five GRK samples. 11,077 are in both, 791 only in WT and 27 only in GRK. The circles are drawn to scale, so near-total overlap looks like one circle inside another — which is itself the finding. Read the counts rather than the areas.

> **NOTE**
>
> This figure is drawn at phosphosite level on purpose. At protein level the same comparison gives 7,457 shared proteins against 3 and 30 unique ones: the two circles coincide exactly, the wedges are too thin to see and the two set names print on top of each other. A Venn is worth drawing when the sets differ; when they do not, the count of shared features is the whole answer and the table below says it more clearly.

45.1.1 Reading it

- A region: the number of features quantified in exactly that combination of selections — the centre of a three-set figure is the features all three have.
- A set name under each circle: the sample or condition name, or the alias if one was typed.
- A wedge with no overlap: features only that selection measured.
- The title says which mode produced the figure, Venn Diagram (samples) or Venn Diagram (conditions), because the same three names can mean either.

> **EASY TO GET WRONG**
>
> A set is a set of *identifiers*, not of rows. Two rows carrying the same protein name or the same collapse key are one element of the set, so the total under a selection can be smaller than the number of rows it has values in. Missing identifiers are excluded; literal blank strings or malformed identifiers need cleaning rather than being treated as biological features. The Tables page counts rows; this page counts identifiers, and the two are equal only when the identifier column is unique.

45.2 Controls

Controls on this page

Control	What it does	Default	Where
Dataset level	Which dataset to compare: Protein, Phospho or Phosphoprotein. Only the levels actually uploaded are offered.	Protein	page
Mode	Single makes one set per sample. Condition makes one set per condition, pooling its replicates.	Single	page
Sample 1 / Condition 1	The first set. The label follows **Mode**.	the first sample or condition of the annotation	page
Sample 2 / Condition 2	The second set. It may not repeat the first.	the first one that differs	page
Sample 3 (optional) / Condition 3 (optional)	The third set. Left at None the figure has two circles; chosen, it has three and must differ from both others.	None	page
Alias 1, Alias 2, Alias 3 (optional)	Replace the set names under the circles and in the table. The sets themselves are unaffected.	empty	page

Controls on this page (continued)

Control	What it does	Default	Where
Width, Height, DPI	The size of the *downloaded* figure. The browser scales the rendered figure to fit the page. Width and height are held between 1 and 60 cm, the resolution between 72 and 600.	15 cm × 12 cm, 300	page
Download format	PNG, JPG, WEBP, SVG or PDF.	PNG	page

In the plot appearance panel

Control	What it does	Default	Where
Venn sets	One colour per set, applied at half opacity so that the overlaps darken. The figure uses blue #1f77b4, orange #ff7f0e and green #2ca02c until a swatch is changed.	as drawn	appearance
Title	Shown or hidden, and its text, size, weight, colour and position.	shown	appearance
X axis, Y axis	A Venn diagram has no axes and carries no axis text; anything typed here is drawn beside the circles.	empty	appearance
Frame and grid	Which of the four spines are drawn, and the grid behind the circles. This figure opens with neither.	no spines, no grid	appearance
Font	One face for the whole figure, or one per text.	plot default	appearance
Background	Theme, paper colour and transparency.	Default, white, opaque	appearance

WORTH KNOWING

The second table is the plot appearance panel, which is hidden until advanced mode is turned on (**Ctrl** + **Shift** + **E**). Everything it contains is described in Part XI; only the entries particular to this plot are repeated here. The per-set colour pickers the earlier manual listed beside the plot are the **Venn sets** card here.

45.3 How it is computed

HOW IT IS COMPUTED

For each selection, a set of identifiers is built from the loaded matrix. In Single mode, a feature is in the set of sample j when x_{ij} is a number and not missing:

$$S_j = \{\,\mathrm{id}_i \;:\; x_{ij} \neq \mathrm{NA}\,\}.$$

In Condition mode the test is taken over the condition's replicates and satisfied by any one of them:

$$S_c = \{\,\mathrm{id}_i \;:\; \exists\, j \in c,\; x_{ij} \neq \mathrm{NA}\,\}.$$

The identifier is the dataset's own: the protein-name column at protein level, the collapse key or the phosphoprotein name at the phospho levels. The regions are then the ordinary set arithmetic — $S_1 \setminus S_2$, $S_1 \cap S_2$, and for three sets $S_1 \cap S_2 \setminus S_3$ and the rest — and each region's count is printed in it.

ON THE GEOMETRY

The circles are drawn area-weighted: their sizes and, as far as the geometry allows, their overlaps follow the counts. For two sets this is exact; for three it cannot always be satisfied at once and the drawing is an approximation. Measure nothing off the figure — the numbers in the regions and in the table are the exact values, and a very small region may be drawn too small to hold its own label.

45.3.1 Edge cases

- Leaving the third selection empty is the normal two-set case, not an error. Repeating a selection is an error, and the page says so rather than drawing a figure in which one circle contains another.
- A feature missing everywhere is in no set and appears in no region; the totals under the set names therefore do not sum to the number of rows in the dataset.
- Two selections that measured exactly the same features give two concentric circles with an empty outer region, which is what identical coverage looks like here.
- A zero in a linear intensity became a missing value when the data were loaded, so a feature reported as zero counts as absent — the same convention the completeness pages use.

45.3.2 Interpreting overlap

Comparable replicates often share many features, but there is no universal acceptable unique fraction. A large unique region is a reason to inspect coverage, missingness and sample identity, not proof of a failed run. Between conditions, a feature just above a detection threshold

on one side and below it on the other can appear unique without evidence of regulation. Use single-feature views to inspect its measurements. A volcano can test features with enough values in both groups; an entirely one-sided feature does not become statistically significant merely because it occupies a unique Venn region.

45.4 The table below the plot

One row per region, then one row per set total. With three sets there are seven regions and three totals; with two, three regions and two totals.

Columns in this table

Column	What it holds
Region	The region, written from the set names or aliases: A only, A & B, A & B & C, and Total A for the whole of a set.
Count	How many identifiers fall in that region. The Total rows are the size of each set, which is the sum of every region that set takes part in.

45.5 In the notebook

A notebook starting point, using the frames introduced in *The function reference*. These compact examples do not encode every UI option; use the plot's notebook ZIP, where available, for its saved settings and preprocessing recipe:

```python
data_by_protein = data.set_index("ProteinNames", drop=False)
fig = overlap_plot_condition(data_by_protein, meta, "WT", "GRK", data_scale="log2")
```

```r
data_by_protein <- data
rownames(data_by_protein) <- data_by_protein$ProteinNames
fig <- overlap_plot_condition(data_by_protein, meta, "WT", "GRK", data_scale = "log2")
```

The indexing line is not decoration. The sets are built from the frame's *index*, so a frame passed in as it was read gives sets of row numbers, every one of them distinct, and a diagram whose regions are counts of rows rather than of identifiers. Setting the index to the identifier

column first — ProteinNames at protein level, PTM_Collapse_key and Phosphoprotein at the two phospho levels — is what makes the notebook figure the one the page draws. overlap_plot_single is the Single mode counterpart and takes sample column names in place of the annotation and the two conditions; in either, a third selection is the optional argument that follows the second. The notebook titles the figure Overlap rather than naming the mode.

45.6 Exporting

The download reproduces the figure at the width, height and resolution set on the page, which is why those three controls do not change what is on screen. A Venn diagram is line art with a few labels, so PDF is the sensible choice wherever it can be used: the circles stay sharp at any size. The table underneath downloads separately as CSV, and it is the only complete record of the counts — a region too small to carry its label on the figure still has its row there.

Behind this page

Function	What it produces
overlap_plot_single	The two- or three-set diagram for individual samples
overlap_plot_condition	The same with the replicates of each condition pooled
venn_png	The figure behind the page, with the set colours and the mode in its title
venn_table	The region and total counts under it

PART X

Summary and reporting



46
The summary tables

The annotation the whole session rests on: one row per sample, one block per level, downloadable as it is stored.

AT A GLANCE

Where	Summary → Tables
Shows	Two blocks — **Protein Metadata** and **Phospho Metadata**. Nothing else.
Data levels	Protein and phosphosite. Phosphoprotein, peptide and precursor have no block here.
Needs	An uploaded dataset and a generated annotation for that level. Without the annotation the block says Metadata not available yet.
Source	The stored annotation metadata — the sample table, not the measurement matrix. No filtering, imputation or data source enters this page.
Download	One CSV per block.
In viewer mode	This is the only Summary → tab that exists.

46.1 What is collected here

Every plot in the session labels its samples condition_n and colours them by condition, and every statistical test picks its columns by condition. All of that comes from one table per level, built on the annotation page and stored with the dataset. This page is that table, printed as it stands.

There are two blocks, in this order:

- **Protein Metadata** — the annotation of the protein dataset.
- **Phospho Metadata** — the annotation of the phosphosite dataset.

Each block carries a count line above the table, reading Rows: r | Columns: c. The rows are *samples*, not features: Rows is the number of annotated sample columns at that level, and Columns is the width of the metadata table, which grows when a metadata file with extra grouping columns was uploaded.

> **EASY TO GET WRONG**
>
> This page is not the session's collected results. The coverage, boxplot, coefficient-of-variation and completeness tables each stay below the plot that produces them, and each downloads from there; the volcano and ANOVA result tables stay on Statistics →. What is gathered here is the annotation, because that is the one table every other page depends on and the one a reader of a finished analysis has to be able to check.

46.1.1 What a row holds

Columns of the metadata table

Column	What it holds
sample	The original sample name — the header of that column in the uploaded data file, exactly as it was spelled there.
condition	The condition the sample was assigned to. This is what the plots group and colour by, and what the condition selectors on every other page offer.
pair_id	Present when supplied in metadata or saved through a pairing panel. Two samples that share a pair_id represent the intended matched experimental unit in two conditions, and a paired test matches them by it.

Any further column of an uploaded metadata file is kept and shown as well, under a name made unique if the file repeated one. A metadata file with no column literally named condition has its first non-sample column copied into condition, so the original column and the condition column can both appear, holding the same values.

46.1.2 When a block is empty

- No dataset of that level is loaded, or its annotation has not been generated: the block reads Metadata not available yet.
- The annotation exists but its metadata table is empty: the same message. An empty metadata table means no sample was assigned, so no plot at that level will draw either.
- The table exists but has no columns the browser can render: No table rows available.
- Phosphoprotein, peptide and precursor never get a block, however many of them are loaded. A phosphoprotein dataset produced by a collapse inherits the phosphosite annotation, which is the block above it.

Both blocks are read once, when the page is opened. Re-annotating, re-uploading or collaps-

ing while the page is on screen does not refresh them; leaving the tab and coming back does.

46.2 Controls

Controls on this page

Control	What it does	Default	Where
Download Protein Metadata	Writes the protein annotation metadata to protein_metadata_full.csv. Shown only when the block has a table.	action	page
Download Phospho Metadata	Writes the phosphosite annotation metadata to phospho_metadata_full.csv. Shown only when the block has a table.	action	page

While one download is running both buttons are disabled and the one in progress reads Downloading..., so two requests cannot overlap.

> NOTE
>
> The page draws no figure, so the plot appearance panel is not shown on it, and neither are the width, height, resolution and format controls that belong to the download of an image.

46.3 Downloading

The download is a comma-separated UTF-8 file with a header row and no index column: one line per sample, the columns in the order the table shows them. It is the stored metadata frame written out directly, so what arrives on disk is what every plot and every test in the session read.

Two differences from the screen are worth knowing, because both make the file the better copy:

- On screen a value that is not a whole number is printed to four decimals, and a true/false value as true or false. The CSV holds the stored value, unrounded.
- On screen a missing value is drawn as an empty cell. In the CSV it is an empty field, which

reads back as NA rather than as the text nan.

A block whose table is scrolled on screen still downloads whole: the fixed-height window is a property of the box the table sits in, not of the data behind it.

Behind this page

Function	What it produces
summary_tables	Both blocks of this page: availability, row and column counts, column names and the table itself
download_current_metadata_annotation	The CSV behind either download button
get_annotation	The stored annotation a block is read from

47
The methods text

What the report says in words: the account the tool writes from what was actually done, and the paragraphs the user writes around it.

AT A GLANCE

Where	Summary → Text
What it is	The prose of the generated report. Two kinds: fixed commentary the tool supplies for each plot, and text blocks written here.
Where it appears	Only in the report built on Summary → Report. Nothing written here changes a plot or a table.
Numbers	Filtering, normalization, imputation, data source, features used and the significant counts are written by the tool from the session, not typed.
Stored	With the browser session. Kept in a saved state file; lost when the tab is closed without saving.
In viewer mode	This tab is not shown.

47.1 What the page is for

A report has to say what was done to the data before it says what the data showed, and most of that is already known to the tool: the filter threshold, the normalization, the imputation parameters, the test, the thresholds and the counts they selected. The tool writes those sentences itself, with the numbers filled in from the session. What it cannot know is the experiment — the material, the question, and what a particular figure is being shown for — and that is what this page is for. Text written here is attached to one plot and to one side of it, and appears only in the generated document.

47.2 Controls

Controls on this page

Control	What it does	Default	Where
Select Plot	Which section of the report the text belongs to. Introduction is the block before the table of contents; every other entry is one plot. Only the plots of the levels actually loaded are offered.	Introduction	page
Position	Whether the text is printed above the figure or below it. Not shown while Introduction is selected.	Above	page
Introduction Text / Text	The text itself. Line breaks are kept; the box is plain text, and anything that looks like markup is printed as characters rather than interpreted.	empty	page
Add / Update	Stores the box under the current plot and position, replacing what was there. A box holding only whitespace stores nothing and clears nothing.	action	page
Delete	Removes the entry for the current plot and position and empties the box.	action	page
Show stored text entries	Prints every stored entry, with its key, so the whole set can be checked at once.	off	page

> NOTE
>
> An entry is keyed by the plot *and* the position, so one plot can carry two blocks — one above the figure and one below it — and changing **Position** moves to the other entry rather than moving the text. The editor always shows the entry for the current combination; switching **Select Plot** without pressing **Add / Update** discards what was typed.

47.2.1 Which plots can carry text

The list is exactly the set of sections the report can contain, and no others: coverage, missing values, retention time, modification and missed cleavage at peptide level; coverage, missing values, histogram intensity, boxplot intensity, coefficient of variation, PCA, abundance, correlation and volcano at protein level; those same nine at phosphosite level, with the phosphosite overview added and coverage split into (Number) and (Quality). Only the levels actually loaded are offered. An entry written for a section the report does not end up containing — a volcano text with no comparison in the report, a protein text in a phospho-only session — is kept but never printed.

47.3 What the tool writes for you

Four things are written without being typed. They appear in this order inside a plot section: the fixed commentary, then the text block for Above, then the figure with its own provenance lines, then the text block for Below.

THE FIXED COMMENTARY

One to four paragraphs under each plot heading, the same words in every report. They describe what the plot is and what a healthy dataset looks like in it; they contain no numbers from the session. They are read from report_general_comments.json beside the application, and re-read every time a report is built, so editing that file changes the commentary of the next report. A plot with no entry in the file gets no commentary paragraph at all. The file is part of the source, so it can be edited by anybody who hosts the application or builds it themselves, and the next report is written with the edited text. It is not editable from the interface.

FILTERING AND NORMALIZATION

Once under each level heading, and again beside every figure that names a data source. Two lines:

- Filtering: either Rows with at least n valid values or Rows with at least p% completeness, followed by the scope — in every condition group, in at least one condition group, or across all annotated samples, independent of condition groups — and were retained (X of Y rows), where Y is the number of rows of the uploaded matrix and X the number that survived the filter. With a threshold of zero, which is the default, the line reads No completeness filter was applied (all Y rows retained).
- Normalization: one of three fixed sentences, for median-centring, quantile normalization (cited as Bolstad et al. 2003) or VSN (cited as Huber et al. 2002). With no normalization the line is left out. All three methods run *before* filtering. Median and quantile methods work on log values; VSN is fitted to linear intensities and returns a log-like analysis scale, and its sentence adds that the CV and abundance plots use the unnormalized data, because VSN output is not on a linear intensity scale.

The whole block is omitted for a level whose annotation carries no filter configuration.

IMPUTATION

Once under each level heading, and again beside every figure that names a data source. With no imputation configured, or with one that was configured and refused, the sentence is that none was applied. Otherwise two sentences. The first is the draw, according to which mode was used: for the default method, a normal distribution centred on the q quantile of the observed values with the adjusted standard deviation times their standard deviation, cited as Lazar et al. 2016; for numerical, a shifted normal distribution stated as a downshift in standard deviations and a width, cited as Tyanova et al. 2016 (the Perseus downshift). The second says whether the parameters were taken sample-wise or across the whole dataset. No random seed is reported.
Beside a figure the wording turns on the figure's own source: a figure whose source is not Imputed Data is described as using the non-imputed data, whatever imputation the level holds.

DATA SOURCE, FEATURES USED, STATISTICS, RESULT

The PCA sections and every volcano comparison carry labelled lines. Data source names the matrix the figure below was drawn from: Data, Filtered Data or Imputed Data. Features used, on the PCA only, reads U of T features (P%) were complete across every sample and entered the projection, and, when $U < T$, says how many were dropped and why. Statistics, on a volcano only, gives the test, the two thresholds and the correction, citing Welch 1947 for the Welch test and Benjamini and Hochberg 1995 for the adjustment; Result, also volcano only, gives how many entities the thresholds selected, named by entity — proteins, phosphosites — rather than *features*.

47.4 Which number comes from which matrix

This is the part to read before quoting a sentence. The generated prose mixes counts taken from three different matrices, and it labels each one, but the labels are easy to read past.

- *Y, the source rows.* The uploaded matrix, before the $\log_2$ transform and before any filter. It is the only place this number appears.
- *X, the filtered rows.* The matrix after filtering and normalization. It appears in the filtering sentence and nowhere else unless a figure names Filtered Data.
- *Every quality-control figure and its table* — coverage, missing values, histogram, boxplot, coefficient of variation, abundance, correlation, and the peptide and phosphosite plots — is drawn from the matrix *as uploaded,* $\log_2$*-transformed and normalized*: unfiltered, unimputed. Quality control has no data source choice by design, since it exists to describe what the instrument produced. The coefficient-of-variation and abundance plots read that same matrix back on the linear scale, which is the scale those statistics are defined on; after VSN

they read the unnormalized data instead, because VSN output is not on a linear intensity scale.

- *The PCA* is the one exception, and takes the source named on its own line. Its Features used count is smaller again: a projection cannot use a feature with a missing value, and a feature with no variance across the samples contributes nothing, so U counts the complete *and* varying rows of the selected matrix.
- *The volcano counts* come from the matrix named in that comparison's own Data source line, which is the one stored with the comparison when it was added.

EASY TO GET WRONG

The filtering sentence stands above figures that were not filtered. A report can say … were retained (6,214 of 8,806 rows) and, two paragraphs below, draw a coverage plot whose bars count the 8,806. Both are correct and both are labelled; the sentence describes what filtering did to the level, not what the figure under it shows. When a count is quoted in a paper, quote it with the matrix it came from.

47.4.1 What the volcano result line counts

HOW IT IS COMPUTED

The comparison is condition$_1$ − condition$_2$: a feature called *upregulated* is higher in the first condition named, and the sentence says so in words as well as by the label.

- *Tested* is not the number of rows in the matrix. A row enters the test only if it has at least two non-missing values in *each* of the two conditions, and, for a paired test, at least two usable pairs. Rows below that are dropped before the test and are in none of the counts.
- *Upregulated* is adjusted p below the threshold *and* $\log_2$ FC above the fold-change threshold; *downregulated* is adjusted p below the threshold and $\log_2$ FC below minus the threshold. Both comparisons are strict.
- *Not significant* is everything else *except* rows without a p-value, which are classed separately and counted in none of the three. A row gets no p-value when its p-value is not finite or when it has no variation (every compared condition constant; for a paired test, identical differences). When there are such rows the line ends with their count, *k* without a p-value, so the four counts add up to the tested total.
- With **Use uncorrected p-values** on, the threshold is applied to the raw p-value and the line says p rather than adjusted p.

FIXED PROSE, NOT GENERATED

The volcano commentary gives an adjusted p-value of 0.05 and an absolute $\log_2$ fold change of 1 as commonly used examples, not as a claim about the current comparison.

> The generated Statistics line states the actual test, correction and thresholds. Quote that line when describing the analysis, and check any text you added yourself against it.

47.5 What the text cannot say

> **NOT A METHODS SECTION**
>
> What the tool writes is an account of the processing, not of the experiment. Nothing in it names the instrument, the gradient, the search engine, the database or its version, the enzyme, the modifications, the FDR applied during the search, the number of biological replicates or what they are replicates of. None of that reaches the tool, so none of it can be written; a methods section built on the generated text alone is a methods section missing its first half. The Introduction block is the place to put it.

47.5.1 Edge cases

- A text block is stored under the plot's internal key. Unloading a level removes its plots from **Select Plot**, but the blocks written for them are kept and reappear when the level is loaded again.
- Text entries survive a save and restore of the session state. They are not cleared when a new dataset is uploaded — unlike the logged volcano comparisons, which are.
- If the comments file cannot be read when the report is built, the report is still built, with the commentary from the last successful read. The warning is shown in the **Report Warnings** panel on Summary → Report only; warnings never appear in the report.

Behind this page

Function	What it produces
report_function	Assembles the report, and with it every sentence described here
_filter_description	The filtering, row-count and normalization sentences
_imputation_paragraphs	The imputation sentences, shared by the level preamble and every figure that names a source
_data_provenance_html	The Data source, Filtering, Imputation and Features used lines
_pca_completeness_description	The count of features that entered the projection
_volcano_statistics_html	The test, thresholds and correction line
_volcano_counts_html	The result line, counted with the same routine the statistics page reports

48
Building a report

One document holding the report's supported figures, their parameters and its provenance, redrawn from the data as it stands when the document is built.

AT A GLANCE

Where	Summary → Report
Format	One self-contained HTML file. Static figures are embedded as images inside it; interactive ones as working plots. There is no PDF or Word output.
What it holds	A sample mapping, then one numbered block per loaded level: its fixed set of figures, three summary tables at protein and phosphosite level, and any logged volcano comparisons.
Chosen by the user	The title, the author, the text blocks, and which volcano comparisons are included. Nothing else is selectable.
Figures	Redrawn at build time, never stored. See *Captured or redrawn.*
In viewer mode	This tab is not shown.

48.1 What a report is

The report brings together a fixed selection of figures from the data that is loaded, each under a heading, each with the prose that belongs to it, in one file that can be sent to somebody who does not have the tool. It is not a figure collection assembled by hand. With one exception — the volcano — there is no list of items to add: a level that is loaded and annotated contributes its whole block, and a level that is not contributes nothing.

1. Upload the datasets and generate their annotations. This alone determines which blocks the report has.
2. Leave every page whose figure will be in it set the way it should appear. The report reads the last state of each page.
3. On Statistics → Volcano Plot, set up each comparison and press **Add Volcano To Report**. Comparisons not added do not appear.

4. On Summary → Text, write the introduction and any text that belongs above or below a figure.
5. Here, fill in **Report Title** and **Author**, press **Generate HTML Report**, and read the preview below.
6. Press **Download HTML Report** to save it.

48.2 Controls

Controls on this page

Control	What it does	Default	Where
Report Title	The heading of the document, its browser tab title, and the name of the downloaded file: everything that is not a letter, digit, hyphen or underscore becomes an underscore. Left blank, the document is titled Untitled Report.	empty	page
Author	Printed under the title. Left blank, it reads Unknown Author.	empty	page
Generate HTML Report	Builds the document. Every figure in it is drawn at this moment. Reads Generating… while it runs.	action	page
Download HTML Report	Saves the document that is currently in the preview. Appears only after a successful build, and saves that build — changing a setting afterwards does not change the file until the report is generated again.	action	page

On the volcano page

Control	What it does	Default	Where
Add Volcano To Report	Adds the comparison as it is currently configured. Refuses while the two conditions are not both chosen and different.	action	page
Remove Volcano From Report	Removes every logged comparison that matches the one currently configured.	action	page

The line beside the two buttons reads Saved entries for <level> and counts the comparisons

logged at the level currently selected, not in the whole session. Both buttons are hidden while **Source** is set to Volcano Control.

> NOTE
>
> The report has no appearance controls of its own. Figure sizes, resolutions, colours and titles all come from the pages that produce the figures, and from the plot appearance panel described in Part XI.

48.3 Captured or redrawn

> THE RULE
>
> No figure is ever stored. Every figure in the report is drawn when **Generate HTML Report** is pressed, from the data as it stands at that moment, with the settings as they stand at that moment. The only thing an item carries from the time it was added is the set of parameters listed below, and that applies to the volcano alone.

What follows from the rule, stated the other way round:

- Change a control on the coverage page, or a colour in the plot appearance panel, then build: the report's figures change, the volcano added a week ago included.
- Re-run the filter, change the imputation, re-annotate, or upload a different file: every number in the report changes, because every figure and every count is computed again from the current matrices. A logged volcano keeps its thresholds but not its results.
- Change a volcano threshold and build without pressing **Add Volcano To Report** again: the report is unaffected. The thresholds it uses are the ones stored with the entry.
- Turning advanced mode off hides its controls; stored appearance overrides still apply to the report. Reset an override to remove it.

> EASY TO GET WRONG
>
> Because the settings are read live, only the *last* state of a page reaches the report. Two coverage plots of the same level with different options cannot both be in one report, and neither can two differently scaled PCAs. The volcano is the exception: each logged comparison carries its own parameters, and several can sit side by side in one section.

What a volcano entry stores

Column	What it holds
Level	Protein or phosphosite. The block the comparison is filed under.
Data Source	Data, Filtered Data or Imputed Data. An unrecognised stored value is read as Data rather than as the most processed matrix.
Condition 1, Condition 2	The comparison, in that order: $\log_2$ FC is condition 1 minus condition 2, so a feature reported as upregulated is higher in condition 1.
Label column	Which identifier labels the points.
Test type	Unpaired pooled, unpaired Welch, or paired.
P-value threshold, log2 FC threshold	As set, and whether **Use uncorrected p-values** was on — that is, whether the first threshold applies to the raw or to the Benjamini–Hochberg adjusted value.
Highlight terms	The list, whitespace-trimmed, empties dropped.
Type	Normal or Interactive — per comparison, so one report can hold a print figure and a hoverable one.
Label top up-regulated, Label top down-regulated	Stored only for a Normal entry; an Interactive entry stores zero for both, because only the static renderer draws labels.

Adding a comparison that matches one already logged in every one of those fields does nothing and says so. Removing works the same way round: it removes every entry matching the current configuration and reports how many. There is no list of logged comparisons on screen and no way to remove one except by reconstructing its settings, which is why the counter beside the buttons is worth watching.

> **EASY TO GET WRONG**
>
> The log lives with the browser session, and it is cleared automatically whenever the loaded datasets change: a new upload, a different data package, a different metadata profile, or a file whose row or column count differs. That is deliberate — a comparison logged against one dataset would be redrawn against another — but it means the log is empty again after a re-upload, without announcement. It is written into a saved state file and comes back with it.

48.4 What the document contains

In order, from the top:

- The title, the author, the date the report was built, and a version line. It is the version of the application that built the report, which is the version on this manual's title page.
- Introduction — only if a text block was written for it.

- Table of Contents, listing the blocks and their numbered sections. A volcano section is listed only when that level has a logged comparison.
- Sample Mapping — one table per level giving, for every sample, its plot label condition_n, its original name in the uploaded file, its condition, and its pair when the annotation has pairing. Two levels with identical mappings share one table. This is the key to every axis label in the document.
- The level blocks, numbered from 1 in the order peptide, protein, phosphosite, with only the loaded ones present.

The document has no warnings section. Warnings are shown in the application only, in the **Report Warnings** panel on this page, and never written into the report.

Each level block opens with its filtering, normalization and imputation paragraphs, then gives its sections in a fixed order. A section is a heading, the fixed commentary, the Above text block, the figure, and the Below text block; the protein and phosphosite coverage, boxplot and coefficient-of-variation sections carry their summary table under the figure, and the PCA and volcano sections carry their provenance and statistics lines between the text block and the figure. Those sentences cite the published methods they describe — the imputation, normalization, test and correction, and the Class I localization threshold — in the text. All of that prose is described in *The methods text*.

> **WORTH KNOWING**
>
> A figure or table that cannot be drawn does not stop the build: a red box saying that it could not be rendered takes its place in the document, and the **Report Warnings** panel on this page names it as <part> could not be rendered: <reason>. The reason appears only in that panel, never in the report, so read the panel before sending the file.

48.4.1 The file

Beside **Download HTML Report**, the notebook download provides one ZIP for the plots and tables included in that report. It has numbered calculation sections, required helpers and resources, and input-file placeholders. Where a section uses imputed data, the exact matrix is supplied. Where it does not, the recipe starts from the original input and reproduces the required preprocessing. It does not add analyses that the report omitted.

The archive belongs to the built report snapshot, not to a later state of the session. If the application cannot construct a complete recipe, it reports that separately; an HTML report may still be available. Check both the **Report Warnings** panel and the notebook availability before handing over a reproducible analysis.

The document is a single HTML file with nothing beside it: static figures are images encoded into the file, interactive ones are complete plots in sandboxed frames that keep their hover, zoom and legend behaviour offline, and a content policy in the file forbids it from fetching anything over a network. It can therefore be opened on a machine that has never seen the tool; browser and print layout can still differ. The file can be large, because every figure is inside it. To obtain a PDF, open it in a browser and print it; an interactive figure prints as

whatever it currently shows, so set it first, or log that comparison as Normal instead.

48.5 What a report cannot include

- *Any level but the three.* There is no phosphoprotein block and no precursor block. A volcano logged at phosphoprotein level is stored and counted, and then never appears.
- *Control-corrected volcanoes.* The two buttons are hidden while **Source** is Volcano Control, and a control-corrected entry is skipped at build time.
- *Any statistical output but the volcano.* No ANOVA, no over-representation analysis, no pathway heatmap, no replicate and variance projection (a what-if projection, not a power calculation).
- *Whole sections of the tool.* No Venn diagram or Pearson comparison; no single-protein figure; no phosphosite distribution, phosphoprotein regulation or Kinase Library; no missing-value heatmap; no distribution, imputation or identifier translation output.
- *Tables other than its own.* It builds the sample mapping and, for the protein and phosphosite blocks, a coverage, boxplot and coefficient-of-variation summary. Nothing else, and neither table of *The summary tables*.
- *A second version of any figure, or a chosen subset.* One report holds at most one of each — the volcano excepted — and a figure of a loaded level cannot be left out.

48.5.1 Edge cases

- A dataset loaded without an annotation contributes no block, and the **Report Warnings** panel on this page says so rather than the build failing; the document itself does not mention it.
- The peptide block uses the protein annotation for its sample mapping whenever a protein dataset is loaded, and its own peptide metadata only in a peptide-only session. Its coverage plot is always drawn at peptide level; the precursor variant has no place in the document.
- The missing-value plot is styled once for all three levels: the completeness page holds one set of options, and all three use it.
- A page never visited in this session contributes its figure with the built-in defaults, not with whatever it showed last time the tool was opened.

Behind this page

Function	What it produces
report_function	Builds the whole document and returns it with its file name and, separately, the warnings shown on this page
addVolcanoReportEntry	The two buttons: the duplicate test and the stored entry
removeVolcanoReportEntry	
getSummaryReportContext	Everything read from the session at build time: page settings, logged comparisons, appearance settings
syncSummaryReportState WithDatasets	Clears the log when the loaded datasets change
run_volcano	Re-runs each logged comparison for its counts
volcano_png / volcano_html	The two renderers, chosen by the entry's Type
sample_label_frame	The sample mapping tables

PART XI

The plot appearance panel



49
The plot appearance panel

One panel that restyles any figure in the tool, and where the controls that used to sit beside each plot went.

AT A GLANCE

Opens with	Advanced mode, **Ctrl** + **Shift** + **E**. The same switch is **Advanced plot option mode** under Settings → Plot Options.
Where	Under the plot options, on every page that draws a figure
Applies to	The plot in front of you, and nothing else
Scope	One set of settings per plot and per tab
Persistence	Held for the browser session, written into a save state and restored with it
Never changes	The data, the annotation, or any number in the table under the plot
New	Not covered by any previous manual

CHANGED IN 1.0.0

Several controls that older manuals show beside a plot are now in this panel: **Bar Color**, **Class I Color**, **Not Class I Color**, **Dot Color**, **Dot Size**, **Text Size**, and the title and legend switches. A reader following an older page will look for them in the options row and not find them.

49.1 Turning it on

Ctrl + **Shift** + **E** turns advanced mode on, and the same keystroke turns it off again; on a Mac **Cmd** does the same as **Ctrl**. The shortcut is read from the physical key, so it is the same keystroke on a QWERTZ or AZERTY keyboard. While advanced mode is on, every page that draws a figure shows the panel between its options and the figure itself. Text-only sequence coverage and table pages have no figure to restyle. The replicate and variance projection (a what-if projection, not a power calculation) and imputation diagnostics also do not expose

this panel; their presence is not determined simply by whether a page contains an image.

Advanced mode only shows and hides the panel. Switching it on changes no figure: an untouched panel sends nothing. Switching it off discards nothing and undoes nothing — the settings made in the panel keep applying to their figures, on screen and in the report, until they are changed back.

Edited fields show their overrides. Untouched fields can display panel fallbacks; those are not always the renderer's own defaults. Frame and grid fields use the defaults supplied by the current plot.

49.1.1 Reading it

- One card per part of the figure. A card is closed until it is opened, and opening one puts every control for that part in front of you.
- The line beside a card's name says what the card changes, so the list of things that *could* be changed stays one line each.
- A colour field is the operating system's colour picker. The condition colours are the exception: they take a #RRGGBB value typed into the field beside the swatch as well.
- A numeric field left empty means *inherit* where inheriting is possible — in the per-condition rows — and reverts to the value shown everywhere else.

49.2 One set of settings per plot

The panel styles the plot in front of you. Its settings are held per plot and per tab: the coverage plot and the CV plot each keep their own, and moving between tabs moves between two independent sets rather than carrying one along. A plot that is drawn at two data levels keeps one set for both — the settings follow the tab, not the level.

Nothing is applied to a figure by being typed. Three hundred and fifty milliseconds after the last change, the settings are reduced to the keys that actually say something, serialised as JSON and sent to the renderer as one `advanced` parameter; the figure comes back drawn from them. The reduction keeps a zero and a `false`, because a fully transparent point, an outline of width zero and a legend that should not be drawn are instructions rather than absences.

> **WORTH KNOWING**
>
> The number a field shows is the panel's own value, not necessarily what the plot is drawing. Sizes in particular differ per plot — the Points card shows 6 pt while the PCA, the abundance curve and the retention-time plot each draw their own default — and the plot keeps its own until the field is edited. Editing a field is what sends it.

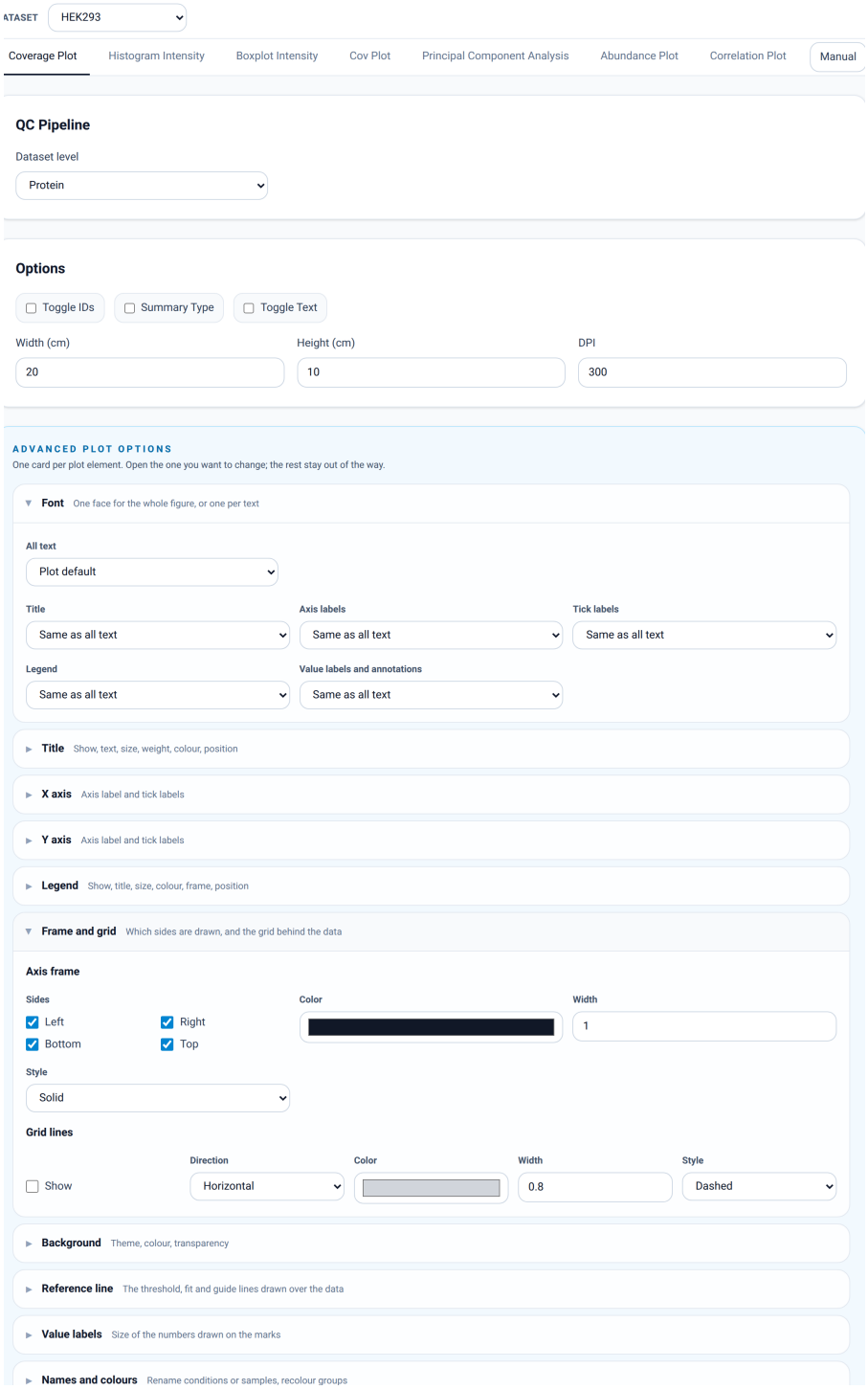


Figure 49.1. The panel under the coverage plot with advanced mode on. **Font** and **Frame and grid** are open; the other eight cards are collapsed to their header and the one-line summary of what they hold. Every control visible here is at its default, which is why the figure above it looks as it does.

49.2.1 What survives what

- Settings live in the browser session. Closing the tab loses them unless the session has been saved.
- A save state carries advanced mode and every plot's settings, and restoring one puts both back, so a restored session draws the figures it was saved with. Settings saved by an older panel are converted automatically: the single axis-line switch becomes the four **Sides** boxes, and the value-label offsets, which no longer have a control, are dropped.
- The quality control and completeness pages keep theirs inside the page state rather than beside the other plots', which is the same save file by a different route.
- The figures in the summary report are drawn from the same settings as the figures on screen.
- In viewer mode the panel works as it does here, but its settings start empty on every visit rather than being carried over.
- Every setting keeps applying with advanced mode switched off; the mode decides only whether the panel is on screen.
- A setting changes what it names and nothing else. Recolouring the grid does not switch it on, hiding one side of the frame leaves the other three as they were, and editing the title touches neither.

49.3 Which cards appear

A card appears when the renderer of the plot in front of you reads the settings that card writes. The panel is therefore short on a Venn diagram and long on a volcano, and an option missing from one plot is missing because that plot has no such part rather than because it was forgotten.

- **Font** — every plot. One face for all text, or one per element. Part XI, *Fonts*.
- **Title**, **X axis**, **Y axis** — every plot. The axis cards hold both the axis label and that axis's tick labels; the phosphosite plot, which draws no x tick labels, shows no tick controls for them.
- **Legend** — plots that draw a legend. The coverage plot in summary mode draws none, so it has no legend card; the heatmaps, the Venn diagram and the two-sample scatter plots have none either.
- **Frame and grid** — every plot except the volcano tile drawn in the browser on the analysis canvas.
- **Background** — the same.

- **Reference line** — the coverage plot, the volcano, the Kinase Library volcano, the phosphoprotein regulation plot, the Q-Q plot, the retention-time plot and the Pearson correlation.
- **Points** — plots with one class of dot: the PCA, the abundance curve, the coverage plot in summary mode, the Pearson correlation, the Q-Q plot, the retention-time plot, the single-protein line plot, the Kinase Library volcano and the phosphoprotein regulation plot.
- **Volcano points** — the volcano, in both renderers, and the volcano tile on the analysis canvas.
- **Boxes and their points** — the intensity boxplot, the CV plot and the single-protein boxplot.
- **Bars** — the phosphosite plot, the phosphosite distribution and the STY plot.
- **Site classes** — the phosphosite coverage plot.
- **Venn sets** — the Venn diagram.
- **Value labels** — plots that print numbers on their marks: the coverage plot, the intensity boxplot, the CV plot, the phosphosite coverage plot and the two peptide bar plots.
- **Colour scale** — the correlation matrix and the two heatmaps.
- **Names and colours** — plots that group by condition or draw one mark per sample. Condition names and colours appear where the plot has conditions, sample names where it has samples.

49.4 Controls

The panel itself

Control	What it does	Default	Where
Ctrl + Shift + E	Turns advanced mode on and off. With it off, the panel is hidden; the settings made in it still apply.	off	keyboard
Advanced plot option mode	The same switch, under Settings → Plot Options.	off	page
Card headers	Each card opens and closes on its own; the state is not saved, so every card starts closed.	closed	appearance

EASY TO GET WRONG

The first change made in the panel also writes the frame and grid values the **Frame**

and grid card is showing, whether or not that card was opened. On most plots those values are what was already being drawn and nothing moves. On the static volcano, which draws only its left and bottom spines, they are all four sides, so the first change of any setting gives it a full box. Set the frame the way you want it in the same session rather than assuming it was left alone.

49.5 The figures in this manual

Every figure in this book was drawn by the application, from the example project that ships inside the package, with the panel and nothing else. The settings are the same on every one, so a reader who wants figures that look like these can type them once and keep them:

The house style used throughout this manual

Column	What it holds
Theme	Minimal
Font	Helvetica / Arial, title 13, axis labels 11, ticks 9, legend 9
Frame	Left and bottom only; #0f172a at 0.9 pt
Grid	Horizontal only, #cbd5e1, solid, 0.6 pt
Legend	No frame
Condition colours	#2563eb and #ff4b5c
Points	Size 7, opacity 0.85, white outline 0.4 pt — smaller and more transparent on the plots that draw thousands of them
Reference line	#ff4b5c, dashed, 1.2 pt
Size	20 by 10.5 cm at 200 dpi for a wide figure, 15 by 13 cm for a square one

Nothing in that list is unavailable to a reader: it is the panel's own vocabulary, and the figures are evidence of what the panel can do rather than of what a designer did to them afterwards.

49.6 Where each card is documented

- *Titles, labels and legends* — the **Title**, **X axis**, **Y axis**, **Legend**, **Value labels** and **Names and colours** cards.
- *Fonts* — the **Font** card.

- *Points, boxes and bars* — the **Points**, **Volcano points**, **Boxes and their points**, **Bars**, **Site classes**, **Venn sets** and **Colour scale** cards.
- *Reference lines* — the **Reference line** card.
- *Axes, grid and background* — the **Frame and grid** and **Background** cards.
- *Figure size, resolution and format* — what the panel adds to the size and format controls on the page.

Behind this panel

Function	What it produces
parse_advanced_plot_settings	Reads the panel's JSON out of the plot request; anything unreadable is treated as no settings at all
_apply_advanced_axes	Applies every text, tick, legend, frame, grid and background setting to a static figure
_advanced_plotly_layout	The same settings, as the layout of an interactive figure
compactAdvancedPlotSettings	Reduces the panel's state to the keys that are sent, keeping zero and false

These are the application's own renderer rather than functions of the proteomicscopylot package: the panel styles the figures the interface draws, and a figure drawn in a notebook is styled by the plotting library directly.

50
Titles, labels and legends

Every piece of text in a figure: what it says, how big it is, and where it sits.

AT A GLANCE

Cards	**Title**, **X axis**, **Y axis**, **Legend**, **Value labels**, **Names and colours**
Sizes	Points, as the plotting library counts them, and absolute against the figure's width and height in centimetres
Offsets	Pixels. On a static figure a title or axis-label offset is taken at the figure's DPI; the legend offset is converted to a fraction of the figure
Empty means	Keep whatever the plot chose for itself
Renaming	For this figure only. The annotation is untouched.

50.1 Title and axis labels

The **Title** card and the two axis cards hold the same six controls for one text each. **Text** overwrites what the plot wrote; left empty, the plot's own title or axis label stands, which is usually the one built from the data level and the comparison and is therefore the one that is correct. **Colour**, **Size**, **Bold** and **Italic** are what they say. **X offset** and **Y offset** move the text from where the plot put it, in pixels, positive to the right and upwards.

On a static figure the offset is added to the text's own position, in pixels at the figure's DPI, so the layout still places the label clear of its tick labels and then moves it by that amount; the same +20 moves a title by the same distance on a 10 cm and a 20 cm figure. On an interactive figure the title's **Y offset** raises or lowers it relative to the top of the plot area, and the canvas grows so that a raised title stays visible; lowering it past the default gap moves it into the plot area. The title's **X offset** there is converted to a fraction of the figure width and stops at the edge. Zero leaves the text where the layout put it, which is what every plot starts with.

Show title is a switch above the card's fields. Turned off, the title is cleared — on every plot, through one setting, rather than through each plot's own header parameter.

Title, X axis and Y axis

Control	What it does	Default	Where
Show title	Clears the title when off. **Title** card only.	on	appearance
Text	Overwrites the plot's own text. Empty keeps it.	empty	appearance
Colour	The colour of that one text.	#111827	appearance
Size	Point size. A size of zero or an empty field leaves the plot's own.	12 for the title, 10 for an axis label	appearance
X offset, Y offset	Moves the text, in pixels, right and up.	0, 0	appearance
Bold, Italic	Weight and slant of that one text.	off, off	appearance

50.2 Tick labels

Tick labels are styled inside the axis card they belong to, so the **X axis** card holds the x-axis label and the labels under the x ticks together. **Size** and **Colour** apply to every tick label on that axis; **Rotation** turns them, in degrees, counter-clockwise from horizontal. Setting a tick colour paints that axis's tick marks as well, and the axis label is put back to its own colour afterwards so that it does not follow the ticks by accident.

The two axes are independent: a plot with long sample names under it can have its x tick labels at 90 degrees and 7 pt while the y axis keeps 10.

Tick labels, in each axis card

Control	What it does	Default	Where
Size	Point size of the tick labels on that axis.	10	appearance
Rotation (deg)	Rotates them; 90 stands them on end.	0	appearance
Colour	Colour of the tick labels, and of that axis's tick marks.	#111827	appearance

> NOTE
>
> The phosphosite plot clears its x tick labels, so its **X axis** card offers the axis label

> and no tick controls. Several plots also set a tick size of their own — the phosphosite coverage plot uses 7 for its sample names — and a size set here replaces it.

50.3 The legend

Show legend removes a legend; it cannot add one. A plot that drew no legend has no handles to put in one, so the switch has an effect in one direction only. **Legend title** overwrites the heading above the entries, which is Condition on most plots and Cluster on a PCA coloured by cluster. **Text size** and **Text colour** apply to the entries and to that heading together.

Draw a frame is deliberately three-state. Untouched, each plot keeps the choice it made — the static volcano draws none, most other static plots do — and touching the switch takes the choice over for that plot.

Offset x / y (px) moves the whole legend from its anchor, in pixels, converted to a fraction of the figure. The anchor itself is the plot's own — most static plots put the legend just outside the top right corner of the axes, a few at the middle of the right edge — and the offsets move it from there.

Legend

Control	What it does	Default	Where
Show legend	Off removes the legend. It cannot create one.	on	appearance
Legend title	Overwrites the heading over the entries.	the plot's own	appearance
Text size	Point size of the entries and the heading.	10	appearance
Text colour	Colour of both.	#111827	appearance
Draw a frame	A box around the legend. Untouched, each plot keeps its own choice.	the plot's own	appearance
Offset x / y (px)	Moves the legend from its anchor, right and up.	0, 0	appearance

50.4 Value labels

Several plots can print a number on each mark — the count on a coverage bar, the median on a box, the percentage on a peptide bar — under a **Toggle Text** switch on the page. **Label size** in the panel is the size of those numbers, in points, for every plot that draws them. It replaces the per-page **Text Size** field older manuals describe, which was bounded differently on each page; the one setting has a floor of 4 and no ceiling.

Value labels

Control	What it does	Default	Where
Label size	Point size of the numbers printed on bars and boxes. Values below 4 are raised to 4.	8	appearance

The card appears only on plots that print numbers and read this size for them. The phosphosite coverage plot prints none and does not show the card; the S/T/Y plot prints a percentage over each bar at a fixed size of 9, which no control changes, and shows none either.

The missing value plot on the completeness page reads both this size and the bar colour, and shows the **Value labels** and **Bars** cards accordingly.

50.5 Renaming conditions and samples

The **Names and colours** card lists the conditions and the samples of the plot in front of you, one field each, and the name typed into a field replaces that condition or sample **in this figure only**. Nothing is written back: the annotation keeps its names, every other plot keeps its names, and the table under the plot keeps its names. Renaming a condition in Data → Annotation is the opposite act — it changes the condition everywhere, for every plot and every statistic — and the two are not alternatives to each other. Use this card for a figure caption's wording; use the annotation for the experiment's.

A rename reaches the legend entry, the tick label and any label built from the condition name. Where a tick reads Control_1, which is a condition name with a replicate number pasted after it, the condition part is replaced and the number kept. A sample's own new name wins over its condition's, because it names one column rather than a group.

Condition colors sits in the same card and overrides the session palette for this plot: a swatch, or a #RRGGBB value typed beside it. The palette set in Data → Colors is what every plot uses until a condition is given a colour here.

Names and colours

Control	What it does	Default	Where
Condition names	A display name for one condition, in this figure only.	the condition's own name	appearance
Sample names	A display name for one sample. Wins over a condition rename for that sample.	the sample ID	appearance
Condition colors	Overrides the session palette for this plot, as a swatch or a hexadecimal value.	the session palette	appearance

> **EASY TO GET WRONG**
>
> A renamed condition is a figure-level label, so a figure and the CSV under it can disagree. Download the table before renaming, or state in the caption which name means which condition; a reader who has both in front of them cannot tell that Treated and Cond_B are the same column.

A condition renamed here is used everywhere the figure names that condition, including the static volcano's title and its fold-change axis label.

Behind these cards

Function	What it produces
_apply_advanced_axes	Applies the title, axis label, tick and legend settings to a static figure
_style_legend	Font, size, colour, heading and frame of a legend that exists
_advanced_label	Resolves one condition or sample name against the rename maps
_rename_composite_sample_label	Renames the condition part of a condition_n tick label
advanced_label_size	The size of the numbers printed on bars and boxes

51
Fonts

Setting the typeface of every text in a figure, or of one of them.

AT A GLANCE

Card	**Font**, the first card in the panel, on every plot
Faces	Helvetica / Arial, Times New Roman, Courier, DejaVu Sans, DejaVu Serif
Scope	One face for all text, with overrides for the title, the axis labels, the tick labels, the legend and the value labels
Untouched	No family is set at all, and every plot keeps the face the plotting library gives it
New	Added in 1.0.0

51.1 Choosing a face

All text sets one face for the figure. The five fields beside it — **Title**, **Axis labels**, **Tick labels**, **Legend**, **Value labels and annotations** — each override it for one kind of text, and each starts on Same as all text, which is not a face but an instruction to follow the global one.

The order is short: an element's own choice, then the global one, then nothing. *Nothing* is the important case, and it is not a default face. With the global field on Plot default and every override on Same as all text, no family is set anywhere and each plot is drawn in whatever face the plotting library uses — which is what every figure in this tool looked like before this card existed. Only an explicit choice changes anything, and it changes only the texts it names.

Value labels and annotations is the widest of the five: it covers the numbers printed on bars and boxes, the *r* and *N* box on a correlation plot, and the feature labels beside the top hits of a volcano — every text a plot drew itself rather than as a title, a label or a legend.

Font

Control	What it does	Default	Where
All text	The face for every text in the figure. Plot default sets none.	Plot default	appearance
Title	Overrides **All text** for the title.	Same as all text	appearance
Axis labels	Overrides it for both axis labels.	Same as all text	appearance
Tick labels	Overrides it for the tick labels of both axes.	Same as all text	appearance
Legend	Overrides it for the legend entries and their heading.	Same as all text	appearance
Value labels and annotations	Overrides it for every text the plot drew itself.	Same as all text	appearance

51.2 What each face is

A face named in the panel is a list of families, tried in order, and the first one installed in the image is the one the figure is drawn with.

The five faces and what they resolve to

Column	What it holds
Helvetica / Arial	Liberation Sans, then Helvetica, then Arial, then DejaVu Sans.
Times New Roman	Liberation Serif, then Times New Roman, then Times, then DejaVu Serif.
Courier	Liberation Mono, then Courier New, then Courier, then DejaVu Sans Mono.
DejaVu Sans	DejaVu Sans, which ships with the plotting library.
DejaVu Serif	DejaVu Serif, likewise.

The faces actually installed in the standard image are the Liberation family. They provide metrically compatible alternatives for common office fonts. Still inspect the exported figure at its final size: glyph coverage, browser fallback and the publication's typesetting can affect appearance.

WHEN A FACE IS NOT INSTALLED

The list is resolved once, against the fonts the image actually has, and the first match is used. Installing Arial does not override an earlier installed family in that list; the selector names a fallback family group rather than a guarantee that one specific font file is used.

If none of a list's names is installed, the last name is used. That last name is always a DejaVu face, which is shipped with the plotting library and is therefore present wherever the tool runs, so the substitution is the one that would have happened anyway rather than a failure in the middle of a render.

The image is checked at build time: every face the panel offers is resolved to a file, and a build whose fonts are missing fails with a message naming them rather than producing an image that offers five faces and draws one.

51.2.1 Interactive plots

An interactive plot is drawn by the reader's browser, so its face is sent as a list of CSS families — Helvetica, Arial, 'Liberation Sans', sans-serif and its equivalents — and the browser picks the first one that machine has. A reader whose machine has none of them sees the generic fallback at the end of the list. This is the one place where two people can look at the same figure in two different faces; a static figure carries its face in the file.

The global face reaches every text of an interactive figure, including the hover labels. The four element overrides reach the title, the axis labels, the tick labels and the legend.

The **Value labels and annotations** override reaches static figures only. In an interactive plot the annotations follow the global face, and the card says so.

The volcano tile drawn in the browser on the analysis canvas shows no **Font** card: that tile is drawn with a fixed sans-serif face.

51.3 Which face to choose

Journals ask for a sans-serif face at a stated minimum size, most often Arial or Helvetica, which is **Helvetica / Arial** here. Times New Roman is worth choosing when a figure has to match body text set in a serif face. Courier is a monospace face: it is the right choice for a figure whose labels are sequences or accessions, where the eye is comparing characters by position, and the wrong one for everything else.

Set the face once, in **All text**. The five overrides exist for the cases that genuinely need two faces — a serif title over sans-serif data labels, a monospace tick label under a sans-serif axis — and a figure that uses more than two faces is harder to read than the same figure in one.

EASY TO GET WRONG

The face applies to the figure and to nothing else. It does not change the interface, the tables, the report, or any text on the page around the plot. In a PNG, JPG or WEBP download the text is pixels, so the face is correct but no longer text; in a PDF the text stays text, with the font embedded in the file, which is what an editor or a typesetter needs in order to restyle or re-flow it.

Behind this card

Function	What it produces
PLOT_FONTS	The five faces, each as a family list for the static renderer and a CSS stack for the interactive one
resolve_font	The element's choice, then the global one, then nothing
installed_family	The first of a list that is installed, or its last name
mpl_font_family	The family a static text artist is given
plotly_font_family	The CSS stack an interactive figure is given
verify_fonts_installed	The build-time check that every offered face exists

52
Points, boxes and bars

The marks themselves: size, shape, colour, opacity and outline.

AT A GLANCE

Cards	**Points**, **Volcano points**, **Boxes and their points**, **Bars**, **Site classes**, **Venn sets**, **Colour scale**
Sizes	A diameter in points, so one number means the same mark in the static figure and in the interactive plot beside it
Per condition	Point settings can be set for all points at once, or for one condition at a time
Opacity	0 to 1. Zero is a value: it draws an invisible mark.
Outlines	Off unless asked for — width 0 means no outline

52.1 Points

The **Points** card styles the one class of dot a scatter draws.

Points

Control	What it does	Default	Where
Dot colour	One colour for every point. Ignored by plots that colour their points themselves — see below.	#1f77b4	appearance
Dot size (pt)	Diameter in points. Values below 0.5 are raised to 0.5.	6	appearance
Dot opacity (0-1)	0 is invisible, 1 is solid.	0.8	appearance
Dot outline color	The ring around each point, drawn only when the width is above zero.	#000000	appearance
Dot outline width (0 = none)	Width of that ring, in points.	0	appearance

Points (continued)

Control	What it does	Default	Where
Dot shape	Circle, Square, Diamond, Triangle, Triangle (down), Cross, X, Star, Pentagon or Hexagon.	Circle	appearance

A size is a diameter in points, in every card and in both renderers. Doubling it doubles the width of the mark and quadruples its area: the static renderer wants an area and is given the square of the diameter, $s = d^2$, so $\sqrt{s}$ is always the mark's width in points. Before this was unified each plot had its own scale, and the same number drew a different dot on each page.

The card's numbers are the panel's, not the plot's. Until a field is edited, each plot draws its own default: 2.5 pt on the Q-Q plot, 1 pt on the retention-time plot, 2.3 pt on the abundance curve, size 10 on the Kinase Library volcano. Editing the field is what replaces them.

52.1.1 Per condition

Where a plot draws its points one condition at a time — the PCA, the abundance curve, the points over a single-protein boxplot — the card gains one row per condition, with **Shape**, **Size**, **Opacity** and **Outline width**. An empty field there means *use the setting above*.

The rule is worth stating once: changing a general setting clears every per-condition override of it. Set the general size and all conditions become that size; set one condition and only it moves; set the general size again and they snap back together. The volcano's three classes behave the same way, with fixed names instead of conditions.

> EASY TO GET WRONG
>
> **Dot colour** has no effect on a plot that has already decided each point's colour: the PCA, the abundance curve, the points over a boxplot and the coverage plot in summary mode take their colour from the condition palette, and the phosphoprotein regulation plot takes it from a colour scale. On those plots the colour is changed in **Condition colors**, in the **Names and colours** card, or not at all.

Dot colour is offered only where it decides the colour. On the plots that colour their points from the condition palette or from a colour scale the field is not shown, because the palette or the scale overrules it.

52.2 Volcano points

A volcano has three classes of point rather than one, so it gets its own card: **All classes** at the top, then **Up-regulated**, **Down-regulated** and **Not significant** on their own rows.

Volcano points

Control	What it does	Default	Where
All classes: Size	Sets all three class sizes and clears their individual values.	6	appearance
All classes: Opacity	The same for opacity.	0.85	appearance
All classes: Shape	One shape for all three classes.	Circle	appearance
Up-regulated	Size, opacity and colour of the up class.	6, 0.85, #b2182b	appearance
Down-regulated	The same for the down class.	6, 0.85, #2166ac	appearance
Not significant	The same for the rest. Drawn smaller and fainter on purpose: they are the backdrop the eye judges the two tails against.	5, 0.55, #bdbdbd	appearance
Highlight size	Diameter of the ring drawn around a highlighted feature.	10	appearance
Highlight outline	Colour of that ring.	#22c55e	appearance
Hit label size (static plot)	Point size of the feature labels beside the top hits. Values below 4 are raised to 4.	7	appearance

The three colours the card shows are the static figure's palette. The interactive volcano starts from a more saturated screen palette — #dc2626 up, #2563eb down, #94a3b8 the rest — so the two renderers do not look identical until a colour is chosen here, at which point both use it.

52.3 Boxes and their points

Four different marks are easy to confuse on a boxplot, and the card keeps them apart: the box itself, its median line, the data points drawn over it, and the outlier fliers beside the whiskers. The fliers are the values the jittered points deliberately exclude, so they are styled separately.

Boxes and their points

Control	What it does	Default	Where
Box border color	The box, its whiskers and its caps.	#000000	appearance
Median color	The line across the box.	#b22222	appearance
Data dot size (pt)	Diameter of the jittered points drawn over the box.	4.5	appearance
Data dot opacity	Opacity of those points.	0.8	appearance
Data dot outline	Colour of their outline.	#000000	appearance
Data dot outline width (0 = none)	Width of that outline. A thin dark edge by default, because a filled dot in a condition's own colour sits on a box of that same colour.	0.4	appearance
Outlier dot color	Fill of the fliers.	#ff0000	appearance
Outlier outline color	Their edge.	#000000	appearance
Outlier size (pt)	Their diameter. Values below 1 are raised to 1.	5	appearance

The four data-dot controls appear only where there are data points over the box, which is the single-protein boxplot; that plot's per-condition rows sit under them and override them condition by condition. The three outlier controls appear where the plot is drawing fliers: always on the single-protein boxplot, and on the intensity boxplot and the CV plot when that page's **Toggle Outliers** is on. The box fill is not a control here — it is the condition's colour, from the palette or from **Condition colors**.

52.4 Bars, classes and sets

Bars, Site classes and Venn sets

Control	What it does	Default	Where
Bar colour	One colour for every bar. Read by the phosphosite plot and the phosphosite distribution, whose own default is #87CEEB.	#2563eb	appearance
Class I	Colour of the Class I part of each bar.	#2563eb	appearance

Bars, Site classes and Venn sets (continued)

Control	What it does	Default	Where
Not Class I	Colour of the rest of it. The plot's own default is #f59e0b, so this swatch shows a colour the figure is not using until it is chosen.	#94a3b8	appearance
Set 1, Set 2, Set 3	One colour per Venn set. The third is drawn only in a three-set diagram.	#2563eb, #f97316, #16a34a	appearance

The phosphosite coverage plot draws a third segment, Unclassified, wherever sites fall into neither class. It is fixed at #94a3b8 and has no control.

The **Bars** card is not offered on the S/T/Y plot, whose three bars are coloured by residue from a fixed palette.

52.5 Colour scale

The **Colour scale** card holds one control, for the plots that paint a matrix rather than drawing marks: the correlation matrix and the two heatmaps.

Colour scale

Control	What it does	Default	Where
Correlation gradient	Coolwarm, Red/blue, Spectral, Pink/green, Purple/green, Brown/teal, Viridis or Cividis.	Coolwarm, and each plot's own until one is chosen	appearance

The eight entries are a shortlist rather than a limit: the renderer accepts any colour map name the plotting library knows, and a name it does not know is ignored in favour of the plot's own rather than replacing the figure with an error. Until a choice is made, the correlation matrix draws coolwarm and the two heatmaps draw plasma — which the card, showing Coolwarm, does not say.

Diverging scales — Coolwarm, Red/blue, Spectral — are the right choice for a correlation matrix, where the midpoint means *no correlation* and the two directions mean opposite things. Sequential scales — Viridis, Cividis — are the right choice for a quantity with one direction, and are the two here that stay legible when printed in greyscale or read by a colour-blind reader.

Behind these cards

Function	What it produces
resolve_marker	A mark's appearance: the plot's default, then the general setting, then this condition's
marker_area	Diameter in points to the area the static renderer wants, $s = d^2$
MARKER_SHAPES	The ten shapes, and the spelling each renderer wants for them
marker_style	The resolved mark, as static scatter arguments
plotly_marker	The same, for an interactive plot
_volcano_marker_style	Size, opacity and colour of the three significance classes
advanced_color	A colour the panel may override, falling back to the plot's own
advanced_color_map	The colour scale the panel asks for, or the plot's own if the name is unknown

53 Reference lines

The threshold, fit and guide lines drawn over the data, and how to change or remove them.

AT A GLANCE

Card	**Reference line**, on the plots listed below
Controls	**Show**, **Colour**, **Style**, **Width**
Applies to	Every line a plot draws *over* its data rather than *from* it
Scope	One setting per plot, covering every such line on that plot
Both renderers	The same four settings reach the static figure and the interactive one
New	The colour existed on two plots; **Show**, **Style** and **Width** are new in 1.0.0

53.1 The four settings

Reference line

Control	What it does	Default	Where
Show	Off draws no line at all. The rest of the figure is unchanged. The coverage plot has no **Show** here: its line is switched by **Total Line** in the plot's own options, and the card appears only while the line is on.	on	appearance
Colour	The line's colour. The swatch shows red until a colour is chosen; each plot's own colour is used until then.	the plot's own	appearance
Style	Solid, Dashed, Dotted or Dash dot.	the plot's own, dashed on most	appearance
Width	Line width. The field shows 1.5; each plot's own is used until it is edited.	the plot's own	appearance

All four are resolved in one place, for both renderers, so a line set here looks the same in the figure and in the interactive plot beside it. That was not true before: two plots read a colour from the panel while their interactive twins ignored it, and the same comparison changed appearance when the renderer was switched.

One setting covers every guide on the page. A volcano has three threshold lines and they take one colour between them; the imputation fit plot draws two curves and they do too.

53.2 Which lines these are

Reference lines, and what each one means

Column	What it holds
Coverage	A horizontal line at the union of features quantified across the samples: a reference for per-sample detection. Red, dashed, 1.5. The peptide and precursor coverage plots draw the same line at the number of peptides or precursors.
Volcano	Two vertical lines at ± the fold-change threshold and one horizontal line at $-\log_{10}$ of the selected p-value threshold. A significant up or down result must pass both criteria. The static figure draws them in #9ca3af, dashed, 0.8; the interactive one in black, dashed, 1.5.
Kinase Library	The volcano drawn from the uploaded Kinase Library results is interactive only and draws the p-value line only, in green, dashed, 1.5.
Q-Q plot	The least-squares fit of the sample quantiles on the theoretical quantiles. Normal samples tend to follow a straight line with sampling variation; they need not lie exactly on it. Red, solid, 2.0.
Imputation fit	Two curves over the histogram: the normal fit in red, solid, 1.5, and the area the imputed values are drawn from in cyan, solid, 2.5. Both read this one setting, so a colour chosen here makes them the same colour.
Retention time	The identity line $y = x$, where predicted and observed retention time agree. Drawn only with the page's **Add Line** switch on. Red, dashed, 1.5 on the static plot and 2.0 on the interactive one.
Pearson correlation	The least-squares fit through the points, in #475569, solid, 2.0. Drawn by the interactive version, from two points or more.

> **NOTE**
>
> Reference lines have different meanings. The Q-Q and Pearson lines are fitted to the points; coverage is a count derived from the data; volcano and Kinase Library thresholds reflect the chosen cutoffs. Hiding any of these changes the drawing, not the calculation. Explain the relevant reference in the caption when it is omitted.

On the Pearson correlation page the card is offered only with **Type** set to Interactive, which

is the renderer that fits and draws the line; the static figure draws none.

The phosphoprotein regulation plot is not offered this card: neither of its renderers draws a threshold line.

The imputation tab has no appearance panel, by design: its figures are diagnostics of the imputation rather than figures to publish, so nothing reaches the reference-line settings its fit plot would otherwise read.

53.3 Why you might turn one off

A threshold line earns its place on screen, where it is how a reader tells the four quadrants of a volcano apart while looking for a feature. In a figure for print it is often the noisiest object in the plot: the caption already states the thresholds, the colour of the points already says which class each one is in, and the line crosses the densest part of the cloud. **Show** off leaves the classification and the colours exactly as they are, and removes only the lines.

The opposite case is a figure that will be read at a small size, where the colours are hard to separate and the line is what makes the boundary legible. There, widen it rather than removing it: a dashed line at 1.5 survives a two-column reduction that a 0.8 line does not.

> **WORTH KNOWING**
>
> A dotted or dash-dot line survives greyscale printing better than a coloured solid one, because its identity is its pattern rather than its hue. Where a figure has both a fit and a threshold, giving them different patterns is not possible — one setting covers both — so distinguish them in the caption.

Behind this card

Function	What it produces
ReferenceLine	The resolved line: whether it is drawn, its colour, its dash pattern and its width
resolve_reference_line	The panel's choice, else this plot's own
ReferenceLine.matplotlib	The line as static plotting arguments
ReferenceLine.plotly	The same line for an interactive figure

Reference lines

54
Axes, grid and background

The frame around the data and the paper behind it.

AT A GLANCE

Cards	**Frame and grid**, **Background**
Where	Every plot except the volcano tile drawn in the browser on the analysis canvas
Starting values	The plot's own, which differ between the static and the interactive renderer
Not here	Tick label size, rotation and colour, which are in the two axis cards; and axis limits, which no control sets

54.1 The axis frame

Sides is four checkboxes — **Left**, **Right**, **Bottom**, **Top** — one per spine. **Color**, **Width** and **Style** apply to the spines that are drawn. The tick marks follow the frame's colour; with every side cleared the ticks lose their length as well, so an axis switched off leaves its labels and nothing else.

Left and bottom spines are often enough to carry a chart's scale; check the target publication's figure guidance rather than treating this as a universal journal rule. Four sides are worth keeping where the figure is a matrix or a grid of panels and the box is what separates them.

Axis frame

Control	What it does	Default	Where
Sides	One checkbox per spine.	all four on a static plot, none on an interactive one	appearance
Color	Colour of the spines and of the tick marks.	#111827	appearance
Width	Width of the spines.	1	appearance

Axis frame (continued)

Control	What it does	Default	Where
Style	Solid, Dashed, Dotted or Dash dot.	Solid	appearance

54.1.1 What each plot starts from

Starting values for the frame and the grid

Column	What it holds
Static plots	All four spines, no grid. The grid, when it is switched on, is horizontal, dashed, 0.8.
Interactive plots	No spines and a grid in both directions, in #EBF0F8, solid, width 1 — the interactive renderer's own look. The volcano, the retention-time plot and the Pearson correlation each have both renderers, and the panel follows the one on screen.
Single-protein line plot, static Pearson	All four spines and a grid in both directions, solid.
Missing value plot	All four spines and a horizontal grid in #b0b0b0, dashed, 0.8.
Phosphosite plot	Left and bottom spines only, no grid.
STY plot	Left and bottom spines, with a horizontal solid grid.
Heatmaps and the Venn diagram	No spines and no grid: there is no scale to carry.

EASY TO GET WRONG

An untouched setting leaves the renderer's own choice in place. Changing the title or a font does not also impose a new frame. Frame and grid settings are independent: change the part you intend to change and inspect the exported figure at its intended size.

54.2 The grid

Show draws the grid or removes it. **Direction** chooses which lines: Horizontal draws them across the y values, Vertical across the x values, Both draws both. **Color**, **Width** and **Style** are the lines themselves.

Changing a grid colour or line style does not itself turn a hidden grid on. Use **Show** to choose visibility; the other controls style it.

Grid lines

Control	What it does	Default	Where
Show	Draws the grid.	off on a static plot, on in an interactive one	appearance
Direction	Both, Vertical or Horizontal.	Horizontal	appearance
Color	Colour of the grid lines.	#d1d5db	appearance
Width	Their width.	0.8	appearance
Style	Solid, Dashed, Dotted or Dash dot.	Dashed	appearance

A grid is read against the marks over it, so it has to be the fainter of the two: light grey, thin, and in one direction only. The direction that helps is the one that carries the quantity — horizontal lines on a bar or box plot, where the reader is comparing heights — and the other direction is usually noise.

54.3 Background and themes

Theme sets the paper, the text, the axis and the grid colours together; **Color** sets the paper alone and wins over a theme applied before it; **Transparent** removes the paper entirely.

The six themes

Column	What it holds
Default	No theme. Every colour stays as the plot drew it, and nothing on this card is applied but what is set explicitly.
White	White paper, near-black text and axes, a light grey grid. The neutral choice for a figure going into a document.
Minimal	The same, with a paler grid — close to a figure with no grid at all.
Soft gray	An off-white page behind a white plot area, slate text and a mid-grey grid: the plot area is separated from the page around it.
Presentation	White, with the darkest text and axes of the six. For a figure that will be projected.
Dark	Dark paper, light text, a dark grey grid — for a dark slide or a dark interface, and not for print.

Background

Control	What it does	Default	Where
Theme	Default, White, Minimal, Soft gray, Presentation or Dark.	Default	appearance
Color	The paper colour, overriding the theme's.	#ffffff	appearance
Transparent	Draws no paper at all, so whatever the figure is placed on shows through.	off	appearance

EASY TO GET WRONG

A transparent background is right on a slide of any colour and wrong wherever the export cannot carry it. PNG, WEBP and PDF keep the transparency; a JPG has no transparency to keep and is flattened onto white, which is the same as not having asked. A dark theme with a transparent background is the trap: light text on no paper, dropped onto a white page, is unreadable.

54.4 Limits and ticks

The panel sets no axis limits. Each plot chooses its own range from the data it draws — the coverage plot, for instance, ends its y axis a tenth above the tallest bar or the reference line, whichever is higher — and the panel styles that range without changing it. Where a range is a scientific choice rather than a cosmetic one, the control for it is on the page: the correlation plot's **Use Full Range** switch is one.

Tick *labels* — their size, rotation and colour — are in the **X axis** and **Y axis** cards, beside the axis label they belong to, and are described with the other text settings. Tick *marks* have no controls of their own: they follow the frame's colour, and they disappear with it.

When comparing exported panels, check their axis ranges rather than assuming that equal-sized pictures use equal scales.

Behind these cards

Function	What it produces
_apply_advanced_ matplotlib_style	Theme, background, grid and spines of a static figure
_advanced_plotly_layout	The same settings as an interactive figure's layout
_advanced_theme	The six themes and the colours each one sets
_advanced_axis_sides	Which of the four spines are drawn

__mpl__grid__line__style	Dash pattern names to the static renderer's own

Axes and background

55
Figure size, resolution and format

Getting a figure out of the tool at the size the destination wants.

AT A GLANCE

On the page	**Width**, **Height**, **DPI**, the format selector and **Download Plot**
In the panel	Everything that decides what is inside the canvas, and the background the file is written on
Formats	PNG, JPG, WEBP, SVG, PDF
Limits	1–60 cm per side, 72–600 DPI, and at most 24 megapixels in total
Interactive plots	Opened in their own tab rather than downloaded

55.1 Size and resolution

Width and **Height** are in centimetres and **DPI** is the resolution the figure is rendered at; all three are on the page, beside the plot, and not in the appearance panel. Together they are the canvas: a 20 cm by 10 cm figure at 300 DPI is 2362 by 1181 pixels.

Each is bounded. A side is held between 1 and 60 cm and the resolution between 72 and 600 DPI, and the resolution is capped a second time so that width times height times DPI stays under 24 megapixels — so asking for 600 DPI on a 60 cm figure quietly gives a lower resolution rather than a request nothing can serve.

The figure on screen is drawn at exactly these numbers and then scaled by the browser to the width of the panel it sits in, which is why changing the width appears to do nothing on screen and changes everything in the file. What it changes is the *proportion* between the canvas and the text: sizes in the appearance panel are in points, which are absolute, so widening a figure from 10 to 20 cm halves the relative size of every label on it.

EASY TO GET WRONG

Set the size first and the text sizes afterwards. A figure that is readable at 20 cm and then reduced to a single column is a figure whose 10 pt axis labels are now 5 pt; the

same figure rendered at 10 cm with 10 pt labels is correct. The order matters because the panel's sizes do not scale with the canvas.

WHEN THE LABELS DO NOT FIT

The renderer measures labels and adjusts the layout, enlarging a crowded figure within safety limits when necessary. Saved dimensions can therefore exceed the requested starting size. If labels still cannot fit, the page reports an error instead of silently clipping them. Long sample names, large text and an outside legend are common causes: shorten display labels, increase the size or reduce text size, then inspect the result at its intended publication size.

55.2 What the panel adds

The panel holds nothing that sets the canvas, and everything that decides what is drawn on it. Three of its settings change the file rather than only the picture:

- **Transparent**, in the **Background** card, writes the figure with no paper behind it.
- **Color** and **Theme**, in the same card, decide the colour of the margin around the axes as well as the plot area, so they decide what a figure pasted onto a white page or a dark slide sits on.
- The **Font** card decides the face the text is drawn in, which matters most in a PDF, where the text stays text.

Everything else — sizes, offsets, colours, marks, lines, frame, grid — reaches the download because the download is the same render as the figure on screen. There is no separate export appearance to keep in step.

55.3 Which format

55.3.1 Downloading reproducible code

Eligible plots and processed tables offer a notebook ZIP beside their download or open control. It contains a Python notebook, the helper functions needed for that result, and required resources. Extract the whole ZIP before opening the notebook. Enter the path to the original

input file at the first placeholder; its filename is supplied as a reminder, but the original local path is not exported. Run the cells in order in the documented environment.

For a result that actually uses imputed values, the archive includes the exact matrix used and records the imputation settings and tool version. The documented imputation call explains the method; the saved matrix avoids depending on a fresh random draw. A plot and its associated processed table offer the same recipe, including both outputs.

This is not a blanket export of every screen. The distribution Q-Q plot and imputation diagnostics have no downloads; the replicate and variance projection (a what-if projection, not a power calculation) and the analysis canvas have no notebook export. The Kinase Library notebook recreates the prepared input table, not an enrichment performed by an external website. A full-report notebook is described in the Report chapter. Filenames and sample names remain in the archive, so removing paths does not anonymise the scientific data.

The five download formats

Column	What it holds
PNG	The default, and the same image as the one on screen. Lossless, with transparency if it was asked for. The right choice for a slide or a document that will not be typeset again.
JPG	Compressed at quality 95 and flattened onto white, since the format has no transparency. Smaller, and the only one of the five that loses information. Use it where a size limit forces it.
WEBP	Lossless, with transparency, and smaller than the same PNG. Useful on the web, and refused by some journal submission systems.
SVG	Drawn as SVG rather than traced from an image: every mark is an object a drawing program can select, and the text is text. The right choice for a figure that needs one last edit by hand.
PDF	Drawn as a PDF rather than converted from an image: the marks are vector, the text is text, and the fonts are embedded in the file. The right choice for a figure that will be scaled or printed.

Choosing SVG or PDF therefore does more than change the extension: the figure is rendered again, as vector, and the DPI becomes a resolution for anything inside it that has to stay raster. A vector figure can be enlarged without becoming coarse; a PNG at the same size cannot.

> **NOTE**
>
> An interactive plot has no download format at all: it is a web page, not an image, and the server has no way to turn one into a picture. Its own toolbar saves a PNG. The volcano tile on the analysis canvas is the exception — it is drawn by the browser and offers **Download SVG** of its own.

55.4 How a downloaded figure differs from the one on screen

- It is the same render. The image on screen was requested at the width, height and resolution on the page and is merely displayed smaller; the file is that image at full size.
- It carries the format's own properties: the transparency PNG, WEBP, SVG and PDF keep and JPG does not, and the vector marks SVG and PDF have and the raster formats do not.
- It carries no interactivity. An interactive plot has no download button at all — **Open Interactive Plot** opens it in its own tab, where the plotting library's own camera button saves a PNG of the current view. To download a publication figure of an interactive plot, switch that plot to its static mode first.
- The height of an interactive plot on the page follows the page's **Height** in centimetres, converted at 96 pixels per inch with room for its toolbar, and has nothing to do with any export.
- The table under a plot is downloaded separately, as CSV, and is unaffected by everything in this chapter — including the condition and sample names renamed in the appearance panel, which are a property of the figure and not of the data.

Behind the export

Function	What it produces
to_png_bytes	Writes the figure in the requested format, checks that its labels fit the canvas, and keeps transparency when it was asked for
cm_to_inch	Centimetres to the inches the renderer works in
cm_to_px	The same for the pixel size of an interactive figure
normalizePlotSizeParams	Holds width, height and DPI inside their bounds and caps the total pixel count
_convert_png_bytes	Converts a rendered PNG to JPG or WEBP

PART XII

Assistant and external tools



56
The assistant

What the assistant can do to the interface, what it will not do at all, and what leaves the machine when it answers.

AT A GLANCE

Where	A **Chatbot** button at the bottom right, in an assistant-enabled build of either application. Other builds have no such button.
Providers	A local model through Ollama, the default; a local server with the OpenAI-compatible API (llama.cpp, vLLM, LM Studio); or OpenAI, Anthropic or Google Gemini with your own API key.
Can change	The view: the open page, a plot control, the appearance panel, canvas tiles. At most eight changes in one request, and one that affects a result has to be approved first.
Cannot change	The data or the processing. No action uploads, filters, imputes, collapses, saves, restores or exports.
Answers	Assembled by the application from evidence it computed. The model chooses which sentences are used; it does not write the numbers in them.
New	Not covered by any previous manual.

56.1 Where it lives

The button at the bottom right opens a panel down the right-hand side of the window, holding the conversation, the model in use, a capability indicator for it and **AI Settings**, collapsed until it is asked for. **Enter** sends, **Shift** + **Enter** starts a new line, and **Send** becomes **Stop** while a request runs — stopping leaves whatever had already been applied in place. The conversation belongs to the panel, not to the analysis: it is not written into a saved state, and changing the provider, the model or the endpoint clears it.

It reads the controls of the page in front of it, and only the analysis sections register them: **QC Pipeline**, **Completeness**, **Statistical Analysis**, **Peptide Level**, **Single Protein**, **Phospho-specific**, **Comparison**, **Analysis** and Data → Distribution. Elsewhere the panel still opens and still answers, but sees no controls and changes nothing on that page.

TRY A SHORT CONVERSATION

Start with "What does this plot show?" Then ask "Make the legend smaller" or "Show protein P12345", replacing the identifier with one in your dataset. Check the actual view after each request. A description of an action is not evidence that the action happened; use the receipts and the changed plot to confirm it. Ask for supporting results when an answer makes a numerical claim.

56.2 What it can do to the interface

Five kinds of change exist, and there are no others.

- **Open a page** that is in this build and this session; the destinations are the sidebar as it currently stands.
- **Set one control** the page has registered, to a value that control accepts: a number inside its own minimum and maximum, a six-digit hex colour, an option that exists in that list.
- **Open the plot appearance panel** — what a request for a rotation, border, font or colour produces while it is closed. It also opens by hand with **Ctrl** + **Shift** + **E**.
- **Search a long option list.** A list of thousands of features is never sent whole: it is searched for a fragment, and at most forty matches come back with the number of matches there were. The view does not change.
- **Add or remove a canvas tile**, in **Analysis** only. A request for everything on one feature is built by the application itself: it resolves the identifier against the uploaded data, keeps existing tiles, and draws a highlighted volcano, a boxplot per feature and a shared lineplot.

The first, third, fourth and fifth of those each stand alone as a step, because the next step has to see the interface the previous one produced; only plain control changes are batched, and one control is never set twice in a batch. A request is bounded at eight applied changes, each reported above the answer in the tool's own words, so a request that half succeeded says which half.

56.2.1 What it asks you first

A change to a setting that affects a result opens a confirmation listing every such change as *control: old value → new value*, stating that results may be recalculated and that uploaded data and processing settings will not be changed. Nothing is applied until it is approved, declining ends the request with the applied changes listed, and if the view moves between the listing and the approval the request is abandoned rather than applied to a page that has changed.

Choosing *what* is shown rather than how — the protein or site, the conditions, the samples, the dataset level, the label column — is a selection rather than a setting and is applied directly, as

are the four actions that change no result. Adding canvas tiles is applied directly, including a request for several at once, though pressing the same add button a second time in one request asks; removing a tile always asks, by name.

56.3 What it will not do

- **Anything to the data.** No action exists for uploading, filtering, imputing, collapsing, translating, annotating, saving, restoring, exporting or running an analysis. The only buttons it can press are the add and remove buttons of the canvas tiles.
- **A scientific choice on your behalf.** Test type, data source, thresholds, pairing and transformations change only when you ask for that exact setting and value.
- **Improve a result.** A request to make the results more significant is refused before any model sees it: it will change the presentation, but it will not choose statistical settings to make results look more significant. These safeguards are not a guarantee of error-free interpretation: check the reported actions and the underlying results, especially before publication.
- **Rename anything real.** Renaming a sample means its display label; uploaded identities are untouched.
- **Invent an identifier.** One that matches nothing at the requested level is reported as matching nothing; one that matches several is answered with the first few and a request to choose. A protein is never silently expanded into its sites, and one data level is never substituted for another.
- **Answer and change at once.** A question is answered with no actions. A request that mixes a change with an explanation is done in that order and explained from the view *after* the change; asking for the explanation first produces a question about which order you meant.

56.4 Controls

The panel itself has four controls and none of them is a setting: **Chatbot** opens and closes the panel, **Send** sends and becomes **Stop** while a request runs, **Supporting evidence** under an answer opens what that answer was assembled from, and **AI Settings** shows the table below. Everything that is remembered between requests is in that table.

In AI Settings

Control	What it does	Default	Where
Select chatbot provider	Unlabelled list: Local (Ollama), Local server (OpenAI-compatible), OpenAI (ChatGPT), Anthropic (Claude), Google (Gemini). Only the providers the server allows are offered.	Local (Ollama)	page
Ollama endpoint	Where the local provider runs. Empty means Ollama on this machine; an address is a machine you nominate, and whoever deployed the instance can restrict which are allowed.	empty	page
Select chatbot model	Unlabelled list of the models that endpoint, or the local server, reports.	the first model reported	page
Model Management	A model name, **Install** and **Remove**, for the endpoint in use. Installing or removing a model changes the whole server, so this block is hidden where the server does not allow it — by default on any deployment with tool passwords, a viewer or a viewer portal.	—	page
GPU Acceleration	On if the machine reports a usable CUDA device. Locked off for a nominated remote endpoint, whose acceleration this machine cannot know.	off unless a device is detected	page
Assistant level	Auto follows the model's level, shown in brackets; Basic, Standard and Advanced override it for this session. See *Assistant levels* below.	Auto	page
Local AI server address	Local server only. The address of a server that speaks the OpenAI-compatible API, such as http://192.168.1.20:8080/v1; pre-filled when the deployment names one. On a desktop installation any address works; on a shared deployment only the addresses its administrator allowed.	the deployment's server, or empty	page
Local AI server API key	Local server only, and only if that server requires a key.	empty	page
API key	Cloud providers only. Used in the provider's authentication header and nowhere else; never part of what is sent as context.	empty	page
Model	Cloud providers only. Free text, pre-filled with a default for the chosen provider.	the provider's default	page
Revoke consent	Cloud providers only, once consent has been given.	—	page

56.4.1 Assistant levels

The line above the settings reads Assistant level: Basic, Standard or Advanced, with where the level comes from: your choice, configured (the administrator's COPYLOT_AI_MODEL_LEVELS), observed (a model seen in this project's real conversations), inferred (from the model's family, generation, size, context window and quantization) or unknown. Each level gives a model the part of the toolbox it handles well; the rules below are the same at every level.

- **Basic.** Small models (up to about 6 billion parameters, older generations up to 14) and models that cannot be recognised. One small step at a time: two earlier messages, up to four control changes per step and eight per request, a styling list narrowed to the parts the question names, and answers assembled from the application's own sentences. Journal styles and "make it look better" are applied by the application, not designed by the model.
- **Standard.** Mid-sized models such as qwen3.5:9b, ministral-3:8b and gemma4:12b. More conversation and evidence, the same safe answer sentences.
- **Advanced.** Large models (about 60 billion parameters and more, the providers' flagship cloud models, glm5). Up to 30 visual changes in one step and 40 per request; the full styling list with current values; the journal presets; its own design within colour-blind-safe palettes; read-only lookups before it plans; its own written answers, whose numbers, names, colours and comparisons the application checks; and, for a model that reads images, a picture of the plot as it is drawn, with one look at the result after its changes to fix what it sees. On a server that supports tool calls it uses the assistant as tools; with a large context window it keeps up to twelve earlier messages and reads up to 40,000 characters of evidence per source.

Models that think before answering (glm, gpt-oss, DeepSeek-R1, OpenAI's o-series and GPT-5, Gemini 2.5) are given extra room for that thinking, since it counts against the length of their reply; a reply that still runs out is reported as such and nothing is changed. For a styling request an Advanced model is asked to think at high rather than its maximum effort where its server allows it, and while a slow model works the chat shows how far it is: thinking, the tokens so far and the seconds elapsed.

> **WHAT NO LEVEL CAN DO**
>
> How the data are produced — filtering, normalization, imputation, transformation and the phosphosite localization cutoff — is never shown to a model and never changed by the assistant, with or without your approval, whether the model plans in one reply or calls tools. The application removes these settings from every request, refuses any change that names one, and the browser refuses it once more. Choosing which processed matrix a plot shows (its data source) stays a view choice that asks for your approval. Asked to change the pipeline, the assistant says so and leaves it to you.

That boundary is deliberate. The assistant changes what a plot looks like and where you are, which is cosmetic and reversible; it does not touch the pipeline — upload, annotation, filtering, imputation, translation and restore are the user's choices and stay so.

56.4.2 Journal styles and design

"Nature style" or "format this for a Cell paper" applies a preset built from the journal's own figure guide, and the answer names the source and which parts were set: sans-serif text, the guide's text sizes, its single-column width (double or 1.5 columns when you say so) with the height scaled to match, at least 300 dpi, and, for Nature, no background gridlines and a colour-blind-safe palette. Panel letters are added when a figure is assembled and are not set here. A journal without a checked guide (Science, PNAS, ACS, ...) is named as such rather than guessed; you are offered Nature, Cell Press or a general publication style, which is labelled as a suggestion. "Make it look its best" gives a Basic or Standard model that general style; an Advanced model designs the figure itself, chooses colours only from the colour-blind-safe palettes unless you name colours, keeps the figure size unless you mention it, and explains its choices in a sentence or two.

"Save this style as lab figure" keeps the open plot's look under that name for this session; "use my lab figure style" applies it to the open plot and "apply this style to all plots" (or "make all my plots Nature style") to every plot: those already styled get it now, the others start from it when opened. Only appearance moves: titles, axis texts and legend titles stay with their plots, and no data source, selection or setting of an analysis is part of a style.

56.5 How an answer is put together

An answer about your results is not written by the model. It is assembled by the application from evidence the application computed, and the model's part is to choose which of it answers the question.

THE FOUR STEPS

1. The request is classified into one kind of work: explain the current plot, review the quality diagnostics together, look up one feature, describe what is loaded, answer a method question without touching the data, change the view, or ask you something back. The choices are narrowed by the wording first, so a question cannot be classified as a change.
2. The evidence for that work is computed by the adapter of the page in question, against the loaded data. A page with no adapter says so rather than being answered from a neighbouring one, and a page answered with its default settings rather than your selections says that in the answer.
3. The evidence becomes numbered sentences, and the model is asked for a list of their identifiers rather than for prose, so every number, unit, identifier and condition name in the rendered answer comes from the computation. An Advanced

model writes its answer itself instead; every sentence whose numbers or identifiers do not appear in the computed evidence, the question or the visible settings is removed, and the answer says how many were. When nothing survives, the numbered sentences are used.

4. Coverage is checked. If the question was about the direction of a contrast, the scale, the axes or causality, a sentence on that topic must be present: where one exists it is added, and where none exists the answer states that the evidence does not resolve it.

WHAT THE MODEL MAY WRITE, AND WHAT IT MAY NOT

One thing: a hypothesis, at most two, and only where the question invites one. It may contain no number and no reference to a value, and is printed with a prefix stating that it is one possible explanation and is not established by these results. A comparison between two values is not free text either: the two must be the same metric, from the same source, under the same settings, for two different subjects, and both finite. A reply that does not fit is asked for once more with the problem stated, and a second failure produces a plain summary of the evidence that says so in its first line.

56.5.1 Edge cases

- Some requests never reach a model: a plain request for a number of canvas tiles is parsed by the application, a question about how to build a canvas is answered with fixed text, a greeting or "what can you do?" gets a short answer, and "make the bars black" sets every condition colour the bars use. Styling applies whether or not the appearance panel is open.
- At most eight canvas tiles are inspected, each on its own terms. Tiles are never combined into one conclusion, and a collapsed tile is reported as collapsed rather than read.
- An enrichment result or an uploaded external Kinase Library result is evidence only while its stored result lives — one hour, and only until the analysis changes — after which the answer asks for it to be run again rather than quoting stale numbers.
- Peptide data has evidence adapters only for the peptide pages, the completeness pages and coverage. Another page asked about peptide data reports that it has none rather than answering from the protein matrix.
- An analysis that changes while a request is running abandons it with a message. Where only part of the evidence fitted the budget, the answer says it is not an exhaustive assessment.

EASY TO GET WRONG

The numbers in an answer are the tool's own, but the *selection* of them is the model's, and a small model selects badly more often than a large one: it can answer a narrower question than you asked, or give the method where you wanted the result. Read the

> receipts first — a control changed two steps earlier means the explanation describes a different figure from the one you asked about — then the evidence disclosure, which shows what was available and what was used. Anything that will go into a paper should be read off the plot and its table.

56.6 What leaves the machine

Ollama processes requests on its configured server. A blank endpoint field does not prove that this server is your computer: a deployment can set its default endpoint centrally. When the server really is local, AI inference does not require sending the context to a cloud model provider. Filling in an endpoint sends the same context to the machine named there; it is not a cloud provider, so no dialogue appears, and a nominated host that is not yours is a third party all the same.

A local server with the OpenAI-compatible API is treated the same way: it receives the full context and no dialogue appears. On a shared deployment only addresses the administrator allowed are reached, so a visitor cannot send its data elsewhere. Such a server fixes its context size when it is started; the assistant reads it where the server reports it (llama.cpp does) and otherwise uses the deployment's setting. GPU use and the installed models belong to that server, so neither control is shown for it.

The first request to OpenAI, Anthropic or Google Gemini in a browser session stops and asks, before anything is sent, with two lists side by side.

- **Sent:** your prompt and the visible conversation history; generated page and project context, including dataset names and dimensions; relevant protein or site identifiers, sample and condition labels, and calculated summaries; the available tabs and plot controls, their selected values, and the receipts for requested view changes.
- **Not attached:** uploaded files and the full raw or processed abundance matrices; local file paths and unrelated datasets; the API key, which is used only in the provider's authentication header.

The dialogue names the provider, states that it processes the context under its own data terms, and offers three ways out: confirm and send, cancel, or switch to the local provider. Consent is given per provider and per browser session, withdrawn with **Revoke consent**, and asked for again rather than carried over if the wording of the disclosure changes.

What is sent is the described interface and the computed summaries, not the dataset. An identifier, a condition name and a column label are *part* of that description, so a study whose sample labels or feature identifiers are themselves confidential needs an approved provider and deployment. Choosing the label *Ollama* alone does not establish where that provider runs or who can access it.

Behind this page

Function	What it produces
assistant_step	One step of a request: plan, validate, act or answer
plan_request	Builds the planner turn, with the schema of legal actions derived from the live controls
scientific_evidence	Computes the evidence for the chosen kind of work
answer_catalogue	Turns evidence into the numbered sentences the model may choose from
assemble_answer	Renders the chosen sentences, checks coverage, appends the qualifications
classify_model	The capability estimate and the budgets that follow from it
cloud_ai_consent_status	What is sent, what is not, and whether consent has been given
runAssistant	The browser side: reads the view, asks for approval, applies changes, collects the receipts

57 GONetView

A separate web application for drawing Gene Ontology term networks, and what does and does not pass between it and this tool.

AT A GLANCE

Where	External Tools → GONetView. In the DataViewer it is the only page of that section.
What it is	A web application in its own right, maintained separately from this tool and published with a DOI and an open source repository. It is not part of the analysis, and nothing passes between the two on its own.
Sends	Nothing. This page holds a description, a picture and three links; it reads no dataset and transmits no result.
Needs	A gene list, which you take there yourself.
Returns	Nothing to this session. What you make there you save there.
New	Not covered by any previous manual.

57.1 What the page is

It is a card, not a tool. The page describes GONetView — an application for visualising Gene Ontology term networks and for seeing how genes connect to biological processes, molecular functions and cellular components — lists what it is good for, shows a picture of it, and offers three links: **Open GONetView**, **DOI** and **GitHub**. The picture is a link to the same application. Under them is the citation to use if a figure made there ends up in a paper, with the software's version and its DOI.

Each link opens in a new browser tab and tells the destination nothing about where it was opened from. Nothing else happens: no request carries your data, because the page has none — it does not read the uploaded datasets, the annotation or any result.

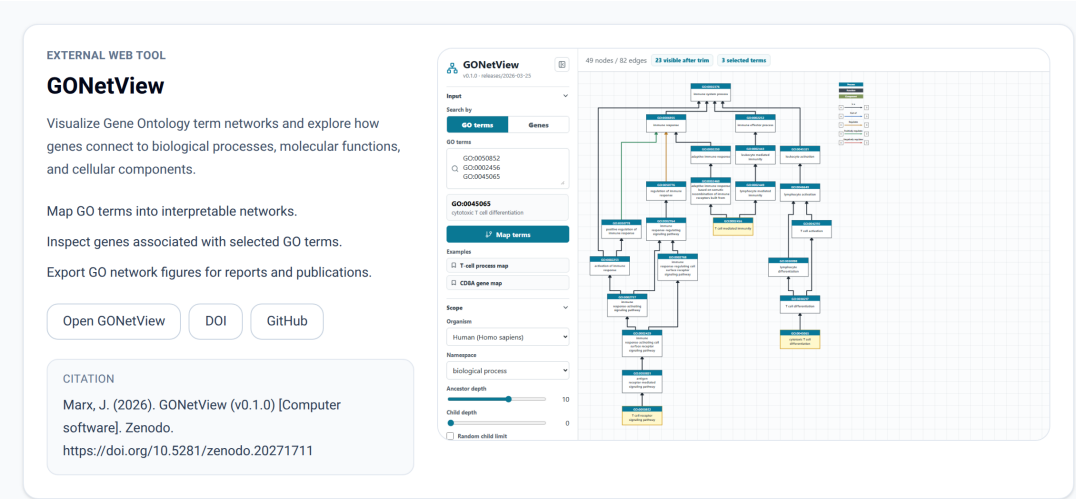


Figure 57.1. The GONetView page: what the tool does, the three links out to it, the citation to use, and a picture of the application itself — which is also a link. Nothing on this page sends anything anywhere; opening a link is what starts a session with the other application.

> **NOTE**
>
> The line between the two applications runs through your browser, not through the tool. Opening the link starts a session with somebody else's application: what you paste into it is sent to whoever hosts it, under their terms, and this tool is neither party to that nor aware of it.

57.2 Taking a list there

Nothing is handed over automatically, so the list has to be carried.

1. Produce the genes in the tool. The usual source is Statistics → ORA, which downloads its up and down gene lists as plain text and its full enrichment table — terms, sizes, overlaps, p-values and the intersecting genes themselves — as CSV.
2. Open GONetView from this page. The tool goes on running behind it; nothing in the session changes, then or later.
3. Paste or upload the list there, and work in that application.
4. Save what you want to keep from it. A figure made there is a file on your disk; it does not come back into this analysis, and no page of this tool will show it.

EASY TO GET WRONG

The enrichment and the network answer different questions. ORA tests whether a term is over-represented in one list against one background, with a one-sided hypergeometric test and a Benjamini–Hochberg correction, and its p-values depend on which background was chosen. A network drawing of terms and their genes does no test at all: an impressive-looking network of terms can be drawn from a gene list that is not enriched for any of them. Take the statistics from the ORA table and the picture from the network, never the other way round.

57.3 If it cannot be reached

The page itself is part of the tool and always opens, with its text, links and citation intact. Two things depend on the outside:

- The picture is fetched from the repository that hosts GONetView. Without a network, or behind a firewall that blocks that host, the frame stays empty. Nothing else on the page depends on it.
- The links are handled by the browser. If the application cannot be reached, the browser reports that, in its own words, in the new tab. The tool learns nothing about it and reports nothing.

No analysis, no plot and no download in this tool depends on GONetView being reachable. An instance installed on a machine with no internet access works exactly as it does otherwise, minus the picture and the links.

GONetView takes Gene Ontology terms, GO identifiers or a gene list.

Carrying the list across by hand is the intended route: GONetView is a separate tool, and this page is a pointer to it rather than an integration.

58
Peptide collapse

Turning a Spectronaut precursor report into the wide phosphosite matrix the rest of the tool expects, without touching the session.

AT A GLANCE

Where	External Tools → Peptide Collapse. Not in the DataViewer, which keeps only **GONetView** in this section.
Takes	One Spectronaut report, tab-separated, chosen from disk. It is not one of the four uploaded datasets and it does not become one.
Gives	One tab-separated file, named after the input with `_collapsed.tsv` in place of its extension, offered as a download as soon as the run finishes.
Runs	Inside the tool, on the machine running it. Nothing is sent anywhere, and nothing is kept: the server's working copy is deleted when the download is served.
Changes	Nothing in the session. The collapsed file re-enters the analysis only when you upload it yourself, as the phospho level.
One setting	**Localization cutoff**, 0 by default. 0.75 is the Class I convention.

58.1 What it is for

Spectronaut writes one row per precursor per run, and a phosphosite is not a precursor: the same site is measured through several charge states, several missed cleavages and several co-modified forms, each of them a separate row. Every phospho page in this tool reads a wide matrix whose rows are sites, so something has to decide which precursors are the same site and what one number for that site in one run is. That is this page, and the reason it sits under **External Tools** rather than under **Data** is that it works on a file rather than on the loaded analysis: it reads nothing from the session and writes nothing to it.

The collapse is therefore a step *before* the tool, run from inside it. The output is an ordinary file on your disk, and it becomes data only at Data → Upload.

KEEP THE TWO STEPS SEPARATE

Choose the precursor report, set the localisation cutoff, and run the collapse. Download the resulting site table and inspect its identifiers and sample columns. Upload that table as phosphosite data to begin the analysis; finishing the collapse does not itself replace a loaded dataset.

NOTE

This is the collapse that produces phosphosites from precursors. Collapsing phosphosites into phosphoproteins is a different operation on an already uploaded dataset, with its own four methods, and is at Data → Annotation under **Aggregate to Phosphoprotein**.

58.2 Running it

1. Choose the report under **Spectronaut TSV file**. The file dialogue is restricted to .tsv and .txt; the file is read as tab-separated whichever of the two it is. Its size is bounded by the same upload limit as a dataset, which whoever deployed the instance can raise.
2. Set **Localization cutoff**. It is a probability between 0 and 1; anything outside that range is clamped as you type and refused by the server if it arrives anyway.
3. Press **Run Peptide Collapse**. The button reads Running… until the file comes back.
4. The download starts by itself. The same file stays available on the page as a link until you choose another input.

Controls on this page

Control	What it does	Default	Where
Spectronaut TSV file	The report to collapse. Choosing a new one clears the previous result.	none	page
Localization cutoff	The minimum site localization probability a row must reach to survive. 0 keeps every localized site; 0.75 is the Class I convention.	0	page
Run Peptide Collapse	Runs the collapse and downloads the result. Disabled until a file is chosen.	—	page

58.3 How it is computed

THE COLUMNS IT REQUIRES

Seven columns must be present, by these exact names: R.FileName, EG.PrecursorId, EG.TotalQuantity (Settings), PEP.PeptidePosition, EG.PTMAssayProbability, PG.Genes and PG.ProteinGroups. Each one is reported as present or absent in turn, and the first absent one stops the run by name. No column is guessed from a synonym here — this is the search engine's own report, not the tool's canonical naming.

FROM A PRECURSOR TO A SITE

Rows whose EG.PrecursorId carries no [Phospho (STY)] marker are dropped, and their count is reported. The precursor identifier is then reduced to a bare sequence: the charge suffix, every bracketed annotation and every underscore are removed.
A precursor carrying *n* phospho markers becomes *n* rows, one per modified residue. The position of each within the peptide follows from the lengths of the pieces between the markers, and the residue at that position — S, T or Y — is read off the bare sequence. The peptide's own start in the protein comes from PEP.PeptidePosition; where that field lists several positions, separated by semicolons or commas, the first is used. The site position in the protein is then

$$\text{site} = \text{position in peptide} + \text{peptide start} - 1.$$

Multiplicity is the number of phospho markers on the precursor, capped at 3: a precursor with four sites is counted as M3.

THE COLLAPSE KEY

Every derived row gets the key that defines the site:

PG.ProteinGroups~PG.Genes_<residue><position>_M<multiplicity>

which for the first site of a singly phosphorylated peptide reads P00001~GENEA_S123_M1. Rows sharing a key are the same site measured through different precursors, and they are what is collapsed.
The underscore is a separator of that key, so a gene name containing one would split it. Such names are rewritten with # before the key is built and rewritten back to _ afterwards, in both the key and the gene column; the count of affected names is reported.

THE QUANTITY

EG.TotalQuantity (Settings) is pivoted to one column per run, taking the first value for each precursor-and-position. Every column is then coerced to a number, so a cell Spectronaut wrote as Filtered becomes NA rather than stopping the run; a cell written as 0 is replaced by NA as well, because it is a sentinel for a measurement that was not made. Only then are the values taken to $\log_2$.

The order matters in both directions. Taking $\log_2$ first would turn a sentinel 0 into $-\infty$, which must not be treated as an observation; deleting zeros *after* the transform would delete a genuine intensity of 1, whose $\log_2$ is 0.

The collapsed value is then the *median of the* $\log_2$ *values* of the precursors sharing a key, per run — not the $\log_2$ of their median. A run in which no precursor of that site was measured keeps its column and holds NA; a run in which nothing at all was measured still appears, as an all-NA column.

LOCALIZATION AND THE CUTOFF

EG.PTMAssayProbability is reduced twice: first to the maximum across runs, then to the maximum across the precursors sharing a key. The site's reported PTM_localization is therefore the best evidence any run produced for it, not an average and not the value in the run being read.

Rows are kept where that value is at least **Localization cutoff**. A row whose probability is missing everywhere fails the comparison and is dropped, at every cutoff including 0.

58.3.1 Edge cases

- A row is discarded before collapsing if its peptide position is absent, its gene is empty, its protein group is empty, or the residue at the modified position could not be read. The count is reported as phosphopeptides filtered for insufficient information.
- Sample columns are named after R.FileName and are ordered as the runs first appear in the report. Renaming samples is the annotation's work, later.
- The page always collapses to protein groups. The function underneath can also collapse to individual proteins, splitting a group into one row per accession, but nothing in the interface selects that.
- Nothing is written back: the input copy and the result are made in a temporary directory that is removed once the download has been served, and both are gone if the run fails.

That is intended: a site with no localization probability anywhere has nothing to be confident about, so it is dropped at any cutoff, including zero.

Collapsing per accession rather than per protein group is not a supported route and the page does not offer it.

58.4 The file it produces

Columns in the collapsed file

Column	What it holds
PTM_Collapse_key	The site: protein group, gene, residue, position and multiplicity, as built above. This is the feature identifier every phospho page uses.
Protein_group	The first accession of the protein group.
Gene_group	The first gene name of the group, with any underscore restored.
UPD_seq	The bare peptide sequence with the modified residue lower-cased and marked with *, so the site can be read in sequence context.
PTM_localization	The highest localization probability observed for that site in any run.
One column per run	The collapsed $\log_2$ intensity of that site in that run, named after R.FileName. NA where nothing was measured.

> **WORTH KNOWING**
>
> Those five annotation columns are exactly the five that **Aggregate to Phosphoprotein** requires before it will collapse phosphosites into phosphoproteins. A collapsed file uploaded unchanged carries them; a file that has been edited in a spreadsheet in between often does not, and the collapse then refuses it by name.

58.5 Getting the result into the analysis

1. Open Data → Upload.
2. Choose Phospho as the dataset type and the _collapsed.tsv file.
3. Upload, then annotate as usual: the run columns are the samples, and the five annotation columns above are not.

The values are already $\log_2$, so tick **Dataset is already log2 transformed** when the annotation is generated. The detector usually reaches the same conclusion from the value range, but the file states it and you know it.

58.6 Exporting

There is nothing to export afterwards — the run *is* the export. The file is tab-separated and written the same way every time, so two collapses of the same report can be compared line for line. The report has to be uploaded; there is no route that collapses a file named by its path on the server. In the DataViewer the whole page is absent.

Behind this page

Function	What it produces
peptideCollapse	Precursor rows to one $\log_2$ row per phosphosite, with the collapse key, the localization maximum and the cutoff
process_and_export	Reads the tab-separated report, collapses it and writes the result beside it
run_peptide_collapse	The page's call: resolves the output name, runs the collapse and reports rows, columns and the log

PART XIII

Administration



59 Settings

Every setting a user can change from inside the application, what it affects, and where it is kept.

AT A GLANCE

Where	The **Settings** entry, last in the sidebar. It is present in the analysis tool and in the viewer alike.
Three cards	**Theme**, **Plot Options** and **R Compatibility**. The first two are settings; the third is a report on the machine the application is running on.
Where it is stored	Appearance preferences belong to the browser. The R status card queries the application server. Browser session storage is not a saved analysis.
Scope	Per browser tab. A second tab, a second browser and a second person each start from the defaults again, and one user's choice is never visible to another.
Not here	Access control, session isolation and the viewer project are set outside the application by whoever started it. See Administration → Access and Administration → The DataViewer.

59.1 What the page is for

Two of the three cards change how the interface behaves for the person looking at it, and nothing else: no analysis result depends on either, and neither is part of a save state. The third card answers a question that is not about preference at all — whether this machine has an R installation the application can call — and it is on this page because that is the only place the answer is reported.

Controls on this page

Control	What it does	Default	Where
Theme	Two buttons, light and dark. The choice is applied to the whole interface at once by setting a class on the document; it changes no plot, because a figure is rendered by the backend and arrives as an image.	light	settings
Advanced plot option mode	The switch in the **Plot Options** card. On, every plot page grows the plot appearance panel described in Part XI. The same switch is bound to **Ctrl / Cmd + Shift + E** from anywhere in the application, and the two stay in step: the setter broadcasts, so the switch on this page follows a keystroke made on a plot page.	off	settings
Refresh	In the **R Compatibility** card. Discards the cached detection result and probes again. Use it after installing R, or after installing the vsn package, without restarting the application.	—	settings

59.2 Where each setting is kept

The distinction that matters here is between a setting that belongs to the installation and a setting that belongs to the browser. Nothing on this page is the former.

- **Theme** is not stored at all. It is applied to the live document and forgotten: the application sets the theme to light every time it initialises, so a reload, a new tab and a restored save state all come back light whatever was chosen before. The theme is deliberately not remembered: every load starts light, and a change to it lasts as long as the tab is open.
- **Advanced plot option mode** is written to the browser's per-tab storage under the key copylot:advancedPlotMode. It therefore survives a reload of that tab, and it does not reach a second tab, a second browser or a second person. In the viewer this key is cleared on every load, so the viewer always opens with the panel hidden.
- The settings *inside* the appearance panel are kept separately, per plot, under copylot:advancedPlotSettingsByKey.v2 — also per tab. Unlike the switch itself, these are part of a save state, so a styled figure can be handed on; see Getting started → Sessions.
- **R Compatibility** stores nothing. It reports what the backend found, and the backend caches that result in memory until **Refresh** or a restart.

> **NOTE**
>
> Per-tab storage is cleared by the browser when the tab is closed, and is not shared between tabs even of the same window. That is the same boundary the analysis itself uses: each tab has its own session on the server, so two tabs can hold two different projects at once.

59.3 R compatibility

The card shows a badge — R available or R not found, and Checking… while the probe runs — and four rows underneath.

- **Version** is the version number parsed out of the probe output, or Unknown if the output did not contain one.
- **Executable** is the full path of the Rscript that answered, or Not detected.
- **Detected from** is how it was found: env from an environment variable, path from the system search path, or common_path from the usual install locations for this operating system.
- **Version output** is what the probe printed, verbatim.

When nothing answers, a panel under the rows gives the reason: each candidate that was tried and why it failed, or Rscript was not found. if there were no candidates at all.

> **HOW THE RUNTIME IS DETECTED**
>
> Candidates are collected in a fixed order and the list is then reduced to the first occurrence of each path, compared without regard to case:
>
> 1. The environment variables COPYLOT_RSCRIPT, COPYLOT_R_BINARY and RSCRIPT, in that order — reported as env.
> 2. Rscript, then Rscript.exe, as resolved on the system search path — reported as path.
> 3. A fixed list of usual install locations for the operating system — reported as common_path.
>
> Each candidate is then run once with --version and a five-second timeout. The first one that exits with status zero is the answer; the version is read from its output as the first number of the form version X.Y.Z. A candidate given as an absolute path that is not a file is recorded as not found without being run.
> The result is cached for the lifetime of the process and is recomputed only by **Refresh**.

59.3.1 What R is actually used for

One thing: variance-stabilising normalisation. That method is delegated to the Bioconductor vsn package through the detected Rscript, and it is offered on the annotation page only when *both* the executable and the package are present — the application asks R whether the package is installed and hides the option otherwise, rather than letting it be chosen and then fail. The three methods that need no R — none, median and quantile — are always available. **Refresh** clears that package probe as well as the runtime detection, which is why installing vsn and pressing **Refresh** is enough to make the option appear.

The R layer is there for anybody who wants to write their own analysis against the R functions; inside the application VSN normalization is its only use, and it is the only thing the check is required for.

> **EASY TO GET WRONG**
>
> R available is not the same statement as vsn being installed. The badge reports only that an Rscript answered its version probe. If VSN is still missing from the normalisation choices while the badge is green, the package is what is missing, not R.

59.4 Settings this page does not hold

- **How a plot looks.** Title, axes, fonts, marks, lines and colours are in the plot appearance panel, per plot, and are described in Part XI. This page only decides whether that panel is shown.
- **The condition colours.** They belong to the dataset, are shared by every plot in the session, and are set in Data → Colors.
- **The size and format of a download.** Those are on each plot page, because they belong to the figure rather than to the application.
- **Anything an administrator decides.** Whether a password is required, whether sessions are isolated, which viewer project is served, how large an upload may be: all of these are read from the environment when the application starts and cannot be changed from inside it. The two chapters that follow describe them.

Behind this page

Function	What it produces
detect_r_runtime	Finds and probes an Rscript; returns the availability, path, version, raw output and detection source shown in the card
clear_r_runtime_cache	What **Refresh** calls to force a fresh probe
vsn_availability	Asks the detected R whether the Bioconductor vsn package is installed, and gives the reason when it is not

Behind this page

Function	What it produces
vsn_reference_version	The vsn version the result was obtained with

60
Access and passwords

Who can open the analysis tool, how that is configured by whoever runs the instance, and what happens to a session when somebody logs in.

AT A GLANCE

Applies to	Any instance that is not a single person on one machine. The gate is off by default and a loopback-only run does not need it.
Turned on by	The environment the application is started in, not by anything inside the application.
Two credentials	A password, typed into the login card; and an access token carried in the fragment of a link. Both are optional and both are configured in the same file.
Stored as	A salted PBKDF2-HMAC-SHA-256 hash. The file holds the hash, never the password.
Granularity	One password for the whole analysis tool. The viewer is different: there, one password selects one project — see Administration → The DataViewer.
Session	Per browser tab, discarded on reload, and replaced with a fresh identifier the moment a login succeeds.

60.1 What the gate is, and what it is not

The gate is a single check in front of the whole application: with it enabled, every request to the backend other than the health check and the login endpoints themselves is answered 401 until the session has authenticated. It is not a user account system. There are no user names, no roles and no per-user data: everybody who knows the password sees the same application, and each of them gets their own isolated session inside it.

That isolation is not optional when the gate is on. A password-protected instance refuses to start unless session isolation is also enabled, because without it every browser would share one set of uploaded data — which is exactly what a shared instance must not do. For the same reason the launcher refuses to listen on anything but the loopback interface until authentication and isolation are both configured.

Set by whoever runs the instance

Control	What it does	Default	Where
COPYLOT_TOOL_ACCESS_ENABLED	Turns the password gate on for the analysis tool.	off	environment
COPYLOT_TOOL_ACCESS_FILE	Where to read the access file from. Without it the standard locations are searched.	unset	environment
COPYLOT_SESSION_ISOLATION	One session, one set of data. Required whenever the gate is on.	on	environment
COPYLOT_SESSION_COOKIE_NAME	The name of the session cookie.	copylot_session_id	environment
COPYLOT_SESSION_COOKIE_SECURE	Marks the session cookie Secure, so the browser sends it only over an encrypted connection.	off	environment
COPYLOT_SESSION_IDLE_SECONDS	How long a session is kept before it is discarded.	86400	environment
COPYLOT_MAX_SESSIONS	How many sessions are kept at once; beyond it empty sessions are discarded first, then the least recently used.	128	environment
COPYLOT_SERVER_MEMORY_MB	Memory budget, in MB, for all sessions' data; above it, stale sessions are removed.	60 % of RAM	environment
COPYLOT_SESSION_PROTECT_SECONDS	A session active within this time is never removed to make room, whatever the two limits above say.	1800	environment
COPYLOT_AI_ALLOWED_PROVIDERS	The assistant providers this server allows, comma separated: ollama, openai_compatible, openai, anthropic, gemini. The AI settings offer only these.	all	environment
COPYLOT_AI_ALLOWED_MODELS	Entries of the form provider:model, comma separated. A provider listed here may use only its listed models; the others stay unrestricted.	unrestricted	environment
COPYLOT_OLLAMA_MODEL_MANAGEMENT	Whether users may install or remove Ollama models, which is a server-wide change. When it is off, the AI settings hide **Model Management**.	on for a local run; off with tool access, viewer or viewer portal	environment
COPYLOT_LOCAL_AI_URL	A local server with the OpenAI-compatible API (llama.cpp, vLLM, LM Studio), such as http://192.168.1.20:8080/v1. It is pre-filled in the AI settings and allowed.	none	environment

Set by whoever runs the instance (continued)

Control	What it does	Default	Where
COPYLOT_LOCAL_AI_ ALLOWED_ ENDPOINTS	Further such servers users may enter on a deployment with tool passwords, a viewer or a viewer portal, as http://host:port, comma separated. These servers receive the full analysis context, so a deployment reaches only listed addresses. A desktop run uses any address its user enters.	none	environment
COPYLOT_LOCAL_AI_ API_KEY, ...MODEL, ...CONTEXT_TOKENS	That server's key, preselected model and context window (read from llama.cpp's /props when not set).	none; first model; 8192	environment
COPYLOT_AI_ MODEL_LEVELS	The assistant level per model, as model=basic, standard or advanced, comma separated (glm5=advanced). Replaces the automatic classification; a user's own choice in the AI settings still wins.	none	environment
COPYLOT_AI_ CALL_TIMEOUT, COPYLOT_LOCAL_AI_ TIMEOUT	Seconds one model reply may take, for Ollama and cloud providers and for a local OpenAI-compatible server. A large model that thinks first needs more than a minute.	300; 300	environment
COPYLOT_LOCAL_AI_ REASONING_EFFORT	low, medium or high for every request, or server to never send one; sent only to a local server whose model supports it. Lower effort answers sooner; too low and the model skips work.	high for styling by Advanced models, otherwise the server's own	environment
COPYLOT_AI_ NATIVE_TOOLS	Whether Advanced models on servers with tool calling use the assistant as tools. Off uses the one-reply plan for all models; the same checks apply either way.	on	environment
COPYLOT_ PROVIDER_ TIMEOUT_SECONDS	Seconds one assistant step may take in total: planning, lookups and the answer.	600	environment
COPYLOT_TOOL_ LOGIN_RATE_LIMIT_ ENABLED	The limit on failed login attempts, with its ...MAX_ATTEMPTS and ...WINDOW_SECONDS companions.	on, 12, 300	environment

NOTE

The tool gate and the viewer portal cannot be enabled in the same process: one is an analysis instance behind one password, the other a reader's instance where the password chooses the project. An instance configured for both refuses to serve the API and says so.

60.2 The access file

The file is JSON. The repository-root examples document its shape:

- tool_access.example.json for the analysis tool;
- viewer_access.example.json for the viewer.

They contain nothing but zeroes where a digest belongs, and the application refuses to start on a file that still does, so copying an example and forgetting to fill it in fails loudly rather than quietly admitting everybody.

Four places are tried, in order, and the first file that exists wins: the path named by COPYLOT_TOOL_ACCESS_FILE, then local/tool_access.local.json, tool_access.local.json and tool_access.json beside the installation. The viewer looks for viewer_access.local.json and viewer_qr_access.local.json in the same three places, with COPYLOT_VIEWER_ACCESS_FILE and COPYLOT_VIEWER_QR_ACCESS_FILE in front.

Fields of the tool access file

Column	What it holds
password_pbkdf2_sha256	The password, as a salted PBKDF2 hash in the form below. The preferred field; when it is present it is the only one consulted.
password_sha256	Retired. An unsalted password digest is no longer accepted as a replacement for the required PBKDF2 hash.
tokens	A list of access tokens, each a plain SHA-256 digest — either the 64-character string on its own, or an object with a token_sha256 key. An empty list means no link access.

Fields of the viewer access file

Column	What it holds
credentials	A list, one entry per project. Each entry names a project and carries one credential for it.
project	The project directory the password opens. Letters, numbers, dot, underscore and hyphen, at most 128 characters, and not made of dots alone.
label	What the project is called on screen once it is open. Optional; the project name is used when it is absent.
password_pbkdf2_sha256	As above, and preferred.
password_sha256, password	Retired unsalted and plaintext credentials. They are refused; use the required PBKDF2 field.

The viewer's token file has the same shape with tokens in place of credentials, each entry carrying token_sha256, project and label.

Every project named in either viewer file must have a configuration of its own, or the instance refuses to start: a password that opens nothing is treated as a configuration error rather than as a login that will fail later.

60.3 How a password is stored

A password gate does not encrypt the connection. Use HTTPS for a network deployment, protect the configuration files and backups, and share access links as credentials. Read-only access prevents changing the project; it does not prevent an authorised reader from copying information they see.

THE PBKDF2 HASH

A hash is produced once, when the password is chosen, and only the hash is written to the file:

1. A salt of 16 bytes is drawn from the operating system's cryptographic random source, new for every password.
2. The password is encoded as UTF-8 and put through PBKDF2 with HMAC-SHA-256 over that salt, for 600000 iterations by default. No derived-key length is given, so the result is the 32-byte output of SHA-256.
3. The three parts are written as one string, separated by dollar signs: iterations$salt$digest, with the salt and the digest in lowercase hexadecimal.

Verification reads the iteration count back out of the stored string rather than assuming it, so a hash made with a different count keeps working. The candidate password is put through the same computation with the stored salt and the stored count, and the two digests are compared with a constant-time comparison.
A stored hash is rejected outright, before any password is tried, unless it has exactly three parts, its salt and digest are valid hexadecimal, its digest is 32 bytes, and its iteration count is at least 100000 and at most 2000000. At startup one further check is made: a hash whose salt or digest is entirely zero is the example placeholder, and is refused.

Access tokens are not passwords and are not treated as such. A token is compared as a plain SHA-256 digest of the string, with no salt and no iterations, because it is a long random secret rather than something a person chose. It is a bearer credential: whoever holds the link is authenticated by it, for as long as it is listed.

EASY TO GET WRONG

A token belongs in the *fragment* of a link — the part after #, written #tool=SECRET for the tool and #qr=SECRET for the viewer. A token in the query string is ignored, and is stripped from the address before anything else happens, because a query string is sent to the server and recorded in logs and proxies while a fragment is not. Links written for an older version with ?tool= or ?qr= no longer work, and any secret that was exposed that way should be replaced.

60.4 What a user sees

Before anything else the application asks the backend whether a gate exists. If none does, no login card ever appears; while the question is being asked the screen says Preparing secure tool access..., or Preparing secure project access... in the viewer.

- Not authenticated: the login card. One field, labelled **Password**, and one button — **Open Tool** in the analysis tool, **Open Viewer** in the viewer, reading Opening... while the attempt is in flight.
- An empty field never reaches the server: the card answers Please enter a password.
- A wrong password is answered Invalid tool password., or Invalid viewer password., in red under the field — the same message whether the password was wrong or the file lists no credential like it.
- Too many attempts from one address — twelve within five minutes, by default — are answered Too many login attempts. Please wait and try again. Password attempts and link attempts are counted separately.
- If the access file is missing or cannot be read: Tool access configuration is unavailable. Contact the administrator. Nothing about the file is shown to the user; the reason goes to the log.
- If the backend cannot be reached at all, the card turns red, prints what went wrong and offers **Retry**.
- A request that reaches the API without an authenticated session is answered 401 with Tool password required. or Viewer password required. That is what a user sees when a session ends while a page is open: the figures on screen stay, and the next action fails.

60.5 The session

The identifier of a session is kept in a cookie that the browser cannot read from script, scoped to the path of that tab's own part of the API, so two tabs of one browser hold two independent sessions and can be logged in to two different things at once.

- **Loading the page logs out.** Before the gate asks whether authentication is needed, it ends whatever session this tab had. A reload therefore always returns to the login card, and the work in that tab is gone with it. This is deliberate on a shared instance, and it is the single most common reason a user believes they have been locked out.
- **A successful login replaces the identifier.** The one the browser sent is discarded together with everything held under it, a fresh one is minted, and the grant is attached to that. An identifier planted in somebody's browser before they log in therefore never becomes an authenticated one.
- **Logging out discards the grant** for that session, and in the viewer also unloads the project.
- **A session is discarded after a day.** The default is 86400 seconds. On a protected instance the clock runs from the login rather than from the last action, so a long-running session ends a day after it began however busy it was.
- **The number of sessions is capped.** When a new login is granted, sessions beyond 128 are discarded oldest first, along with their data.

> NOT A LOCKOUT
>
> Four things look like a lost account and are not. A reload returning to the login card is the reset above, not an expired password. A 429 clears itself once the five-minute window passes — there is nothing to unlock and nothing an administrator has to reset. An access link keeps working after it has been used, but the secret is stripped from the address bar the moment it is consumed, so a bookmark made *after* opening the page will not contain it: keep the original link. And if the login card accepts the password but every request afterwards answers 401, the session cookie is not coming back — the usual cause is a cookie marked Secure on an instance being reached over an unencrypted connection.

There is no logout control, by design: a session ends by being left alone until it expires, or by reloading the interface.

Behind this page

Function	What it produces
hash_password	Produces the stored iterations$salt$digest string from a password
verify_password	Recomputes the hash of a candidate password and compares it in constant time
validate_password_hash	The startup check on a stored hash, including the refusal of the example placeholder
authenticate_tool_password	Checks a typed password and grants a rotated session

Behind this page

Function	What it produces
authenticate_tool_token	The same for a token carried in a link
validate_tool_access_configuration	The startup validation of the access file
authenticate_viewer_session	Matches a password against the viewer credentials and selects its project
authenticate_viewer_qr_session	The same for a viewer token

61 The DataViewer

The read-only companion: a finished project, handed to somebody who needs to read it rather than to analyse it.

AT A GLANCE

What it is	The same application, built in viewer mode and put in front of data that is already on the server. Same backend, same plotting functions, same figures.
Where	Wherever the viewer was deployed; the address is set by whoever deployed it.
Differs from the tool	Nothing may be uploaded, changed, imputed, collapsed, translated or exported as a session. The sections that did those things are not in the sidebar, and the endpoints behind them are refused.
Where the data come from	Files named by a `viewer_config.json`. Nobody using the viewer chooses them.
Who may open it	Everybody, if the portal is off. One project per password, if it is on — see Administration → Access.
Processing	Log2, normalisation, filtering and imputation are carried out at load time by the same functions the analysis tool calls. There is no second pipeline.

61.1 What the viewer is for

A finished project is usually handed on to somebody who did not run it: a collaborator, a supervisor, the group that supplied the samples. What they need is every figure, every table under it and the annotation that explains what was done — and what they do not need is the ability to re-filter, re-impute or replace the data underneath. The viewer is that division drawn in the application itself rather than in an instruction not to touch anything.

It uses the same calculation and rendering code for the plot pages it offers. Some modules are deliberately unavailable, as listed below. With the same matrices and settings, a shared plot has the same interpretation as in the analysis tool; a different loading or imputation recipe

can still produce different input values.

What selects a project

Control	What it does	Default	Where
--viewer	Starts the launcher in viewer mode. The viewer entry point adds it, and port 8001, when they are not given.	off	launcher
--project NAME	Which project under viewer_data to serve. Also accepted as --project=NAME. With exactly one project present it is selected on its own.	none	launcher
COPYLOT_ VIEWER_MODE	The same switch through the environment, which is how a container sets it.	off	environment
COPYLOT_ VIEWER_PROJECT	The project name, as an alternative to --project.	unset	environment
COPYLOT_ VIEWER_CONFIG	An explicit path to one configuration file, overriding the search.	unset	environment
COPYLOT_VIEWER_ PORTAL_ENABLED	Password-to-project routing. Requires viewer mode and session isolation.	off	environment

EASY TO GET WRONG

The interface is a built bundle, and the mode is fixed when it is built: a bundle built for analysis serves the full analysis interface even when the backend is in viewer mode. The backend still refuses every write, so nothing breaks — but the sidebar offers pages that cannot work. The launcher checks the two against each other and refuses to start on a definite mismatch, naming the command to rebuild with.

61.2 How it differs from the analysis tool

Two layers do this, and they agree: a page that is not offered is removed from the sidebar *and* its route redirects, so a restored interface state naming a hidden page cannot reach it. Behind both, the backend refuses the requests outright.

- **Data** keeps only **Upload** and **Annotation**. **Upload** becomes a read-only preview — a level chooser and the note that data is loaded from viewer_config.json — and **Annotation** shows what the configuration did: the normalisation, the filter rule in words, and whether the configured imputation was applied. **Imputation**, **Distribution**, **ID Translator**, **Colors**

and **Restore** are gone.

- **Statistical Analysis** loses **Replicate & Variance Projection** (a what-if projection, not a power calculation).
- **Summary** keeps only **Tables**: no **Text** and no **Report**.
- **External Tools** keeps only **GONetView**; peptide collapse is not offered.
- **Phospho-specific** loses **Kinase Library**, and the whole section is absent unless the project supplied phosphosite or phosphoprotein data.
- **Peptide Level** loses **Sequence Coverage**, and the whole section is absent unless the project supplied a peptide report.
- **Single Protein** loses **Phosphosites on Protein Heatmap** unless the project supplied phosphosite data.
- Quick save does nothing and the browser is not asked to warn about unsaved work: there is no analysis to lose. The appearance panel is deliberately kept — **Ctrl / Cmd + Shift + E** works, so a reader can style a figure for a talk — but the switch is reset on every load.

Every request that would change something is answered 403 with `This operation is disabled in read-only viewer mode.` That covers uploads of data and of metadata, creating or renaming a dataset or a metadata profile, generating or deleting an annotation, pairing, phosphoprotein aggregation and upload, running *or downloading* an imputation, running or downloading an identifier translation, installing or removing an assistant model, writing condition colours, every external-tool run, and both halves of save and restore. Exporting a session is refused for the same reason as importing one.

Of the downloads, only the notebook exports are blocked: the notebook buttons are not shown and the notebook ZIP requests are refused, because every notebook ZIP carries the exact input matrix. Data tables are not blocked: viewers may see and download them — the source and filtered matrices and the per-sample values — as well as the plots. This is an editing restriction, not a guarantee against extraction: readers can retain displayed information and any downloads the viewer offers. Reading the condition colours is allowed, because the figures have to be drawn in the colours the project was saved with.

61.3 Publishing a project

1. Put the data files where the instance can read them: a directory named for the project under `viewer_data`. A project meant to travel with the installation goes in `viewer_data/<project>/`; one that is not to be committed goes in `viewer_data/local/<project>/`; `viewer_data/projects/` is searched too. The name may contain letters, numbers, dot, underscore and hyphen, up to 128 characters, and may not be made of dots alone.
2. Write a `viewer_config.json` in that directory naming the files and stating the processing.

A viewer_config.local.json beside it takes precedence, which is how a local variant is kept out of the published one.

3. If the instance routes by password, add an entry to the viewer access file naming this project. Every project named there must have a configuration, or the instance refuses to start.

4. Open it. A single project is selected on its own; otherwise name it with --project.

> **NOTE**
>
> Publishing does not rewrite anything. The data files are read exactly as whatever produced them wrote them, and the settings in viewer_config.json are read exactly as they were typed: neither is generated from the other, and the two have their own pipelines. What the configuration describes — log2, normalisation, filtering, imputation — is carried out in memory when the project is loaded, and is discarded with the session. A published project can therefore be corrected by editing its configuration alone, and re-exported from the analysis tool without its configuration being touched.

61.4 The project configuration

The file is a JSON object with one key, datasets, holding at least one named package. The name is what that dataset is called in the interface; a project with two packages offers both in the dataset chooser. Every key below sits inside one package.

The files

Column	What it holds
protein_file	The protein matrix, in wide format.
phospho_file	The phosphosite matrix.
phosprot_file	The phosphoprotein matrix — already collapsed; see below.
peptide_file	The peptide or precursor report, in long format. Stored as a path rather than read into memory.
metadata_profiles	Named annotation profiles, each with a meta_file and a meta_phospho_file. A profile given as a plain string is protein metadata only. A configuration that instead names meta_file at the top level gets one profile called Metadata.

A path is taken relative to the configuration file unless it is absolute. The string none means *not provided*, and so do an empty string, a path that ends in a separator and a path that turns out to be a directory — the pipeline that would have used it then stays out of the sidebar. A path that is named but missing is reported as a warning and the rest of the project still loads.

The processing

Column	What it holds
filter_method_*	at_least_n or percentage.
filter_min_*	How many valid values a feature needs, for at_least_n. Default 0.
filter_percentage_*	The percentage for the other method. Default 70, clamped to 0–100.
filter_mode_*	per_group, in at least one group or overall. Default per_group.
log2_transform_*	Whether the matrix still has to be transformed. Default true.
normalization	none, median, quantile or vsn, for every level at once; normalization_protein and normalization_phospho override it per level.
imputation	A mode on its own, or an object with mode, q_value, adjust_std, down_shift, width, seed and sample_wise. It may also be given per level.
collapse_mode	How the phosphoprotein file was collapsed *before* it was published. Provenance, not an instruction.

The three level suffixes are protein, phospho and phosphoprotein. Phosphoprotein normalisation is always none whatever the file says, because a phosphoprotein matrix is collapsed from phosphosite values that were normalised already, and normalising twice would be wrong.

WHAT HAPPENS WHEN A PROJECT IS LOADED

Each package is made active in turn and its files are read. The annotation for each level is then computed by the same function the Data → Annotation page calls, with the filter, the log2 flag and the normalisation from the configuration; the imputation, if one is asked for, is run by the tool's own imputation function, with the tool's own rule for which matrix it draws from and the tool's own refusals — phosphoprotein is not imputable, and unfiltered data is never quietly substituted when a filter kept nothing. Any parameter the configuration does not name takes the value the tool itself would have used.

Nothing is collapsed. collapse_mode only records how the supplied phosphoprotein file was made, and it matters because robust_site_median centres each site on its own median across samples: those values are ratios rather than abundances, and the axis labels follow.

An unrecognised value — a misspelt normalisation, filter mode, imputation mode or collapse mode — falls back to the safe default and is reported as a warning on the annotation rather than stopping the load. One bad key must not cost a shared project.

61.4.1 Edge cases

- A configuration with no datasets object, or with an empty one, is an error: the project does not load.
- An imputation that was configured but could not run keeps its mode and is marked as not applied, and the reason appears in the annotation warnings. Nothing silently falls back to non-imputed values.

- Editing a configuration and reloading the page picks the change up: the file's modification time is part of what identifies it, so an edited project is re-read rather than served from the copy already in memory.
- With the portal on, logging in loads that password's project into that session alone, and logging out unloads it again.

Behind this page

Function	What it produces
load_viewer_state	Reads one configuration and loads every package, profile and annotation it describes
resolve_viewer_project_config	Finds a named project's configuration file
bootstrap_current_viewer_session	Loads the configured project into the current session
compute_annotation_from_metadata	The annotation, filter and normalisation — the tool's own function
impute_values_with_diagnostics	The imputation — also the tool's own
choose_best_source	Which matrix the imputation draws from, and the refusal when a filter kept nothing

A

Keyboard shortcuts

Everything the application binds to a key, and the keys it deliberately leaves alone.

AT A GLANCE

Modifier	**Ctrl** on Windows and Linux, **Cmd** on macOS. Both are accepted everywhere a shortcut asks for one.
Global	Two: **Ctrl** + **S** saves the session, **Ctrl** + **Shift** + **E** opens and closes the plot appearance panel.
Keyboard layout	Matched on the physical key, not on the letter the layout produces, so both work unchanged on QWERTZ and AZERTY.
In the viewer	**Ctrl** + **Shift** + **E** works. **Ctrl** + **S** does not, and is left to the browser.
Not taken	**Ctrl** + **Shift** + **S** and **Ctrl** + **Shift** + **A** are the browser's, and the application does not answer either.

Shortcuts

Control	What it does	Default	Where
Ctrl + **S**	Writes the session to the save state file, without a dialog if one has already been chosen. Analysis mode only.	—	global
Ctrl + **Shift** + **E**	Turns advanced mode on and off, and with it the plot appearance panel under every plot.	—	global
Enter	In a number or text field: commits what has been typed, exactly as clicking away from the field would.	—	any field
Enter	In the assistant's message box: sends the message.	—	assistant
Shift + **Enter**	In the assistant's message box: a line break, without sending.	—	assistant

Shortcuts (continued)

Control	What it does	Default	Where
Ctrl-click, Cmd-click, Shift-click	On a point of a canvas volcano: adds that feature to the selection, or removes it, instead of replacing the selection.	—	analysis canvas
Ctrl-click, Cmd-click	In a multiple-choice list — the condition and protein pickers — selects and deselects individual entries.	—	list fields

A.1 The two global shortcuts

Both are answered by one handler on the window, so both work from any page and from inside a text field. Three conditions decide whether the handler does anything at all: **Alt** must not be held, **Ctrl** or **Cmd** must be, and a key held down long enough to repeat fires once rather than once per repeat.

A.1.1 Ctrl + S — save the session

Ctrl + S is the quick save described in the Sessions chapter. It builds the same state file as the button on Data → Restore and writes it to the file already chosen, so after the first save the key is the whole operation and no dialog appears. Where the browser cannot write to a file directly the state is downloaded instead, and the application then asks you to confirm that the download finished — until you do, the session is still counted as unsaved and the warning on leaving the page stays.

The shortcut does not exist in a published viewer. A save state packages the whole session, matrices included, which a viewer has no business taking away, and the route behind it refuses the request in any case; the key is therefore left to the browser rather than bound to something that fails.

A.1.2 Ctrl + Shift + E — the plot appearance panel

Ctrl + Shift + E is the same switch as **Advanced plot option mode** on the Settings page, and setting it either way moves the other: the panel opens under whichever plot is on screen, and closes again on the second press. It works in the viewer on purpose, because the appearance panel is how somebody makes a publication-ready figure out of a project that was published for them.

> **WHY E AND NOT A**
>
> Earlier builds opened the panel with **Ctrl** + **Shift** + **A**. Chrome answers that combination with its own tab search before the page ever sees the key, so on the most common browser the shortcut appeared to do nothing at all. The binding moved to **E**, and the guard that rejects **A** is held in place by a test rather than by memory.

> **EASY TO GET WRONG**
>
> **Shift** is part of the identity of a shortcut here, not a modifier to be ignored. **Ctrl** + **S** fires only without **Shift**, so **Ctrl** + **Shift** + **S** reaches the browser and still does whatever the browser does with it — a screenshot, or Save As. A handler that swallowed it would take that away silently, which is how a user loses a screenshot they meant to take.

A.2 Keys inside a field

Every number field in the tool is typed into freely and committed on leaving it, which is what makes a half-typed number harmless: nothing is recomputed from $0.$ on the way to 0.05. **Enter** is the keyboard form of leaving the field. It commits the draft and triggers exactly the same refresh that clicking elsewhere would, and it never submits anything else.

- A comma is accepted as a decimal separator and replaced with a point as you type, so $0{,}05$ and 0.05 are the same entry.
- A value that is not a number is not committed; the field returns to the value it held.
- In the assistant's message box the rule is inverted, because the box holds prose: **Enter** sends and **Shift** + **Enter** starts a new line.

A.3 Keys used with the mouse

Two places read a modifier held during a click rather than a key on its own.

- On a volcano tile of the analysis canvas, a plain click selects the feature under the pointer and replaces whatever was selected before. Holding **Ctrl**, **Cmd** or **Shift** toggles that one feature in and out of the selection instead, so a list can be built up point by point. The canvas also has a **Multi-select mode** button, which draws a rectangle over many points

at once; while that mode is on, the modifier click is not read.

- In the multiple-choice lists — selecting conditions, proteins or sites — **Ctrl**-click and **Cmd**-click add and remove single entries, which is the browser's own behaviour for such a list and is stated under each one.

A.4 What the application does not bind

There are no shortcuts for moving between sections or tabs, for running an analysis, for downloading a figure or for opening the assistant: each of those is a click, and the application defines nothing that would shadow the browser's own keys for finding, printing, reloading or switching tabs. Two combinations are left alone deliberately rather than by omission — **Ctrl** + **Shift** + **S**, as above, and **Ctrl** + **Shift** + **A**, which Chrome answers first and which therefore cannot be a shortcut this application has.

Behind these keys, in the interface

Function	What it produces
downloadRestoreState	Builds the state file that **Ctrl** + **S** writes
saveRestoreStateBlob	Writes it to the chosen file, or downloads it where that is not possible
setAdvancedPlotMode	The switch behind **Ctrl** + **Shift** + **E**, shared with the Settings page
getAdvancedPlotMode	Reports the current position of that switch to every page listening for it

B
When something does not work

The refusals the tool actually issues, what each one means, and what to do about it.

AT A GLANCE

Arranged by	Symptom, because that is what the reader has.
Quoted text	Each entry names the sentence the application prints. Where a message carries a name or a number, it is written here in braces: {level}, {file}.
First rule	Nothing is silently substituted. A refusal means the tool declined to answer the question asked, not that it answered a different one.
Second rule	A message naming a page names the page to go to. Almost every refusal upstream of a plot is settled at Data ⟶ Annotation.

B.1 How to read this appendix

A matrix quietly substituted for the one that was asked for produces real numbers under the wrong label, which is worse than an error, so a request the tool cannot answer comes back as a sentence rather than as a figure. Those sentences are what follows.

A table, a run or a download prints its refusal where the result would have been. A plot is fetched as an image, so when it fails the page asks the same address again to read the reason and prints that; only when the second request yields nothing usable does the generic sentence appear.

B.2 Upload

B.2.1 The file is refused before anything is read

Unsupported file type: {ext} means the extension is not one of the five that are read: csv, tsv, txt, xlsx, parquet. An .xls workbook, a .zip or a .raw is refused at the extension, before the contents matter. Export the table again in one of the five.

B.2.2 The file parses but is not a table

Three checks run on the parsed table and each has its own sentence.

- {file} has no data rows. Check the export finished and that the header is the first line of the file. — the file has a header and nothing under it, or the export was truncated.
- {file} has only 1 column. A report table needs an identifier column and at least one measurement column — check the file is not a saved web page and that the separator is correct. — almost always a tab-separated file read as comma-separated, or a saved HTML page with a table-like name.
- Could not read {file}: … — the parser's own failure, rewritten. The four it recognises are an empty file, a file that is not valid UTF-8 text, rows that do not all have the same number of columns, and something named .xlsx that is not a workbook. Re-saving as UTF-8 settles the second; the third is usually a stray separator inside a text field.

B.2.3 The file is too large to hold

Four budgets guard the upload and each names the number it enforced: the bytes as they arrive, the parsed shape in cells, the materialised table in megabytes, and the column count. A compressed file can pass the first and fail the third, which is why the messages differ. A column-count refusal is a reason to check the table orientation, selected export fields and separator. Large reports can also be legitimate. The limits are set by whoever deployed the instance; raising them requires sufficient memory.

B.3 Annotation, levels and data sources

B.3.1 A page says the annotation is missing

Failed to load conditions. Generate annotation first. on a page, and No annotation metadata available for {level}. Generate annotation first. from a request, both mean the same thing: a dataset has been uploaded but Data → Annotation has not been run for that level. Nothing downstream draws before the annotation exists, because until then no column is known to be a sample and no sample is known to belong to a condition.

No {level} dataset is currently loaded. Upload data first. is one step earlier: there is no dataset

at that level at all. If you uploaded one and this appears anyway, the session has been replaced — see *The session ended* below.

B.3.2 The filter removed everything

No features passed the selected filter. Adjust the filter in Data > Annotation and generate annotation again to continue. Unfiltered data has not been substituted. The completeness filter is strict enough that no row survived it. Loosen it on the annotation page and generate the annotation again. The final sentence is the important one: the unfiltered matrix was not quietly put in its place, so nothing downstream is running on data you did not ask for.

B.3.3 The phosphoprotein level is not there

Phosphoprotein is produced, not uploaded — unless you upload one directly — and until it has been produced it does not exist as a level. Three sentences mark the three stages.

- Generate phospho annotation first to enable aggregation. — on Data → Annotation. The collapse takes its metadata, its filter and its normalization from the phospho annotation, so that annotation has to exist first. The same refusal from a request reads Phospho annotation is required first. Generate phospho annotation to share metadata with phosphoprotein.
- No phosphoprotein dataset loaded yet. — the phospho annotation exists and the collapse has not been run. Press **Aggregate to Phosphoprotein**.
- The requested data level is unavailable. — something asked for a level that is not in this session, most often a restored canvas tile or a saved link built when the level did exist.

B.3.4 The data source is not available

A named data source never falls back. Imputed {level} data is not available. Run imputation successfully first. means imputation has not been run for that level, or the run did not finish; the page says the same thing as Run Imputation, then select Imputed Data here. Filtered {level} data is not available. Generate annotation/filtering first. is the equivalent one step earlier. On the analysis canvas the short form is The selected data source is unavailable.

Stored imputed {level} data is invalid because sample columns are missing: {samples}. Run imputation again. is the one case where an apparently working session holds a matrix that does not match its own metadata: the imputed matrix exists but was built before the annotation changed. Running imputation again rebuilds it against the current one.

B.4 Plots

B.4.1 A peptide or precursor plot asks for an abundance column

Peptide and precursor plots are drawn from the long-format report, which carries several numeric columns and only some of them are quantities. The column is therefore chosen by name, never guessed from the numbers, and there are three outcomes.

- Peptide dataset is missing a usable abundance column. Expected a column named 'Precursor.Quantity' or one containing quantity, intensity, abundance, area, LFQ, or iBAQ. Columns found: … — no column in the report carries an abundance word in its name. Rename the quantity column in the export, or export a report that includes one.
- Multiple peptide abundance columns are available. Select one explicitly in the plot options. Choices: … — several credible columns and no Precursor.Quantity among them to settle it. Choose one in **Abundance Column** on the page; until you do, no plot is drawn.
- Selected peptide abundance column '{name}' is not a supported abundance column in this dataset. Supported columns: … — usually a restored setting naming a column that the current report does not have.

A column holding charge, m/z, a retention time, a q-value, a CV, a score or a standard deviation is never offered, whatever its dtype: those are numeric too, and a plot drawn from one of them would be a perfectly good picture of the wrong quantity.

B.4.2 A plot fails and names no reason

Failed to load plot image. Check backend logs for details. is the fallback, printed only when the second request returned nothing the page could read as a reason — the process stopped, or the connection did. Every refusal the application knows how to name is printed instead of this sentence, so seeing it means the failure was not one of them.

B.4.3 A PCA refuses, or loses one ellipse

A projection is complete-case: a feature missing in any sample cannot enter it. Need at least {n} samples for {n}D projection. and Need at least {n} non-constant features for {n}D PCA. mean there are not enough samples, or not enough features that vary, for the requested number of components; All features have zero variance; dimensionality reduction cannot be computed. means every retained feature holds the same value in every sample.

A missing ellipse is a notice under the plot rather than an error: No ellipse was drawn for group {name}… The minimum enclosing ellipse needs the group's points to span the plane, and replicates that fall on a line, or too few distinct points, give it nothing to fit. It is a descriptive boundary, not a confidence region. The rest of the figure is drawn and the points themselves are unaffected.

B.4.4 A phospho result is empty rather than wrong

No phosphosites pass the selected significance and fold-change thresholds is a figure, not a refusal, and both the static and the interactive renderer answer with the same sentence. It

means the thresholds were strict, not that the plot broke — which is exactly the distinction an error would have hidden.

B.5 Statistics

B.5.1 A comparison returns nothing: too few replicates

Every feature needs at least two finite values in *each* of the two conditions before it can be tested, because a variance cannot be estimated from one measurement. When no feature clears that bar in both conditions, the comparison comes back as `No rows have enough replicate values for the selected comparison.`

Read it as a statement about the pair of conditions. A condition with a single replicate can never produce a test, and a condition of three in which most features are missing in two of them fails the same way; the completeness plot and the missing-value heatmap show which of the two it is.

ANOVA states the same requirement per condition and names the offenders: `Each ANOVA condition needs at least two sample columns. Missing: …`, or `ANOVA needs at least two conditions.` when fewer than two were selected. The page phrases it in advance as `At least two conditions with replicate samples are required to run ANOVA.` Where the design is large enough but the data are not, the refusal is `No rows have enough replicate values for ANOVA.`

B.5.2 The post-hoc panel is not offered

`A post-hoc needs at least three conditions.` With two, the ANOVA and the single pairwise comparison test the same hypothesis. Nothing has gone wrong: with two conditions there is only one pair, and the ANOVA has already tested it.

B.5.3 A paired test refuses

Pairing is structure, and it comes from the metadata rather than from the order of the columns.

- `Paired tests require a 'pair_id' metadata column with one stable subject identifier per sample.` — the uploaded metadata has no `pair_id` column. An unpaired test needs none; a paired one cannot be run without it.
- `Paired tests require at least two matched pair IDs.` — the column exists, but fewer than two subjects appear in both conditions.
- `Each sample in condition '{name}' can be paired only once.` — one subject identifier is used twice inside the same condition, so which of the two samples is the partner is undefined.
- `Paired tests require two different conditions.` — a condition cannot be paired with itself.

B.5.4 An enrichment or pathway page asks for gene names

Gene names are required for ORA. Please generate or load gene names first. and its pathway-heatmap equivalent. Over-representation works on gene symbols, so a dataset identified only by accession has nothing to look up. Data → ID translator produces the column.

B.6 Restore

B.6.1 The state file restores, but the datasets are missing

An ordinary save state records where the uploaded files were read from rather than the measurements themselves, so restoring it on another machine, or after the files have been moved or renamed, cannot find them. The restore stops before touching anything and a **Missing Files** panel appears, one entry per dataset, each carrying the package name, the level, the saved path and the saved file name:

- Saved path for {level} dataset in package '{name}' does not exist. — the path was recorded and nothing is there now.
- No saved absolute path was available for {level} dataset in package '{name}'. Upload a replacement file to continue restore. — a portable state, which records no paths by design.

Choose a replacement file for every entry and press **Restore** again; the button stays disabled until each one has a file. The page summarises it as Some saved dataset paths are missing. Upload the listed files to finish restore.

This is the case portable save states exist for: a portable state never claims to know where the data are and always asks, which makes it the one to send to a colleague.

B.7 The session

B.7.1 The session ended

A session lives in the memory of the running application, one per browser tab, and three things end it: refreshing the tab, stopping the application, and leaving it untouched past the idle limit — twenty-four hours unless whoever deployed the instance set another. When a session is discarded, its datasets, annotations, uploaded metadata and condition colours go with it.

There is no message announcing the expiry. What you see is the state of a fresh session: `No {level} dataset is currently loaded. Upload data first.` on a page that was working, and empty selectors where there were conditions. A save state restored from Data → Restore brings all of it back, which is the reason to take one before leaving an analysis overnight.

> **EASY TO GET WRONG**
>
> Two tabs are two sessions and always were. An analysis opened in a second tab does not see the first tab's data, and saving from one does not save the other. This is deliberate — it is what lets two projects be open side by side — but it is also the most common reason for a dataset that seems to have disappeared.

B.7.2 An instance asks for a password

Where the installation was configured with a password, every request outside the login route is answered with `Tool password required.` or `Viewer password required.` until one has been entered, and the login screen is shown. `Too many login attempts. Please wait and try again.` is the rate limit; it clears on its own. `Tool access configuration is unavailable. Contact the administrator.` is not about your password at all — the configuration file the instance was started with is missing or malformed, and only whoever deployed it can fix it.

B.7.3 A viewer cannot do something the manual describes

A published viewer is for reading a project, not for processing one. Upload and annotation are a read-only view of what was published, and the data-processing pages, the replicate and variance projection (a what-if projection, not a power calculation), the external peptide collapse and the phospho and peptide pipelines are absent rather than present and failing. Three writes are refused even when reached directly: exporting a save state, downloading an imputation, and saving condition colours. Notebook exports are blocked too; data tables and plots can still be downloaded. **Ctrl + S** is not bound in a viewer for the same reason.

Where these refusals come from

Function	What it produces
read_dataframe	Reads an uploaded table and issues the upload refusals above
select_matrix	Resolves a level and a data source, or refuses — the only way an analysis gets a matrix
resolve_peptide_quantity_column	Chooses the abundance column by name, or asks for one
inspect_missing_files	Builds the **Missing Files** list before a restore touches anything
ellipse_skip_message	The notice under a PCA whose group has no ellipse
expire_idle_session	Discards a session that has passed the idle limit

C
Glossary

The terms this manual uses precisely, one or two sentences each.

AT A GLANCE

Defined at length	In Getting started ⟶ Concepts, for the terms that carry the most weight: feature, sample, condition, annotation, level, data source, and the boxplot.
Here	The same definitions in one line, plus the statistical and proteomics vocabulary the chapters use in passing.
Order	Alphabetical, abbreviations under the abbreviation.
Scale	Unless a term says otherwise, values are $\log_2$ and missing is NA.

C.1 About these definitions

Each of these words means something slightly different in each laboratory, and a number this tool produces only means anything against the definition the tool used. What follows is that definition and nothing broader — where a term has a general meaning and a narrower one inside the application, the narrower one is what is written, because that is the one the numbers on screen obey.

C.2 Terms

- **Annotation** — The table that says which columns are samples and which condition each belongs to, produced at Data ⟶ Annotation together with the $\log_2$ scale, the normalization and the completeness filter. Nothing downstream draws before it exists.
- **Appearance panel** — The styling panel under every plot, shown while advanced mode is on (**Ctrl** + **Shift** + **E**). It changes how a figure looks and never what it is drawn from.

- **BH** — Benjamini–Hochberg: the procedure that turns the p-values of one comparison into adjusted p-values controlling the false discovery rate under its statistical assumptions. It is applied across the features of that comparison and over its finite p-values only.
- **Class I** — A phosphosite whose localization probability is at least 0.75. Below that a site is Not Class I, and a site with no localization value at all is Unclassified.
- **Collapse** — Summarising the phosphosites of one protein into one phosphoprotein value per sample. Four methods exist: robust_site_median (the default), mean, sum_ignore_na and sum_propagate_na.
- **Condition** — A group of samples the annotation treats as replicates of one another. Conditions come from the annotation, not from the file, and their colours are shared by every plot in the session.
- **CV** — Coefficient of variation: the standard deviation of a feature's replicates divided by their mean, as a percentage, CV = (SD/mean) × 100. It is computed on linear intensities, within one condition, and needs at least two samples.
- **Data source** — How far along the pipeline a matrix is: Data as annotated, Filtered Data, Imputed Data. A source that does not exist is refused rather than substituted.
- **FDR** — False discovery rate: the expected proportion of false positives among the features called significant. It is what the BH adjustment controls, and it is a property of the list, not of any one feature in it.
- **Feature** — A row of the matrix, whatever the matrix is: a protein group, a phosphosite, a phosphoprotein, a peptide or a precursor. Used wherever a sentence is true of all of them.
- **Fold change** — On the $\log_2$ scale a difference, taken as condition 1 minus condition 2, so a positive $\log_2$ FC means higher in the condition named first. On an ordinary log2 intensity scale, a difference of 1 corresponds to a twofold geometric-mean ratio; VSN is only log-like, especially at low intensity.
- **IQR** — Interquartile range, $Q_3 - Q_1$: the span of the central half of a distribution, drawn as the box of a boxplot. The whiskers reach the most extreme observed values still within 1.5×IQR of the box.
- **Level** — Which matrix is on screen: protein, phosphosite, phosphoprotein, peptide or precursor. Changing the level changes what a row is; changing the data source does not.
- **Localization probability** — The confidence, between 0 and 1, that a modification sits on the residue it was assigned to. Read from PTM_localization, and 0.75 is the Class I threshold.
- **Missed cleavage** — A cleavable residue inside a peptide that the protease did not cut. Counted for trypsin (after K and R, both blocked by a following proline) or for trypsin/Lys-C, where K is cut even before a proline and R is not.
- **NA** — The only kind of missing this tool recognises. There is no sentinel value and no zero that means absent: in a linear upload a zero in a sample column becomes NA before the transform, and in a $\log_2$ upload zero is a real measurement and is kept.
- **Paired test** — A test that matches each sample of one condition to a sample of the other and tests their differences, using the pair_id metadata column to say which matches which. It needs at least two matched pair identifiers and each identifier once per condition.

- **Phosphoprotein** — One row per protein, produced by collapsing its phosphosites or uploaded directly. Under robust_site_median the values are ratios to each site's own median rather than abundances, and the axes say so.
- **Phosphosite** — One row per localized modification site, identified by its PTM_Collapse_key. Labelled Phospho in the **Dataset level** selector.
- **Precursor** — One charge state of one modified peptide, as the long-format report lists it under Precursor.Id. The peptide level counts distinct Stripped.Sequence values instead, so multiple precursors can contribute to one peptide sequence.
- **Report entry** — One volcano comparison added to the report list. The entry stores its level, data source, conditions, controls, identifier, test, thresholds and render type, so the report redraws that comparison exactly as it was added and two entries differing in any of those stay two entries.
- **Session** — Everything the running application knows about your analysis, held in memory for one browser tab. Two tabs are two sessions; a session ends on refresh, on shutdown, or after the idle limit the deployment sets.
- **Unpaired test** — A test that compares the two conditions as independent groups, in either of two variance models: pooled, which assumes the two variances are equal, or Welch, which does not. Both need at least two finite values per feature in each condition.
- **z-score** — A row rescaled across the samples drawn, by subtracting the row's mean and dividing by its standard deviation, so that features at different abundances can be compared by shape. A row with no spread has no defined z-score; its handling is stated by the relevant plot rather than interpreted as a meaningful standardized pattern.

> **WORTH KNOWING**
>
> Two pairs are worth keeping apart. *Level* and *data source* are different questions — which features, and how far along — and sit next to each other in the same card. *Phosphosite* and *phosphoprotein* are different levels, not two names for the phospho data: the second is produced from the first by a collapse, and it does not exist until that collapse has been run.

Where these terms are decided

Function	What it produces
log2_transform_data	Zeros to NA, then $\log_2$ — what makes NA the only missing
filter_data	The completeness filter that produces Filtered Data
transform_phosprot	The collapse, and the four methods it accepts
resolve_test_type	Turns a test name into a design and a variance model: paired or not, pooled or Welch
volcano_data_f	Where the fold change is taken, as condition 1 minus condition 2

D
The functions behind the pages

The principal notebook functions, what they produce, and the chapter that uses it, so a figure can be reproduced in a notebook.

AT A GLANCE

Packages	proteomicscopylot for Python, installed with pip install ".[bio]" (-e for development; the bio extra is needed only for reading FASTA files). notebook_functions.R for R: a script loaded with source(), not an installable R package.
Names	Identical in Python and R. A plain name is in both; a [py] marks one the Python package has and the R functions do not.
Groups	In the order of the manual's parts: data, quality control, completeness, statistics, peptide, single protein, phosphoproteomics, comparison, then the shared helpers.
Arguments	Not listed here. Each function's own docstring or help page carries them, and the chapter named on the line states the defaults the application sends.
What they do not do	None of them touches the session. A notebook call reads the frame you hand it and returns a figure, a frame or a number.

D.1 How to use this appendix

The application and notebooks share scientific conventions, but the UI also has wrappers for selection, styling, provenance and export. A short notebook call demonstrates the calculation; it is not a promise of pixel-identical output with every advanced UI setting. Use the plot's notebook ZIP for the complete supported replay recipe and resources.

Two frames are wanted almost everywhere. data is the wide matrix, one row per feature and one column per sample, on the $\log_2$ scale unless the function says otherwise. meta is the annotation: one row per sample, with at least a sample and a condition column, and a pair_id column where a paired test is wanted.

READING A ROW

A name written plainly exists under that name in both Python and R. A name carrying py exists only in the Python package — the R functions reach the same behaviour through the language's own facilities or through a private helper, and there is nothing to call. Where the two packages share a name but not a signature, the name appears twice, once for each.

D.2 Reading and preparing data

Part II — Data

Function	What it produces
read_data	Reads a CSV, TSV, TXT, XLSX or parquet table and normalises its column names; any other extension is refused. *Upload*
rename_cols	The column-name map behind read_data: the common Spectronaut, DIA-NN and MaxQuant headers to the names used throughout, with reverse hits and site-only identifications dropped. *Upload*
log2_transform_data	Sets zeros in the sample columns to NA, then takes $\log_2$ of every sample column. *Annotation*
inverse_log2_transform_data	The reverse, returning linear intensities. *Annotation*
normalize_data	Applies one global normalization — none, median centering or quantile normalization — to the sample columns of a $\log_2$ matrix. VSN is available through the R package and application, not this standalone Python helper. *Annotation*
median_center	Shifts each sample so that all samples share one median. *Annotation*
quantile_normalize	Forces every sample onto a common intensity distribution. *Annotation*
filter_data	The completeness filter: a count or a percentage of valid values, per condition, in at least one condition, or overall. *Filtering*
transform_phosprot	Collapses phosphosites to one phosphoprotein value per sample, in one of robust_site_median, mean, sum_ignore_na or sum_propagate_na. *Collapse*
impute_values	Fills missing values with draws from a downshifted normal, sample-wise or over the whole matrix. *Imputation*
qqnorm_plot	The quantile-quantile plot of the annotated values against a normal, on a random sample of 100,000 values where there are more. *Distribution*
sample_id_map	Maps an extracted sample identifier back to the original sample name, refusing where two names share an identifier. *Annotation*
strip_sample_measurement_suffix	Removes the path, the file extension and the acquisition suffixes from a raw-file name. *Annotation*
extract_sample_id	The trailing number of a stripped sample name, which is what matches a report row to a metadata row. *Annotation*
extract_id_or_number	An alias of extract_sample_id, kept so that older notebooks keep working. *Annotation*

D.3 Quality control

Part III — Quality control

Function	What it produces
coverage_plot	Features quantified per sample, for a wide matrix: protein, phosphosite or phosphoprotein. *Coverage*
coverage_plot_summary	The same counts aggregated to one bar per condition, with the standard deviation and the individual samples. *Coverage*
coverage_plot_pep	Coverage from the long-format report, counted by stripped sequence or by precursor identifier. *Coverage*
histo_int	The distribution of $\log_2$ intensities, one curve per condition over each feature's mean across that condition's replicates. *Histogram*
boxplot_int	One box per condition, over the mean intensity of each feature in that condition. *Boxplot*
boxplot_int_single	One box per sample, over the intensities of that sample. *Boxplot*
cov_plot	One box per condition, over the coefficient of variation of each feature in that condition, computed on linear intensities. *Coefficient of variation*
pca_plot	The static principal component projection, in two or three dimensions, with optional minimum-enclosing ellipses (not confidence regions). *PCA*
pca_plot_interactive	The same projection as a hoverable figure. *PCA*
pca_scores	The coordinates pca_plot draws, as a table: one row per sample, with the explained variance of each component. *PCA*
resolve_feature_scaling	The scaling a choice and a method resolve to: centred, or centred and scaled to unit variance. *PCA*
minimum_enclosing_ellipse	Khachiyan's smallest enclosing ellipse for a set of points. *PCA*
fitted_group_ellipse	One condition's ellipse, or nothing where its points do not span the plane. *PCA*
ellipse_skip_message	The notice naming the conditions that got no ellipse and why. *PCA*
abundance_plot	Features ranked by their mean abundance, on $\log_{10}$. *Abundance*
interactive_abundance_plot	The same ranking as a hoverable figure, with a search. *Abundance*
abundance_axis_label	The y-axis label for that plot, which differs for a centred phosphoprotein matrix. *Abundance*
corr_plot	The sample-by-sample correlation matrix, as a matrix or as an ellipse plot. *Correlation*
sample_correlation_matrix	The Pearson correlations between samples that corr_plot colours, as a table in metadata order. *Correlation*

D.4 Completeness

Part IV — Completeness

Function	What it produces
missing_value_plot	One bar per missing-value count: how many features are missing in exactly that many samples. *Missing values*
missing_value_plot_pep	The same count from the long-format report, by stripped sequence. *Missing values*
missing_value_plot_prec	The same count from the long-format report, by precursor. *Missing values*
missing_value_heatmap	Present against missing, one cell per feature and sample. *Heatmap*
missing_protein_table	Missing values per feature, as a table. *Tables*
missing_sample_table	Missing values per sample, with the percentage of the matrix each accounts for. *Tables*
missing_value_outlier	The samples whose missing percentage is above a threshold. *Tables*
all_miss_val	The percentage of the whole matrix that is missing, to two decimals. *Tables*

D.5 Statistics

Part V — Statistics

Function	What it produces
volcano_data_f[py]	Differential abundance between two conditions: the fold change as condition 1 minus condition 2, the p-value, its BH adjustment and the significance class. *Volcano*
volcano_data_f[R]	The same table. The R version takes two further arguments after label_column, control1_columns and control2_columns, which the Python package reaches through volcano_data_f_control instead. The first eleven positional arguments match. *Volcano*
volcano_data_f_control	The control-corrected form: each condition is first referred to its own control, and the two corrected sets are then compared. *Volcano*
volcano_plot	The volcano figure from that table, with the two threshold lines. *Volcano*
volcano_plot_h	The same figure with named features highlighted. *Volcano*
volcano_plot_sim	A what-if projection, not a power calculation: the observed mean fold changes are held fixed and the pooled t-test is recomputed under a modified variance and replicate count. *Replicate and Variance Projection*
resolve_test_type	Turns a test name into the design and the variance model it stands for: unpaired pooled, unpaired Welch, or paired. *Volcano*
variance_model	The variance model alone, for the same names. *Volcano*
paired_test_enabled	Reads the older Paired/Unpaired argument, refusing anything else. *Volcano*

Part V — Statistics

Function	What it produces
paired_sample_columns	Orders the two conditions' sample columns so that partners line up, from the pair_id column. *Volcano*
paired_t_test [py]	The paired t-test over a matrix of per-feature differences, returning the mean difference, the number of pairs and the p-value. *Volcano*
negative_log10 [py]	$-\log_{10}$ of a vector, which is the volcano's y-axis. *Volcano*
condition_mean	The mean across one condition's replicates, blanked where too few were observed. *Volcano*
constant_within_group	Marks the features whose observed replicates in a group all hold one value, which have no variance to test. *Volcano*
anova_data_f	One-way ANOVA across the selected conditions, with BH adjustment. *ANOVA*
anova_posthoc_data	Tukey HSD for every pair of conditions following that ANOVA. *ANOVA*
anova_cluster_data	Clusters the significant features into a chosen number of groups. *ANOVA*
anova_plot	The heatmap of significant features, optionally as row z-scores and with dendrograms. *ANOVA*
anova_plot_cluster	The same heatmap annotated by cluster membership. *ANOVA*
different_genes	The up-, down- and background gene sets a volcano defines, ready for enrichment. *ORA*
different_genes_control	The same sets from a control-corrected volcano. *ORA*
enrichment_data	The over-representation table: one row per GO term, with its overlap, its hypergeometric p-value and its BH adjustment. *ORA*
enrichment_analysis	The dot plot of that table. *ORA*
load_go_db	Loads the bundled GO term-to-gene table the enrichment searches. *ORA*
proteins_in_pathway	The heatmap of the features of one pathway across conditions or samples. *Pathway heatmap*

D.6 Peptide level

Part VI — Peptide level

Function	What it produces
rt_vs_pred_rt_plot	Observed retention time against predicted, as a hexbin or a scatter, with the identity line. *RT Plot*
modification_plot	Distinct modified sequences per sample, one series per tracked modification. *Modification Plot*
missed_cleavage_plot	Peptides per sample by their number of missed cleavages. *Missed Cleavage Plot*

Part VI — Peptide level

Function	What it produces
count_missed_cleavages	Counts the internal cleavable residues of one peptide sequence, for the chosen protease. The first argument is sequence in both languages. *Missed Cleavage Plot*
normalize_protease	Accepts the spellings of trypsin and trypsin_lysc and refuses anything else. *Missed Cleavage Plot*
fasta_to_dataframe	Reads a FASTA file into a frame of names and sequences. *Sequence Coverage*
calculate_coverage	The percentage of one protein's sequence covered by its observed peptides. *Sequence Coverage*
vis_coverage	The same sequence marked up for display, in fixed-width chunks. *Sequence Coverage*

D.7 Single protein

Part VII — Single protein

Function	What it produces
boxplot_int_single_prot	One protein's intensities, one box per condition, optionally with the individual samples over it. *Protein boxplot*
single_protein_values	The values that boxplot draws for one protein, as a table of condition, sample and value. *Protein boxplot*
compare_prot_line	One or more features across the samples of the selected conditions, as lines. *Protein lineplot*
phossite_on_prot_regulation	The heatmap of every phosphosite belonging to one protein. *Heatmap*

D.8 Phosphoproteomics

Part VIII — Phosphoproteomics

Function	What it produces
simple_phos_site_plot	Phosphoproteins, phosphopeptides and phosphosites as three counts, above a localization cutoff. *Phosphosite plot*

Part VIII — Phosphoproteomics

Function	What it produces
phossite_coverage_plot	Sites quantified per sample, split into Class I and Not Class I. *Phosphosite coverage*
phossite_coverage_plot_summary	The same counts aggregated to one pair of bars per condition. *Phosphosite coverage*
phossite_dist	How many sites each phosphoprotein contributed, above a cutoff. *Phosphosite distribution*
sty_plot	Modified serine, threonine and tyrosine counts, with their percentages. *S/T/Y*
convert_to_centered	Rewrites a modified sequence as the seven residues either side of the modified one, padded where the peptide is shorter. *Kinase Library*
kinase_library_data_f	A volcano result reshaped into the input the Kinase Library's enrichment expects. The older name ksea_data_f remains as an alias. *Kinase Library*
kinase_library_data_f_control	The same, from a control-corrected volcano; alias ksea_data_f_control. *Kinase Library*
kinase_library_volcano	The kinase volcano built from the two returned result files; alias ksea_volcano. *Kinase Library*
kinase_tree_data_pval	Kinase against $-\log_{10}$ adjusted p-value, for a kinase tree. *Kinase Library*
kinase_tree_data_fc	Kinase against $\log_2$ frequency factor, for the same tree. *Kinase Library*
phosprot_regulation_data	One row per protein: how many of its sites went up, how many down, and by how much. *Phosprot regulation*
phosprot_regulation	The scatter of that table. *Phosprot regulation*

D.9 Comparison

Part IX — Comparison

Function	What it produces
correlation_xy_intensity_single	Two samples against each other, one point per feature quantified in both. *Pearson correlation*
correlation_xy_intensity	Two conditions against each other, using each condition's mean. *Pearson correlation*
overlap_plot_single	The Venn diagram of the features quantified in two or three samples. *Venn diagram*
overlap_plot_condition	The same overlap taken across two or three conditions. *Venn diagram*

D.10 Shared helpers

These are called by the functions above rather than by a page, and are public because a notebook doing its own preparation needs them.

Used everywhere

Function	What it produces
info	Prints what the package is.
make_columns_unique	Renames duplicated column names so that a frame can be indexed by name at all.
sanitize_dataframe	Makes the columns unique and casts every non-numeric column to text, which is what the table renderers assume.
number_to_str	Renders a number the same way in both packages: no trailing decimal point on a whole number, and no scientific notation below 10^{15}.
bool_to_str	TRUE, FALSE or empty, so that the two packages write a flag identically.
format_value	Renders any single value for a table cell, through the two above.
extract_number	The first run of digits in a string, or −1 where there is none.
row_zscore	Row-wise z-scores with the sample standard deviation; a row with no spread is centred and left at zero. The one definition every heatmap uses.
load_example_data	Reads the bundled HEK293 example by name: protein, phospho, phosphoprotein, protein_metadata or phospho_metadata.
example_data_path	The path of that bundled example file, for reading it another way.

D.11 Names from the previous manuals that are not functions here

CHANGED IN version 1.0.0

sum_mean_impute and sum_mean are described as collapse methods in the older phosphoprotein manual. Neither is a function and neither is a mode: transform_phosprot accepts exactly robust_site_median, mean, sum_ignore_na and sum_propagate_na, in both packages, and refuses any other name with mode must be 'robust_site_median', 'sum_propagate_na', 'sum_ignore_na', or 'mean'.

To fill missing sites before summing — which is what sum_mean_impute was described as doing — impute at phosphosite level first and then collapse with the imputed sites as the source. The filling is then the imputation you chose and can state, rather than an unconditional condition mean nobody selected.

Where a viewer configuration still names one of them in its collapse_mode field, the value is not recognised, the loader warns, and the phosphoprotein values are described as intensities. The effect is on the axis label, not on the numbers: the viewer never

collapses anything, and `collapse_mode` there is a record of how the file it was handed had already been produced.
Record the actual aggregation used; a legacy name is not evidence that that method was run.

E
References

The published methods the application implements, and where each is used.

AT A GLANCE

What	Every outside method the application, the report and this manual cite, with the publication it comes from.
Where	The report's methods sentences cite the same entries.
Identifiers	References are given by their DOI where they have one; look a DOI up at any DOI resolver.

E.1 Methods and their sources

- **Quantile normalization** — Normalization across samples. Bolstad et al. (2003), doi:10.1093/bioinformatics/19.2.185.
- **Variance-stabilizing normalization** — Normalization delegated to Bioconductor vsn. Huber et al. (2002), doi:10.1093/bioinformatics/18.suppl_1.S96.
- **Low-quantile normal imputation (default)** — Missing values drawn around a low quantile of the observed values, in the spirit of MinProb. Lazar et al. (2016), doi:10.1021/acs.jproteome.5b00981.
- **Downshifted-normal imputation** — Missing-value imputation, numerical method (Perseus defaults). Tyanova et al. (2016), doi:10.1038/nmeth.3901.
- **Student's t-test** — Pooled-variance and paired volcano tests. Student (1908), doi:10.1093/biomet/6.1.1.
- **Welch's t-test** — Unequal-variance volcano test. Welch (1947), doi:10.1093/biomet/34.1-2.28.
- **Benjamini-Hochberg false discovery rate** — Multiple-testing correction. Benjamini & Hochberg (1995), doi:10.1111/j.2517-6161.1995.tb02031.x.
- **Tukey-Kramer post-hoc test** — Pairwise comparisons after ANOVA. Tukey (1949); Kramer (1956), Biometrics 5:99-114; Biometrics 12:307-310.
- **Hypergeometric over-representation test** — GO over-representation analysis. Rivals

et al. (2007), doi:10.1093/bioinformatics/btl633.

- **Gene Ontology** — Term annotations for over-representation analysis. Ashburner et al. (2000); Gene Ontology Consortium (2023), doi:10.1038/75556; doi:10.1093/genetics/iyad031.
- **Class I phosphosite localization (probability ≥ 0.75)** — Phosphosite coverage classes. Olsen et al. (2006), doi:10.1016/j.cell.2006.09.026.
- **Kinase-substrate specificity atlases** — Kinase Library input preparation and result visualization. Johnson et al. (2023); Yaron-Barir et al. (2024), doi:10.1038/s41586-022-05575-3; doi:10.1038/s41586-024-07407-y.
- **UMAP** — Sample embedding (optional). McInnes et al. (2018), arXiv:1802.03426.
- **t-SNE** — Sample embedding (optional). van der Maaten & Hinton (2008), J. Mach. Learn. Res. 9:2579-2605.

F
Version and reproducibility

What to record with a result so that it can be understood and reproduced later.

AT A GLANCE

Version	This manual describes CoPYlot 1.0.0.
Before the first release	The release changelog is a placeholder, not an authoritative history of every development build.
Keep together	Original input files, metadata, settings, output tables, and the software version used to produce them.
Random operations	Keep the realised imputed matrix, not just a seed.

F.1 A version number is not the whole analysis

Two figures labelled 1.0.0 can differ because their inputs, conditions, normalization, filtering, data sources or plotting options differ. During development, two builds can also share that version label. For a result that must be reproduced, record the build or commit identifier when available, as well as the version shown by the application.

There is no reliable rule that every earlier development build used an opposite fold-change sign or that a particular update left every p-value unchanged. Establish those facts from the actual settings and result tables, not from the age of an image.

F.2 A small reproducibility checklist

1. Keep the original measurement file and metadata. A screenshot cannot recover the values used in the calculation.
2. Record the data level, declared scale, normalization, filter, imputation and selected data source. For a comparison, also keep condition order, test, pairing and thresholds.
3. Save a session and export the numerical tables. Check what a portable save omits before moving it to another computer.

4. Where available, keep the notebook ZIP with the figure or report. It records the recipe and resources; plots using imputation retain the actual imputed values needed for that replay.
5. Keep the environment information. A seed alone does not promise identical random values across R, Python and library versions.
6. When reopening an uncertain old result, reproduce the calculation and compare its numbers. Do not repair an unexplained discrepancy by changing only the caption.

F.3 Check the meaning before comparing figures

In a two-condition volcano, a positive log2 fold change means higher in condition 1 than condition 2. In a row-standardized heatmap, red and blue show a feature's values relative to its own row, not its abundance relative to another feature. For a centred phosphoprotein summary, the values are relative site patterns, not measurements of total protein abundance. These distinctions remain important even when the figures look alike.

> **WORTH KNOWING**
>
> A reproducible result is one whose input, calculation and interpretation can be recovered. Keeping the table and recipe alongside the attractive figure is usually more useful than keeping the figure alone.